

# Package ‘asremlPlus’

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**Version** 4.4.62

**Date** 2026-05-19

**Title** Augments 'ASReml-R' in Fitting Mixed Models and Packages  
Generally in Exploring Prediction Differences

**Depends** R (>= 3.5.0)

**Imports** dae, devtools, doParallel, dplyr, foreach, ggplot2, graphics,  
grDevices, methods, nloptr, parallel, qqplotr, RColorBrewer,  
reshape2, rlang, stats, sticky, stringr, tryCatchLog, utils

**Suggests** emmeans (>= 1.8.8), lattice, lmerTest, pbkrtest, R.rsp,  
testthat, tictoc

**Enhances** asreml

**VignetteBuilder** R.rsp

**SystemRequirements** asreml

**LazyData** true

**Description** Assists in automating the selection of terms to include in mixed models when 'asreml' is used to fit the models. Procedures are available for choosing models that conform to the hierarchy or marginality principle, for fitting and choosing between two-dimensional spatial models using correlation, natural cubic smoothing spline and P-spline models. A history of the fitting of a sequence of models is kept in a data frame. Also used to compute functions and contrasts of, to investigate differences between and to plot predictions obtained using any model fitting function. The content falls into the following natural groupings: (i) Data, (ii) Model modification functions, (iii) Model selection and description functions, (iv) Model diagnostics and simulation functions, (v) Prediction production and presentation functions, (vi) Response transformation functions, (vii) Object manipulation functions, and (viii) Miscellaneous functions (for further details see 'asremlPlus-package' in help). The 'asreml' package provides a computationally efficient algorithm for fitting a wide range of linear mixed models using Residual Maximum Likelihood. It is a commercial package and a license for it can be purchased from 'VSNi' <<https://vsni.co.uk/>> as 'asreml-R', who will supply a zip file for local installation/updating (see <<https://asreml.kb.vsn.co.uk/>>). It is not needed for functions that are methods for 'alldiffs' and 'data.frame' objects. The package 'asremlPlus' can also be installed from <<http://chris.brien.name/rpackages/>>.

**License** MIT + file LICENSE

**URL** <http://chris.brien.name>

**BugReports** <https://github.com/briencj/asremlPlus/issues>

**NeedsCompilation** no

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|                    |                                                                                                               |
|--------------------|---------------------------------------------------------------------------------------------------------------|
| asremlPlus-package | <i>Augments 'ASReml-R' in Fitting Mixed Models and Packages Generally in Exploring Prediction Differences</i> |
|--------------------|---------------------------------------------------------------------------------------------------------------|

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## Description

Assists in automating the selection of terms to include in mixed models when 'asreml' is used to fit the models. Procedures are available for choosing models that conform to the hierarchy or marginality principle, for fitting and choosing between two-dimensional spatial models using correlation, natural cubic smoothing spline and P-spline models. A history of the fitting of a sequence of models is kept in a data frame. Also used to compute functions and contrasts of, to investigate differences between and to plot predictions obtained using any model fitting function. The content falls into the following natural groupings: (i) Data, (ii) Model modification functions, (iii) Model selection and description functions, (iv) Model diagnostics and simulation functions, (v) Prediction production and presentation functions, (vi) Response transformation functions, (vii) Object manipulation functions, and (viii) Miscellaneous functions (for further details see 'asremlPlus-package' in help). The 'asreml' package provides a computationally efficient algorithm for fitting a wide range of linear mixed models using Residual Maximum Likelihood. It is a commercial package and a license for it can be purchased from 'VSNi' <<https://vsni.co.uk/>> as 'asreml-R', who will supply a zip file for local installation/updating (see <<https://asreml.kb.vsn.co.uk/>>). It is not needed for functions that are methods for 'alldiffs' and 'data.frame' objects. The package 'asremlPlus' can also be installed from <<http://chris.brien.name/rpackages/>>.

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Note that many of the function below are S3 methods so that the suffix can be omitted. Of course, whether or not the suffix is omitted, the object supplied to the first argument must be of the class specified by the suffix. For example `getFormulae.asreml` is a `getFormulae` method for an `asreml` object and so `.asreml` can be omitted and the object supplied to the first argument must be of class `asreml`.

### (i) Data

[Oats.dat](#)

[Wheat.dat](#)

[WaterRunoff.dat](#)

### (ii) Model modification

[addSpatialModel.asrtests](#)

[addSpatialModelOnIC.asrtests](#)

[changeTerms.asrtests](#)

`iterate.asrtests`

`newfit.asreml`

`reparamSigDevn.asrtests`

`rmboundary.asrtests`

`setvarianceterms.call`

### (iii) Model selection and description

`addto.test.summary`

`changeModelOnIC.asrtests`

`chooseModel.asrtests`

`chooseModel.data.frame`

`chooseSpatialModelOnIC.asrtests`

`getTestPvalue.asrtests`

`infoCriteria.asreml`

`infoCriteria.list`

`R2adj.asreml`

`recalcWaldTab.asrtests`

`REMLRT.asreml`

`bootREMLRT.asreml`

`testranfix.asrtests`

`testresidual.asrtests`

`testswapan.asrtests`

## (iv) Model diagnostics and simulation

`plotVariofaces``variofaces.asreml``estimateV.asreml``simulate.asreml`

## (v) Prediction production and presentation

`addBacktransforms.alldiffs``allDifferences.data.frame``exploreLSDs``findLSDminerrors``linTransform.alldiffs``pairdiffsTransform.alldiffs``pickLSDstatistics``plotLSDerrors.data.frame``plotLSDerrors.alldiffs``plotLSDs.data.frame``plotLSDs.alldiffs``plotPredictions.data.frame``plotPvalues.alldiffs``plotPvalues.data.frame``predictPlus.asreml`

Forms the predictions for a term, their pairwise differences and associated statistics. A factor having parallel values may o

`ratioTransform.alldiffs`

`recalcLSD.alldiffs`

`redoErrorIntervals.alldiffs`

`renewClassify.alldiffs`

`sort.alldiffs`

`subset.alldiffs`

`sort.predictions.frame`

(vi) Response transformation

`angular`

`angular.mod`

`powerTransform`

(vii) Object manipulation

`as.alldiffs`

`asrtests`

`as.asrtests`

`as.predictions.frame`

`convAsremlobj.asrem1`

`convEffectNames2DataFrame.asrem1`

`facCombine.alldiffs`

`facRecast.alldiffs`

`facRename.alldiffs`

`getFormulae.asrem1`

`is.alldiffs`

```

is.asrtests

is.predictions.frame

makeTPPSplineMats.data.frame

print.alldiffs
print.asrtests
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print.test.summary
print.wald.tab
printFormulae.asreml
sort.alldiffs

subset.alldiffs

subset.list

validAlldiffs
validAsrtests
validPredictionsFrame

(viii) Miscellaneous

getASRemlVersionLoaded

isCompoundSymmetric

loadASRemlVersion
num.recode

permute.square
permute.to.zero.lowertri

```

The functions whose names end in 'alldiffs' utilize an [alldiffs.object](#) that stores: (i) a [predictions.frame](#), being a data frame containing predicted values, variables indexing them and their standard errors and estimability status; the lower and upper limits of error intervals will be included when these are requested, (ii) optionally, square matrices containing all pairwise differences, the standard errors and p-values of the differences, and a data.frame containing LSD values and their summary statistics, (iii) optionally, the variance matrix of the predictions, and (iv) if the response was transformed for analysis, a data frame with backtransforms of the predicted values.

The functions whose names end in 'asrtests', which are most of the model functions, utilize an [asrtests.object](#) that stores: (i) the currently fitted model in `asreml.obj`, (ii) the table of test statistics for the fixed effects in `wald.tab`, and (iii) a data frame that contains a history of the changes made to the model in `test.summary`.



**Author(s)**

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**References**

Butler, D. G., Cullis, B. R., Gilmour, A. R., Gogel, B. J. and Thompson, R. (2023). *ASReml-R Reference Manual Version 4.2*. VSN International Ltd, <https://asreml.kb.vsnl.co.uk/>

**See Also**

asreml

**Examples**

```
## Not run:
## Analyse wheat dat using asreml and asremlPlus (see the WheatSpatial Vignette for details)
## Set up for analysis
library(dae)
library(asreml)
library(asremlPlus)
## use ?Wheat.dat for data set details
data(Wheat.dat)

# Add row and column covariates for the spatial modelling
tmp.dat <- within(Wheat.dat,
  {
    cColumn <- dae::as.numfac(Column)
    cColumn <- cColumn - mean(unique(cColumn))
    cRow <- dae::as.numfac(Row)
    cRow <- cRow - mean(unique(cRow))
  })

# Fit an initial model - Row and column random
current.asr <- do.call(asreml,
  list(yield ~ Rep + WithinColPairs + Variety,
    random = ~ Row + Column,
    residual = ~ Row:Column,
    data = tmp.dat))

# Initialize a model sequence by loading the current fit into an asrtests object
current.asrt <- as.asrtests(current.asr, NULL, NULL, ICLikelihood = "full",
  label = "Initial model")

# Check for and remove any boundary terms and print a summary of the fit in the asrtests object
current.asrt <- rmboundary(current.asrt)
print(current.asrt)

## Compare a series of information criteria to select a linear mixed model for the data

# Check the need for the term for within Column pairs (a post hoc factor)
current.asrt <- changeModelOnIC(current.asrt, dropFixed = "WithinColPairs",
  label = "Try dropping withinColPairs", ICLikelihood = "full")
print(current.asrt)
```

```

# Fit an ar1 model for local spatial variation
spatial.ar1.asrt <- addSpatialModelOnIC(current.asrt, spatial.model = "corr",
                                     row.covar = "cRow", col.covar = "cColumn",
                                     row.factor = "Row", col.factor = "Column",
                                     ICLikelihood = "full")

spatial.ar1.asrt <- rmboundary(spatial.ar1.asrt)
infoCriteria(list(nonspatial = current.asrt$asreml.obj,
                 ar1 = spatial.ar1.asrt$asreml.obj))
print(spatial.ar1.asrt)

# Choose a model for local spatial variation from several potential models
suppressWarnings(
  spatial.asrts <- chooseSpatialModelOnIC(current.asrt,
                                     row.covar = "cRow", col.covar = "cColumn",
                                     row.factor = "Row", col.factor = "Column",
                                     dropRandom = "Row + Column",
                                     rotateX = TRUE, ngridangles = NULL,
                                     asreml.option = "grp", return.asrts = "all"))

# Output the results
print(spatial.asrts$spatial.IC)
print(R2adj(spatial.asrts$asrts$TPNCSS$asreml.obj, include.which.random = ~ .))
print(spatial.asrts$best.spatial.mod)
print(spatial.asrts$asrts$TPNCSS)
printFormulae(spatial.asrts$asrts$TPNCSS$asreml.obj)

## Diagnosing checking using residual plots and variofaces

# Get current fitted asreml object and update to include standardized residuals
current.asr <- spatial.asrts$asrts$TPNCSS$asreml.obj
current.asr <- update(current.asr, aom=TRUE)
Wheat.dat$res <- residuals(current.asr, type = "stdCond")
Wheat.dat$fit <- fitted(current.asr)

# Do residuals-versus-fitted values plot
with(Wheat.dat, plot(fit, res))

# Plot variofaces
variofaces(current.asr, V=NULL, units="addtores",
           maxiter=50, update = FALSE,
           ncores = parallel::detectCores())

# Plot normal quantile plot
ggplot(data = Wheat.dat, mapping = aes(sample = res)) +
  stat_qq_band(bandType = "ts") + stat_qq_line() + stat_qq_point() +
  labs(x = "Theoretical Quantiles", y = "Sample Quantiles",
       title = "Normal probability plot") +
  theme(plot.title = element_text(size = 12, face = "bold")) + theme_bw()

## Prediction production and presentation

# Get Variety predictions and all pairwise prediction differences and p-values
Var.diffs <- predictPlus(classify = "Variety",
                       asreml.obj=current.asr,
                       error.intervals="halfLeast",
                       wald.tab=current.asrt$wald.tab,
                       sortFactor = "Variety",
                       tables = "predictions")

```

```
# Plot the Variety predictions, with halfLSD intervals, and the p-values
plotPredictions(Var.diffs$predictions,
                 classify = "Variety", y = "predicted.value",
                 error.intervals = "half")
plotPvalues(Var.diffs)

## End(Not run)
```

---

```
addBacktransforms.alldiffs
```

*Adds or recalculates the backtransforms component of an `alldiffs.object`.*

---

## Description

Given an `alldiffs.object`, adds or recalculate its backtransforms component. The values of `transform.power`, `offset`, `scale` and `transform.function` from the backtransforms component will be used, unless this component is NULL when the values supplied in the call will be used.

## Usage

```
## S3 method for class 'alldiffs'
addBacktransforms(alldiffs.obj,
                  transform.power = 1, offset = 0, scale = 1,
                  transform.function = "identity", ...)
```

## Arguments

`alldiffs.obj` An `alldiffs.object`.

`transform.power`

A **numeric** specifying the power of a transformation, if one has been applied to the response variable. Unless it is equal to 1, the default, back-transforms of the predictions will be obtained and presented in tables or graphs as appropriate. The back-transformation raises the predictions to the power equal to the reciprocal of `transform.power`, unless it equals 0 in which case the exponential of the predictions is taken. Confidence limits and Least Significant limits will be similarly backtransformed. Approximate standard errors of the backtransformed predictions will also be calculated.

`offset`

A **numeric** that has been added to each value of the response after any scaling and before applying any power transformation.

`scale`

A **numeric** by which each value of the response has been multiplied before adding any offset and applying any power transformation.

`transform.function`

A **character** giving the name of a function that specifies the scale on which the predicted values are defined. This may be the result of a transformation of the data using the function or the use of the function as a link function in the fitting of a generalized linear (mixed) model (GL(M)M). The possible transform.functions are identity, log, inverse, sqrt, logit, probit, and cloglog. The predicted.values and error.intervals, if not `StandardError` intervals, will be back-transformed using the inverse function of the `transform.function`.

The standard.error column will be set to NA, unless (i) asreml returns columns named transformed.value and approx.se, as well as those called predicted.values and standard.error (such as when a GLM is fitted) and (ii) the values in transformed.value are equal to those obtained by backtransforming the predicted.values using the inverse function of the transform.function. Then, the approx.se values will be saved in the standard.error column of the backtransforms component of the returned alldiffs.obj. Also, the transformed.value and approx.se columns are removed from both the predictions and backtransforms components of the alldiffs.obj. Note that the values that end up in the standard errors column are approximate for the backtransformed values and are not used in calculating error.intervals.

... Provision for passing arguments to functions called internally - not used at present.

## Details

The backtransformed prediction ( $y$ ) is computed from the prediction ( $z$ ) from the transformed data as follows: (i) for transform.power equal to 0  $y = (\exp(z) - \text{offset})/\text{scale}$ , and (ii) for transform.power  $p$  not equal to 0 or 1,  $y = (z^{1/p} - \text{offset})/\text{scale}$ . The approximate standard error of  $y$  is calculated using the delta method as follows: (i)  $se(y) = \exp(z)/\text{scale} * se(z)$ , and (ii)  $se(y) = z^{1/p}/(p * z * \text{scale}) * se(z)$ .

## Value

An `alldiffs.object` with components predictions, vcov, differences, p.differences, sed, LSD and backtransforms.

The backtransforms component will have the attributes (i) LSDtype, LSDby and LSDstatistic added from the predictions component and (ii) transform.power, offset, scale, and link.

## Author(s)

Chris Brien

## See Also

`asremlPlus-package`, `as.alldiffs`, `sort.alldiffs`, `subset.alldiffs`, `print.alldiffs`, `renewClassify.alldiffs`, `redoErrorIntervals.alldiffs`, `plotPredictions.data.frame`, `predictPlus.asreml`, `predictPresent.asreml`

## Examples

```
##Subset WaterRunoff data to reduce time to execute
data(WaterRunoff.dat)
tmp <- subset(WaterRunoff.dat, Date == "05-18" & Benches != "3")

##Use asreml to get predictions and associated statistics

## Not run:
asreml.options(keep.order = TRUE) #required for asreml-R4 only
current.asr <- asreml(fixed = log.Turbidity ~ Benches + (Sources * (Type + Species)),
                     random = ~ Benches:MainPlots,
                     keep.order=TRUE, data= tmp)
current.asrt <- as.asrttests(current.asr, NULL, NULL)
TS.diffs <- predictPlus(classify = "Sources:Type",
```

```

asreml.obj = current.asr,
wald.tab = current.asrt$wald.tab,
present = c("Sources", "Type", "Species"))

## End(Not run)

##Use lmeTest and emmeans to get predictions and associated statistics

if (requireNamespace("lmerTest", quietly = TRUE) &&
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(log.Turbidity ~ Benches + (Sources * (Type + Species)) +
                          (1|Benches:MainPlots),
                          data=tmp)
  TS.emm <- emmeans::emmeans(m1.lmer, specs = ~ Sources:Species)
  TS.preds <- summary(TS.emm)
  den.df <- min(TS.preds$df, na.rm = TRUE)
  ## Modify TS.preds to be compatible with a predictions.frame
  TS.preds <- as.predictions.frame(TS.preds, predictions = "emmean",
                                  se = "SE", interval.type = "CI",
                                  interval.names = c("lower.CL", "upper.CL"))

  ## Form an all.diffs object and check its validity
  TS.vcov <- vcov(TS.emm)
  TS.diffs <- allDifferences(predictions = TS.preds, classify = "Sources:Species",
                           vcov = TS.vcov, tdf = den.df)
  validAlldiffs(TS.diffs)
}

## Recalculate the back-transforms of the predictions obtained using asreml or lmerTest
if (exists("TS.diffs"))
{
  TS.diffs <- addBacktransforms.alldiffs(TS.diffs, transform.power = 0)
}

```

---

addSpatialModel.asrtests

*Adds, to a supplied model, a spatial model that accounts for local spatial variation.*

---

## Description

Adds either a correlation, two-dimensional tensor-product natural cubic smoothing spline (TP-NCSS), or a two-dimensional tensor-product penalized P-spline model (TPPS) to account for the local spatial variation exhibited by a response variable measured on a potentially irregular grid of rows and columns of the units. The data may be arranged in sections, for each of which there is a grid and for which the model is to be fitted separately. Also, the rows and columns of a grid are not necessarily one observational unit wide. For TPPS models for which the order of differencing the penalty matrix is two, the an optimal rotation of the null-space eigenvectors of the penalty matrix can be investigated.

No hypothesis testing or comparison of information criteria is made. To use information criteria to decide whether to change the model use [chooseSpatialModelOnIC.asrtests](#).

The model fit supplied in the `asrtests.obj` should not include terms that will be included in the local spatial model. All spatial model terms are fitted as fixed or random. Consequently, the residual model does not have to be iid.

One or more rows is added for each section to the `test.summary.data.frame`. Convergence and the occurrence of fixed correlations in fitting the model is checked and a note included in the action if there was not. All components of the `asrtests.object` are updated for the new model.

## Usage

```
## S3 method for class 'asrtests'
addSpatialModel(asrtests.obj, spatial.model = "TPPS",
  sections = NULL,
  row.covar = "cRow", col.covar = "cCol",
  row.factor = "Row", col.factor = "Col",
  corr.funcs = c("ar1", "ar1"), corr.orders = c(0, 0),
  row.corrFitfirst = TRUE,
  allow.corrJointFit = TRUE, nugget.variance = TRUE,
  dropFixed = NULL, dropRandom = NULL,
  nsegs = NULL, nestorder = c(1,1),
  degree = c(3,3), difforder = c(2,2),
  usRandLinCoeffs = TRUE,
  rotateX = FALSE, ngridangles = NULL,
  which.rotacriterion = "AIC", nrotacores = 1,
  asreml.option = "grp", tpps4mbf.obj = NULL,
  allow.unconverged = TRUE, allow.fixedcorrelation = TRUE,
  checkboundaryonly = FALSE, update = TRUE, trace = FALSE,
  maxit = 30, IClikelihood = "full", which.IC = "AIC", ...)
```

## Arguments

- |                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <code>asrtests.obj</code>  | An <code>asrtests.object</code> containing the components (i) <code>asreml.obj</code> , (ii) <code>wald.tab</code> , and (iii) <code>test.summary</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>spatial.model</code> | A single character string nominating the type of spatial model to fit. Possible values are <code>corr</code> , <code>TPNCSS</code> and <code>TPPS</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>sections</code>      | A single character string that specifies the name of the column in the <code>data.frame</code> that contains the <code>factor</code> that identifies different sections of the data to which separate spatial models are to be fitted. Note that, for other terms that involve sections in the random formula, there should be separate terms for each level of sections. For example, in a blocked experiment involving multiple sites, there should be the sum of separate terms for the Blocks at each Site i.e. a formula that contains terms like <code>at(Site, i):Block</code> for each site and these are separated by '+'. Otherwise, the combined term (e.g. <code>Site:Block</code> ) will impact on the fitting of the local spatial models for the different Sites. Similarly, a separate residual variance for each of the sections should be fitted, unless there is a need to fit a different variance structure to the residual, e.g. heterogeneous residual variances depending on treatments. Separate residual variances for sections can be achieved using the <code>asreml</code> functions <code>dsum</code> or <code>idh</code> . Because, unlike random terms, terms for residual variances are not removed from the model, compound residual terms can be used to include them in the model, e.g. terms with <code>idh</code> or <code>dsum</code> with multiple levels in the <code>list</code> or leaving levels out altogether. In addition to allowing the independent fitting of models to the sections, sepa- |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | rate residual variance terms allows a nugget variance to be fitted in a correlation model for each of the sections.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| row.covar          | A single character string nominating a <a href="#">numeric</a> that contains the values of a centred covariate indexing the rows of a grid. The <a href="#">numeric</a> must be a column in the <a href="#">data.frame</a> stored in the <code>asrem1.obj</code> that is a component of the supplied <code>asrtests.obj</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| col.covar          | A single character string nominating a <a href="#">numeric</a> that contains the values of a centred covariate indexing the columns of a grid. The <a href="#">numeric</a> must be a column in the <a href="#">data.frame</a> stored in the <code>asrem1.obj</code> that is a component of the supplied <code>asrtests.obj</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| row.factor         | A single character string nominating a <a href="#">factor</a> that indexes the rows of a grid that are to be one dimension of a spatial correlation model. The <a href="#">factor</a> must be a column in the <a href="#">data.frame</a> stored in the <code>asrem1.obj</code> that is a component of the supplied <code>asrtests.obj</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| col.factor         | A single character string nominating a <a href="#">factor</a> that indexes the columns of a grid that are to be one dimension of a spatial correlation model. The <a href="#">factor</a> must be a column in the <a href="#">data.frame</a> stored in the <code>asrem1.obj</code> that is a component of the supplied <code>asrtests.obj</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| corr.funcs         | A single character string of length two that specifies the <code>asrem1</code> one-dimensional correlation or variance model function for the row and column dimensions of a two-dimensional separable spatial correlation model to be fitted when <code>spatial.model</code> is <code>corr</code> ; the two-dimensional model is fitted as a random term. If a correlation or variance model is not to be investigated for one of the dimensions, specify "" for that dimension. If the correlation model is <code>corb</code> , the values of <code>corr.orders</code> are used for its order argument (b).                                                                                                                                                                                                                                                                                                                                               |
| corr.orders        | A numeric of length two that specifies the order argument (b) values for the row and column dimensions of a two-dimensional separable spatial correlation model when <code>spatial.model</code> is <code>corr</code> and the <code>corr.funcs</code> for a dimension is <code>corb</code> , the <code>asrem1</code> banded correlation model. If one of the dimensions does not involve an order argument, set the value of <code>corr.orders</code> for that dimension to zero. For a dimension for which the <code>corr.funcs</code> is <code>corb</code> and <code>corr.orders</code> is zero, a model with a single band, the correlation between immediate neighbours, will be fitted and then further bands, up to a maximum of 10 bands, will be added until the addition of an extra band does not reduce the information criterion nominated using <code>which.IC</code> . Note that the two-dimensional spatial model is fitted as a random term. |
| row.corrFitfirst   | A <a href="#">logical</a> . If TRUE then, in fitting the model for <code>spatial.model</code> set to <code>corr</code> , the row correlation or variance function is fitted first, followed by the addition of the column correlation or variance function. If FALSE, the order of fitting is reversed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| allow.corrJointFit | A <a href="#">logical</a> which, if TRUE, will allow the simultaneous fitting of correlation functions for the two dimensions of the grid when separate fits have failed to fit any correlation functions. This argument is available for when a joint fit hangs the system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| nugget.variance    | A <a href="#">logical</a> which, if TRUE, will result in an attempt to fit a nugget or unit-specific variance. Otherwise, a nugget or unit-specific variance will not be fitted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| dropFixed          | A single character string or a character vector of strings with an element for each level of sections in the same order as the sections levels. Each                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

string, which if it is not NA and after the addition of " . ~ . -" and conversion to a formula that is then expanded, specifies the sum of a set of terms to be dropped from the fixed formula in fitting splines (TPPS and TPNCSS). The result is that the fitted model supplied in the `asrtests.obj`, that includes these terms, will be compared with the fitted model that has had them removed and a spatial model added.

An element that is NA indicates that no term pertaining to the corresponding sections level is to be removed. If `sections` is not NULL and a single character string has been supplied, the terms specified by the string are taken to be terms that are independent of the sections and will be removed when adding the spatial model for the first sections.

The terms must match those in the `wald.tab` component of the `asrtests.obj`. The fixed terms will be reordered so that single-variable terms come first, followed by two-variable terms and so on. Note also that multiple terms specified using a single `asreml::at` function can only be dropped as a whole. If the term was specified using an `asreml::at` function with a single level, then it can be removed and either the level itself or its `numeric` position in the levels returned by the `levels` function can be specified.

dropRandom

A single character string or a character vector of strings with an element for each level of sections in the same order as the sections levels. Each string, which if it is not NA and after the addition of " . ~ . -" and conversion to a formula that is then expanded, specifies the sum of a set of terms to be dropped from the random formula in fitting splines (TPPS and TPNCSS). The result is that the fitted model supplied in the `asrtests.obj`, that includes these terms, will be compared with the fitted model that has had them removed and a spatial model added.

An element that is NA indicates that no term pertaining to the corresponding sections level is to be removed. If `sections` is not NULL and a single character string has been supplied, the terms specified by the string are taken to be terms that are independent of the sections and will be removed when adding the spatial model for the first sections.

The terms must match those in the `vparameters` component of the `asreml.obj` component in the `asrtests.obj`. Note also that multiple terms specified using a single `asreml::at` function can only be dropped as a whole. If the term was specified using an `asreml::at` function with a single level, then it can be removed and either the level itself or its `numeric` position in the levels returned by the `levels` function can be specified.

nsegs

A pair of `numeric` values giving the number of segments into which the column and row ranges are to be split, respectively, for fitting a P-spline model (TPPS) (each value specifies the number of internal knots + 1). If not specified, then (number of unique values - 1) is used in each dimension; for a grid layout with equal spacing, this gives a knot at each data value. If `sections` is not NULL and the grid differs between the sections, then `nsegs` will differ between the sections.

nestorder

A `numeric` of length 2. The order of nesting for column and row dimensions, respectively, in fitting a P-spline model (TPPS). A value of 1 specifies no nesting, a value of 2 generates a spline with half the number of segments in that dimension, etc. The number of segments in each direction must be a multiple of the order of nesting.

degree

A `numeric` of length 2. The degree of polynomial spline to be used for column and row dimensions respectively, in fitting a P-spline (TPPS).



|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| difforder           | A <a href="#">numeric</a> of length 2. The order of differencing for column and row dimensions, respectively, in fitting a P-spline (TPPS).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| usRandLinCoeffs     | A <a href="#">logical</a> which, if TRUE and a P-spline (TPPS) is being fitted, will attempt to fit an unstructured variance model to the constant and linear terms in the interactions for constant and linear terms in one grid dimension interacting with smooth terms in the second grid dimension. The unstructured variance model can only be fitted if both the constant and linear interaction terms have been retained in the fitted model. This argument can be used to omit the attempt to fit an unstructured variance model when the attempt results in a system error.                                                                                                                                                                                                                            |
| rotateX             | A <a href="#">logical</a> indicating whether to rotate the eigenvectors of the penalty matrix, as described by Piepho, Boer and Williams (2022), when fitting a P-spline (TPPS). Setting rotateX to TRUE results in a search for an optimized rotation under a model that omits the random spline interaction terms. If ngridangles is set to NULL, the optimal rotation is found using an optimizer (nloptr::bobyqa). Otherwise, the optimal rotation is found by exploring the fit over a two-dimensional grid of rotation angle pairs. The optimization seeks to optimize the criterion nominated in which.rotacriterion. Rotation of the eigenvectors is only relevant for difforder values greater than 1 and has only been implemented for difforder equal to 2.                                          |
| ngridangles         | A <a href="#">numeric</a> of length 2. If NULL (the default), the optimal pair of angles for rotating the eigenvectors of the penalty matrix of a P-spline (TPPS) will be determined using a nonlinear optimizer (nloptr::bobyqa). Otherwise, its two values specify the numbers of angles between 0 and 90 degrees for each of the row and column dimensions to be used in determining the optimal pair of angles. Specifying factors of 90 will result in integer-valued angles. The number of grid points, and hence re-analyses will be the product of the values of (ngridangles + 1).                                                                                                                                                                                                                     |
| which.rotacriterion | A single character string nominating which of the criteria, out of the deviance, the likelihood, the AIC and the BIC, is to be used in determining the optimal rotation of the eigenvectors of the penalty matrix. The deviance uses the REML value computed by asreml; the other criteria use the full likelihood, evaluated using the REML estimates, that is computed by <a href="#">infoCriteria.asreml</a> .                                                                                                                                                                                                                                                                                                                                                                                               |
| nrotacores          | A <a href="#">numeric</a> specifying the number of cores to deploy for running the analyses required to search the two-dimensional grid of rotation angles when rotateX is TRUE. Parallel processing has been implemented for analyzing, for each column angle, the set of angles to be investigated for the row dimension. The default value of one means that parallel processing will not be used. The value chosen for nrotacores needs to be balanced against the other processes that are using parallel processing at the same time.                                                                                                                                                                                                                                                                     |
| asreml.option       | A single character string specifying whether the grp or mbf methods are to be used to supply externally formed covariate matrices to asreml when fitting a P-spline (TPPS). Compared to the mbf method, the grp method is somewhat faster, but creates large <a href="#">asrtests.objects</a> for which the time it takes to save them can exceed any gains in execution speed. The grp method adds columns to the <a href="#">data.frame</a> containing the data. On the other hand, the mbf method adds only the fixed covariates to data and stores the random covariates in the environment of the internal function that calls the spline-fitting function; there are three smaller <a href="#">data.frames</a> for each section that are not stored in the asreml.object resulting from the fitted model. |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>tps4mbf.obj</code>            | An object made with <code>makeTPPSplineMats.data.frame</code> that contains the spline basis information for fitting P-splines. The argument <code>tps4mbf.obj</code> only needs to be set when the <code>mbf</code> option of <code>asreml.option</code> is being used and it is desired to use <code>mbf</code> <code>data.frames</code> that have been created and stored prior to calling <code>addSpatialModel.asrtests</code> . If <code>tps4mbf.obj</code> is NULL, <code>makeTPPSplineMats.data.frame</code> will be called internally to produce the required <code>mbf</code> <code>data.frames</code> . |
| <code>allow.unconverged</code>      | A logical indicating whether to accept a new model even when it does not converge. If FALSE and the fit of the new model does not converge, the supplied <code>asrtests.obj</code> is returned. Also, if FALSE and the fit of the new model has converged, but that of the old model has not, the new model will be accepted.                                                                                                                                                                                                                                                                                      |
| <code>allow.fixedcorrelation</code> | A logical indicating whether to accept a new model even when it contains correlations in the model whose values have been designated as fixed, bound or singular. If FALSE and the new model contains correlations whose values have not been able to be estimated, the supplied <code>asrtests.obj</code> is returned. The fit in the <code>asreml.obj</code> component of the supplied <code>asrtests.obj</code> will also be tested and a warning issued if both fixed correlations are found in it and <code>allow.fixedcorrelation</code> is FALSE.                                                           |
| <code>checkboundaryonly</code>      | If TRUE then boundary and singular terms are not removed by <code>rmboundary.asrtests</code> ; a warning is issued instead. Note that, for correlation models, the fitting of each dimension and the test for a nugget term are performed with <code>checkboundaryonly</code> set to TRUE and its supplied setting only honoured using a call to <code>rmboundary.asrtests</code> immediately prior to returning the final result of the fitting.                                                                                                                                                                  |
| <code>update</code>                 | If TRUE, then <code>newfit.asreml</code> is called to fit the model to be tested, using the values of the variance parameters stored in the <code>asreml.object</code> , that is stored in <code>asrtests.obj</code> , as starting values. If FALSE, then <code>newfit.asreml</code> will not use the stored variance parameter values as starting values when fitting the new model, the only modifications being (i) to add the terms for the spatial models and (ii) those specified via . . .                                                                                                                  |
| <code>trace</code>                  | If TRUE then the stages in fitting a correlation model are displayed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>maxit</code>                  | A <code>numeric</code> specifying the maximum number of iterations that <code>asreml</code> should perform in fitting a model.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>IClikelihood</code>           | A character that controls both the occurrence and the type of likelihood for information criterion in the <code>test.summary</code> of the new <code>asrtests.object</code> . If none, none are included. Otherwise, if REML, then the AIC and BIC based on the Restricted Maximum Likelihood are included; if full, then the AIC and BIC based on the full likelihood, evaluated using REML estimates, are included. (See also <code>infoCriteria.asreml</code> .)                                                                                                                                                |
| <code>which.IC</code>               | A character specifying the information criterion to be used in selecting the best model. Possible values are AIC and BIC. The value of the criterion for supplied model must exceed that for changed model for the changed model to be returned. (For choosing the rotation angle of the eigenvectors of the penalty matrix, see <code>which.rotacriterion</code> .)                                                                                                                                                                                                                                               |
| <code>...</code>                    | Further arguments passed to <code>changeModelOnIC.asrtests</code> , <code>newfit.asreml</code> , <code>asreml</code> and <code>tpsmbb</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

## Details

The model to which the spatial models is to be added is supplied in the `asrtests.obj`. It should not include terms that will be included in the local spatial model. All spatial model terms are fitted as fixed or random. Consequently, the residual model does not have to be iid. The improvement in the fit resulting from the addition of a spatial model to the supplied model is evaluated. Note that the data must be in the order that corresponds to the residual argument with a variable to the right of another variable changes levels in the data frame faster than those of the other variable e.g. Row:Column implies that all levels for Column in consecutive rows of the data.frame with a single Row level.

For the corr spatial model, the default model is an autocorrelation model of order one (ar1) for each dimension. However, any of the single dimension correlation/variance models from `asreml` can be specified for each dimension, as can no correlation model for a dimension; the models for the two dimensions can differ. Using a forward selection procedure, a series of models are tried, without removing boundary or singular terms, beginning with the addition of row correlation and followed by the addition of column correlation or, if the `row.corrFitfirst` is set to FALSE, the reverse order. If the fitting of the first-fitted correlation did not result in a model change because the fitting did not converge or correlations were fixed, but the fit of the second correlation was successful, then adding the first correlation will be retried. If one of the metric correlation functions is specified (e.g. exp), then the `row.covar` or `col.covar` will be used in the spatial model. However, because the correlations are fitted separately for the two dimensions, the `row.factor` and `col.factor` are needed for all models and is used for a dimension that does not involve a correlation/variance function for the fit being performed. Also, the correlation models are fitted as random terms and so the correlation model will include a variance parameter for the grid even when ar1 is used to specify the correlation model, i.e. the model fitted is a variance model and there is no difference between ar1 and ar1v in fitting the model. The variance parameter for this term represents the spatial variance and the fit necessarily includes a nugget term, this being the residual variance. If any correlation is retained in the model, for a section if sections is not NULL, then the need for a nugget term is assessed by fixing the corresponding residual variance to one, unless there are multiple residual variances and these are not related to the sections. Once the fitting of the correlation model has been completed, the `rmboundary` function will be executed with the `checkboundaryonly` value supplied in the `addSpatialModel.asrtests` call. Finally, checking for bound and singular random terms associated with the correlation model and residual terms will be carried out when there are correlation terms in the model and `checkboundaryonly` has been set to FALSE; as many as possible will be removed from the fitted model, in some cases by fixing variance terms to one.

The tensor-product natural-cubic-smoothing-spline (TPNCSS) spatial model is as described by Verbyla et al. (2018), the tensor-product penalized-cubic-spline (TPPSC2) model with second-order differencing of the penalty is similar to that described by Rodriguez-Alvarez et al. (2018), and the tensor-product, first-difference-penalty, linear spline (TPPSL1) model is amongst those described by Piepho, Boer and Williams (2022). The fixed terms for the spline models are `row.covar + col.covar + row.covar:col.covar` and the random terms are `spl(row.covar) + spl(col.covar) + dev(row.covar) + dev(col.covar) + spl(row.covar):col.covar + row.covar:spl(col.covar) + spl(row.covar):spl(col.covar)`, except that `spl(row.covar) + spl(col.covar)` is replaced with `spl(row.covar):int(col.covar) + int(row.covar):spl(col.covar)` in the TPPSC2 model, where `int(.)` indicates an intercept or constant value specific to its argument. For TPPSL1 models, the terms `spl(row.covar):col.covar + row.covar:spl(col.covar)` are omitted. The supplied model should not include any of these terms. However, any fixed or random main-effect Row or Column term that has been included as an initial model for comparison with a spatial model can be removed prior to fitting the spatial model using `dropFixed` or `dropRandom`. For the P-spline models with second-order differencing, the model matrices used to fit the pairs of random terms (i) `spl(row.covar):int(col.covar)` and `spl(row.covar):col.covar` and (ii) `int(row.covar):spl(col.covar)` and `row.covar:spl(col.covar)` are transformed using the

spectral decomposition of their penalty matrices. An unstructured variance model is tried for each of these pairs. For TPPSC2, it is also possible to optimize the rotation of the null-space eigenvectors of the penalty matrix for each of these random-term pairs (for more information see Piepho, Boer and Williams, 2022). The optimization is achieved either using an optimizer or takes the form of a search over a grid of rotation angles for a reduced model; the fit of the full model with rotation using the optimal rotation angles will be returned.

The TPPCS and TPP1LS models are fitted using functions from the R package TPSbits authored by Sue Welham (2022). There are two methods for supplying the spline basis information produced by tpsmb to asreml. The grp method adds it to the data.frame supplied in the data argument of the asreml call. The mbf method creates smaller data.frames with the spline basis information in the same environment as the internal function that calls the spline-fitting function. If it is desired to use in a later session, an asreml function, or asrtests function that calls asreml, (e.g. predict.asreml, predictPlus.asreml, or changeTerms.asrtests) on an asreml.object created using mbf terms, then the mbf data.frames will need to be recreated using makeTPPSplineMats.data.frame in the new session, supplying, if there has been rotation of the penalty matrix eigenvectors, the theta values that are returned as the attribute theta.opt of the asreml.obj.

All models utilize the function changeTerms.asrtests to fit the spatial model. Arguments from tpsmb and changeTerms.asrtests can be supplied in calls to addSpatialModel.asrtests and will be passed on to the relevant function through the ellipses argument (...).

The data for experiment can be divided sections and the same spatial model fitted separately to each. The fit over all of the sections is assessed. For more detail see sections above.

Each combination of a row.coords and a col.coords does not have to specify a single observation; for example, to fit a local spatial model to the main units of a split-unit design, each combination would correspond to a main unit and all subunits of the main unit would have the same combination.

## Value

An asrtests.object containing the components (i) asreml.obj, possibly with attribute theta.opt, (ii) wald.tab, and (iii) test.summary for the model that includes the spatial model, unless the spatial model fails to be fitted when allow.unconverged and/or allow.fixedcorrelation is set to FALSE. If the asrtests.object is the result of fitting a TPPCS model with an exploration of the rotation of the eigenvectors of the penalty matrix for the linear components, then the asreml.obj will have an attribute theta.opt that contains the optimal rotation angles of the eigenvectors.

## Author(s)

Chris Brien

## References

- Piepho, H.-P., Boer, M. P., & Williams, E. R. (2022). Two-dimensional P-spline smoothing for spatial analysis of plant breeding trials. *Biometrical Journal*, **64**, 835-857.
- Rodriguez-Alvarez, M. X., Boer, M. P., van Eeuwijk, F. A., & Eilers, P. H. C. (2018). Correcting for spatial heterogeneity in plant breeding experiments with P-splines. *Spatial Statistics*, **23**, 52-71.
- Verbyla, A. P., De Faveri, J., Wilkie, J. D., & Lewis, T. (2018). Tensor Cubic Smoothing Splines in Designed Experiments Requiring Residual Modelling. *Journal of Agricultural, Biological and Environmental Statistics*, **23**(4), 478-508.
- Welham, S. J. (2022) TPSbits: Creates Structures to Enable Fitting and Examination of 2D Tensor-Product Splines using ASReml-R. Version 1.0.0.

**See Also**

[as.asrtests](#), [makeTPPSplineMats.data.frame](#), [addSpatialModelOnIC.asrtests](#), [chooseSpatialModelOnIC.asrtests](#), [changeModelOnIC.asrtests](#), [changeTerms.asrtests](#), [rmboundary.asrtests](#), [testranfix.asrtests](#), [testresidual.asrtests](#), [newfit.asreml](#), [reparamSigDevn.asrtests](#), [changeTerms.asrtests](#), [infoCriteria.asreml](#)

**Examples**

```
## Not run:

data(Wheat.dat)

#Add row and column covariates
Wheat.dat <- within(Wheat.dat,
  {
    cColumn <- dae::as.numfac(Column)
    cColumn <- cColumn - mean(unique(cColumn))
    cRow <- dae::as.numfac(Row)
    cRow <- cRow - mean(unique(cRow))
  })

#Fit initial model
current.asr <- asreml(yield ~ Rep + WithinColPairs + Variety,
  random = ~ Row + Column,
  data=Wheat.dat)

#Create an asrtests object, removing boundary terms
current.asrt <- as.asrtests(current.asr, NULL, NULL,
  label = "Random Row and Column effects")
current.asrt <- rmboundary(current.asrt)

#Create an asrtests object with a P-spline spatial variation model
spatial.asrt <- addSpatialModel(current.asrt, spatial.model = "TPPS",
  row.covar = "cRow", col.covar = "cColumn",
  dropRowterm = "Row", dropColterm = "Column",
  asreml.option = "grp")
infoCriteria(current.asrt$asreml.obj)

#Create an asrtests object with a P-spline spatial variation model
#that includes rotation of the eigenvectors of the penalty matrix
spatial.asrt <- addSpatialModel(current.asrt, spatial.model = "TPPS",
  row.covar = "cRow", col.covar = "cColumn",
  dropRowterm = "Row", dropColterm = "Column",
  rotateX = TRUE,
  which.rotacriterion = "dev",
  nrotacores = parallel::detectCores(),
  asreml.option = "mbf")
infoCriteria(current.asrt$asreml.obj)

## End(Not run)
```

---

addSpatialModelOnIC.asrtests

*Uses information criteria to decide whether to add a spatial model to account for local spatial variation.*

---

## Description

Adds either a correlation, two-dimensional tensor-product natural cubic smoothing spline (TP-NCSS), or a two-dimensional tensor-product penalized P-spline model (TPPS) to account for the local spatial variation exhibited by a response variable measured on a potentially irregular grid of rows and columns of the units. The data may be arranged in sections for each of which there is a grid and for which the model is to be fitted separately. Also, the rows and columns of a grid are not necessarily one observational unit wide. The spatial model is only added if the information criterion of the supplied model is decreased with the addition of the local spatial model. For TPPS models for which the order of differencing the penalty matrix is two, the improvement in the fit from rotating the eigenvectors of the penalty matrix can be investigated; if there is no improvement, the unrotated fit will be returned.

A row is added for each section to the test.summary data.frame of the `asrtests.object` stating whether or not the new model has been swapped for a model in which the spatial model has been add to the supplied model. Convergence and the occurrence of fixed correlations in fitting the model is checked and a note included in the action if there was not. All components of the `asrtests.object` are updated to exhibit the differences between the supplied and the new model, if a spatial model is added.

## Usage

```
## S3 method for class 'asrtests'
addSpatialModelOnIC(asrtests.obj, spatial.model = "TPPS",
                    sections = NULL,
                    row.covar = "cRow", col.covar = "cCol",
                    row.factor = "Row", col.factor = "Col",
                    corr.funcs = c("ar1", "ar1"), corr.orders = c(0, 0),
                    row.corrFitfirst = TRUE,
                    allow.corrJointFit = TRUE, nugget.variance = TRUE,
                    dropFixed = NULL, dropRandom = NULL,
                    nsegs = NULL, nestorder = c(1,1),
                    degree = c(3,3), difforder = c(2,2),
                    usRandLinCoeffs = TRUE,
                    rotateX = FALSE, ngridangles = NULL,
                    which.rotacriterion = "AIC", nrotacores = 1,
                    asreml.option = "grp", tpps4mbf.obj = NULL,
                    allow.unconverged = TRUE, allow.fixedcorrelation = TRUE,
                    checkboundaryonly = FALSE, update = TRUE, trace = FALSE,
                    maxit = 30, ICLikelihood = "full", which.IC = "AIC", ...)
```

## Arguments

- |                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| <code>asrtests.obj</code>  | An <code>asrtests.object</code> containing the components (i) <code>asreml.obj</code> , (ii) <code>wald.tab</code> , and (iii) <code>test.summary</code> .                                                                                                                                                                                                                                                                                                                                                            |
| <code>spatial.model</code> | A single character string nominating the type of spatial model to fit. Possible values are <code>corr</code> , <code>TPNCSS</code> and <code>TPPS</code> .                                                                                                                                                                                                                                                                                                                                                            |
| <code>sections</code>      | A single character string that specifies the name of the column in the <code>data.frame</code> that contains the <code>factor</code> that identifies different sections of the data to which separate spatial models are to be fitted. Note that, for other terms that involve sections in the random formula, there should be separate terms for each level of sections. For example, in a blocked experiment involving multiple sites, there should be the sum of separate terms for the Blocks at each Site i.e. a |

formula that contains terms like `at(Site, i):Block` for each site and these are separated by '+'. Otherwise, the combined term (e.g. `Site:Block`) will impact on the fitting of the local spatial models for the different Sites. Similarly, a separate residual variance for each of the sections should be fitted, unless there is a need to fit a different variance structure to the residual, e.g. heterogeneous residual variances depending on treatments. Separate residual variances for sections can be achieved using the `asreml` functions `dsum` or `idh`. Because, unlike random terms, terms for residual variances are not removed from the model, compound residual terms can be used to include them in the model, e.g. terms with `idh` or `dsum` with multiple levels in the `list` or leaving levels out altogether. In addition to allowing the independent fitting of models to the sections, separate residual variance terms allows a nugget variance to be fitted in a correlation model for each of the sections.

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <code>row.covar</code>        | A single character string nominating a <a href="#">numeric</a> that contains the values of a centred covariate indexing the rows of a grid. The <a href="#">numeric</a> must be a column in the <a href="#">data.frame</a> stored in the <code>asreml.obj</code> that is a component of the supplied <code>asrtests.obj</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>col.covar</code>        | A single character string nominating a <a href="#">numeric</a> that contains the values of a centred covariate indexing the columns of a grid. The <a href="#">numeric</a> must be a column in the <a href="#">data.frame</a> stored in the <code>asreml.obj</code> that is a component of the supplied <code>asrtests.obj</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <code>row.factor</code>       | A single character string nominating a <a href="#">factor</a> that indexes the rows of a grid that are to be one dimension of a spatial correlation model. The <a href="#">factor</a> must be a column in the <a href="#">data.frame</a> stored in the <code>asreml.obj</code> that is a component of the supplied <code>asrtests.obj</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>col.factor</code>       | A single character string nominating a <a href="#">factor</a> that indexes the columns of a grid that are to be one dimension of a spatial correlation model. The <a href="#">factor</a> must be a column in the <a href="#">data.frame</a> stored in the <code>asreml.obj</code> that is a component of the supplied <code>asrtests.obj</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>corr.funcs</code>       | A single character string of length two that specifies the <code>asreml</code> one-dimensional correlation or variance model function for the row and column dimensions of a two-dimensional separable spatial correlation model to be fitted when <code>spatial.model</code> is <code>corr</code> ; the two-dimensional model is fitted as a random term. If a correlation or variance model is not to be investigated for one of the dimensions, specify "" for that dimension. If the correlation model is <code>corb</code> , the values of <code>corr.orders</code> are used for its order argument (b).                                                                                                                                                                                                                                                                                                                                               |
| <code>corr.orders</code>      | A numeric of length two that specifies the order argument (b) values for the row and column dimensions of a two-dimensional separable spatial correlation model when <code>spatial.model</code> is <code>corr</code> and the <code>corr.funcs</code> for a dimension is <code>corb</code> , the <code>asreml</code> banded correlation model. If one of the dimensions does not involve an order argument, set the value of <code>corr.orders</code> for that dimension to zero. For a dimension for which the <code>corr.funcs</code> is <code>corb</code> and <code>corr.orders</code> is zero, a model with a single band, the correlation between immediate neighbours, will be fitted and then further bands, up to a maximum of 10 bands, will be added until the addition of an extra band does not reduce the information criterion nominated using <code>which.IC</code> . Note that the two-dimensional spatial model is fitted as a random term. |
| <code>row.corrFitfirst</code> | A <a href="#">logical</a> . If TRUE then, in fitting the model for <code>spatial.model</code> set to <code>corr</code> , the row correlation or variance function is fitted first, followed by the addition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |



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|                    | of the column correlation or variance function. If FALSE, the order of fitting is reversed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| allow.corrJointFit | A <b>logical</b> which, if TRUE, will allow the simultaneous fitting of correlation functions for the two dimensions of the grid when separate fits have failed to fit any correlation functions. This argument is available for when a joint fit hangs the system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| nugget.variance    | A <b>logical</b> which, if TRUE, will result in an attempt to fit a nugget or unit-specific variance. Otherwise, a nugget or unit-specific variance will not be fitted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| dropFixed          | <p>A single character string or a character vector of strings with an element for each level of sections in the same order as the sections levels. Each string, which if it is not NA and after the addition of " ~ . -" and conversion to a formula that is then expanded, specifies the sum of a set of terms to be dropped from the fixed formula in fitting splines (TPPS and TPNCSS). The result is that the fitted model supplied in the <code>asrtests.obj</code>, that includes these terms, will be compared with the fitted model that has had them removed and a spatial model added.</p> <p>An element that is NA indicates that no term pertaining to the corresponding sections level is to be removed. If sections is not NULL and a single character string has been supplied, the terms specified by the string are taken to be terms that are independent of the sections and will be removed when adding the spatial model for the first sections.</p> <p>The terms must match those in the <code>wald.tab</code> component of the <code>asrtests.obj</code>. The fixed terms will be reordered so that single-variable terms come first, followed by two-variable terms and so on. Note also that multiple terms specified using a single <code>asreml : at</code> function can only be dropped as a whole. If the term was specified using an <code>asreml : at</code> function with a single level, then it can be removed and either the level itself or its <b>numeric</b> position in the levels returned by the <code>levels</code> function can be specified.</p> |
| dropRandom         | <p>A single character string or a character vector of strings with an element for each level of sections in the same order as the sections levels. Each string, which if it is not NA and after the addition of " ~ . -" and conversion to a formula that is then expanded, specifies the sum of a set of terms to be dropped from the random formula in fitting splines (TPPS and TPNCSS). The result is that the fitted model supplied in the <code>asrtests.obj</code>, that includes these terms, will be compared with the fitted model that has had them removed and a spatial model added.</p> <p>An element that is NA indicates that no term pertaining to the corresponding sections level is to be removed. If sections is not NULL and a single character string has been supplied, the terms specified by the string are taken to be terms that are independent of the sections and will be removed when adding the spatial model for the first sections.</p> <p>The terms must match those in the <code>vparameters</code> component of the <code>asreml.obj</code> component in the <code>asrtests.obj</code>. Note also that multiple terms specified using a single <code>asreml : at</code> function can only be dropped as a whole. If the term was specified using an <code>asreml : at</code> function with a single level, then it can be removed and either the level itself or its <b>numeric</b> position in the levels returned by the <code>levels</code> function can be specified.</p>                                                                          |
| nsegs              | A pair of <b>numeric</b> values giving the number of segments into which the column and row ranges are to be split, respectively, for fitting a P-spline model (TPPS)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |



|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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|                     | (each value specifies the number of internal knots + 1). If not specified, then (number of unique values - 1) is used in each dimension; for a grid layout with equal spacing, this gives a knot at each data value. If sections is not NULL and the grid differs between the sections, then nsegs will differ between the sections.                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| nestorder           | A <a href="#">numeric</a> of length 2. The order of nesting for column and row dimensions, respectively, in fitting a P-spline model (TPPS). A value of 1 specifies no nesting, a value of 2 generates a spline with half the number of segments in that dimension, etc. The number of segments in each direction must be a multiple of the order of nesting.                                                                                                                                                                                                                                                                                                                                                                                                          |
| degree              | A <a href="#">numeric</a> of length 2. The degree of polynomial spline to be used for column and row dimensions respectively, in fitting a P-spline (TPPS).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| difforder           | A <a href="#">numeric</a> of length 2. The order of differencing for column and row dimensions, respectively, in fitting a P-spline (TPPS).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| usRandLinCoeffs     | A <a href="#">logical</a> which, if TRUE and a P-spline (TPPS) is being fitted, will attempt to fit an unstructured variance model to the constant and linear terms in the interactions for constant and linear terms in one grid dimension interacting with smooth terms in the second grid dimension. The unstructured variance model can only be fitted if both the constant and linear interaction terms have been retained in the fitted model. This argument can be used to omit the attempt to fit an unstructured variance model when the attempt results in a system error.                                                                                                                                                                                   |
| rotateX             | A <a href="#">logical</a> indicating whether to rotate the eigenvectors of the penalty matrix, as described by Piepho, Boer and Williams (2022), when fitting a P-spline (TPPS). Setting rotateX to TRUE results in a search for an optimized rotation under a model that omits the random spline interaction terms. If ngridangles is set to NULL, the optimal rotation is found using an optimizer (nloptr::bobyqa). Otherwise, the optimal rotation is found by exploring the fit over a two-dimensional grid of rotation angle pairs. The optimization seeks to optimize the criterion nominated in which.rotacriterion. Rotation of the eigenvectors is only relevant for difforder values greater than 1 and has only been implemented for difforder equal to 2. |
| ngridangles         | A <a href="#">numeric</a> of length 2. If NULL (the default), the optimal pair of angles for rotating the eigenvectors of the penalty matrix of a P-spline (TPPS) will be determined using a nonlinear optimizer (nloptr::bobyqa). Otherwise, its two values specify the numbers of angles between 0 and 90 degrees for each of the row and column dimensions to be used in determining the optimal pair of angles. Specifying factors of 90 will result in integer-valued angles. The number of grid points, and hence re-analyses will be the product of the values of (ngridangles + 1).                                                                                                                                                                            |
| which.rotacriterion | A single character string nominating which of the criteria out of the deviance, the likelihood, the AIC and the BIC in determining the optimal rotation of the eigenvectors of the penalty matrix. The deviance uses the REML value computed by asreml; the other criteria use the full likelihood, evaluated using the REML estimates, that is computed by <a href="#">infoCriteria.asreml</a> .                                                                                                                                                                                                                                                                                                                                                                      |
| nrotacores          | A <a href="#">numeric</a> specifying the number of cores to deploy for running the analyses required to search the two-dimensional grid of rotation angles when rotateX is TRUE. Parallel processing has been implemented for analyzing, for each column angle, the set of angles to be investigated for the row dimension. The default                                                                                                                                                                                                                                                                                                                                                                                                                                |

value of one means that parallel processing will not be used. The value chosen for `nrotacores` needs to be balanced against the other processes that are using parallel processing at the same time.

- `asreml.option` A single character string specifying whether the `grp` or `mbf` methods are to be used to supply externally formed covariate matrices to `asreml` when fitting a P-spline (TPPS). Compared to the `mbf` method, the `grp` method is somewhat faster, but creates large `asrtests.objects` for which the time it takes to save them can exceed any gains in execution speed. The `grp` method adds columns to the `data.frame` containing the data. On the other hand, the `mbf` method adds only the fixed covariates to data and stores the random covariates in the environment of the internal function that calls the spline-fitting function; there are three smaller `data.frames` for each section that are not stored in the `asreml.object` resulting from the fitted model.
- `tps4mbf.obj` An object made with `makeTPPSplineMats.data.frame` that contains the spline basis information for fitting P-splines. The argument `tps4mbf.obj` only needs to be set when the `mbf` option of `asreml.option` is being used and it is desired to use `mbf data.frames` that have been created and stored prior to calling `addSpatialModelOnIC.asrtests`. If `tps4mbf.obj` is NULL, `makeTPPSplineMats.data.frame` will be called internally to produce the required `mbf data.frames`.
- `allow.unconverged` A logical indicating whether to accept a new model even when it does not converge. If FALSE and the fit of the new model does not converge, the supplied `asrtests.obj` is returned. Also, if FALSE and the fit of the new model has converged, but that of the old model has not, the new model will be accepted.
- `allow.fixedcorrelation` A logical indicating whether to accept a new model even when it contains correlations in the model whose values have been designated as fixed, bound or singular. If FALSE and the new model contains correlations whose values have not been able to be estimated, the supplied `asrtests.obj` is returned. The fit in the `asreml.obj` component of the supplied `asrtests.obj` will also be tested and a warning issued if both fixed correlations are found in it and `allow.fixedcorrelation` is FALSE.
- `checkboundaryonly` If TRUE then boundary and singular terms are not removed by `rmboundary.asrtests`; a warning is issued instead. Note that, for correlation models, the fitting of each dimension and the test for a nugget term are performed with `checkboundaryonly` set to TRUE and its supplied setting only honoured using a call to `rmboundary.asrtests` immediately prior to returning the final result of the fitting.
- `update` If TRUE, then `newfit.asreml` is called to fit the model to be tested, using the values of the variance parameters stored in the `asreml.object`, that is stored in `asrtests.obj`, as starting values. If FALSE, then `newfit.asreml` will not use the stored variance parameter values as starting values when fitting the new model, the only modifications being (i) to add the terms for the spatial models and (ii) those specified via ...
- `trace` If TRUE then the stages in fitting a correlation model are displayed.
- `which.IC` A character specifying the information criterion to be used in selecting the best model. Possible values are AIC and BIC. The value of the criterion for supplied model must exceed that for changed model for the changed model to be returned. (For choosing the rotation angle of the eigenvectors of the penalty matrix, see `which.rotacriterion`.)

|              |                                                                                                                                                                                                                                                                                                                                          |
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| maxit        | A <a href="#">numeric</a> specifying the maximum number of iterations that <code>asreml</code> should perform in fitting a model.                                                                                                                                                                                                        |
| IClikelihood | A character specifying whether Restricted Maximum Likelihood (REML) or the full likelihood, evaluated using REML estimates, ( <code>full</code> ) are to be used in calculating the information criteria to be included in the <code>test.summary</code> of an <a href="#">asrtests.object</a> or to be used in choosing the best model. |
| ...          | Further arguments passed to <a href="#">changeModelOnIC.asrtests</a> , <code>asreml</code> and <code>tpsmmb</code> .                                                                                                                                                                                                                     |

## Details

A fitted spatial model is only returned if it improves the fit over and above that of achieved with the model fit supplied in the `asrtests.obj`. To fit the spatial model without any hypotheses testing or comparison of information criteria use [addSpatialModel.asrtests](#). The model fit supplied in the `asrtests.obj` should not include terms that will be included in the local spatial model. All spatial model terms are fitted as fixed or random. Consequently, the residual model does not have to be iid. Note that the data must be in the order that corresponds to the `residual` argument with a variable to the right of another variable changes levels in the data frame faster than those of the other variable e.g. `Row:Column` implies that all levels for `Column` in consecutive rows of the data frame with a single `Row` level.

For the `corr` spatial model, the default model is an autocorrelation model of order one (`ar1`) for each dimension. However, any of the single dimension correlation/variance models from `asreml` can be specified for each dimension, as can no correlation model for a dimension; the models for the two dimensions can differ. Using a forward selection procedure, a series of models are tried, without removing boundary or singular terms, beginning with the addition of row correlation and followed by the addition of column correlation or, if the `row.corrFitfirst` is set to `FALSE`, the reverse order. If the fitting of the first-fitted correlation did not result in a model change because the fitting did not converge or correlations were fixed, but the fit of the second correlation was successful, then adding the first correlation will be retried. If one of the metric correlation functions is specified (e.g. `exp`), then the `row.covar` or `col.covar` will be used in the spatial model. However, because the correlations are fitted separately for the two dimensions, the `row.factor` and `col.factor` are needed for all models and is used for a dimension that does not involve a correlation/variance function for the fit being performed. Also, the correlation models are fitted as random terms and so the correlation model will include a variance parameter for the grid even when `ar1` is used to specify the correlation model, i.e. the model fitted is a variance model and there is no difference between `ar1` and `ar1v` in fitting the model. The variance parameter for this term represents the spatial variance and the fit necessarily includes a nugget term, this being the residual variance. If any correlation is retained in the model, for a section if sections is not `NULL`, then the need for a nugget term is assessed by fixing the corresponding residual variance to one, unless there are multiple residual variances and these are not related to the sections. Once the fitting of the correlation model has been completed, the `rmboundary` function will be executed with the `checkboundaryonly` value supplied in the `addSpatialModelOnIC.asrtests` call. Finally, checking for bound and singular random terms associated with the correlation model and residual terms will be carried out when there are correlation terms in the model and `checkboundaryonly` has been set to `FALSE`; as many as possible will be removed from the fitted model, in some cases by fixing variance terms to one.

The tensor-product natural-cubic-smoothing-spline (TPNCSS) spatial model is as described by Verbyla et al. (2018), the tensor-product penalized-cubic-spline (TPPSC2) model with second-order differencing of the penalty is similar to that described by Rodríguez-Alvarez et al. (2018), and the tensor-product, first-difference-penalty, linear spline (TPPSL1) model is amongst those described by Piepho, Boer and Williams (2022). The fixed terms for the spline models are `row.covar + col.covar + row.covar:col.covar` and the random terms are `spl(row.covar) + spl(col.covar) + dev(row.covar) + dev(col.covar) + spl(row.covar):col.covar + row.covar:spl(col.covar) + spl(row.covar):spl(col.covar)`, except that `spl(row.covar) + spl(col.covar)` is replaced

with `spl(row.covar):int(col.covar) + int(row.covar):spl(col.covar)` in the TPPSC2 model, where `int(.)` indicates an intercept or constant value specific to its argument. For TPPSL1 models, the terms `spl(row.covar):col.covar + row.covar:spl(col.covar)` are omitted. The supplied model should not include any of these terms. However, any fixed or random main-effect Row or Column term that has been included as an initial model for comparison with a spatial model can be removed prior to fitting the spatial model using `dropFixed` or `dropRandom`. For the P-spline models with second-order differencing, the model matrices used to fit the pairs of random terms (i) `spl(row.covar):int(col.covar)` and `spl(row.covar):col.covar` and (ii) `int(row.covar):spl(col.covar)` and `row.covar:spl(col.covar)` are transformed using the spectral decomposition of their penalty matrices. An unstructured variance model is tried for each of these pairs and retained if it improves the fit. For TPPSC2, it is also possible to optimize the rotation of the null-space eigenvectors of the penalty matrix for each of these random-term pairs (for more information see Piepho, Boer and Williams, 2022). The optimization is achieved either using an optimizer or takes the form of a search over a grid of rotation angles for a reduced model; the fit of the full model with rotation using the optimal rotation angles will only be returned if it improves on the fit of the full, unrotated model.

The TPPCS and TPP1LS models are fitted using functions from the R package `TPSbits` authored by Sue Welham (2022). There are two methods for supplying the spline basis information produced by `tpsmmb` to `asreml`. The `grp` method adds it to the `data.frame` supplied in the `data` argument of the `asreml` call. The `mbf` method creates smaller `data.frames` with the spline basis information in the same environment as the internal function that calls the spline-fitting function. If it is desired to use in a later session, an `asreml` function, or `asrtests` function that calls `asreml`, (e.g. `predict.asreml`, `predictPlus.asreml`, or `changeTerms.asrtests`) on an `asreml` object created using `mbf` terms, then the `mbf data.frames` will need to be recreated using `makeTPPSplineMats.data.frame` in the new session, supplying, if there has been rotation of the penalty matrix eigenvectors, the `theta` values that are returned as the attribute `theta.opt` of the `asreml.obj`.

All models utilize the function `changeModelOnIC.asrtests` to assess the model fit, the information criteria used in assessing the fit being calculated using `infoCriteria`. Any bound terms are removed from the model. Arguments from `tpsmmb` and `changeModelOnIC.asrtests` can be supplied in calls to `addSpatialModelOnIC.asrtests` and will be passed on to the relevant function through the ellipses argument (`...`).

The data for experiment can be divided sections and the same spatial model fitted separately to each. The fit over all of the sections is assessed. For more detail see sections above.

Each combination of a `row.coords` and a `col.coords` does not have to specify a single observation; for example, to fit a local spatial model to the main units of a split-unit design, each combination would correspond to a main unit and all subunits of the main unit would have the same combination.

## Value

An `asrtests.object` containing the components (i) `asreml.obj`, possibly with attribute `theta.opt`, (ii) `wald.tab`, and (iii) `test.summary` for the model whose fit has the smallest information criterion between the supplied and spatial model. The values of the degrees of freedom and the information criteria in the `test.summary` are differences between those of the changed model and those of the model supplied to `addSpatialModelOnIC`. If the `asrtests.object` is the result of fitting a TPPCS model with an exploration of the rotation of the eigenvectors of the penalty matrix for the linear components, then the `asreml.obj` will have an attribute `theta.opt` that contains the optimal rotation angles of the eigenvectors.

## Author(s)

Chris Brien

## References

- Piepho, H.-P., Boer, M. P., & Williams, E. R. (2022). Two-dimensional P-spline smoothing for spatial analysis of plant breeding trials. *Biometrical Journal*, **64**, 835-857.
- Rodriguez-Alvarez, M. X., Boer, M. P., van Eeuwijk, F. A., & Eilers, P. H. C. (2018). Correcting for spatial heterogeneity in plant breeding experiments with P-splines. *Spatial Statistics*, **23**, 52-71.
- Verbyla, A. P., De Faveri, J., Wilkie, J. D., & Lewis, T. (2018). Tensor Cubic Smoothing Splines in Designed Experiments Requiring Residual Modelling. *Journal of Agricultural, Biological and Environmental Statistics*, **23**(4), 478-508.
- Welham, S. J. (2022) TPSbits: *Creates Structures to Enable Fitting and Examination of 2D Tensor-Product Splines using ASReml-R*. Version 1.0.0.

## See Also

[as.asrtests](#), [makeTPPSplineMats.data.frame](#), [addSpatialModel.asrtests](#), [chooseSpatialModelOnIC.asrtests](#), [changeModelOnIC.asrtests](#), [changeTerms.asrtests](#), [rmboundary.asrtests](#), [testtranfix.asrtests](#), [testresidual.asrtests](#), [newfit.asreml](#), [reparamSigDevn.asrtests](#), [changeTerms.asrtests](#), [infoCriteria.asreml](#)

## Examples

```
## Not run:

data(Wheat.dat)

#Add row and column covariates
Wheat.dat <- within(Wheat.dat,
  {
    cColumn <- dae::as.numfac(Column)
    cColumn <- cColumn - mean(unique(cColumn))
    cRow <- dae::as.numfac(Row)
    cRow <- cRow - mean(unique(cRow))
  })

#Fit initial model
current.asr <- asreml(yield ~ Rep + WithinColPairs + Variety,
  random = ~ Row + Column,
  data=Wheat.dat)

#Create an asrtests object, removing boundary terms
current.asrt <- as.asrtests(current.asr, NULL, NULL,
  label = "Random Row and Column effects")
current.asrt <- rmboundary(current.asrt)

current.asrt <- addSpatialModelOnIC(current.asrt, spatial.model = "TPPS",
  row.covar = "cRow", col.covar = "cColumn",
  dropRowterm = "Row", dropColterm = "Column",
  asreml.option = "grp")

infoCriteria(current.asrt$asreml.obj)

## End(Not run)
```

---

|                    |                                                 |
|--------------------|-------------------------------------------------|
| addto.test.summary | <i>Adds a row to a test.summary data.frame.</i> |
|--------------------|-------------------------------------------------|

---

## Description

A row that summarizes the result of a proposed change to a model is added to a `test.summary` data.frame. Only the values of those arguments for which there are columns in `test.summary` will be included in the row.

## Usage

```
addto.test.summary(test.summary, terms, DF = 1, denDF = NA,
                   p = NA, AIC = NA, BIC = NA,
                   action = "Boundary")
```

## Arguments

|                           |                                                                                                                                                                                                                                                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>test.summary</code> | A <a href="#">data.frame</a> whose columns are a subset of <code>terms</code> , <code>DF</code> , <code>denDF</code> , <code>p</code> , <code>AIC</code> , <code>BIC</code> and <code>action</code> . Each row summarizes the results of proposed changes to the fitted model. See <a href="#">asrtests.object</a> for more information. |
| <code>terms</code>        | A <a href="#">character</a> giving the name of a term that might be added to or removed from the model or a label indicating a change that might be made to the model.                                                                                                                                                                   |
| <code>DF</code>           | A <a href="#">numeric</a> giving the numerator degrees of freedom for a Wald F-statistic or the number of variance parameters in the current model minus the number in the proposed model.                                                                                                                                               |
| <code>denDF</code>        | A <a href="#">numeric</a> giving the denominator degrees of freedom for a Wald F-statistic.                                                                                                                                                                                                                                              |
| <code>p</code>            | A <a href="#">numeric</a> giving the p-value for a Wald F-statistic or REML ratio test.                                                                                                                                                                                                                                                  |
| <code>AIC</code>          | A <a href="#">numeric</a> giving Akaike Information Criterion (AIC) for a model or the difference between the AIC values for the current and proposed models.                                                                                                                                                                            |
| <code>BIC</code>          | A <a href="#">numeric</a> giving Bayesian (Schwarz) Information Criterion for a model or the difference between the AIC values for the current and proposed models.                                                                                                                                                                      |
| <code>action</code>       | A <a href="#">character</a> giving what action was taken with respect to the proposed change. See <a href="#">asrtests.object</a> for more information.                                                                                                                                                                                  |

## Value

A [data.frame](#).

## Author(s)

Chris Brien

## See Also

[asremlPlus-package](#), [asrtests.object](#), [print.test.summary](#)

**Examples**

```

## Not run:
data(Wheat.dat)

## Fit an autocorrelation model
ar1.asr <- asreml(yield ~ Rep + WithinColPairs + Variety,
  random = ~ Row + Column + units,
  residual = ~ ar1(Row):ar1(Column),
  data=Wheat.dat)
ar1.asrt <- as.asrtests(ar1.asr, NULL, NULL,
  label = "Autocorrelation model")
ar1.asrt <- rmboundary.asrtests(ar1.asrt)

## Fit a tensor spline
Wheat.dat <- within(Wheat.dat,
  {
    cRow <- dae::as.numfac(Row)
    cRow <- cRow - mean(unique(cRow))
    cColumn <- dae::as.numfac(Column)
    cColumn <- cColumn - mean(unique(cColumn))
  })
ts.asr <- asreml(yield ~ Rep + cRow + cColumn + WithinColPairs +
  Variety,
  random = ~ spl(cRow) + spl(cColumn) +
    dev(cRow) + dev(cColumn) +
    spl(cRow):cColumn + cRow:spl(cColumn) +
    spl(cRow):spl(cColumn),
  residual = ~ Row:Column,
  data=Wheat.dat)
ts.asrt <- as.asrtests(ts.asr, NULL, NULL,
  label = "Tensor spline model")
ts.asrt <- rmboundary.asrtests(ts.asrt)
ar1.ic <- infoCriteria(ar1.asrt$asreml.obj)
ts.ic <- infoCriteria(ts.asrt$asreml.obj)
if (ar1.ic$AIC < ts.ic$AIC)
{
  ic.diff <- ar1.ic - ts.ic
  new.asrt <- ar1.asrt
  new.asrt$test.summary <- addto.test.summary(ar1.asrt$test.summary,
    terms = "Compare ar1 to ts",
    DF = ic.diff$varDF,
    AIC = ic.diff$AIC, BIC = ic.diff$BIC,
    action = "Chose ar1")
} else
{
  ic.diff <- ts.ic - ar1.ic
  new.asrt <- ts.asrt
  new.asrt$test.summary <- addto.test.summary(ts.asrt$test.summary,
    terms = "Compare ar1 to ts",
    DF = ic.diff$varDF,
    AIC = ic.diff$AIC, BIC = ic.diff$BIC,
    action = "Chose ts")
}

## End(Not run)

```

---

```
allDifferences.data.frame
```

*Using supplied predictions and standard errors of pairwise differences or the variance matrix of predictions, forms all pairwise differences between the set of predictions, and p-values for the differences.*

---

## Description

Uses supplied predictions and standard errors of pairwise differences, or the variance matrix of predictions to form, in an `alldiffs.object`, for those components not already present, (i) a table of all pairwise differences of the predictions, (ii) the p-value of each pairwise difference, and (iii) the minimum, mean, maximum and accuracy of LSD values. Predictions that are aliased (or inestimable) are removed from the predictions component of the `alldiffs.object` and standard errors of differences involving them are removed from the sed component.

If necessary, the order of the columns of the variables in the predictions component are changed to be the initial columns of the `predictions.frame` and to match their order in the `classify`. Also, the rows of predictions component are ordered so that they are in standard order for the variables in the `classify`. That is, the values of the last variable change with every row, those of the second-last variable only change after all the values of the last variable have been traversed; in general, the values of a variable are the same for all the combinations of the values to the variables to its right in the `classify`. The `sortFactor` or `sortOrder` arguments can be used to order of the values for the `classify` variables, which is achieved using `sort.alldiffs`.

Each p-value is computed as the probability of a t-statistic as large as or larger than the absolute value of the observed difference divided by its standard error. The p-values are stored in the `p.differences` component. The degrees of freedom of the t-distribution is the degrees of freedom stored in the `tdf` attribute of the `alldiffs.object`. This t-distribution is also used in calculating the LSD statistics stored in the LSD component of the `alldiffs.object`.

## Usage

```
## S3 method for class 'data.frame'
allDifferences(predictions, classify, vcov = NULL,
               differences = NULL, p.differences = NULL, sed = NULL,
               LSD = NULL, LSDtype = "overall", LSDsupplied = NULL,
               LSDby = NULL, LSDstatistic = "mean",
               LSDaccuracy = "maxAbsDeviation",
               retain.zeroLSDs = FALSE,
               zero.tolerance = .Machine$double.eps ^ 0.5,
               backtransforms = NULL,
               response = NULL, response.title = NULL,
               term = NULL, tdf = NULL,
               x.num = NULL, x.fac = NULL,
               level.length = NA,
               pairwise = TRUE, alpha = 0.05,
               transform.power = 1, offset = 0, scale = 1,
               transform.function = "identity",
               inestimable.rm = TRUE,
               sortFactor = NULL, sortParallelToCombo = NULL,
               sortNestingFactor = NULL, sortOrder = NULL,
               decreasing = FALSE, ...)
```



**Arguments**

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| predictions   | A <a href="#">predictions.frame</a> , or a <code>data.frame</code> , beginning with the variables classifying the predictions and also containing columns named <code>predicted.value</code> , <code>standard.error</code> and <code>est.status</code> ; each row contains a single predicted value. It may also contain columns for the lower and upper limits of error intervals for the predictions. Note that the names <code>standard.error</code> and <code>est.status</code> have been changed to <code>std.error</code> and <code>status</code> in the <code>pvals</code> component produced by <code>asreml-R4</code> ; if the new names are in the <code>data.frame</code> supplied to <code>predictions</code> , they will be returned to the previous names.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| classify      | A character string giving the variables that define the margins of the multiway table that has been predicted. Multiway tables are specified by forming an interaction type term from the classifying variables, that is, separating the variable names with the <code>:</code> operator.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| vcov          | A matrix containing the variance matrix of the predictions; it is used in computing the variance of linear transformations of the predictions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| differences   | A matrix containing all pairwise differences between the predictions; it should have the same number of rows and columns as there are rows in <code>predictions</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| p.differences | A matrix containing p-values for all pairwise differences between the predictions; each p-value is computed as the probability of a t-statistic as large as or larger than the observed difference divided by its standard error. The degrees of freedom of the t distribution for computing it are computed as the denominator degrees of freedom of the F value for the fixed term, if available; otherwise, the degrees of freedom stored in the attribute <code>tdf</code> are used; the matrix should be of the same size as that for <code>differences</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| sed           | A matrix containing the standard errors of all pairwise differences between the predictions; they are used in computing the p-values.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| LSD           | An <a href="#">LSD.frame</a> containing the mean, minimum and maximum LSD for determining the significance of pairwise differences, as well as an assigned LSD and a measure of the accuracy of the LSD. If <code>LSD</code> is <code>NULL</code> then the <a href="#">LSD.frame</a> stored in the <code>LSD</code> component will be calculated and the values of <code>LSDtype</code> , <code>LSDby</code> and <code>LSDstatistic</code> added as attributes of the <a href="#">alldiffs.object</a> . The LSD for a single prediction assumes that any predictions to be compared are independent; this is not the case if residual errors are correlated.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| LSDtype       | A <a href="#">character</a> string that can be <code>overall</code> , <code>factor.combinations</code> , <code>per.prediction</code> or supplied. It determines whether the values stored in a row of a <a href="#">LSD.frame</a> are the values calculated (i) <code>overall</code> from the LSD values for all pairwise comparisons, unless there is only one, possibly repeated, prediction, when a notional LSD is calculated, (ii) the values calculated from the pairwise LSDs for the levels of each <code>factor.combination</code> , unless there is only one prediction for a level of the <code>factor.combination</code> , when a notional LSD is calculated, (iii) <code>per.prediction</code> , being based, for each prediction, on all pairwise differences involving that prediction, or (iv) as supplied values of the LSD, specified with the <code>LSDsupplied</code> argument; these supplied values are to be placed in the <code>assignedLSD</code> column of the <a href="#">LSD.frame</a> stored in an <a href="#">alldiffs.object</a> so that they can be used in LSD calculations.<br><br>See <a href="#">LSD.frame</a> for further information on the values in a row of this <code>data.frame</code> and how they are calculated. |
| LSDsupplied   | A <a href="#">data.frame</a> or a named <a href="#">numeric</a> containing a set of LSD values that correspond to the observed combinations of the values of the <code>LSDby</code> variables in                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | <p>the <code>predictions.frame</code> or a single LSD value that is an overall LSD. If a <code>data.frame</code>, it may have (i) a column for the LSDby variable and a column of LSD values or (ii) a single column of LSD values with rownames being the combinations of the observed values of the LSDby variables. Any name can be used for the column of LSD values; assignedLSD is sensible, but not obligatory. Otherwise, a <code>numeric</code> containing the LSD values, each of which is named for the observed combination of the values of the LSDby variables to which it corresponds. (Applying the function <code>dae::fac.combine</code> to the predictions component is one way of forming the required combinations for the (row) names.) The values supplied will be incorporated into assignedLSD column of the <code>LSD.frame</code> stored as the LSD component of the <code>alldiffs.object</code>.</p>                                                                                               |
| LSDby           | <p>A <code>character</code> (vector) of variables names, being the names of the <code>factors</code> or <code>numerics</code> in the <code>classify</code>; for each combination of their levels and values, there will be or is a row in the <code>LSD.frame</code> stored in the LSD component of the <code>alldiffs.object</code> when LSDtype is <code>factor.combinations</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| LSDstatistic    | <p>A <code>character</code> nominating one or more of minimum, q10, q25, mean, median, q75, q90 or maximum as the value(s) to be stored in the assignedLSD column in an <code>LSD.frame</code>; the values in the assignedLSD column are used in computing halfLeastSignificant error.intervals. Here q10, q25, q75 and q90 indicate the sample quantiles corresponding to probabilities of 0.1, 0.25, 0.75 and 0.9 for the group of LSDs from which a single LSD value is calculated. The function <code>quantile</code> is used to obtain them. The mean LSD is calculated as the square root of the mean of the squares of the LSDs for the group. The median is calculated using the <code>median</code> function. Multiple values are only produced for LSDtype set to <code>factor.combinations</code>, in which case LSDby must not be NULL and the number of values must equal the number of observed combinations of the values of the variables specified by LSDby. If LSDstatistic is NULL, it is reset to mean.</p> |
| LSDaccuracy     | <p>A <code>character</code> nominating one of maxAbsDeviation, maxDeviation, q90Deviation or RootMeanSqDeviation as the statistic to be calculated as a measure of the accuracy of assignedLSD. The option q90Deviation produces the sample quantile corresponding to a probability of 0.90. The deviations are the differences between the LSDs used in calculating the LSD statistics and each assigned LSD and the accuracy is expressed as a proportion of the assigned LSD value. The calculated values are stored in the column named accuracyLSD in an <code>LSD.frame</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| retain.zeroLSDs | <p>A <code>logical</code> indicating whether to retain or omit LSDs that are zero when calculating the summaries of LSDs.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| zero.tolerance  | <p>A <code>numeric</code> specifying the value such that if an LSD is less than it, the LSD will be considered to be zero.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| backtransforms  | <p>A <code>data.frame</code> containing the backtransformed values of the predicted values that is consistent with the predictions component, except that the column named predicted.value is replaced by one called backtransformed.predictions. Any error.interval values will also be the backtransformed values. Each row contains a single predicted value.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| response        | <p>A character specifying the response variable for the predictions. It is stored as an attribute to the <code>alldiffs.object</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| response.title  | <p>A character specifying the title for the response variable for the predictions. It is stored as an attribute to the <code>alldiffs.object</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| term            | <p>A <code>character</code> string giving the variables that define the term that was fitted using <code>asreml</code> and that corresponds to <code>classify</code>. It only needs to be specified when</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | it is different to <code>classify</code> ; it is stored as an attribute of the <code>alldiffs.object</code> . It is likely to be needed when the fitted model includes terms that involve both a <code>numeric</code> covariate and a <code>factor</code> that parallel each other; the <code>classify</code> would include the covariate and the term would include the factor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>tdf</code>                | an integer specifying the degrees of freedom of the standard error. It is used as the degrees of freedom for the t-distribution on which p-values and confidence intervals are based. It is stored as an attribute to the <code>alldiffs.object</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>x.num</code>              | A <code>character</code> string giving the name of the numeric covariate that (i) is potentially included in terms in the fitted model and (ii) is the x-axis variable for plots. Its values will not be converted to a <code>factor</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>x.fac</code>              | A <code>character</code> string giving the name of the factor that (i) corresponds to <code>x.num</code> and (ii) is potentially included in terms in the fitted model. It should have the same number of levels as the number of unique values in <code>x.num</code> . The levels of <code>x.fac</code> must be in the order in which they are to be plotted - if they are dates, then they should be in the form <code>yyyymmdd</code> , which can be achieved using <code>as.Date</code> . However, the levels can be non-numeric in nature, provided that <code>x.num</code> is also set.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>level.length</code>       | The maximum number of characters from the levels of factors to use in the row and column labels of the tables of pairwise differences and their p-values and standard errors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>pairwise</code>           | A logical indicating whether all pairwise differences of the predictions and their standard errors and p-values are to be computed and stored. If <code>FALSE</code> , the components <code>differences</code> and <code>p.differences</code> will be <code>NULL</code> in the returned <code>alldiffs.object</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>alpha</code>              | A <code>numeric</code> giving the significance level for LSDs or one minus the confidence level for confidence intervals. It is stored as an attribute to the <code>alldiffs.object</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>transform.power</code>    | A <code>numeric</code> specifying the power of a transformation, if one has been applied to the response variable. Unless it is equal to 1, the default, back-transforms of the predictions will be obtained and stored in the <code>backtransforms</code> component of the <code>alldiffs.object</code> . The back-transformation raises the predictions to the power equal to the reciprocal of <code>transform.power</code> , unless it equals 0 in which case the exponential of the predictions is taken. Confidence limits and Least Significant limits will be similarly backtransformed. Approximate standard errors of the backtransformed predictions will also be calculated.                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>offset</code>             | A <code>numeric</code> that has been added to each value of the response after any scaling and before applying any power transformation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>scale</code>              | A <code>numeric</code> by which each value of the response has been multiplied before adding any offset and applying any power transformation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>transform.function</code> | A <code>character</code> giving the name of a function that specifies the scale on which the predicted values are defined. This may be the result of a transformation of the data using the function or the use of the function as a link function in the fitting of a generalized linear (mixed) model (GL(M)M). The possible <code>transform.functions</code> are <code>identity</code> , <code>log</code> , <code>inverse</code> , <code>sqrt</code> , <code>logit</code> , <code>probit</code> , and <code>cloglog</code> . The <code>predicted.values</code> and <code>error.intervals</code> , if not <code>StandardError</code> intervals, will be back-transformed using the inverse function of the <code>transform.function</code> . The <code>standard.error</code> column will be set to <code>NA</code> , unless (i) <code>asreml</code> returns columns named <code>transformed.value</code> and <code>approx.se</code> , as well as those called <code>predicted.values</code> and <code>standard.error</code> (such as when a GLM is fitted) and (ii) the values in |

transformed.value are equal to those obtained by backtransforming the predicted.values using the inverse function of the transform.function. Then, the approx.se values will be saved in the standard.error column of the backtransforms component of the returned alldiffs.obj. Also, the transformed.value and approx.se columns are removed from both the predictions and backtransforms components of the alldiffs.obj. Note that the values that end up in the standard errors column are approximate for the backtransformed values and are not used in calculating error.intervals.

- inestimable.rm A logical indicating whether rows for predictions that are not estimable are to be removed from the components of the `alldiffs.object`.
- sortFactor A `character` containing the name of the factor that indexes the set of predicted values that determines the sorting of the components. If there is only one variable in the classify term then sortFactor can be NULL and the order is defined by the complete set of predicted values. If there is more than one variable in the classify term then sortFactor must be set. In this case the sortFactor is sorted in the same order within each combination of the values of the sortParallelToCombo variables: the classify variables, excluding the sortFactor. There should be only one predicted value for each unique value of sortFactor within each set defined by a combination of the values of the classify variables, excluding the sortFactor factor. The order to use is determined by either sortParallelToCombo or sortOrder.
- sortParallelToCombo A `list` that specifies a combination of the values of the factors and numerics, excluding sortFactor, that are in classify. Each of the components of the supplied `list` is named for a classify variable and specifies a single value for it. The combination of this set of values will be used to define a subset of the predicted values whose order will define the order of sortFactor. Each of the other combinations of the values of the factors and numerics will be sorted in parallel. If sortParallelToCombo is NULL then the first value of each classify variable, except for the sortFactor factor, in the predictions component is used to define sortParallelToCombo. If there is only one variable in the classify then sortParallelToCombo is ignored.
- sortNestingFactor A `character` containing the name of the factor that defines groups of the sortFactor within which the predicted values are to be ordered. If there is only one variable in the classify then sortNestingFactor is ignored.
- sortOrder A character vector whose length is the same as the number of levels for sortFactor in the predictions component of the `alldiffs.object`. It specifies the desired order of the levels in the reordered components of the `alldiffs.object`. The argument sortParallelToCombo is ignored.
- The following creates a sortOrder vector levs for factor f based on the values in x:
- ```
levs <- levels(f)[order(x)].
```
- decreasing A logical passed to order that determines whether the order for sorting the components of the `alldiffs.object` is for increasing or decreasing magnitude of the predicted values.
- ... provision for passing arguments to functions called internally - not used at present.

**Value**

An `alldiffs.object` with components `predictions`, `vcov`, `differences`, `p.differences` `sed`, and `LSD`.

The name of the response, the `response.title`, the term, the `classify`, `tdf`, `alpha`, `sortFactor` and the `sortOrder` will be set as attributes to the object. Note that the `classify` in an `alldiffs.object` is based on the variables indexing the predictions, which may differ from the `classify` used to obtain the original predictions (for example, when the `alldiffs.objects` stores a linear transformation of predictions).

Also, see `predictPlus.asreml` for more information.

**Author(s)**

Chris Brien

**See Also**

`asremlPlus-package`, `as.alldiffs`, `as.predictions.frame`, `sort.alldiffs`, `subset.alldiffs`, `print.alldiffs`, `renewClassify.alldiffs`, `redoErrorIntervals.alldiffs`, `recalcLSD.alldiffs`, `pickLSDstatistics.alldiffs`, `plotPredictions.data.frame`, `predictPlus.asreml`, `predictPresent.asreml`

**Examples**

```
data(Oats.dat)

## Use asreml to get predictions and associated statistics

## Not run:
m1.asr <- asreml(Yield ~ Nitrogen*Variety,
                 random=~Blocks/Wplots,
                 data=Oats.dat)
current.asrt <- as.asrtests(m1.asr)
Var.pred <- asreml::predict.asreml(m1.asr, classify="Nitrogen:Variety",
                                   sed=TRUE)
if (getASRemlVersionLoaded(nchar = 1) == "3")
  Var.pred <- Var.pred$predictions
Var.preds <- Var.pred$pvals
Var.sed <- Var.pred$sed
Var.vcov <- NULL
wald.tab <- current.asrt$wald.tab
den.df <- wald.tab[match("Variety", rownames(wald.tab)), "denDF"]

## End(Not run)

## Use lmerTest and emmeans to get predictions and associated statistics
if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(Yield ~ Nitrogen*Variety + (1|Blocks/Wplots),
                           data=Oats.dat)
  Var.emm <- emmeans::emmeans(m1.lmer, specs = ~ Nitrogen:Variety)
  Var.preds <- summary(Var.emm)
  den.df <- min(Var.preds$df)
  ## Modify Var.preds to be compatible with a predictions.frame
  Var.preds <- as.predictions.frame(Var.preds, predictions = "emmean",
```

```

se = "SE", interval.type = "CI",
interval.names = c("lower.CL", "upper.CL"))

Var.vcov <- vcov(Var.emm)
Var.sed <- NULL
}

## Use the predictions obtained with either asreml or lmerTest
if (exists("Var.preds"))
{
  ## Order the Varieties in decreasing order for the predictions values in the
  ## first N level
  Var.diffs <- allDifferences(predictions = Var.preds,
                             classify = "Nitrogen:Variety",
                             sed = Var.sed, vcov = Var.vcov, tdf = den.df,
                             sortFactor = "Variety", decreasing = TRUE)
  print.alldiffs(Var.diffs, which="differences")

  ## Change the order of the factors in the alldiffs object and reorder components
  Var.reord.diffs <- allDifferences(predictions = Var.preds,
                                   classify = "Variety:Nitrogen",
                                   sed = Var.sed, vcov = Var.vcov, tdf = den.df)
  print.alldiffs(Var.reord.diffs, which="predictions")
}

```

---

|                 |                                          |
|-----------------|------------------------------------------|
| alldiffs.object | <i>Description of an alldiffs object</i> |
|-----------------|------------------------------------------|

---

## Description

An object of S3-class `alldiffs` that stores the predictions for a model, along with supplied statistics for all pairwise differences. While `alldiffs.object` can be constructed by defining a list with the appropriate components, it can be formed by passing the components to `as.alldiffs`, or from a predictions data.frame using `allDifferences.data.frame`.

`as.alldiffs` is function that assembles an object of this class from supplied components.

`is.alldiffs` is the membership function for this class; it tests that an object is of class `alldiffs`.

`validAlldiffs(object)` can be used to test the validity of an object with this class.

`allDifferences.data.frame` is the function that constructs an object of this class by calculating components from statistics supplied via its arguments and then using `as.alldiffs` to make the object.

## Value

A list with classes `alldiffs` and `list` containing the following components: `predictions`, `vcov`, `differences`, `p.differences`, `sed`, `LSD` and `backtransforms`. Except for `predictions`, the components are optional and can be set to `NULL`.

An `alldiffs.object` also has attributes `response`, `response.title`, `term`, `classify`, `tdf`, `alpha`, `sortFactor` and `sortOrder`, which may be set to `NULL`.

The details of the components are as follows:

1. `predictions`: A `predictions.frame`, being a `data.frame` beginning with the variables classifying the predictions, in the same order as in the `classify`, and also containing columns named `predicted.value`, `standard.error` and `est.status`; each row contains a single predicted value. The number of rows should equal the number of unique combinations of the `classify` variables and will be in standard order for the `classify` variables. That is, the values of the last variable change with every row, those of the second-last variable only change after all the values of the last variable have been traversed; in general, the values of a variable are the same for all the combinations of the values to the variables to its right in the `classify`. The `data.frame` may also include columns for the lower and upper values of error intervals, either standard error, confidence or half-LSD intervals. The names of these columns will consist of three parts separated by full stops: 1) the first part will be lower or upper; 2) the second part will be one of Confidence, StandardError or halfLeastSignificant; 3) the third component will be limits.  
Note that the names `standard.error` and `est.status` have been changed to `std.error` and `status` in the `pvals` component produced by `asreml-R4`; if the new names are in the `data.frame` supplied to `predictions`, they will be returned to the previous names.
2. `differences`: A matrix containing all pairwise differences between the predictions; it should have the same number of rows and columns as there are rows in `predictions`.
3. `p.differences`: A matrix containing p-values for all pairwise differences between the predictions; each p-value is computed as the probability of a t-statistic as large as or larger than the observed difference divided by its standard error. The degrees of freedom of the t distribution for computing it are computed as the denominator degrees of freedom of the F value for the fixed term, if available; otherwise, the degrees of freedom stored in the attribute `tdf` are used; the matrix should be of the same size as that for `differences`.
4. `sed`: A matrix containing the standard errors of all pairwise differences between the predictions; they are used in computing the p-values in `p.differences`.
5. `vcov`: A matrix containing the variance matrix of the predictions; it is used in computing the variance of linear transformations of the predictions.
6. `LSD`: An `LSD.frame` containing (i) `c`, the number of pairwise predictions comparisons for each LSD value and the mean, minimum, maximum and assigned LSD, (ii) the column `accuracyLSD` that gives a measure of the accuracy of the assigned LSD, given the variation in LSD values, and (iii) the columns `false.pos` and `false.neg` that contain the number of false positives and negatives if the assignedLSD value(s) is(are) used to determine the significance of the pairwise predictions differences. The LSD values in the `assignedLSD` column is used to determine the significance of pairwise differences that involve predictions for the combination of levels given by a row name. The value in the `assignedLSD` column is specified using the `LSDstatistic` argument.
7. `backtransforms`: When the response values have been transformed for analysis, a `data.frame` containing the backtransformed values of the predicted values is added to the `alldiffs.object`. This `data.frame` is consistent with the `predictions` component, except that the column named `predicted.value` is replaced by one called `backtransformed.predictions`. Any `error.interval` values will also be the backtransformed values. Each row contains a single predicted value.

The details of the attributes of an `alldiffs.object` are:

1. `response`: A character specifying the response variable for the predictions.
2. `response.title`: A character specifying the title for the response variable for the predictions.
3. `term`: A character giving the variables that define the term that was fitted using `asreml` and that corresponds to `classify`. It is often the same as `classify`.



4. `classify`: A character giving the variables that define the margins of the multiway table used in the prediction. Multiway tables are specified by forming an interaction type term from the classifying variables, that is, separating the variable names with the `:` operator.
5. `tdf`: An integer specifying the degrees of freedom of the standard error. It is used as the degrees of freedom for the t-distribution on which p-values and confidence intervals are based.
6. `alpha`: An integer specifying the significance level. It is used as the significance level calculating LSDs.
7. `LSDtype`: If the LSD component is not NULL then `LSDtype` is added as an attribute. A character nominating the type of grouping of seds to be used in combining LSDs.
8. `LSDby`: If the LSD component is not NULL then `LSDby` is added as an attribute. A character vector containing the names of the factors and numerics within whose combinations the LSDs are to be summarized.
9. `LSDstatistic`: If the LSD component is not NULL then `LSDstatistic` is added as an attribute. A character nominating what statistic to use in summarizing a set of LSDs.
10. `LSDaccuracy`: If the LSD component is not NULL then `LSDaccuracy` is added as an attribute. A character nominating the method of calculating a measure of the accuracy of the LSDs stored in the `assignedLSD` column of the `LSD.frame`.
11. `sortFactor`: factor that indexes the set of predicted values that determined the sorting of the components.
12. `sortOrder`: A character vector that is the same length as the number of levels for `sortFactor` in the predictions component of the `alldiffs.object`. It specifies the order of the levels in the reordered components of the `alldiffs.object`.

The following creates a `sortOrder` vector `levs` for factor `f` based on the values in `x`:  
`levs <- levels(f)[order(x)].`

See [predictPlus.asreml](#) for more information.

### Author(s)

Chris Brien

### See Also

[is.alldiffs](#), [as.alldiffs](#), [validAlldiffs](#), [allDifferences.data.frame](#)

### Examples

```
data(Oats.dat)

## Use asreml to get predictions and associated statistics

## Not run:
m1.asr <- asreml(Yield ~ Nitrogen*Variety,
                random=~Blocks/Wplots,
                data=Oats.dat)
current.asrt <- as.asrtests(m1.asr)
Var.pred <- asreml::predict.asreml(m1.asr, classify="Nitrogen:Variety",
                                  sed=TRUE)
if (getASRemlVersionLoaded(nchar = 1) == "3")
  Var.pred <- Var.pred$predictions
Var.preds <- Var.pred$pvals
Var.sed <- Var.pred$sed
```



```

Var.vcov <- NULL

## End(Not run)

## Use lmerTest and emmeans to get predictions and associated statistics
if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(Yield ~ Nitrogen*Variety + (1|Blocks/Wplots),
                          data=Oats.dat)
  Var.emm <- emmeans::emmeans(m1.lmer, specs = ~ Nitrogen:Variety)
  Var.preds <- summary(Var.emm)
  den.df <- min(Var.preds$df)
  ## Modify Var.preds to be compatible with a predictions.frame
  Var.preds <- as.predictions.frame(Var.preds, predictions = "emmean",
                                   se = "SE", interval.type = "CI",
                                   interval.names = c("lower.CL", "upper.CL"))

  Var.vcov <- vcov(Var.emm)
  Var.sed <- NULL
}

## Use the predictions obtained with either asreml or lmerTest
if (exists("Var.preds"))
{
  ## Form an all.diffs object
  Var.diffs <- as.alldiffs(predictions = Var.preds, classify = "Nitrogen:Variety",
                          sed = Var.sed, vcov = Var.vcov, tdf = den.df)

  ## Check the class and validity of the alldiffs object
  is.alldiffs(Var.diffs)
  validAlldiffs(Var.diffs)
}

```

angular

*Applies the angular transformation to proportions.***Description**

Applies the angular transformation to numeric values. It is given by  $\sin^{-1}(\sqrt{\text{proportions}})$

**Usage**

```
angular(proportions, n)
```

**Arguments**

|             |                                    |
|-------------|------------------------------------|
| proportions | The proportions.                   |
| n           | The divisor(s) for each proportion |

**Value**

A numeric.

**Author(s)**

Chris Brien

**See Also**[angular.mod](#), [powerTransform](#).**Examples**

```
n <-25
y <- rbinom(10, n, 0.5)
y <- c(y,0,n)
p <- y/n
p.ang <- angular(p, n)
```

angular.mod

*Applies the modified angular transformation to a vector of counts.***Description**

Applies the angular transformation to a vector of counts. A modified transformation is used that is appropriate when  $N < 50$  and the proportion is not between 0.3 and 0.7. The transformation is given by  $\sin^{-1} \frac{\text{count} + 0.375}{n + 0.75}$ .

**Usage**

```
angular.mod(count, n)
```

**Arguments**

|       |                                                                      |
|-------|----------------------------------------------------------------------|
| count | The numeric vector of counts.                                        |
| n     | The number(s) of observations from which the count(s) were obtained. |

**Value**

A numeric vector.

**Author(s)**

Chris Brien

**See Also**[angular](#), [powerTransform](#).**Examples**

```
n <-25
y <- rbinom(10, n, 0.5)
y <- c(y,0,n)
p.ang.mod <- angular.mod(y, n)
```

---

|             |                                                                                                                                                                                                      |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| as.alldiffs | <i>Forms an <a href="#">alldiffs.object</a> from the supplied predictions, along with those statistics, associated with the predictions and their pairwise differences, that have been supplied.</i> |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

---

## Description

Creates an [alldiffs.object](#) that consists of a list containing the following components: predictions, vcov, differences, p.differences, sed, LSD and backtransforms. Predictions must be supplied to the function while the others will be set only if they are supplied; those not supplied are set to NULL. It also has attributes response, response.title, term, classify, tdf, tdf, alpha, sortFactor and sortOrder. which will be set to the values supplied or NULL if none are supplied.

## Usage

```
as.alldiffs(predictions, vcov = NULL, differences = NULL,
             p.differences = NULL, sed = NULL, LSD = NULL,
             backtransforms = NULL,
             response = NULL, response.title = NULL,
             term = NULL, classify = NULL,
             tdf = NULL, alpha = 0.05,
             sortFactor = NULL, sortOrder = NULL)
```

## Arguments

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| predictions   | A <a href="#">predictions.frame</a> , being a data.frame beginning with the variables classifying the predictions and also containing columns named predicted.value, standard.error and est.status; each row contains a single predicted value. It may also contain columns for the lower and upper limits of error intervals for the predictions. Note that the names standard.error and est.status have been changed to std.error and status in the pvals component produced by asreml-R4; if the new names are in the data.frame supplied to predictions, they will be returned to the previous names. |
| differences   | A matrix containing all pairwise differences between the predictions; it should have the same number of rows and columns as there are rows in predictions.                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| p.differences | A matrix containing p-values for all pairwise differences between the predictions; each p-value is computed as the probability of a t-statistic as large as or larger than the observed difference divided by its standard error. The degrees of freedom of the t distribution for computing it are computed as the denominator degrees of freedom of the F value for the fixed term, if available; otherwise, the degrees of freedom stored in the attribute tdf are used; the matrix should be of the same size as that for differences.                                                                |
| sed           | A matrix containing the standard errors of all pairwise differences between the predictions; they are used in computing the p-values.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| vcov          | A matrix containing the variance matrix of the predictions; it is used in computing the variance of linear transformations of the predictions.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| LSD           | An <a href="#">LSD.frame</a> containing the mean, minimum and maximum LSD for determining the significance of pairwise differences, as well as an assigned LSD and a measure of the accuracy of the LSD. If LSD is NULL then the <a href="#">LSD.frame</a> stored                                                                                                                                                                                                                                                                                                                                         |

in the LSD component will be calculated and the values of `LSDtype`, `LSDby` and `LSDstatistic` added as attributes of the `alldiffs.object`. The LSD for a single prediction assumes that any predictions to be compared are independent; this is not the case if residual errors are correlated.

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>backtransforms</code> | A <code>data.frame</code> containing the backtransformed values of the predicted values that is consistent with the predictions component, except that the column named <code>predicted.value</code> is replaced by one called <code>backtransformed.predictions</code> . Any <code>error.interval</code> values will also be the backtransformed values. Each row contains a single predicted value.                                                                                                                                                                                   |
| <code>response</code>       | A character specifying the response variable for the predictions. It is stored as an attribute to the <code>alldiffs.object</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>response.title</code> | A character specifying the title for the response variable for the predictions. It is stored as an attribute to the <code>alldiffs.object</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>term</code>           | A <code>character</code> string giving the variables that define the term that was fitted using <code>asreml</code> and that corresponds to <code>classify</code> . It only needs to be specified when it is different to <code>classify</code> ; it is stored as an attribute of the <code>alldiffs.object</code> . It is likely to be needed when the fitted model includes terms that involve both a <code>numeric</code> covariate and a <code>factor</code> that parallel each other; the <code>classify</code> would include the covariate and the term would include the factor. |
| <code>classify</code>       | A character string giving the variables that define the margins of the multiway table used in the prediction. Multiway tables are specified by forming an interaction type term from the classifying variables, that is, separating the variable names with the <code>:</code> operator. It is stored as an attribute to the <code>alldiffs.object</code> .                                                                                                                                                                                                                             |
| <code>tdf</code>            | an integer specifying the degrees of freedom of the standard error. It is used as the degrees of freedom for the t-distribution on which p-values and confidence intervals are based. It is stored as an attribute to the <code>alldiffs.object</code> .                                                                                                                                                                                                                                                                                                                                |
| <code>alpha</code>          | A <code>numeric</code> giving the significance level for LSDs or one minus the confidence level for confidence intervals. It is stored as an attribute to the <code>alldiffs.object</code> .                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>sortFactor</code>     | A character containing the name of the factor that indexes the set of predicted values that determined the sorting of the components.                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>sortOrder</code>      | A character vector that is the same length as the number of levels for <code>sortFactor</code> in the predictions component of the <code>alldiffs.object</code> . It specifies the order of the levels in the reordered components of the <code>alldiffs.object</code> .<br>The following creates a <code>sortOrder</code> vector <code>levs</code> for factor <code>f</code> based on the values in <code>x</code> : <code>levs &lt;- levels(f)[order(x)]</code> .                                                                                                                     |

### Value

An S3-class `alldiffs.object`. Also, see `predictPlus.asreml` for more information.

### Author(s)

Chris Brien

### See Also

`asremlPlus-package`, `alldiffs.object`, `is.alldiffs`, `as.alldiffs`, `print.alldiffs`, `sort.alldiffs`, `subset.alldiffs`, `allDifferences.data.frame`, `renewClassify.alldiffs`, `redoErrorIntervals.alldiffs`, `recalcLSD.alldiffs`, `predictPlus.asreml`, `plotPredictions.data.frame`, `predictPresent.asreml`

**Examples**

```

data(Oats.dat)

## Use asreml to get predictions and associated statistics

## Not run:
m1.asr <- asreml(Yield ~ Nitrogen*Variety,
                random=~Blocks/Wplots,
                data=Oats.dat)
current.asrt <- as.asrtests(m1.asr)
Var.pred <- asreml::predict.asreml(m1.asr, classify="Nitrogen:Variety",
                                   sed=TRUE)
if (getASRemlVersionLoaded(nchar = 1) == "3")
  Var.pred <- Var.pred$predictions
Var.preds <- Var.pred$pvals
Var.sed <- Var.pred$sed
Var.vcov <- NULL

## End(Not run)

## Use lmerTest and emmeans to get predictions and associated statistics
if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(Yield ~ Nitrogen*Variety + (1|Blocks/Wplots),
                           data=Oats.dat)
  Var.emm <- emmeans::emmeans(m1.lmer, specs = ~ Nitrogen:Variety)
  Var.preds <- summary(Var.emm)
  den.df <- min(Var.preds$df)
  ## Modify Var.preds to be compatible with a predictions.frame
  Var.preds <- as.predictions.frame(Var.preds, predictions = "emmean",
                                    se = "SE", interval.type = "CI",
                                    interval.names = c("lower.CL", "upper.CL"))

  Var.vcov <- vcov(Var.emm)
  Var.sed <- NULL
}

## Use the predictions obtained with either asreml or lmerTest
if (exists("Var.preds"))
{
  ## Form an alldiffs object
  Var.diffs <- as.alldiffs(predictions = Var.preds, classify = "Nitrogen:Variety",
                           sed = Var.sed, vcov = Var.vcov, tdf = den.df)

  ## Check the class and validity of the alldiffs object
  is.alldiffs(Var.diffs)
  validAlldiffs(Var.diffs)
}

```

as.asrtests

*Forms an asrtests object that stores (i) a fitted asreml object, (ii) a pseudo-anova table for the fixed terms and (iii) a history of changes and hypothesis testing used in obtaining the model.*

## Description

An `asrtests.object` that is a list consisting of the components `asreml.obj`, `wald.tab` and `test.summary`.

A call to `as.asrtests` with `test.summary = NULL` re-initializes the `test.summary` `data.frame`.

If there is no `wald.tab`, `wald.asreml` is called. In all cases, `recalcWaldTab` is called and any changes made as specified by the `recalcWaldTab` arguments supplied via `...`

The `label` argument can be used to include an entry in `test.summary` for the starting model. If a `label` is included, (i) the information criteria calculated using the `asreml.obj` will be added to the `test.summary`, if `IClikelihood` is not set to `none` and (ii) the number of variance parameters is included in the `denDF` column, if `IClikelihood` is set to `none`.

## Usage

```
as.asrtests(asreml.obj, wald.tab = NULL, test.summary = NULL,
            denDF = "numeric", label = NULL,
            IClikelihood = "none", bound.exclusions = c("F", "B", "S", "C"), ...)
```

## Arguments

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>asreml.obj</code>       | an <code>asreml</code> object for a fitted model.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <code>wald.tab</code>         | A <code>data.frame</code> containing a pseudo-anova table for the fixed terms produced by <code>wald.asreml</code> ; it should have 4 or 6 columns. Sometimes <code>wald.asreml</code> returns a <code>data.frame</code> and at other times a list. For example, it may return a list when <code>denDF</code> is used. In this case, the Wald component of the list is to be extracted and stored. It is noted that, as of <code>asreml</code> version 4, <code>wald.asreml</code> has a <code>kenadj</code> argument.                           |
| <code>test.summary</code>     | A <code>data.frame</code> with columns <code>term</code> , <code>DF</code> , <code>denDF</code> , <code>p</code> and <code>action</code> containing the results of previous hypothesis tests.                                                                                                                                                                                                                                                                                                                                                    |
| <code>denDF</code>            | Specifies the method to use in computing approximate denominator degrees of freedom when <code>wald.asreml</code> is called. Can be <code>none</code> to suppress the computations, <code>numeric</code> for numerical methods, <code>algebraic</code> for algebraic methods or <code>default</code> , the default, to automatically choose numeric or algebraic computations depending on problem size. The denominator degrees of freedom are calculated according to Kenward and Roger (1997) for fixed terms in the dense part of the model. |
| <code>label</code>            | A character to use as an entry in the <code>terms</code> column in <code>test.summary</code> to indicate as far as is possible the nature of the model that has been fitted. The <code>action</code> column in <code>test.summary</code> will be <code>Starting model</code> .                                                                                                                                                                                                                                                                   |
| <code>IClikelihood</code>     | A character that controls both the occurrence and the type of likelihood for information criterion in the <code>test.summary</code> of the new <code>asrtests.object</code> . If <code>none</code> , none are included. Otherwise, if <code>REML</code> , then the AIC and BIC based on the Restricted Maximum Likelihood are included; if <code>full</code> , then the AIC and BIC based on the full likelihood, evaluated using REML estimates, are included. (See also <code>infoCriteria.asreml</code> .)                                    |
| <code>bound.exclusions</code> | A character specifying the bound (constraint) codes that will result in a variance parameter being excluded from the count of estimated variance parameters in calculating information criteria. If set to <code>NULL</code> then none will be excluded.                                                                                                                                                                                                                                                                                         |
| <code>...</code>              | further arguments passed to <code>wald.asreml</code> and <code>recalcWaldTab</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

**Value**

An object of S3-class `asrtests` that also inherits S3-class `list`.

**Author(s)**

Chris Brien

**References**

Kenward, M. G., & Roger, J. H. (1997). Small sample inference for fixed effects from restricted maximum likelihood. *Biometrics*, **53**, 983-997.

**See Also**

[asremlPlus-package](#), [is.alldiffs](#), [as.alldiffs](#), [recalcWaldTab](#),  
[testtranfix.asrtests](#), [chooseModel.asrtests](#), [rmboundary.asrtests](#),  
[reparamSigDevn.asrtests](#)

**Examples**

```
## Not run:
data(Wheat.dat)

# Fit initial model
current.asr <- asreml(yield ~ Rep + WithinColPairs + Variety,
                     random = ~ Row + Column + units,
                     residual = ~ ar1(Row):ar1(Column),
                     data=Wheat.dat)

# Load current fit into an asrtests object
current.asrt <- as.asrtests(current.asr, NULL, NULL)

# Check for and remove any boundary terms
current.asrt <- rmboundary(current.asrt)

## End(Not run)
```

---

|                                   |                                                                                                                                 |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <code>as.predictions.frame</code> | <i>Forms a <a href="#">predictions.frame</a> from a <code>data.frame</code>, ensuring that the correct columns are present.</i> |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------|

---

**Description**

Creates a [predictions.frame](#) from a [data.frame](#) by adding the class [predictions.frame](#) to it, and renaming the columns containing the predictions, `se`, `est.status` and `error.intervals`.

**Usage**

```
as.predictions.frame(data, classify = NULL,
                    predictions = NULL, se = NULL, est.status = NULL,
                    interval.type = NULL, interval.names = NULL)
```

**Arguments**

|                |                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data           | A <a href="#">data.frame</a> containing columns giving the variables that uniquely index the predicted values and columns with the predicted values, their standard errors and, optionally, their estimation status (est.status).                                                                                                                                                    |
| classify       | A <a href="#">character</a> string giving the variables that define the margins of the multiway table that was predicted. Multiway tables are specified by forming an interaction type term from the classifying variables, that is, separating the variable names with the : operator. For predicting the overall mean, the classify is set to "(Intercept)".                       |
| predictions    | A <a href="#">character</a> giving the name of the column in data that contains the predicted values. This column will be renamed to predicted.value.                                                                                                                                                                                                                                |
| se             | A <a href="#">character</a> giving the name of the column in data that contains the standard errors of the predicted values. This column will be renamed to standard.error.                                                                                                                                                                                                          |
| est.status     | A <a href="#">character</a> giving the name of the column in data that contains the estimation status of the predicted values. It will have a value Estimable for predicted values that have been estimated and a value Aliased for predicted values that are NA. If a column named est.status is not present in data and est.status is NULL, a column est.status will be generated. |
| interval.type  | A <a href="#">character</a> specifying the type of error.intervals stored in data that require renaming. If NULL, error.intervals will not be renamed, even if they are present. Otherwise, interval.type should be set to one of "CI", "SE" or "halfLSD".                                                                                                                           |
| interval.names | A <a href="#">character</a> specifying the column names of the lower and upper limits stored in data that are to be renamed. The character must be of length two, with the first element being the name of the 'lower' limit and the second element being the name of the 'upper' limit.                                                                                             |

**Value**

An S3-class [predictions.frame](#).

**Author(s)**

Chris Brien

**See Also**

[asremlPlus-package](#), [predictions.frame](#), [is.predictions.frame](#), [predictions.frame](#), [validPredictionsFrame](#)

**Examples**

```
data(Oats.dat)

## Use asreml to get predictions and associated statistics

## Not run:
m1.asr <- asreml(Yield ~ Nitrogen*Variety,
                 random=~Blocks/Wplots,
                 data=Oats.dat)
current.asrt <- as.asrtests(m1.asr)
Var.pred <- asreml::predict.asreml(m1.asr, classify="Nitrogen:Variety",
```



```

                                sed=TRUE)
  if (getASremlVersionLoaded(nchar = 1) == "3")
    Var.pred <- Var.pred$predictions
  #Form predictions.frame changing asreml-R4 names to the standard names, if these are present
  Var.preds <- as.predictions.frame(Var.pred$pvals, se = "std.error",
                                   est.status = "status")

## End(Not run)

## Use lmerTest and emmeans to get predictions and associated statistics
if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(Yield ~ Nitrogen*Variety + (1|Blocks/Wplots),
                           data=Oats.dat)
  Var.emm <- emmeans::emmeans(m1.lmer, specs = ~ Nitrogen:Variety)
  Var.preds <- summary(Var.emm)
  Var.preds <- as.predictions.frame(Var.preds, predictions = "emmean",
                                   se = "SE", interval.type = "CI",
                                   interval.names = c("lower.CL", "upper.CL"))
}

## Check the class and validity of the alldiffs object
if (exists("Var.preds"))
{
  is.predictions.frame(Var.preds)
  validPredictionsFrame(Var.preds)
}

```

---

asremlPlus-deprecated *Deprecated Functions in the Package asremlPlus*

---

## Description

These functions have been renamed and deprecated in asremlPlus:

1. addrm.terms.asreml and addrm.terms.asrtests -> [changeTerms.asrtests](#),
2. alldiffs -> [as.alldiffs](#),
3. asrtests-> [as.asrtests](#),
4. choose.model.asreml and choose.model.asrtests -> [chooseModel.asrtests](#),
5. facRecode and facRecode.alldiffs -> [facRecast.alldiffs](#),
6. info.crit and info.crit.asreml -> [infoCriteria.asreml](#),
7. newrcov.asrtests -> [changeTerms.asrtests](#),
8. plotvariofaces.asreml -> [plotVariofaces.data.frame](#),
9. power.transform -> [powerTransform](#),
10. predictiondiffs.asreml -> [allDifferences.data.frame](#),
11. predictionplot.asreml -> [plotPredictions.data.frame](#),
12. predictparallel.asreml -> [predictPlus.asreml](#),
13. pred.present.asreml -> [predictPresent.asreml](#),

14. `recalc.wald.tab.asreml` and `recalc.wald.tab.asrtests` -> `recalcWaldTab.asrtests`,
15. `reorderClassify` and `reorderClassify.alldiffs` -> `renewClassify.alldiffs`,
16. `reml.lrt` and `reml.lrt.asreml` -> `REMLRT.asreml`,
17. `rmboundary.asreml` -> `rmboundary.asrtests`,
18. `setvarianceterms.asreml` -> `setvarianceterms.call`,
19. `sig.devn.reparam.asreml` and `sig.devn.reparam.asrtests` -> `reparamSigDevn.asrtests`,
20. `testtranfix.asreml` -> `testtranfix.asrtests`,
21. `testrcov.asreml` and `testrcov.asrtests` -> `testresidual.asrtests`,
22. `testswapan.asreml` -> `testswapan.asrtests`

## Usage

```

addrm.terms.asreml(...)
addrm.terms.asrtests(...)
alldiffs(...)
asrtests(...)
choose.model.asreml(...)
choose.model.asrtests(...)
facRecode(...)
facRecode.alldiffs(...)
info.crit(...)
info.crit.asreml(...)
newrcov.asrtests(...)
plotvariofaces.asreml(...)
power.transform(...)
predictiondiffs.asreml(...)
predictionplot.asreml(...)
predictparallel.asreml(...)
pred.present.asreml(...)
recalc.wald.tab.asreml(...)
recalc.wald.tab.asrtests(...)
reml.lrt(...)
reml.lrt.asreml(...)
## S3 method for class 'alldiffs'
reorderClassify(...)
## S3 method for class 'asreml'
rmboundary(...)
setvarianceterms.asreml(...)
sig.devn.reparam.asreml(...)
sig.devn.reparam.asrtests(...)
testtranfix.asreml(...)
testrcov.asreml(...)
testrcov.asrtests(...)
## S3 method for class 'asreml'
testswapan(...)

```

## Arguments

...                      absorbs arguments passed from the old functions of the style `foo.bar()`.

**Author(s)**

Chris Brien

asremlPlusTips

*The randomly-presented, startup tips.***Description**

The intermittent, randomly-presented, startup tips.

**Startup tips**

Need help? The manual is a vignette and is in the vignettes subdirectory of the package's install directory.

Find out what has changed in asremlPlus: enter `news(package = 'asremlPlus')`.

Need help getting started? Enter `vignette(package = 'asremlPlus')`.

To avoid start-up message that ASReml-R is needed, load asreml before asremlPlus.

The methods for `alldiffs` and `data.frame` do not require asreml

Use `suppressPackageStartupMessages()` to eliminate all package startup messages.

To see all the intermittent, randomly-presented, startup tips enter `?asremlPlusTips`.

To install the latest version: go to <http://chris.brien.name/rpackages>.

For versions between CRAN releases (and more) go to <http://chris.brien.name/rpackages>.

**Author(s)**

Chris Brien

asrtests.object

*Description of an asrtests object***Description**

An object of S3-class `asrtests` that contains information derived from the fits of a mixed model using asreml.

`as.asrtests` is function that makes an object of this class.

`is.list` is the membership function for this class; it tests that an object is of class `list`.

`validAsrtests` can be used to test the validity of an `asrtests.object`.

**Value**

A list that contains three components:

1. `asreml.obj`: an object of class `asreml` that contains the fit of a model;
2. `wald.tab`: A `data.frame` containing a pseudo-anova table for the fixed terms produced by `wald.asreml`. It has rownames that correspond to the fixed terms that were fitted and four columns. If denominator degrees of freedom were calculated then the columns are `DF`, `denDF`, `F.inc`, `Pr`; otherwise the columns are `Df`, `Sum of Sq`, `Wald statistic`, and `Pr(Chisq)`.
3. `test.summary`: A `data.frame` with columns `terms`, `DF`, `denDF`, `p`, `AIC`, `BIC` and `action`, each row of which summarizes the results of proposed changes to the fitted model.  
Possible codes for `action` are: `Dropped`, `Retained`, `Swapped`, `Unswapped`, `Unchanged`, `Significant`, `Nonsignificant`, `Absent`, `Added`, `Removed` and `Boundary`. If the either of the models did not converge, `unconverged` will be added to the code. `Unchanged` is used when `allow.unconverged` is `FALSE`. Note that the logical `asreml.obj$converge` also reflects whether there is convergence.  
A row is added to the `test.summary` for each term that is dropped, added or tested or a note that several terms have been added or removed. When values for the `AIC` and `BIC` are included in the row, then the `DF` are the number of fixed parameters in the model and `denDF` are the numbers of variance parameters. When `changeModelOnIC` adds a row then the values of the degrees of freedom and information criteria are differences between those for the model that is supplied and the model changed by `changeModelOnIC`.

**Author(s)**

Chris Brien

**See Also**

[as.asrtests](#), [as.asrtests](#), [validAsrtests](#)

---

|                                |                                                                                                            |
|--------------------------------|------------------------------------------------------------------------------------------------------------|
| <code>bootREMLRT.asreml</code> | <i>Uses the parametric bootstrap to calculate the p-value for a REML ratio test to compare two models.</i> |
|--------------------------------|------------------------------------------------------------------------------------------------------------|

---

**Description**

Extracts the REML log likelihood for two `asreml` objects and forms the observed REML ratio statistic. It assumes that the second `asreml` object is the result of fitting a model that is a reduced version of the model for the first object and is considered to the null model. Using the mean and `V`, `nboot` bootstrap samples of simulated response values are generated in parallel; that is, `ncores` cores are used and each is used to generate and analyse a sample. The full and reduced models are fitted to the data and if either analysis fails to converge another sample is generated and analysed using the current core, with a maximum of `max.retries` attempts to obtain a sample that converges for both analysis. Thus the maximum number of data sets that will be generated is `nboot * max.retries`. If a bootstrap sample converges for both analyses, the REML ratio test statistic is formed for it. The p-value is then calculated as  $(k + 1)/(b + 1)$  where  $k$  is the number of simulated ratio test statistics greater than the observed test statistic and  $s$  is the number of bootstrap samples that were returned.

The function checks that the models do not differ in either their fixed or sparse models. It also check the difference in the number of variance parameters between the two fits to the models, taking into account the `bound.exclusions`.

**Usage**

```
## S3 method for class 'asreml'
bootREMLRT(h0.asreml.obj, h1.asreml.obj,
  nboot = 100, max.retries = 5, seed = NULL,
  means=NULL, V = NULL, extra.matrix = NULL, ignore.terms = NULL,
  fixed.spline.terms = NULL,
  bound.exclusions = c("F","B","S","C"),
  tolerance = 1E-10, update = TRUE, trace = FALSE,
  ncores = 2, ...)
```

**Arguments**

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>h0.asreml.obj</code>      | asreml object containing the fit under the model for the null hypothesis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>h1.asreml.obj</code>      | asreml object containing the fit under the model for the alternative hypothesis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>nboot</code>              | The number of bootstrap samples to be generated.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>max.retries</code>        | The maximum number of attempts to generate a sample whose analyses converge for both models.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>seed</code>               | A single value, interpreted as an integer, that specifies the starting value of the random number generator. The "L'Ecuyer-CMRG" random generator is used and <code>nextRNGStream</code> is used to seed each core from the original seed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>means</code>              | The vector of means to be used in generating simulated bootstrap samples. If it is <code>NULL</code> , the fitted values based on object are used. It must be the same length as the response variable for object.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>V</code>                  | The fitted variance matrix, i.e. having the pattern and values that conform to the model fit stored in the supplied object. If it is <code>NULL</code> , <code>estimateV.asreml</code> is used to estimate the variance matrix for the observations from the variance parameter estimates from the <code>reduced.asreml.obj</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>extra.matrix</code>       | A matrix of order equal to the number of observations that is to be added to the variance matrix, the latter based on the information in <code>asreml.obj</code> . It is assumed that the sigma-parameterized values of the variance parameter estimates, such as is given in the <code>varcomp</code> component of <code>summary.asreml</code> , have been used in calculating <code>extra.matrix</code> ; the values in the <code>vparameters</code> component of <code>G.param</code> and <code>R.param</code> may be either gamma- or sigma-parameterized. The argument <code>extra.matrix</code> can be used in conjunction with <code>ignore.terms</code> as a workaround to include components of the variance matrix for variance functions that have not been implemented in <code>estimateV</code> . |
| <code>ignore.terms</code>       | A character giving terms from either the random or residual models that are to be ignored in that their contributions to the variance is not to be included in the estimated matrix. The term names are those given in the <code>vparameters</code> component of the <code>asreml</code> object or the <code>varcomp</code> component produced by <code>summary.asreml</code> , but only up to the first exclamation mark (!). This can be used in conjunction with <code>estimateV.asreml</code> as a workaround to include components of the variance matrix for variance functions that have not been implemented in <code>estimateV</code> .                                                                                                                                                               |
| <code>fixed.spline.terms</code> | A character vector giving one or more spline terms in the random model that are regarded as fixed and so are to be ignored because they are not regarded as contributing to the variance. The term names are those given in the <code>vparameters</code> component of the <code>asreml</code> object or the <code>varcomp</code> component produced by <code>summary.asreml</code> , but only up to the first exclamation mark (!).                                                                                                                                                                                                                                                                                                                                                                            |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bound.exclusions | A character specifying one or more bound codes that will result in a variance parameter in the random model being excluded from contributing to the variance. If set to NULL then none will be excluded.                                                                                                                                                                                                                           |
| tolerance        | The value such that eigenvalues less than it are considered to be zero.                                                                                                                                                                                                                                                                                                                                                            |
| update           | If TRUE then the arguments R.param and G.param are set to those in the asreml object supplied in object so that the values from the original model are used as starting values. If FALSE then asreml calls are evaluated, the only changes from the previous call being that (i) the model is fitted to simulated data and (ii) modifications specified via ... are made, except that changes cannot be made to any of the models. |
| trace            | If TRUE then partial iteration details are displayed when ASReml-R functions are invoked; if FALSE then no output is displayed.                                                                                                                                                                                                                                                                                                    |
| ncores           | A numeric specifying the number of cores to use in doing the simulations. In choosing a value for ncores, it is necessary to take into account other processes that are using parallel processing at the same time.                                                                                                                                                                                                                |
| ...              | Other arguments that are passed down to the function asreml. Changes to the models are not allowed. Other changes are dangerous and generally should be avoided.                                                                                                                                                                                                                                                                   |

## Value

A list with the following components:

1. **REMLRT**: the observed REML ratio statistic.
2. **p**: the bootstrap p-value for the observed test statistic.
3. **DF**: the calculated difference in DF for the variance parameters in the two models.
4. **totalunconverged**: the total number of unconverged analyses over the simulations.
5. **REMLRT.sim**: a numeric containing the values of the ratio statistics for the simulated data. It has an attribute called na.action that can be retrieved using attr(REMLRT.sim, which = "na.action"); it contains a list of the simulation numbers that were abandoned because max.retries failed to converge for both models.
6. **nunconverged**: the number of unconverged analyses for each bootstrap sample, the maximum being max.retries.

## Note

A bootstrap sample is generated using a multivariate normal distribution with expected value as specified by means and variance matrix given by V. Each simulated sample is analysed according to the reduced model and, provided this analysis converges, according to the full.model. If one of these analyses fails to converge, it is abandoned and another sample is generated for this simulation. As many as max.retries attempts are made to generate a data set for which both analyses converge. If data set that converges for both analyses is not generated for a simulation, NA is returned for that bootstrap sample. Hence, the maximum number of data sets that will be generated is nboot \* max.retries and less than nboot samples will be generated if a data set that converges for both analyses is not obtained within max.retries attempts.

If a bootstrap sample converges for both analyses, the REML ratio test statistic is calculated as  $2(\log(REML)_F - \log(REML)_R)$ .

The DF is calculated from the information in full.asreml.obj and reduced.asreml.obj. The degrees of freedom are computed as the difference between the two models in the number of variance parameters whose estimates do not have a code for bound specified in bound.exclusions.

If ASReml-R version 4 is being used then the codes specified in `bound.exclusions` are not restricted to a subset of the default codes, but a warning is issued if a code other than these is specified. For ASReml-R version 3, only a subset of the default codes are allowed: F (Fixed), B (Boundary), C (Constrained) and S (Singular).

### Author(s)

Chris Brien

### See Also

[REMLRT.asreml](#), [infoCriteria.asreml](#), [newfit.asreml](#), [testranfix.asrtests](#)

### Examples

```
## Not run:
  bootREMLRT(ICV.max, ICV.red, ncores = parallel::detectCores())

## End(Not run)
```

---

changeModelOnIC.asrtests

*Uses information criteria to decide whether to change an already fitted model.*

---

### Description

Uses information criteria to decide whether to change the fitted model stored in the supplied [asrtests.object](#) according to the specified modifications. The function [changeTerms](#) is used to change the model. Thus, the model can be modified using a combination of adding and removing sets of terms from one or both of the fixed or random models, replacing the residual model and changing the bounds and/or initial values of some terms. The model will be unchanged if terms specified in `dropFixed` or `dropRandom` are not in the fitted model.

A row is added to the `test.summary` data.frame of the [asrtests.object](#) using the supplied label and stating whether or not the new model has been swapped for the supplied model. Convergence in fitting the model is checked and a note included in the action if there was not. All components of the [asrtests.object](#) are updated to exhibit the differences between the supplied and new models.

To obtain a list of the information criteria for a set of models use [changeTerms.asrtests](#) with `IClikelihood` set to REML or full, or use [infoCriteria.asreml](#).

### Usage

```
## S3 method for class 'asrtests'
changeModelOnIC(asrtests.obj,
  dropFixed = NULL, addFixed = NULL,
  dropRandom = NULL, addRandom = NULL,
  newResidual = NULL,
  allow.absentDropTerms = FALSE, label = "Changed terms",
  allow.unconverged = TRUE, allow.fixedcorrelation = TRUE,
  checkboundaryonly = FALSE,
```

```

trace = FALSE, update = TRUE, denDF = "numeric",
set.terms = NULL, ignore.suffices = TRUE,
bounds = "P", initial.values = NA,
which.IC = "AIC", IClikelihood = "REML",
fixedDF = NULL, varDF = NULL,
bound.exclusions = c("F", "B", "S", "C"),
...)
```

## Arguments

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| asrtests.obj           | An <a href="#">asrtests.object</a> containing the components (i) <code>asreml.obj</code> , (ii) <code>wald.tab</code> , and (iii) <code>test.summary</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| dropFixed              | A single character string in the form of a formula which, after addition of <code>" ~ . -"</code> and after expansion, specifies the sum of a set of terms to be dropped from the fixed formula. The names must match those in the <code>wald.tab</code> component of the <code>asrtests.obj</code> . The fixed terms will be reordered so that single-variable terms come first, followed by two-variable terms and so on. Note that multiple terms specified using a single <code>asreml::at</code> function can only be dropped as a whole. If the term was specified using an <code>asreml::at</code> function with a single level, then it can be removed and either the level itself or its <a href="#">numeric</a> position in the levels returned by the <a href="#">levels</a> function can be specified. |
| addFixed               | A single character string in the form of a formula which, after addition of <code>" ~ . +"</code> and expansion, specifies the sum of a set of terms to be added to the fixed formula. The fixed terms will be reordered so that single-variable terms come first, followed by two-variable terms and so on.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| dropRandom             | A single character string in the form of a formula which, after addition of <code>" ~ . -"</code> and expansion, specifies the sum of a set of terms to be dropped from the random formula. The names must match those in the <code>vparameters</code> component of the <code>asreml.obj</code> component in the <code>asrtests.obj</code> . Note that multiple terms specified using a single <code>asreml::at</code> function can only be dropped as a whole. If the term was specified using an <code>asreml::at</code> function with a single level, then it can be removed and either the level itself or its <a href="#">numeric</a> position in the levels returned by the <a href="#">levels</a> function can be specified.                                                                                |
| addRandom              | A single character string in the form of a formula which, after addition of <code>" ~ . +"</code> and expansion, specifies the sum of a set of terms to be added to the random formula.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| newResidual            | A single character string in the form of a formula which, after addition of <code>" ~ "</code> , specifies the residual (or <code>rcov</code> ) model. To remove the model, enter <code>"-(.)"</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| allow.absentDropTerms  | A logical indicating whether to change the model when terms specified in <code>dropFixed</code> or <code>dropRandom</code> are not in the fitted model.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| label                  | A character to use as an entry in the terms column in <code>test.summary</code> to indicate as far as is possible the terms that are being manipulated.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| allow.unconverged      | A logical indicating whether to accept a new model even when it does not converge. If <code>FALSE</code> and the fit of the new model does not converge, the supplied <code>asrtests.obj</code> is returned. Also, if <code>FALSE</code> and the fit of the new model has converged, but that of the old model has not, the new model will be accepted.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| allow.fixedcorrelation | A logical indicating whether to accept a new model even when it contains correlations in the model whose values have been designated as fixed, bound                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |



|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | or singular. If FALSE and the new model contains correlations whose values have not been able to be estimated, the supplied <code>asrtests.obj</code> is returned. The fit in the <code>asreml.obj</code> component of the supplied <code>asrtests.obj</code> will also be tested and a warning issued if both fixed correlations are found in it and <code>allow.fixedcorrelation</code> is FALSE.                                                                                                                                                                                                                                                                   |
| <code>checkboundaryonly</code> | If TRUE then boundary and singular terms are not removed by <code>rmboundary.asrtests</code> ; a warning is issued instead.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>trace</code>             | If TRUE then partial iteration details are displayed when ASReml-R functions are invoked; if FALSE then no output is displayed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>update</code>            | If TRUE, and <code>set.terms</code> is NULL, then <code>newfit.asreml</code> is called to fit the model to be tested, using the values of the variance parameters stored in the <code>asreml.object</code> , that is stored in <code>asrtests.obj</code> , as starting values. If FALSE or <code>set.terms</code> is not NULL, then <code>newfit.asreml</code> will not use the stored variance parameter values as starting values when fitting the new model, the only modifications being (i) those specified by this function's arguments and (ii) those specified via . . . .                                                                                    |
| <code>denDF</code>             | Specifies the method to use in computing approximate denominator degrees of freedom when <code>wald.asreml</code> is called. Can be none to suppress the computations, numeric for numerical methods, algebraic for algebraic methods or default, the default, to automatically choose numeric or algebraic computations depending on problem size. The denominator degrees of freedom are calculated according to Kenward and Roger (1997) for fixed terms in the dense part of the model.                                                                                                                                                                           |
| <code>set.terms</code>         | A character vector specifying the terms that are to have bounds and/or initial values set prior to fitting the new model. The names must match those in the <code>vparameters</code> component of the <code>asreml.obj</code> component in the new <code>asrtests.object</code> . The terms in the model do not need to change from those in the model in the supplied <code>asrtests.obj</code> .                                                                                                                                                                                                                                                                    |
| <code>ignore.suffices</code>   | A logical vector specifying whether the suffices of the <code>asreml</code> -assigned names of the variance terms (i.e. the information to the right of an "!", other than "R!") is to be ignored in matching elements of terms. If TRUE for an element of terms, the suffices are stripped from the <code>asreml</code> -assigned names. If FALSE for an element of terms, the element must exactly match an <code>asreml</code> -assigned name for a variance term. This vector must be of length one or the same length as terms. If it is of length one then the same action is applied to the <code>asreml</code> -assigned suffices for all the terms in terms. |
| <code>bounds</code>            | A <code>character</code> vector specifying the bounds to be applied to the terms specified in <code>set.terms</code> . This vector must be of length one or the same length as <code>set.terms</code> . If it is of length one then the same constraint is applied to all the terms in <code>set.terms</code> . If any of the bounds are equal to NA then they are left unchanged for those terms.                                                                                                                                                                                                                                                                    |
| <code>which.IC</code>          | A character specifying the information criterion to be used in selecting the best model. Possible values are AIC and BIC. The value of the criterion for supplied model must exceed that for changed model for the changed model to be returned.                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>IClikelihood</code>      | A character specifying whether Restricted Maximum Likelihood (REML) or the full likelihood, evaluated using REML estimates, ( <code>full</code> ) are to be used in calculating the information criteria for choosing between models when family is set to <code>asr_gaussian</code> . For family set to <code>asr_binomial</code> or <code>asr_poisson</code> and                                                                                                                                                                                                                                                                                                    |

|                  |                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | with dispersion set to 1, the deviance is extracted from object and used to calculate the AIC and BIC (as suggested by Damian Collins); the setting of IClikelihood is ignored and the log-likelihood set to NA. The information criteria are not valid for other settings of family and dispersion.                                                                                         |
| fixedDF          | A numeric giving the number of estimated fixed parameters. If NULL then this is determined from the information in <code>asreml.obj</code> .                                                                                                                                                                                                                                                 |
| varDF            | A numeric giving the number of estimated variance parameters. If NULL then this is determined from the information in <code>asreml.obj</code> . It replaces the <code>varDF</code> argument.                                                                                                                                                                                                 |
| initial.values   | A character vector specifying the initial values for the terms specified in <code>terms</code> . This vector must be of length one or the same length as <code>terms</code> . If it is of length one then the same initial value is applied to all the terms in <code>terms</code> . If any of the <code>initial.values</code> are equal to NA then they are left unchanged for those terms. |
| bound.exclusions | A character specifying the bound (constraint) codes that will result in a variance parameter being excluded from the count of estimated variance parameters in calculating the information criteria. If set to NULL then none will be excluded.                                                                                                                                              |
| ...              | Further arguments passed to <code>asreml</code> , <code>wald.asreml</code> and <a href="#">as.asrtests</a> .                                                                                                                                                                                                                                                                                 |

### Value

An [asrtests.object](#) containing the components (i) `asreml.obj`, (ii) `wald.tab`, and (iii) `test.summary`. The values of the degrees of freedom and the information criteria are differences between those of the changed model and those of the model supplied to `changeModelOnIC`.

### Author(s)

Chris Brien

### See Also

[as.asrtests](#), [rmboundary.asrtests](#), [testranfix.asrtests](#), [testresidual.asrtests](#), [newfit.asreml](#), [reparamSigDevn.asrtests](#), [chooseModel.asrtests](#), [changeTerms.asrtests](#), [infoCriteria.asreml](#)

### Examples

```
## Not run:

data(Wheat.dat)
current.asr <- asreml(yield ~ Rep + WithinColPairs + Variety,
  random = ~ Row + Column + units,
  residual = ~ ar1(Row):ar1(Column),
  data=Wheat.dat)
current.asrt <- as.asrtests(current.asr, NULL, NULL,
  label = "Maximal model")
current.asrt <- rmboundary(current.asrt)

# Drop both Row and Column terms
current.asrt <- changeModelOnIC(current.asrt,
  dropRandom = "Row + Column",
  checkboundaryonly = TRUE,
  which.IC = "AIC", IClikelihood = "full")
```

```

current.asrt <- iterate(current.asrt)

# Add and drop both fixed and random terms
current.asrt <- changeModelOnIC(current.asrt,
                               addFixed = "vRow", dropFixed = "WithinColPairs",
                               addRandom = "spl(vRow)", dropRandom = "units",
                               checkboundaryonly = TRUE,
                               which.IC = "AIC", IClkelihood = "full")

# Replace residual with model without Row autocorrelation
current.asrt <- changeModelOnIC(current.asrt,
                               newResidual = "Row:ar1(Column)",
                               label="Row autocorrelation",
                               IClkelihood = "full")

## End(Not run)

```

---

|                      |                                                                                                                                                                               |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| changeTerms.asrtests | <i>Adds and drops terms from one or both of the fixed or random model, replaces the residual (rcov) model with a new model and changes bounds or initial values of terms.</i> |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

---

## Description

The specified terms are simply added or dropped, without testing, from either the fixed or random model and/or the residual (rcov) model replaced. Also, the bounds and/or initial values of some terms can be changed. No hypothesis testing is performed, but a check is made for boundary or singular terms.

A row is added to the test.summary data.frame of the `asrtests.object` using the supplied label and stating which models have been changed. Information criteria can be included in the row of the test.summary. Convergence in fitting the model is checked and a note included in the action if there was not. All components of the `asrtests.object` are updated.

To only change the terms based on a comparison of information criteria use `changeModelOnIC.asrtests`.

## Usage

```

## S3 method for class 'asrtests'
changeTerms(asrtests.obj,
            dropFixed = NULL, addFixed = NULL,
            dropRandom = NULL, addRandom = NULL,
            newResidual = NULL, label = "Changed terms",
            allow.unconverged = TRUE, allow.fixedcorrelation = TRUE,
            checkboundaryonly = FALSE,
            trace = FALSE, update = TRUE, denDF = "numeric",
            set.terms = NULL, ignore.suffices = TRUE,
            bounds = "P", initial.values = NA,
            IClkelihood = "none", bound.exclusions = c("F","B","S","C"),
            ...)

```

**Arguments**

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| asrtests.obj           | An <a href="#">asrtests.object</a> containing the components (i) <code>asrem1.obj</code> , (ii) <code>wald.tab</code> , and (iii) <code>test.summary</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| dropFixed              | A single character string in the form of a formula which, after addition of <code>" ~ . -"</code> and after expansion, specifies the sum of a set of terms to be dropped from the fixed formula. The names must match those in the <code>wald.tab</code> component of the <code>asrtests.obj</code> . The fixed terms will be reordered so that single-variable terms come first, followed by two-variable terms and so on. Note that multiple terms specified using a single <code>asrem1::at</code> function can only be dropped as a whole. If the term was specified using an <code>asrem1::at</code> function with a single level, then it can be removed and either the level itself or its <a href="#">numeric</a> position in the levels returned by the <a href="#">levels</a> function can be specified. |
| addFixed               | A single character string in the form of a formula which, after addition of <code>" ~ . +"</code> and expansion, specifies the sum of a set of terms to be added to the fixed formula. The fixed terms will be reordered so that single-variable terms come first, followed by two-variable terms and so on.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| dropRandom             | A single character string in the form of a formula which, after addition of <code>" ~ . -"</code> and expansion, specifies the sum of a set of terms to be dropped from the random formula. The terms must match those in the <code>vparameters</code> component of the <code>asrem1.obj</code> component in the <code>asrtests.obj</code> . Note that multiple terms specified using a single <code>asrem1::at</code> function can only be dropped as a whole. If the term was specified using an <code>asrem1::at</code> function with a single level, then it can be removed and either the level itself or its <a href="#">numeric</a> position in the levels returned by the <a href="#">levels</a> function can be specified.                                                                                |
| addRandom              | A single character string in the form of a formula which, after addition of <code>" ~ . +"</code> and expansion, specifies the sum of a set of terms to be added to the random formula.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| newResidual            | A single character string in the form of a formula which, after addition of <code>" ~ "</code> , specifies the residual (or rcov) model. To remove the model, enter <code>"-(.)"</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| label                  | A character to use as an entry in the terms column in <code>test.summary</code> to indicate as far as is possible the terms that are being manipulated.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| allow.unconverged      | A logical indicating whether to accept a new model even when it does not converge. If FALSE and the fit does not converge, the supplied <code>asrtests.obj</code> is returned.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| allow.fixedcorrelation | A logical indicating whether to accept a new model even when it contains correlations in the model whose values have been designated as fixed, bound or singular. If FALSE and the new model contains correlations whose values have not been able to be estimated, the supplied <code>asrtests.obj</code> is returned. The fit in the <code>asrem1.obj</code> component of the supplied <code>asrtests.obj</code> will also be tested and a warning issued if both fixed correlations are found in it and <code>allow.fixedcorrelation</code> is FALSE.                                                                                                                                                                                                                                                           |
| checkboundaryonly      | If TRUE then boundary and singular terms are not removed by <a href="#">rmboundary.asrtests</a> ; a warning is issued instead.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| trace                  | If TRUE then partial iteration details are displayed when ASReml-R functions are invoked; if FALSE then no output is displayed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| update                 | If TRUE, and <code>set.terms</code> is NULL, then <a href="#">newfit.asrem1</a> is called to fit the model to be tested, using the values of the variance parameters stored in the <code>asrem1.object</code> ,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | that is stored in <code>asrtests.obj</code> , as starting values. If <code>FALSE</code> or <code>set.terms</code> is not <code>NULL</code> , then <code>newfit.asreml</code> will not use the stored variance parameter values as starting values when fitting the new model, the only modifications being (i) those specified by this function's arguments and (ii) those specified via . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>denDF</code>            | Specifies the method to use in computing approximate denominator degrees of freedom when <code>wald.asreml</code> is called. Can be <code>none</code> to suppress the computations, <code>numeric</code> for numerical methods, <code>algebraic</code> for algebraic methods or <code>default</code> , the default, to automatically choose numeric or algebraic computations depending on problem size. The denominator degrees of freedom are calculated according to Kenward and Roger (1997) for fixed terms in the dense part of the model.                                                                                                                                                                                                                                                                                                                      |
| <code>set.terms</code>        | A character vector specifying the terms that are to have bounds and/or initial values set prior to fitting the new model. The names must match those in the <code>vparameters</code> component of the <code>asreml.obj</code> component in the new <code>asrtests.object</code> . The terms in the model do not need to change from those in the model in the supplied <code>asrtests.obj</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>ignore.suffices</code>  | A logical vector specifying whether the suffices of the <code>asreml</code> -assigned names of the variance terms (i.e. the information to the right of an "!", other than "R!") is to be ignored in matching elements of terms. If <code>TRUE</code> for an element of terms, the suffices are stripped from the <code>asreml</code> -assigned names. If <code>FALSE</code> for an element of terms, the element must exactly match an <code>asreml</code> -assigned name for a variance term. This vector must be of length one or the same length as <code>terms</code> . If it is of length one then the same action is applied to the <code>asreml</code> -assigned suffices for all the terms in <code>terms</code> .                                                                                                                                           |
| <code>bounds</code>           | A <code>character</code> vector specifying the bounds to be applied to the terms specified in <code>set.terms</code> . This vector must be of length one or the same length as <code>set.terms</code> . If it is of length one then the same constraint is applied to all the terms in <code>set.terms</code> . If any of the bounds are equal to <code>NA</code> then they are left unchanged for those terms.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>initial.values</code>   | A character vector specifying the initial values for the terms specified in <code>terms</code> . This vector must be of length one or the same length as <code>terms</code> . If it is of length one then the same initial value is applied to all the terms in <code>terms</code> . If any of the <code>initial.values</code> are equal to <code>NA</code> then they are left unchanged for those terms.                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>IClikelihood</code>     | A character that controls both the occurrence and the type of likelihood for information criterion in the <code>test.summary</code> of the new <code>asrtests.object</code> . If <code>none</code> , none are included. Otherwise, if <code>REML</code> and <code>family</code> is set to <code>asr_guassian</code> (the default), then the AIC and BIC based on the Restricted Maximum Likelihood are included; if <code>full</code> and <code>family</code> is set to <code>asr_guassian</code> , then the AIC and BIC based on the full likelihood, evaluated using <code>REML</code> estimates, are included. if <code>family</code> is <code>asr_binomial</code> or <code>asr_poisson</code> , with <code>dispersion</code> set to 1, the deviance is extracted from object and used to calculate the AIC and BIC. (See also <code>infoCriteria.asreml</code> .) |
| <code>bound.exclusions</code> | A character specifying the bound (constraint) codes that will result in a variance parameter being excluded from the count of estimated variance parameters in calculating the information criteria. If set to <code>NULL</code> then none will be excluded.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>...</code>              | Further arguments passed to <code>asreml</code> , <code>wald.asreml</code> and <code>as.asrtests</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## Value

An `asrtests.object` containing the components (i) `asreml.obj`, (ii) `wald.tab`, and (iii) `test.summary`.

**Author(s)**

Chris Brien

**References**

Kenward, M. G., & Roger, J. H. (1997). Small sample inference for fixed effects from restricted maximum likelihood. *Biometrics*, **53**, 983-997.

**See Also**

[as.asrtests](#), [rmboundary.asrtests](#), [testranfix.asrtests](#), [testresidual.asrtests](#), [newfit.asreml](#), [reparamSigDevn.asrtests](#), [chooseModel.asrtests](#), [changeModelOnIC.asrtests](#), [infoCriteria.asreml](#)

**Examples**

```
## Not run:
terms <- "(Date/(Sources * (Type + Species)))"
current.asrt <- changeTerms(current.asrt, addFixed = terms)

current.asrt <- changeTerms(current.asrt, dropFixed = "A + B", denDF = "algebraic")

data(Wheat.dat)
current.asr <- asreml(yield ~ Rep + WithinColPairs + Variety,
  random = ~ Row + Column + units,
  residual = ~ ar1(Row):ar1(Column),
  data=Wheat.dat)
current.asrt <- as.asrtests(current.asr, NULL, NULL)
current.asrt <- rmboundary(current.asrt)
# Add and drop both fixed and random terms
current.asrt <- changeTerms(current.asrt,
  addFixed = "vRow", dropFixed = "WithinColPairs",
  addRandom = "spl(vRow)", dropRandom = "units",
  checkboundaryonly = TRUE)
# Replace residual with model without Row autocorrelation
current.asrt <- changeTerms(current.asrt,
  newResidual = "Row:ar1(Column)",
  label="Row autocorrelation")

## End(Not run)
```

ChickpeaEnd.dat

*A large data set comprising the end of imaging data from a chick pea experiment conducted in high-throughput greenhouses*

**Description**

The data collected after imaging had been completed on the 1056 plants in the experiment reported by Atieno et al. (2017). The design employed for the experiment was a split-plot design in which two consecutive carts formed a main plot. The split-plot design assigned 245 genotypes to main plots, the genotypes being unequally replicated 2 or 3 times. Treatments (non-saline, saline) were randomized to the two subplots (carts) within each main plot.

The columns in the `data.frame` are: Smarthouse, Lane, Position, Zone, Mainplot, Subplot, Replicate, xLane, xPosition, Genotypes, Treatments, Biomass, PlantHeight, SenescenceRank, TotalPods, FilledPods, EmptyPods, SeedNo, TotalSeedWt, SeedWt100.

The columns Smarthouse, Lane and Position uniquely identify the rows of observations. Zones are groups of 4 Lanes, Mainplots are the 44 pairs of consecutive Subplots within each Zone, and a Subplot is a cart containing a single plant. The columns xLane and xPosition are numeric covariates for location within a Smarthouse. Genotypes and Treatments indicate the genotype and treatment that each plant was allocated. The response variables are Biomass, PlantHeight, SenescenceRank, TotalPods, FilledPods, EmptyPods, SeedNo, TotalSeedWt and SeedWt100.

### Usage

```
data(ChickpeaEnd.dat)
```

### Format

A `data.frames` with 1056 rows by 20 columns.

### References

Atieno, J., Li, Y., Langridge, P., Dowling, K., Brien, C., Berger, B., Varshney, R. K., and Sutton, T. (2017). Exploring genetic variation for salinity tolerance in chickpea using image-based phenotyping. *Scientific Reports*, 7, 1300. doi:[10.1038/s41598017012117](https://doi.org/10.1038/s41598017012117)

---

|             |                                                                                                                                                                                     |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| chooseModel | <i>Determines the set of significant terms using p-values and records the tests performed in a <code>data.frame</code>, taking into account the marginality relations of terms.</i> |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

---

### Description

Using p-values from hypothesis tests, determines the set of significant terms, taking into account the hierarchy or marginality of terms. In particular, a term will not be tested if it is marginal to (or nested in) one that is significant. For example, if A:B is significant, then neither A nor B will be tested. The tests conducted in choosing selected model are listed in a summary `data.frame`.

### Usage

```
chooseModel(object, ...)
```

### Arguments

|                     |                                                                            |
|---------------------|----------------------------------------------------------------------------|
| <code>object</code> | an object using which p-values can be obtained for use in model selection. |
| <code>...</code>    | further arguments passed to or from other methods.                         |

### Details

`chooseModel` is the generic function for the `chooseModel` method. Use `methods("chooseModel")` to get all the methods for the `chooseModel` generic.

`chooseModel.asrtests` is a method for an `asrtests.object`. It uses `testranfix.asrtests` to conduct tests to determine the p-values used in the model selection.

`chooseModel.data.frame` is a method for a `data.frame`. It uses the p-values stored in the `data.frame` in the model selection.

**Author(s)**

Chris Brien

**See Also**[chooseModel.asrtests](#), [chooseModel.asrtests](#), [changeModelOnIC.asrtests](#), [testranfix.asrtests](#)


---

|                      |                                                                                                                                                                               |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| chooseModel.asrtests | <i>Determines and records the set of significant terms using an <a href="#">asrtests.object</a>, taking into account the hierarchy or marginality relations of the terms.</i> |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

---

**Description**

Performs a series of hypothesis tests on a set of fixed and/or random terms taking into account the marginality of terms. In particular, a term will not be tested if it is marginal to (or nested in) one that is significant. For example, if A:B is significant, then neither A nor B will be tested. For a random term, the term is removed from the model fit, any boundary terms are removed using [rmboundary.asrtests](#) and a REML likelihood ratio test is performed using [REMLRT.asrem1](#). If it is not significant and `drop.ran.ns` is TRUE, the term is permanently removed from the model. Note that if boundary terms are removed, the reduced model may not be nested in the full model in which case the test is not valid. For fixed terms, the Wald tests are performed and the p-value for the term obtained. If it is not significant and `drop.fix.ns` is TRUE, the term is permanently removed from the model. A row that records the outcome of a test is added to `test.summary` for each term that is tested.

**Usage**

```
## S3 method for class 'asrtests'
chooseModel(object, terms.marginality=NULL,
            alpha = 0.05, allow.unconverged = TRUE,
            allow.fixedcorrelation = TRUE,
            checkboundaryonly = FALSE, drop.ran.ns=TRUE,
            positive.zero = FALSE, bound.test.parameters = "none",
            drop.fix.ns=FALSE, denDF = "numeric", dDF.fault = "none",
            dDF.values = NULL, trace = FALSE, update = TRUE,
            set.terms = NULL, ignore.suffices = TRUE,
            bounds = "P", initial.values = NA,
            IClikelihood = "none", ...)
```

**Arguments**

`object` an [asrtests.object](#) containing the components (i) `asrem1.obj`, (ii) `wald.tab`, and (iii) `test.summary`.

`terms.marginality`

A square matrix of ones and zeros with row and column names being the names of the terms to be tested. The names of fixed terms must match those in the `wald.tab` component of the object, while the names of random terms must match those in the `vparameters` component of the `asrem1.obj` component in the object. The diagonal elements of the matrix should be one, indicating that a term is marginal to itself. Elements should be one if the row term is marginal to the column term. All other elements should be zero.



|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| alpha                  | The significance level for the test.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| allow.unconverged      | A logical indicating whether to accept a new model even when it does not converge. If FALSE and a fit when a term is removed does not converge, the term will not be removed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| allow.fixedcorrelation | A logical indicating whether to accept a new model even when it contains correlations in the model whose values have been designated as fixed, bound or singular. If FALSE and the new model contains correlations whose values have not been able to be estimated, the supplied <code>asrtests.obj</code> is returned. The fit in the <code>asreml.obj</code> component of the supplied <code>asrtests.obj</code> will also be tested and a warning issued if both fixed correlations are found in it and <code>allow.fixedcorrelation</code> is FALSE.                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| checkboundaryonly      | If TRUE then boundary and singular terms are not removed by <code>rmboundary.asrtests</code> ; a warning is issued instead.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| drop.ran.ns            | A logical indicating whether to drop nonsignificant random terms from the model.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| positive.zero          | Indicates whether the hypothesized values for the variance components being tested are on the boundary of the parameter space. For example, this is true for positively-constrained variance components that, under the reduced model, are zero. This argument does not need to be set if <code>bound.test.parameters</code> is set.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| bound.test.parameters  | Indicates whether for the variance components being tested, at least some of the hypothesized values are on the boundary of the parameter space. The possibilities are "none", "onlybound" and "one-and-one". The default is "none", although if it is set to "none" and <code>positive.zero</code> is TRUE then <code>bound.test.parameters</code> is taken to be "onlybound". When <code>bound.test.parameters</code> is set to "one-and-one", it signifies that there are two parameters being tested, one of which is bound and the other is not. For example, the latter is true for testing a covariance and a positively-constrained variance component that, under the reduced model, are zero.                                                                                                                                                                                                                                                                                                                |
| drop.fix.ns            | A logical indicating whether to drop a fixed term from the model when it is nonsignificant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| denDF                  | Specifies the method to use in computing approximate denominator degrees of freedom when <code>wald.asreml</code> is called. Can be none to suppress the computations, numeric for numerical methods, algebraic for algebraic methods or default, the default, to automatically choose numeric or algebraic computations depending on problem size. The denominator degrees of freedom are calculated according to Kenward and Roger (1997) for fixed terms in the dense part of the model.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| dDF.fault              | A <a href="#">character</a> specifying the method to use to obtain substitute denominator degrees of freedom. when the numeric or algebraic methods produce faulty values, viz. NA, Inf or less than 0.01. Consistent with when no <code>denDF</code> are available, the default is "residual" and so the residual degrees of freedom from <code>asreml.obj\$nedf</code> are used. If <code>dDF.fault</code> = "none", no substitute denominator degrees of freedom are employed; if <code>dDF.fault</code> = "residual"; if <code>dDF.fault</code> = "maximum", the maximum of those <code>denDF</code> that are available, excluding that for the Intercept, is used; if all <code>denDF</code> are faulty, <code>asreml.obj\$nedf</code> is used. If <code>dDF.fault</code> = "supplied", a vector of values for the denominator degrees of freedom is to be supplied in <code>dDF.values</code> . Any other setting is ignored and a warning message produced. Generally, substituting these degrees of freedom is |

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | anticonservative in that it is likely that the degrees of freedom used will be too large.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| dDF.values      | A vector of values to be used when <code>dDF.fault = "supplied"</code> . Its values will be used when <code>denDF</code> in a test for a fixed effect is NA. This vector must be the same length as the number of fixed terms, including (Intercept) whose value could be NA.                                                                                                                                                                                                                                                                                                                                                                                                                     |
| trace           | If TRUE then partial iteration details are displayed when ASReml-R functions are invoked; if FALSE then no output is displayed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| update          | If TRUE, and <code>set.terms</code> is NULL, then <code>newfit.asreml</code> is called to fit the model to be tested, using the values of the variance parameters stored in the <code>asreml.object</code> , that is stored in <code>asrtests.obj</code> , as starting values. If FALSE or <code>set.terms</code> is not NULL, then <code>newfit.asreml</code> will not use the stored variance parameter values as starting values when fitting the new model, the only modifications being (i) to the terms in the fixed and random models corresponding to terms in <code>terms.marginality</code> and (ii) those specified via ....                                                           |
| set.terms       | A character vector specifying the terms that are to have bounds and/or initial values set prior to fitting. The names must match those in the <code>vparameters</code> component of the <code>asreml.obj</code> component in the new <code>asrtests.object</code> .                                                                                                                                                                                                                                                                                                                                                                                                                               |
| ignore.suffices | A logical vector specifying whether the suffices of the <code>asreml</code> -assigned names of the variance terms (i.e. the information to the right of an "!", other than "R!") is to be ignored in matching elements of terms. If TRUE for an element of terms, the suffices are stripped from the <code>asreml</code> -assigned names. If FALSE for an element of terms, the element must exactly match an <code>asreml</code> -assigned name for a variance term. This vector must be of length one or the same length as <code>terms</code> . If it is of length one then the same action is applied to the <code>asreml</code> -assigned suffices for all the terms in <code>terms</code> . |
| bounds          | A <code>character</code> vector specifying the bounds to be applied to the terms specified in <code>set.terms</code> . This vector must be of length one or the same length as <code>set.terms</code> . If it is of length one then the same constraint is applied to all the terms in <code>set.terms</code> . If any of the bounds are equal to NA then they are left unchanged for those terms.                                                                                                                                                                                                                                                                                                |
| initial.values  | A character vector specifying the initial values for the terms specified in <code>terms</code> . This vector must be of length one or the same length as <code>terms</code> . If it is of length one then the same initial value is applied to all the terms in <code>terms</code> . If any of the <code>initial.values</code> are equal to NA then they are left unchanged for those terms.                                                                                                                                                                                                                                                                                                      |
| IClikelihood    | A character that controls both the occurrence and the type of likelihood for information criterion in the <code>test.summary</code> of the new <code>asrtests.object</code> . If none, none are included. Otherwise, if REML, then the AIC and BIC based on the Restricted Maximum Likelihood are included; if full, then the AIC and BIC based on the full likelihood, evaluated using REML estimates, are included. (See also <code>infoCriteria.asreml</code> .)                                                                                                                                                                                                                               |
| ...             | further arguments passed to <code>asreml</code> , <code>wald.asreml</code> and <code>as.asrtests</code> via <code>testranfix.asrtests</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

## Value

A list containing:

1. `asrtests.obj`: an `asrtests.object` containing the components (i) `asreml.obj`, (ii) `wald.tab`, and (iii) `test.summary`;
2. `sig.tests`: a character vector whose elements are the significant terms amongst those tested.

**Author(s)**

Chris Brien

**References**

Kenward, M. G., & Roger, J. H. (1997). Small sample inference for fixed effects from restricted maximum likelihood. *Biometrics*, **53**, 983-997.

**See Also**

[chooseModel](#), [chooseModel.data.frame](#), [as.asrtests](#), [testranfix.asrtests](#), [testresidual.asrtests](#), [REMLRT.asreml](#), [rmboundary.asrtests](#), [newfit.asreml](#), [changeModelOnIC.asrtests](#), [changeTerms.asrtests](#), [reparamSigDevn.asrtests](#)

**Examples**

```
## Not run:
data(WaterRunoff.dat)
asreml.options(keep.order = TRUE) #required for asreml-R4 only
current.asr <- asreml(log.Turbidity ~ Benches + (Sources * (Type + Species)) * Date,
                    random = ~Benches:MainPlots:SubPlots:spl(xDay),
                    data = WaterRunoff.dat, keep.order = TRUE)
current.asrt <- as.asrtests(current.asr, NULL, NULL)
terms.treat <- c("Sources", "Type", "Species",
               "Sources:Type", "Sources:Species")
terms <- sapply(terms.treat,
               FUN=function(term){paste("Date:",term,sep="")},
               simplify=TRUE)
terms <- c("Date", terms)
terms <- unname(terms)
marginality <- matrix(c(1,0,0,0,0,0, 1,1,0,0,0,0, 1,0,1,0,0,0,
                      1,0,1,1,0,0, 1,1,1,0,1,0, 1,1,1,1,1,1), nrow=6)
rownames(marginality) <- terms
colnames(marginality) <- terms
choose <- chooseModel(current.asrt, marginality)
current.asrt <- choose$asrtests.obj
sig.terms <- choose$sig.terms

## End(Not run)
```

---

chooseModel.data.frame

*Determines the set of significant terms from results stored in a data.frame, taking into account the marginality relations of terms and recording the tests used in a [data.frame](#).*

---

**Description**

Uses the p.values from a set of hypothesis tests that are stored in the supplied [data.frame](#) to choose a model to describe the effects of the terms corresponding to the p-values, taking into account the hierarchy or marginality of terms. In particular, a term will not be tested if it is marginal to (or nested in) one that is significant. For example, if A:B is significant, then neither A nor B will be tested. The tests used in choosing the selected model are listed in the [data.frame](#) `choose.summary`. No change is made to the p.values, the DF and denDF being for information only.

**Usage**

```
## S3 method for class 'data.frame'
chooseModel(object, terms=NULL, p.values = "Pr",
            DF = "Df", denDF = "denDF", omit.DF = FALSE,
            terms.marginality=NULL, alpha = 0.05, ...)
```

**Arguments**

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object            | a <a href="#">data.frame</a> object containing the results of hypothesis tests for a set of terms. Its components should include terms, p.values, and, if not set to NA, DF and denDF.                                                                                                                                                                                                                                                                                                                                                        |
| terms             | A character giving the name of the column in object containing the terms corresponding to the p.values. If NULL, it is assumed that the row names of object give the terms.                                                                                                                                                                                                                                                                                                                                                                   |
| p.values          | A character giving the name of the column in object containing the p-values to use in deciding whether or not terms are significant.                                                                                                                                                                                                                                                                                                                                                                                                          |
| DF                | Can be a character or a numeric that specifies the numerator degrees of freedom for the tests. If it is a character, it must be the name of a column in object containing the numerator degrees of freedom that are to be included in the choose.summary data.frame. if it is a numeric, its length must equal 1 or the number of rows in object. In either case, a column labelled DF will be included in the choose.summary data.frame. It will contain either the replicated single value (which can be NA) or the values supplied.        |
| denDF             | Can be a character or a numeric that specifies the denominator degrees of freedom for the tests. If it is a character, it must be the name of a column in object containing the denominator degrees of freedom that are to be included in the choose.summary data.frame. if it is a numeric, its length must equal 1 or the number of rows in object. In either case, a column labelled denDF will be included in the choose.summary data.frame. It will contain either the replicated single value (which can be NA) or the values supplied. |
| omit.DF           | A logical indicating whether or not both the numerator and denominator degrees of freedom are to be omitted from choose.summary. Doing so will mean that the choose.summary no longer has the same columns as a test.summary from an <a href="#">asrtests.object</a> .                                                                                                                                                                                                                                                                        |
| terms.marginality | A square matrix of ones and zeros with row and column names being the names of the those terms in the terms column of object that are to be tested. The diagonal elements should be one, indicating that a term is marginal to itself. Elements should be one if the row term is marginal to the column term. All other elements should be zero. The names of the rows and columns should match the those elements of terms that are to be tested.                                                                                            |
| alpha             | The significance level for the hypothesis testing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| ...               | Provision for passing arguments to functions called internally - not used at present.                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

**Value**

A list containing:



---

chooseSpatialModelOnIC.asrtests

*Uses information criteria to choose the best fitting spatial model for accounting for local spatial variation.*

---

## Description

For a response variable measured on a potentially irregular grid of rows and columns of the units, uses information criteria (IC) to decide whether the fit and parsimony of the model fitted to a set of data can be improved by adding, to the fitted model stored in the supplied `asrtests.object`, one of the following spatial models to account for the local spatial variation: (i) a two-dimensional first-order autocorrelation model, (ii) a two-dimensional tensor-product natural cubic smoothing spline model (TPNCSS), (iii) a two-dimensional tensor-product penalized P-spline model with second-difference penalties (TPPSC2) model, or (iv) a two-dimensional tensor-product penalized linear spline model with first-difference penalties (TPPSL1). The models from which to select can be reduced to a subset of these four models. For each model, a term from the spatial model is only added to the supplied model if the IC of the supplied model is decreased with the addition of that term. If no model improves the IC when a local spatial variation model is added, then the supplied, nonspatial model will be returned. The data can be arranged in sections, for each of which there is a grid and for which the model is to be fitted separately. Also, the rows and columns of a grid are not necessarily one observational unit wide. For TPPSC2 models, the improvement in the fit from rotating the eigenvectors of the penalty matrix can be investigated; if there is no improvement, the unrotated fit will be returned.

One or more rows is added to the `test.summary.data.frame` of the `asrtests.object`, for each section and each spatial model, stating whether or not the new model has been swapped for a model in which the spatial model has been added to the supplied model. Convergence in fitting the model is checked and a note included in the action if there was not. All components of the `asrtests.object` are updated to exhibit the differences between the supplied and any new model.

## Usage

```
## S3 method for class 'asrtests'
chooseSpatialModelOnIC(asrtests.obj, trySpatial = "all",
  sections = NULL,
  row.covar = "cRow", col.covar = "cCol",
  row.factor = "Row", col.factor = "Col",
  corr.funcs = c("ar1", "ar1"), corr.orders = c(0, 0),
  row.corrFitfirst = TRUE,
  allow.corrJointFit = TRUE, nugget.variance = TRUE,
  dropFixed = NULL, dropRandom = NULL,
  nsegs = NULL, nestorder = c(1,1),
  usRandLinCoeffs = TRUE,
  rotateX = FALSE, ngridangles = NULL,
  which.rotacriterion = "AIC", nrotacores = 1,
  asreml.option = "grp", tpps4mbf.obj = NULL,
  allow.unconverged = TRUE, allow.fixedcorrelation = TRUE,
  checkboundaryonly = FALSE, update = TRUE, trace = FALSE,
  maxit = 30, IClikelihood = "full", which.IC = "AIC",
  return.asrts = "best", ...)
```

**Arguments**

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| asrtests.obj | An <code>asrtests.object</code> containing the components (i) <code>asreml.obj</code> , (ii) <code>wald.tab</code> , and (iii) <code>test.summary</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| trySpatial   | A character string nominating the types of spatial model whose fits are to be assessed. Possible values are <code>none</code> , <code>all</code> , <code>corr</code> , <code>TPNCSS</code> , <code>TPPSC2</code> (or <code>TPPCS</code> ), and <code>TPPSL1</code> (or <code>TPP1LS</code> ). If set to <code>none</code> , then just the supplied nonspatial model and the information about its information criteria will be returned. If <code>all</code> , then <code>corr</code> , <code>TPNCSS</code> , <code>TPPSC2</code> and <code>TPPSL1</code> will be fitted. Which fitted models are returned is controlled by <code>return.asrts</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| sections     | A single character string that specifies the name of the column in the <code>data.frame</code> that contains the <code>factor</code> that identifies different sections of the data to which separate spatial models are to be fitted. Note that, for other terms that involve sections in the random formula, there should be separate terms for each level of sections. For example, in a blocked experiment involving multiple sites, there should be the sum of separate terms for the Blocks at each Site i.e. a formula that contains terms like <code>at(Site, i):Block</code> for each site and these are separated by <code>'+'</code> . Otherwise, the combined term (e.g. <code>Site:Block</code> ) will impact on the fitting of the local spatial models for the different Sites. Similarly, a separate residual variance for each of the sections should be fitted, unless there is a need to fit a different variance structure to the residual, e.g. heterogeneous residual variances depending on treatments. Separate residual variances for sections can be achieved using the <code>asreml</code> functions <code>dsum</code> or <code>idh</code> . Because, unlike random terms, terms for residual variances are not removed from the model, compound residual terms can be used to include them in the model, e.g. terms with <code>idh</code> or <code>dsum</code> with multiple levels in the <code>list</code> or leaving levels out altogether. In addition to allowing the independent fitting of models to the sections, separate residual variance terms allows a nugget variance to be fitted in a correlation model for each of the sections. |
| row.covar    | A single character string nominating a <code>numeric</code> that contains the values of a centred covariate indexing the rows of a grid. The <code>numeric</code> must be a column in the <code>data.frame</code> stored in the <code>asreml.obj</code> that is a component of the supplied <code>asrtests.obj</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| col.covar    | A single character string nominating a <code>numeric</code> that contains the values of a centred covariate indexing the columns of a grid. The <code>numeric</code> must be a column in the <code>data.frame</code> stored in the <code>asreml.obj</code> that is a component of the supplied <code>asrtests.obj</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| row.factor   | A single character string nominating a <code>factor</code> that indexes the rows of a grid that are to be one dimension of a spatial correlation model. The <code>factor</code> must be a column in the <code>data.frame</code> stored in the <code>asreml.obj</code> that is a component of the supplied <code>asrtests.obj</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| col.factor   | A single character string nominating a <code>factor</code> that indexes the columns of a grid that are to be one dimension of a spatial correlation model. The <code>factor</code> must be a column in the <code>data.frame</code> stored in the <code>asreml.obj</code> that is a component of the supplied <code>asrtests.obj</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| corr.funcs   | A single character string of length two that specifies the <code>asreml</code> one-dimensional correlation or variance model function for the row and column dimensions of a two-dimensional separable spatial correlation model to be fitted when <code>spatial.model</code> is <code>corr</code> ; the two-dimensional model is fitted as a random term. If a correlation or variance model is not to be investigated for one of the dimensions, specify <code>""</code> for that dimension. If the correlation model is <code>corb</code> , the values of <code>corr.orders</code> are used for its order argument (b).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>corr.orders</code>        | A numeric of length two that specifies the order argument (b) values for the row and column dimensions of a two-dimensional separable spatial correlation model when <code>spatial.model</code> is <code>corr</code> and the <code>corr.funcs</code> for a dimension is <code>corb</code> , the <code>asreml</code> banded correlation model. If one of the dimensions does not involve an order argument, set the value of <code>corr.orders</code> for that dimension to zero. For a dimension for which the <code>corr.funcs</code> is <code>corb</code> and <code>corr.orders</code> is zero, a model with a single band, the correlation between immediate neighbours, will be fitted and then further bands, up to a maximum of 10 bands, will be added until the addition of an extra band does not reduce the information criterion nominated using <code>which.IC</code> . Note that the two-dimensional spatial model is fitted as a random term.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>row.corrFitfirst</code>   | A <a href="#">logical</a> . If TRUE then, in fitting the model for <code>spatial.model</code> set to <code>corr</code> , the row correlation or variance function is fitted first, followed by the addition of the column correlation or variance function. If FALSE, the order of fitting is reversed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>allow.corrJointFit</code> | A <a href="#">logical</a> which, if TRUE, will allow the simultaneous fitting of correlation functions for the two dimensions of the grid when separate fits have failed to fit any correlation functions. This argument is available for when a joint fit hangs the system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>nugget.variance</code>    | A <a href="#">logical</a> which, if TRUE, will result in an attempt to fit a nugget or unit-specific variance. Otherwise, a nugget or unit-specific variance will not be fitted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>dropFixed</code>          | <p>A single character string or a character vector of strings with an element for each level of sections in the same order as the sections levels. Each string, which if it is not NA and after the addition of "<code> ~ . -</code>" and conversion to a formula that is then expanded, specifies the sum of a set of terms to be dropped from the fixed formula in fitting splines (TPPS and TPNCSS). The result is that the fitted model supplied in the <code>asrtests.obj</code>, that includes these terms, will be compared with the fitted model that has had them removed and a spatial model added.</p> <p>An element that is NA indicates that no term pertaining to the corresponding sections level is to be removed. If <code>sections</code> is not NULL and a single character string has been supplied, the terms specified by the string are taken to be terms that are independent of the sections and will be removed when adding the spatial model for the first sections.</p> <p>The terms must match those in the <code>wald.tab</code> component of the <code>asrtests.obj</code>. The fixed terms will be reordered so that single-variable terms come first, followed by two-variable terms and so on. Note also that multiple terms specified using a single <code>asreml: :at</code> function can only be dropped as a whole. If the term was specified using an <code>asreml: :at</code> function with a single level, then it can be removed and either the level itself or its <a href="#">numeric</a> position in the levels returned by the <a href="#">levels</a> function can be specified.</p> |
| <code>dropRandom</code>         | A single character string or a character vector of strings with an element for each level of sections in the same order as the sections levels. Each string, which if it is not NA and after the addition of " <code> ~ . -</code> " and conversion to a formula that is then expanded, specifies the sum of a set of terms to be dropped from the random formula in fitting splines (TPPS and TPNCSS). The result is that the fitted model supplied in the <code>asrtests.obj</code> , that includes these terms, will be compared with the fitted model that has had them removed and a spatial model added.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |



An element that is NA indicates that no term pertaining to the corresponding sections level is to be removed. If sections is not NULL and a single character string has been supplied, the terms specified by the string are taken to be terms that are independent of the sections and will be removed when adding the spatial model for the first sections.

The terms must match those in the `vparameters` component of the `asreml.obj` component in the `asrtests.obj`. Note also that multiple terms specified using a single `asreml::at` function can only be dropped as a whole. If the term was specified using an `asreml::at` function with a single level, then it can be removed and either the level itself or its `numeric` position in the levels returned by the `levels` function can be specified.

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| nsegs               | A pair of <code>numeric</code> values giving the number of segments into which the column and row ranges are to be split, respectively, for fitting a P-spline model (TPPS) (each value specifies the number of internal knots + 1). If not specified, then (number of unique values - 1) is used in each dimension; for a grid layout with equal spacing, this gives a knot at each data value. If sections is not NULL and the grid differs between the sections, then nsegs will differ between the sections.                                                                                                                                                                                                                                                                                                           |
| nestorder           | A <code>numeric</code> of length 2. The order of nesting for column and row dimensions, respectively, in fitting a P-spline model (TPPS). A value of 1 specifies no nesting, a value of 2 generates a spline with half the number of segments in that dimension, etc. The number of segments in each direction must be a multiple of the order of nesting.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| usRandLinCoeffs     | A <code>logical</code> which, if TRUE and a P-spline (TPPS) is being fitted, will attempt to fit an unstructured variance model to the constant and linear terms in the interactions for constant and linear terms in one grid dimension interacting with smooth terms in the second grid dimension. The unstructured variance model can only be fitted if both the constant and linear interaction terms have been retained in the fitted model. This argument can be used to omit the attempt to fit an unstructured variance model when the attempt results in a system error.                                                                                                                                                                                                                                          |
| rotateX             | A <code>logical</code> indicating whether to rotate the eigenvectors of the penalty matrix, as described by Piepho, Boer and Williams (2022), when fitting a P-spline (TPPS). Setting rotateX to TRUE results in a search for an optimized rotation under a model that omits the random spline interaction terms. If ngridangles is set to NULL, the optimal rotation is found using an optimizer ( <code>nloptr::bobyqa</code> ). Otherwise, the optimal rotation is found by exploring the fit over a two-dimensional grid of rotation angle pairs. The optimization seeks to optimize the criterion nominated in <code>which.rotacriterion</code> . Rotation of the eigenvectors is only relevant for <code>difforder</code> values greater than 1 and has only been implemented for <code>difforder</code> equal to 2. |
| ngridangles         | A <code>numeric</code> of length 2. If NULL (the default), the optimal pair of angles for rotating the eigenvectors of the penalty matrix of a P-spline (TPPS) will be determined using a nonlinear optimizer ( <code>nloptr::bobyqa</code> ). Otherwise, its two values specify the numbers of angles between 0 and 90 degrees for each of the row and column dimensions to be used in determining the optimal pair of angles. Specifying factors of 90 will result in integer-valued angles. The number of grid points, and hence re-analyses will be the product of the values of ( <code>ngridangles</code> + 1).                                                                                                                                                                                                      |
| which.rotacriterion | A single character string nominating which of the criteria, out of the deviance,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

the likelihood, the AIC and the BIC, is to be used in determining the optimal rotation of the eigenvectors of the penalty matrix. The deviance uses the REML value computed by `asreml`; the other criteria use the full likelihood, evaluated using the REML estimates, that is computed by `infoCriteria.asreml`.

- `nrotacores` A [numeric](#) specifying the number of cores to deploy for running the analyses required to search the two-dimensional grid of rotation angles when `rotateX` is TRUE. Parallel processing has been implemented for analyzing, for each column angle, the set of angles to be investigated for the row dimension. The default value of one means that parallel processing will not be used. The value chosen for `nrotacores` needs to be balanced against the other processes that are using parallel processing at the same time.
- `asreml.option` A single character string specifying whether the `grp` or `mbf` methods are to be used to supply externally formed covariate matrices to `asreml` when fitting a P-spline (TPPS). Compared to the `mbf` method, the `grp` method is somewhat faster, but creates large [asrtests.objects](#) for which the time it takes to save them can exceed any gains in execution speed. The `grp` method adds columns to the [data.frame](#) containing the data. On the other hand, the `mbf` method adds only the fixed covariates to data and stores the random covariates in the environment of the internal function that calls the spline-fitting function; there are three smaller [data.frames](#) for each section that are not stored in the `asreml.object` resulting from the fitted model.
- `tpps4mbf.obj` An object made with [makeTPPSplineMats.data.frame](#) that contains the spline basis information for fitting P-splines. The argument `tpps4mbf.obj` only needs to be set when the `mbf` option of `asreml.option` is being used and it is desired to use `mbf` [data.frames](#) that have been created and stored prior to calling `chooseSpatialModelOnIC.asrtests`. If `tpps4mbf.obj` is NULL, [makeTPPSplineMats.data.frame](#) will be called internally to produce the required `mbf` [data.frames](#).
- `allow.unconverged` A logical indicating whether to accept a new model even when it does not converge. If FALSE and the fit of the new model does not converge, the supplied `asrtests.obj` is returned. Also, if FALSE and the fit of the new model has converged, but that of the old model has not, the new model will be accepted.
- `allow.fixedcorrelation` A logical indicating whether to accept a new model even when it contains correlations in the model whose values have been designated as fixed, bound or singular. If FALSE and the new model contains correlations whose values have not been able to be estimated, the supplied `asrtests.obj` is returned. The fit in the `asreml.obj` component of the supplied `asrtests.obj` will also be tested and a warning issued if both fixed correlations are found in it and `allow.fixedcorrelation` is FALSE.
- `checkboundaryonly` If TRUE then boundary and singular terms are not removed by [rmboundary.asrtests](#); a warning is issued instead. Note that, for correlation models, the fitting of each dimension and the test for a nugget term are performed with `checkboundaryonly` set to TRUE and its supplied setting only honoured using a call to [rmboundary.asrtests](#) immediately prior to returning the final result of the fitting.
- `update` If TRUE, and `set.terms` is NULL, then [newfit.asreml](#) is called to fit the model to be tested, using the values of the variance parameters stored in the `asreml.object`, that is stored in `asrtests.obj`, as starting values. If FALSE or `set.terms` is not NULL, then [newfit.asreml](#) will not use the stored variance parameter values as

|              |                                                                                                                                                                                                                                                                                                                                                       |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | starting values when fitting the new model, the only modifications being (i) to fit spatial terms and (ii) those specified via . . . .                                                                                                                                                                                                                |
| trace        | If TRUE then the stages in fitting a correlation model are displayed.                                                                                                                                                                                                                                                                                 |
| which.IC     | A character specifying the information criterion to be used in selecting the best model. Possible values are AIC and BIC. The value of the criterion for supplied model must exceed that for changed model for the changed model to be returned. (For choosing the rotation angle of the eigenvectors of the penalty matrix, see which.rotacriterion. |
| maxit        | A <a href="#">numeric</a> specifying the maximum number of iterations that asreml should perform in fitting a model.                                                                                                                                                                                                                                  |
| IClikelihood | A character specifying whether Restricted Maximum Likelihood (REML) or the full likelihood, evaluated using REML estimates, (full) are to be used in calculating the information criteria to be included in the test.summary of an <a href="#">asrtests.object</a> or to be used in choosing the best model.                                          |
| return.asrts | A character string specifying whether the <a href="#">asrtests.object</a> for the "best" fitting model (smallest AIC or BIC), including the supplied nonspatial model, is returned or the <a href="#">asrtests.objects</a> resulting from the attempted fits of "all" of the models specified using trySpatial are returned.                          |
| ...          | Further arguments passed to <a href="#">changeModelOnIC.asrtests</a> , asreml and tpsmmb.                                                                                                                                                                                                                                                             |

## Details

For each spatial model that is to be fitted, a fitted spatial model is only returned if it improves the fit over and above that achieved with the model fit supplied in the `asrtests.obj`, because terms in the spatial model are not added unless model fit is improved by their addition as measured by an IC. If `return.asrts` is `all`, then this applies to each spatial model specified by `trySpatial`. To force a spatial model to be fitted use `addSpatialModel.asrtests`. The model fit supplied in the `asrtests.obj` should not include terms that will be included in any local spatial model. All spatial model terms are fitted as fixed or random. Consequently, the residual model does not have to be iid. The improvement in the fit resulting from the addition of a spatial model to the supplied model is evaluated. Note that the data must be in the order that corresponds to the residual argument with a variable to the right of another variable changing levels in the data frame faster than those of the preceding variables e.g. `Row:Column` implies that all levels for `Column` are in consecutive rows of the data.frame that have a single `Row` level.

For the `corr` spatial model, the default model is an autocorrelation model of order one (`ar1`) for each dimension. However, any of the single dimension correlation/variance models from `asreml` can be specified for each dimension, as can no correlation model for a dimension; the models for the two dimensions can differ. Using a forward selection procedure, a series of models are tried, without removing boundary or singular terms, beginning with the addition of row correlation and followed by the addition of column correlation or, if the `row.corrFitfirst` is set to `FALSE`, the reverse order. If the fitting of the first-fitted correlation did not result in a model change because the fitting did not converge or correlations were fixed, but the fit of the second correlation was successful, then adding the first correlation will be retried. If one of the metric correlation functions is specified (e.g. `exp`), then the `row.covar` or `col.covar` will be used in the spatial model. However, because the correlations are fitted separately for the two dimensions, the `row.factor` and `col.factor` are needed for all models and are used for any dimension that does not involve a correlation/variance function for the fit being performed. Also, the correlation models are fitted as random terms and so the correlation model will include a variance parameter for the grid even when `ar1` is used to specify the correlation model, i.e. the model fitted is a variance model and there is no difference between `ar1` and `ar1v` in fitting the model. The variance parameter for this term represents the spatial variance and the fit necessarily includes a nugget term, this being the residual variance. If any

correlation is retained in the model, for a section if sections is not NULL, then the need for a nugget term is assessed by fixing the corresponding residual variance to one, unless there are multiple residual variances and these are not related to the sections. Once the fitting of the correlation model has been completed, the `rmboundary` function will be executed with the `checkboundaryonly` value supplied in the `chooseSpatialModelOnIC.asrtests` call. Finally, checking for bound and singular random terms associated with the correlation model and residual terms will be carried out when there are correlation terms in the model and `checkboundaryonly` has been set to FALSE; as many as possible will be removed from the fitted model, in some cases by fixing variance terms to one.

The tensor-product natural-cubic-smoothing-spline (TPNCSS) spatial model is as described by Verbyla et al. (2018), the tensor-product penalized-cubic-spline (TPPSC2) model with second-order differencing of the penalty is similar to that described by Rodriguez-Alvarez et al. (2018), and the tensor-product, first-difference-penalty, linear spline (TPPSL1) model is amongst those described by Piepho, Boer and Williams (2022). The fixed terms for the spline models are `row.covar + col.covar + row.covar:col.covar` and the random terms are `spl(row.covar) + spl(col.covar) + dev(row.covar) + dev(col.covar) + spl(row.covar):col.covar + row.covar:spl(col.covar) + spl(row.covar):spl(col.covar)`, except that `spl(row.covar) + spl(col.covar)` is replaced with `spl(row.covar):int(col.covar) + int(row.covar):spl(col.covar)` in the TPPSC2 model, where `int(.)` indicates an intercept or constant value specific to its argument. For TPPSL1 models, the terms `spl(row.covar):col.covar + row.covar:spl(col.covar)` are omitted. The supplied model should not include any of these terms. However, any fixed or random main-effect Row or Column term that has been included as an initial model for comparison with a spatial model can be removed prior to fitting the spatial model using `dropFixed` or `dropRandom`. For the P-spline models with second-order differencing, the model matrices used to fit the pairs of random terms (i) `spl(row.covar):int(col.covar)` and `spl(row.covar):col.covar` and (ii) `int(row.covar):spl(col.covar)` and `row.covar:spl(col.covar)` are transformed using the spectral decomposition of their penalty matrices. An unstructured variance model is tried for each of these pairs and retained if it improves the fit. For TPPSC2, it is also possible to optimize the rotation of the null-space eigenvectors of the penalty matrix for each of these random-term pairs (for more information see Piepho, Boer and Williams, 2022). The optimization is achieved either using an optimizer or takes the form of a search over a grid of rotation angles for a reduced model; the fit of the full model with rotation using the optimal rotation angles will only be returned if it improves on the fit of the full, unrotated model.

The TPPSC2 and TPPSL1 models are fitted using functions from the R package `TPSbits` authored by Sue Welham (2022). There are two methods for supplying the spline basis information produced by `tpsmb` to `asreml`. The `grp` method adds it to the `data.frame` supplied in the `data` argument of the `asreml` call. The `mbf` method creates smaller `data.frames` with the spline basis information in the same environment as the internal function that calls the spline-fitting function. If it is desired to use in a later session, an `asreml` function, or `asrtests` function that calls `asreml`, (e.g. `predict.asreml`, `predictPlus.asreml`, or `changeTerms.asrtests`) on an `asreml` object created using `mbf` terms, then the `mbf data.frames` will need to be recreated using `makeTPPSplineMats.data.frame` in the new session, supplying, if there has been rotation of the penalty matrix eigenvectors, the `theta` values that are returned as the attribute `theta.opt` of the `asreml.obj`.

All models utilize the function `changeModelOnIC.asrtests` to assess the model fit, the information criteria used in assessing the fit being calculated using `infoCriteria`. Arguments from `tpsmb` and `changeModelOnIC.asrtests` can be supplied in calls to `chooseSpatialModelOnIC.asrtests` and will be passed on to the relevant function through the `ellipses` argument (...).

The data for experiment can be divided into sections and an attempt to fit the same spatial model to each is made. The fit may differ for each of the sections, but the fit over all of the sections is assessed. For more detail see sections above.

Each combination of a row.coords and a col.coords does not have to specify a single observation; for example, to fit a local spatial model to the main units of a split-unit design, each combination would correspond to a main unit and all subunits of the main unit would have the same combination.

### Value

A [list](#) containing four components: (i) asrts, (ii) spatial.IC, (iii) best.spatial.mod, and (iv) best.spatial.IC.

The component asrts itself holds a [list](#) of one or more [asrtests.objects](#), either the best overall out of the supplied model and the spatial models, or, for each spatial model, the best out of the supplied model and that spatial model. Each [asrtests.object](#) contains the components: (i) asreml.obj, (ii) wald.tab, and (iii) test.summary. If the [asrtests.object](#) is the result of fitting a TPPSC2 model with an exploration of the rotation of the eigenvectors of the penalty matrix for the linear components, then the asreml.obj will have an attribute theta.opt that contains the optimal rotation angles of the eigenvectors.

The spatial.IC component holds a [data.frame](#) with summary of the values of the information criteria for the supplied model and those resulting from adding the spatial models to the supplied model. In the case of a spatial correlation model, the information criteria for the selected spatial correlation model is returned. If a spatial model could not be fitted, then all returned values will be NA).

The best.spatial.mod component is a character giving the name of the best spatial model, and best.spatial.AIC gives the value of its AIC.

### Author(s)

Chris Brien

### References

- Piepho, H.-P., Boer, M. P., & Williams, E. R. (2022). Two-dimensional P-spline smoothing for spatial analysis of plant breeding trials. *Biometrical Journal*, **64**, 835-857.
- Rodriguez-Alvarez, M. X., Boer, M. P., van Eeuwijk, F. A., & Eilers, P. H. C. (2018). Correcting for spatial heterogeneity in plant breeding experiments with P-splines. *Spatial Statistics*, **23**, 52-71.
- Verbyla, A. P., De Faveri, J., Wilkie, J. D., & Lewis, T. (2018). Tensor Cubic Smoothing Splines in Designed Experiments Requiring Residual Modelling. *Journal of Agricultural, Biological and Environmental Statistics*, **23**(4), 478-508.
- Welham, S. J. (2022) TPSbits: *Creates Structures to Enable Fitting and Examination of 2D Tensor-Product Splines using ASReml-R*. Version 1.0.0.

### See Also

[as.asrtests](#), [makeTPPSplineMats.data.frame](#), [addSpatialModelOnIC.asrtests](#), [addSpatialModel.asrtests](#), [changeModelOnIC.asrtests](#), [changeTerms.asrtests](#), [rmboundary.asrtests](#), [testranfix.asrtests](#), [testresidual.asrtests](#), [newfit.asreml](#), [reparamSigDevn.asrtests](#), [changeTerms.asrtests](#), [infoCriteria.asreml](#)

### Examples

```
## Not run:

data(Wheat.dat)
```

```

#Add row and column covariates
Wheat.dat <- within(Wheat.dat,
  {
    cColumn <- dae::as.numfac(Column)
    cColumn <- cColumn - mean(unique(cColumn))
    cRow <- dae::as.numfac(Row)
    cRow <- cRow - mean(unique(cRow))
  })

#Fit initial model
current.asr <- asrem1(yield ~ Rep + WithinColPairs + Variety,
  random = ~ Row + Column,
  data=Wheat.dat)

#Create an asrtests object, removing boundary terms
current.asrt <- as.asrtests(current.asr, NULL, NULL,
  label = "Random Row and Column effects")
current.asrt <- rmboundary(current.asrt)

# Choose the best of four models for the local spatial variation
current.asrt <- chooseSpatialModelOnIC(current.asrt,
  row.covar = "cRow", col.covar = "cColumn",
  dropRowterm = "Row", dropColterm = "Column",
  asrem1.option = "grp")

## End(Not run)

```

---

|                      |                                                                                                      |
|----------------------|------------------------------------------------------------------------------------------------------|
| convAsremlobj.asrem1 | <i>Recreates an asrem1 object so that it is compatible with the currently loaded asrem1 version.</i> |
|----------------------|------------------------------------------------------------------------------------------------------|

---

## Description

Recreate an existing asrem1 object so that it is compatible with the currently loaded asrem1 version. It involves refitting the model stored in the asrem1 object.

## Usage

```
## S3 method for class 'asrem1'
convAsremlobj(asrem1.obj, ...)
```

## Arguments

|            |                                                                                                        |
|------------|--------------------------------------------------------------------------------------------------------|
| asrem1.obj | An asrem1 object with a component named call (from a previous call to either asrem1 or update.asrem1). |
| ...        | Provision for passing arguments to functions called internally - not used at present.                  |

## Value

An asrem1 object.

**Author(s)**

Chris Brien

**References**

Butler, D. G., Cullis, B. R., Gilmour, A. R., Gogel, B. J. and Thompson, R. (2023). *ASReml-R Reference Manual Version 4.2*. VSN International Ltd, <https://asreml.kb.vsnl.co.uk/>.

**See Also**

[newfit.asreml](#), [update.asreml](#)

**Examples**

```
## Not run:
m1.asr <- convAsremlobj(m1.asr)

## End(Not run)
```

---

convEffectNames2DataFrame.asreml

*Converts the effects names for a term stored in the component of an asreml object into a [data.frame](#).*

---

**Description**

Converts the effects names for a term stored in the component of an asreml object into a [data.frame](#) that has a column for each factor and variable in the term. It facilitates adding the effects to the [data.frame](#) supplied to asreml for an analysis. This function can only be used with asreml v4.2 or later.

**Usage**

```
## S3 method for class 'asreml'
convEffectNames2DataFrame(asreml.obj, term, use = "design.matrix", sep = ":", ...)
```

**Arguments**

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| asreml.obj | An object resulting from the fitting of a model using asreml v4.2.                                                                                                                                                                                                                                                                                                                                                                                                              |
| term       | A <a href="#">character</a> nominating a single model term. It should be the name of the term in the component, of the asreml.obj, nominated in the use argument. The variables/factors in term must match those in this component. Thus, it will not include asreml variance or correlation functions, but may include functions such as at. Generally, each element of the term, separated from other elements by colons, should involve a single factor or variable.         |
| use        | A <a href="#">character</a> specifying a component in the asreml.obj. The default is the design.matrix. Other possibilities are the random.coeffs or G.aom. For the option design.matrix, the design argument of asreml.options must be set to TRUE, before executing either the asreml or update.asreml call produces the asreml.obj. The option G.aom specifies the G component of the aom component that is included in the asreml.obj if the aom argument is set to TRUE in |

either asreml.options before calling asreml or update.asreml, or in a call to asreml or update.asreml that is involved in producing the asreml.obj.

sep A [character](#) specifying the separator of factors/variables in the term.

... Provision for passing arguments to functions called internally - not used at present.

## Value

A [data.frame](#) with columns for the factors and variables in term. It includes the attribute effect.names that contains the extracted effects names for the term

## Author(s)

Chris Brien

## References

Butler, D. G., Cullis, B. R., Gilmour, A. R., Gogel, B. J. and Thompson, R. (2023). *ASReml-R Reference Manual Version 4.2*. VSN International Ltd, <https://asreml.kb.vsnl.co.uk/>.

## Examples

```
## Not run:
G.dat <- convEffectNames2DataFrame(m1.asr, term = "Row:Column", use = "G.aom")

G.dat <- lapply(c("at(Smarthouse, 'SW'):Lane:Position",
                 "at(Smarthouse, 'SE'):Lane:Position"),
              function(term, asreml.obj)
                tmp <- convEffectNames2DataFrame.asreml(asreml.obj, term = term),
              asreml.obj = m1.asr)
G.dat <- do.call(rbind, G.dat)

## End(Not run)
```

---

|                  |                                                                                                                            |
|------------------|----------------------------------------------------------------------------------------------------------------------------|
| estimateV.asreml | <i>Forms the estimated variance, random or residual matrix for the observations from the variance parameter estimates.</i> |
|------------------|----------------------------------------------------------------------------------------------------------------------------|

---

## Description

Forms the estimated variance (**V**), random (**G**) or (**R**) matrix for the observations, a square symmetric matrix of order equal to the number of observations. The estimates of the variance parameters and the information about the random and residual models for which they were estimated are obtained from the asreml object. This function is not available in ASReml-R version 3.

## Usage

```
## S3 method for class 'asreml'
estimateV(asreml.obj, which.matrix = "V",
          extra.matrix = NULL, ignore.terms = NULL, fixed.spline.terms = NULL,
          bound.exclusions = c("F", "B", "S", "C"), ...)
```



## Arguments

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>asreml.obj</code>         | An asreml object from a call to asreml in which the data argument has been set.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>which.matrix</code>       | A character giving the matrix that is to be formed. It must be one of "V", to produce the variance matrix $\mathbf{V} = \mathbf{G} + \mathbf{R}$ , "G" to produce the matrix $\mathbf{G}$ , corresponding to the random formula, or "R" to produce the matrix $\mathbf{R}$ , corresponding to the residual formula.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>extra.matrix</code>       | A matrix of order equal to the number of observations that is to be added to the matrix specified by <code>which.matrix</code> , the latter based on the information in <code>asreml.obj</code> . It is assumed that the sigma-parameterized values of the variance parameter estimates, such as is given in the <code>varcomp</code> component of <code>summary.asreml</code> , have been used in calculating <code>extra.matrix</code> ; the values in the <code>vparameters</code> component of <code>G.param</code> and <code>R.param</code> may be either gamma- or sigma-parameterized. The argument <code>extra.matrix</code> can be used in conjunction with <code>ignore.terms</code> as a workaround to include components of the variance matrix for variance functions that have not been implemented in <code>estimateV</code> . |
| <code>ignore.terms</code>       | A character giving terms from either the random or residual models that are to be ignored in that their contributions to the variance is not to be included in the estimated matrix. The term names are those given in the <code>vparameters</code> component of the asreml object or the <code>varcomp</code> component produced by <code>summary.asreml</code> , but only up to the first exclamation mark (!). This can be used in conjunction with <code>estimateV.asreml</code> as a workaround to include components of the variance matrix for variance functions that have not been implemented in <code>estimateV</code> .                                                                                                                                                                                                           |
| <code>fixed.spline.terms</code> | A character vector giving one or more spline terms in the random model that are regarded as fixed and so are to be ignored because they are not regarded as contributing to the variance. The term names are those given in the <code>vparameters</code> component of the asreml object or the <code>varcomp</code> component produced by <code>summary.asreml</code> , but only up to the first exclamation mark (!).                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>bound.exclusions</code>   | A character specifying one or more bound codes that will result in a variance parameter in the random model being excluded from contributing to the variance. If set to NULL then none will be excluded.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>...</code>                | Provision for passing arguments to functions called internally - not used at present.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

## Details

The information about the variance parameters in the fitted mixed model are obtained from the `G.param` and `R.param` components of the asreml object. The function can deal with the following variance functions in either the random or residual models: `id`, `diag`, `us`, `ar1`, `ar2`, `ar3`, `sar`, `sar2`, `ma1`, `ma2`, `arma`, `exp`, `gau`, `cor`, `corb` and `corg`. All of these functions, except `us`, can be combined with either `v` or `h`. It will also cope with the following functions in the random model: `at`, `str`, `spl`, `dev`, `grp`, `fa` and `rr`. Additionally, it can deal with the function `dsum` in the residual model. For further information see the ASReml-R User Guide Version 4 (Butler et al., 2023).

## Value

A matrix containing the estimated variance matrix. It has an attribute `missing.termmatrix` (use `attr(x, which = "missing.termmatrix")` to access the attribute). It will be NULL, unless the design matrix could not be obtained for one or more model terms. If it is not NULL, it will be a list

of terms that could not be produced for inclusion in the variance matrix estimate, and NA will be returned for the estimated variance matrix.

### Author(s)

Chris Brien

### References

Butler, D. G., Cullis, B. R., Gilmour, A. R., Gogel, B. J. and Thompson, R. (2023). *ASReml-R Reference Manual Version 4.2*. VSN International Ltd, <https://asreml.kb.vsnl.co.uk/>.

### See Also

asreml, [simulate.asreml](#), [variofaces.asreml](#).

### Examples

```
## Not run:
data(Wheat.dat)
current.asr <- asreml(yield ~ Rep + WithinColPairs + Variety,
                     random = ~ Row + Column + units,
                     residual = ~ ar1(Row):ar1(Column),
                     data=Wheat.dat)
# Form variance matrix based on estimated variance parameters
V <- estimateV(current.asr)

## End(Not run)
```

---

|                      |                                                                                       |
|----------------------|---------------------------------------------------------------------------------------|
| exploreLSDs.alldiffs | <i>Explores the computed LSD values for pairwise differences between predictions.</i> |
|----------------------|---------------------------------------------------------------------------------------|

---

### Description

Given an [alldiffs.object](#) with an sed component, the LSDs are calculated for all pairwise comparisons of predictions. It then calculates (i) a table of frequencies of the LSD values, (ii) the distinct values of the LSDs after rounding, (iii) various statistics from the LSD values, (iv) a measure of the accuracy of each of the LSD statistics, (v) the numbers of false positives and false negatives for each of the LSD statistics if pairwise comparisons are based on the LSD statistic, (vi) the accuracy of each statistic in representing the LSD values for each prediction and (vii) a matrix containing the LSD values for comparing each pair of predictions. Histograms of the frequencies can also be produced.

### Usage

```
## S3 method for class 'alldiffs'
exploreLSDs(alldiffs.obj, LSDtype = "overall", LSDby = NULL,
            LSDaccuracy = "maxAbsDeviation", alpha = 0.05, digits = 3,
            retain.zeroLSDs = FALSE,
            zero.tolerance = .Machine$double.eps ^ 0.5,
            plotHistogram = FALSE, ...)
```

## Arguments

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| alldiffs.obj    | An <code>alldiffs.object</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| LSDtype         | A <code>character</code> string that can be <code>overall</code> or <code>factor.combinations</code> . It determines whether the LSD values that are investigated and stored are (i) the overall minimum, <code>quantile10</code> , <code>quantile25</code> , <code>mean</code> , <code>median</code> , <code>quantile75</code> , <code>quantile90</code> , and maximum of all pairwise LSDs, or (ii) the minimum, <code>quantile10</code> , <code>quantile25</code> , <code>mean</code> , <code>median</code> , <code>quantile75</code> , <code>quantile90</code> , and maximum for the pairwise LSDs for each <code>factor.combination</code> , unless there is only one prediction for a <code>factor.combination</code> , when notional LSDs are calculated. The <code>LSDtype</code> specified here does not have to match that used in the creating the <code>alldiffs.object</code> . See <code>LSD.frame</code> for further information on how the LSD statistics are calculated. |
| LSDby           | A <code>character</code> (vector) of variables names, being the names of the <code>factors</code> or <code>numerics</code> in the <code>classify</code> ; for each combination of the values the of the <code>factors</code> and <code>numerics</code> , the LSD statistics and accuracy are computed, as well histograms plotted, when <code>LSDtype</code> is <code>factor.combinations</code> . The <code>LSDby</code> specified here does not have to match that used in the creating the <code>alldiffs.object</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| LSDaccuracy     | A <code>character</code> nominating one of <code>maxAbsDeviation</code> , <code>maxDeviation</code> , <code>q90Deviation</code> or <code>RootMeanSqDeviation</code> as the statistic to be calculated as a measure of the accuracy of an LSD statistic when its values are used as an approximate LSD. The option <code>q90Deviation</code> produces the sample quantile corresponding to a probability of 0.90. The deviations are the differences between a set of LSDs and an LSD statistic calculated from those LSDs; the accuracy is expressed as a proportion of the value of the LSD statistic.                                                                                                                                                                                                                                                                                                                                                                                   |
| alpha           | A <code>numeric</code> specifying the significance level for an LSD to compare a pair of predictions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| digits          | A <code>numeric</code> specifying the number of significant digits to retain in rounding the LSDs before determining the distinct rounded LSDs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| retain.zeroLSDs | A <code>logical</code> indicating whether to retain or omit LSDs that are zero when calculating the summaries of LSDs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| zero.tolerance  | A <code>numeric</code> specifying the value such that if an LSD is less than it, the LSD will be considered to be zero.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| plotHistogram   | A <code>logical</code> indicating whether or not histograms of the LSD values are to be plotted. The <code>LSDtype</code> argument controls whether one histogram of all LSD values is plotted or histograms are plotted for each combination of the levels of the factors specified by the <code>LSDby</code> argument.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| ...             | Provision for passing arguments to functions called internally - not used at present.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

## Details

The false positives and negatives are computed by comparing, for each pair of predictions within each levels-combination of the `LSDby` variables, the significance of the pair difference determined using (i) the true LSD that is computed from the standard error of differences for the pair and (ii) the approximate LSD that is a statistic computed from the true LSDs for all pairwise difference within each levels-combination of the `LSDby` variables. The number of false positives is the number of pairwise differences for which a difference is declared significant using the approximate LSD, but not using the true LSD. The number of false negatives is the number of pairwise differences for which a difference is declared nonsignificant using the approximate LSD, but significant using the true LSD.

The LSD accuracy for a set of LSDs is a function of the deviations of those LSDs and an LSD statistic calculated from them; the accuracy is expressed as a proportion of the value of the LSD statistic.

### Value

A [list](#) with components `frequencies`, `distinct.vals`, `statistics`, `accuracy`, `per.pred.accuracy` and `LSD`:

1. `frequencies` is a `data.frame` with the frequency distribution of the LSD values;
2. `distinct.vals` is a `list`, each component of which contains the distinct values of the LSDs after rounding;
3. `statistics` is a `data.frame` with the minimum, `quantile10`, `quantile25`, mean, median, `quantile75`, `quantile90`, and maximum of the LSD values;
4. `accuracy` is a `data.frame` with the accuracies of the minimum, `quantile10`, `quantile25`, mean, median, `quantile75`, `quantile90`, and maximum of the LSD values with respect to the values from which these statistics are calculated;
5. `false.pos` is a `data.frame` with the numbers of false positives for the pairwise comparisons within each levels-combination of the LSDby variables when each of the minimum, `quantile10`, `quantile25`, mean, median, `quantile75`, `quantile90`, and maximum of the LSD values is used as an approximate LSD in determining the significance of the pairwise differences;
6. `false.neg` is a `data.frame` with the numbers of false negatives for the pairwise comparisons within each levels-combination of the LSDby variables when each of the minimum, `quantile10`, `quantile25`, mean, median, `quantile75`, `quantile90`, and maximum of the LSD values is used as an approximate LSD in determining the significance of the pairwise differences;
7. `per.pred.accuracy` is a `data.frame` with the accuracies of the minimum, `quantile10`, `quantile25`, mean, median, `quantile75`, `quantile90`, and maximum of the LSD values for a set of predictions when these statistics are used to represent the LSDs for the comparisons amongst the set of predictions;
8. `LSD` is a square matrix containing the LSD values for all pairwise comparisons of the predictions contained in the supplied `alldiffs.obj`.

In the `statistics`, `accuracy`, `false.pos` and `false.neg` `data.frames`, `c` is the number of pairwise comparisons on which the values in the same row are based. The accuracy measure is specified by the `LSDaccuracy` argument.

### Author(s)

Chris Brien

### See Also

[asremlPlus-package](#), [plotLSDs.data.frame](#), [plotLSDs.alldiffs](#),  
[plotLSDerrors.alldiffs](#), [plotLSDerrors.data.frame](#), [recalcLSD.alldiffs](#),  
[pickLSDstatistics.alldiffs](#), [redoErrorIntervals.alldiffs](#)

### Examples

```
data(WaterRunoff.dat)

##Use asreml to get predictions and associated statistics
```

```

## Not run:
asreml.options(keep.order = TRUE) #required for asreml-R4 only
current.asr <- asreml(fixed = pH ~ Benches + (Sources * (Type + Species)),
                    random = ~ Benches:MainPlots,
                    keep.order=TRUE, data= WaterRunoff.dat)
current.asrt <- as.asrtests(current.asr, NULL, NULL)
TS.diffs <- predictPlus(classify = "Sources:Type",
                      asreml.obj = current.asr,
                      wald.tab = current.asrt$wald.tab,
                      present = c("Sources", "Type", "Species"))

## End(Not run)

## Use lmeTest and emmeans to get predictions and associated statistics

if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(pH ~ Benches + (Sources * (Type + Species)) +
                          (1|Benches:MainPlots),
                          data=na.omit(WaterRunoff.dat))
  TS.emm <- emmeans::emmeans(m1.lmer, specs = ~ Sources:Type)
  TS.preds <- summary(TS.emm)
  den.df <- min(TS.preds$df, na.rm = TRUE)
  ## Modify TS.preds to be compatible with a predictions.frame
  TS.preds <- as.predictions.frame(TS.preds, predictions = "emmean",
                                  se = "SE", interval.type = "CI",
                                  interval.names = c("lower.CL", "upper.CL"))

  ## Form an all.diffs object and check its validity
  TS.vcov <- vcov(TS.emm)
  TS.diffs <- allDifferences(predictions = TS.preds, classify = "Sources:Type",
                           vcov = TS.vcov, tdf = den.df)
  validAlldiffs(TS.diffs)
}

## Plot p-values for predictions obtained using asreml or lmerTest
if (exists("TS.diffs"))
{
  ##Explore the LSD values for predictions obtained using asreml or lmerTest
  LSDstat <- exploreLSDs(TS.diffs, LSDtype = "factor.combinations",
                        LSDby = "Sources")
}

```

---

|                     |                                                                                  |
|---------------------|----------------------------------------------------------------------------------|
| facCombine.alldiffs | <i>Combines several factors into one in the components of an alldiffs.object</i> |
|---------------------|----------------------------------------------------------------------------------|

---

## Description

Combines several [factors](#), in the prediction component of object, into one whose levels are the combinations of the used levels of the individual [factors](#). The matching changes are made to the other components and the attributes of the [alldiffs.object](#). If any of the factors to be combined are in LSDby, they are removed from the LSDby, unless the factors to be combined are

exactly those in the LSDby. The levels of the factors are combined using fac.combine from the dae package.

### Usage

```
## S3 method for class 'alldiffs'
facCombine(object, factors, order="standard",
           combine.levels=TRUE, sep="_", level.length = NA, ...)
```

### Arguments

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object         | An <code>alldiffs.object</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| factors        | A <code>character</code> containing the names of <code>factors</code> in the prediction component of object whose levels are to be combined.                                                                                                                                                                                                                                                                                                                                                                                                |
| order          | Either standard or yates. The order in which the levels combinations of the <code>factors</code> are to be considered as numbered when forming the levels of the combined <code>factor</code> ; standard numbers them as if they are arranged in standard order, that is with the levels of the first factor moving slowest and those of the last factor moving fastest; yates numbers them as if they are arranged in Yates order, that is with the levels of the first factor moving fastest and those of the last factor moving slowest. |
| combine.levels | A logical specifying whether the levels labels of the new <code>factor</code> are to be combined from those of the <code>factors</code> being combined. The default is to use the integers from 1 to the product of the numbers of combinations of used levels of the individual <code>factors</code> , numbering the levels according to order.                                                                                                                                                                                            |
| sep            | A character string to separate the levels when <code>combine.levels = TRUE</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| level.length   | The maximum number of characters from the levels of factors to use in the row and column labels of the tables of pairwise differences and their p-values and standard errors.                                                                                                                                                                                                                                                                                                                                                               |
| ...            | Further arguments passed to <code>redoErrorIntervals.alldiffs</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

### Value

A modified `alldiffs.object`.

### Author(s)

Chris Brien

### See Also

`as.alldiffs`, `allDifferences.data.frame`, `print.alldiffs`, `sort.alldiffs`, `renewClassify.alldiffs`; `fac.combine` in package **dae**.

### Examples

```
data("Ladybird.dat")

## Use asrem1 to get predictions and associated statistics

## Not run:
m1.asr <- asrem1(logitP ~ Host*Cadavers*Ladybird,
```

```

        random = ~ Run,
        data = Ladybird.dat)
current.asrt <- as.asrtests(m1.asr)
HCL.pred <- asreml::predict.asreml(m1.asr, classify="Host:Cadavers:Ladybird",
                                sed=TRUE)

HCL.preds <- HCL.pred$pvals
HCL.sed <- HCL.pred$sed
HCL.vcov <- NULL
wald.tab <- current.asrt$wald.tab
den.df <- wald.tab[match("Host:Cadavers:Ladybird", rownames(wald.tab)), "denDF"]

## End(Not run)

## Use lmeTest and emmeans to get predictions and associated statistics
if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(logitP ~ Host*Cadavers*Ladybird + (1|Run),
                          data=Ladybird.dat)
  HCL.emm <- emmeans::emmeans(m1.lmer, specs = ~ Host:Cadavers:Ladybird)
  HCL.preds <- summary(HCL.emm)
  den.df <- min(HCL.preds$df)
  ## Modify HCL.preds to be compatible with a predictions.frame
  HCL.preds <- as.predictions.frame(HCL.preds, predictions = "emmean",
                                   se = "SE", interval.type = "CI",
                                   interval.names = c("lower.CL", "upper.CL"))

  HCL.vcov <- vcov(HCL.emm)
  HCL.sed <- NULL
}

## Use the predictions obtained with either asreml or lmerTest
if (exists("HCL.preds"))
{
  ## Form an all.diffs object
  HCL.diffs <- as.alldiffs(predictions = HCL.preds, classify = "Host:Cadavers:Ladybird",
                          sed = HCL.sed, vcov = HCL.vcov, tdf = den.df)

  ## Check the class and validity of the alldiffs object
  is.alldiffs(HCL.diffs)
  validAlldiffs(HCL.diffs)

  ## Combine Cadavers and Ladybird
  HCL.diffs <- facCombine(HCL.diffs, factors = c("Cadavers", "Ladybird"))

  ## Check the validity of HCL.diffs
  validAlldiffs(HCL.diffs)
}

```

---

facRecast.alldiffs

*Reorders and/or revises the factor levels using the order of old levels in levels.order and the new labels for the levels given in newlabels. The values in levels.order must be unique.*

---

## Description

Reorders and revises the levels and labels of a [factor](#), in the prediction component of an [alldiffs.object](#). The values in the `levels.order` vector should be the same as the levels in the existing [factor](#), but the order can be changed. To revise the levels, specify the new levels in the `newlabels` vector and these will replace the corresponding value in the `levels.order` vector. The matching changes are made to the other components and attributes of the [alldiffs.object](#).

## Usage

```
## S3 method for class 'alldiffs'
facRecast(object, factor, levels.order = NULL, newlabels = NULL, ...)
```

## Arguments

|                           |                                                                                                                                                                                                                                                              |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>object</code>       | An <a href="#">alldiffs.object</a> .                                                                                                                                                                                                                         |
| <code>factor</code>       | A <a href="#">character</a> containing the name of a <a href="#">factor</a> in the prediction component of <code>object</code> whose levels and labels are to be recast.                                                                                     |
| <code>levels.order</code> | A <a href="#">vector</a> of length <code>levels(factor)</code> containing the old levels in the new order for the factor being created. If <code>levels.order</code> is <code>NULL</code> , then the current levels of <code>levels(factor)</code> are used. |
| <code>newlabels</code>    | A <a href="#">vector</a> of length <code>levels(factor)</code> containing values to use in the revision.                                                                                                                                                     |
| <code>...</code>          | Further arguments passed to the <a href="#">factor</a> call creating the new <a href="#">factor</a> .                                                                                                                                                        |

## Value

A modified [alldiffs.object](#).

## Author(s)

Chris Brien

## See Also

[as.alldiffs](#), [allDifferences.data.frame](#), [print.alldiffs](#), [sort.alldiffs](#), [facCombine.alldiffs](#), [facRename.alldiffs](#), [renewClassify.alldiffs](#); `fac.recast` in package **dae**.

## Examples

```
data("Ladybird.dat")

## Use asreml to get predictions and associated statistics

## Not run:
m1.asr <- asreml(logitP ~ Host*Cadavers*Ladybird,
                random = ~ Run,
                data = Ladybird.dat)
current.asrt <- as.asrtests(m1.asr)
HCL.pred <- asreml::predict.asreml(m1.asr, classify="Host:Cadavers:Ladybird",
                                sed=TRUE)

HCL.preds <- HCL.pred$pvals
HCL.sed <- HCL.pred$sed
HCL.vcov <- NULL
```



```

wald.tab <- current.asrt$wald.tab
den.df <- wald.tab[match("Host:Cadavers:Ladybird", rownames(wald.tab)), "denDF"]

## End(Not run)

## Use lmeTest and emmeans to get predictions and associated statistics
if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(logitP ~ Host*Cadavers*Ladybird + (1|Run),
                           data=Ladybird.dat)
  HCL.emm <- emmeans::emmeans(m1.lmer, specs = ~ Host:Cadavers:Ladybird)
  HCL.preds <- summary(HCL.emm)
  den.df <- min(HCL.preds$df)
  ## Modify HCL.preds to be compatible with a predictions.frame
  HCL.preds <- as.predictions.frame(HCL.preds, predictions = "emmean",
                                   se = "SE", interval.type = "CI",
                                   interval.names = c("lower.CL", "upper.CL"))

  HCL.vcov <- vcov(HCL.emm)
  HCL.sed <- NULL
}

## Use the predictions obtained with either asreml or lmerTest
if (exists("HCL.preds"))
{
  ## Form an all.diffs object
  HCL.diffs <- allDifferences(predictions = HCL.preds, classify = "Host:Cadavers:Ladybird",
                             sed = HCL.sed, vcov = HCL.vcov, tdf = den.df)

  ## Check the class and validity of the alldiffs object
  is.alldiffs(HCL.diffs)
  validAlldiffs(HCL.diffs)

  ## Recast the Ladybird and Host factors
  HCL.diffs <- facRecast(HCL.diffs, factor = "Ladybird",
                        newlabels = c("none", "present"))
  HCL.diffs <- facRecast(HCL.diffs, factor = "Ladybird",
                        levels.order = c("present", "none"),
                        newlabels = c("yes", "no"))
  HCL.diffs <- facRecast.alldiffs(HCL.diffs, factor = "Host",
                                 levels.order = c("trefoil", "bean"))

  ## Check the validity of HCL.diffs
  validAlldiffs(HCL.diffs)
}

```

---

|                    |                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------|
| facRename.alldiffs | <i>Renames <b>factors</b> in the prediction component of an <b>alldiffs.object</b>.</i> |
|--------------------|-----------------------------------------------------------------------------------------|

---

## Description

Renames **factors** in the prediction component of an **alldiffs.object**. These changes are propagated to the other components and attributes of the **alldiffs.object**.

**Usage**

```
## S3 method for class 'alldiffs'
facRename(object, factor.names, newnames, ...)
```

**Arguments**

|              |                                                                                                                                         |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| object       | An <code>alldiffs.object</code> .                                                                                                       |
| factor.names | A <code>character</code> containing the names of the <code>factors</code> in the prediction component of object that are to be renamed. |
| newnames     | A <code>character</code> containing the new names of the <code>factors</code> in the prediction component of object.                    |
| ...          | Provision for passing arguments to functions called internally - not used at present.                                                   |

**Value**

A modified `alldiffs.object`.

**Author(s)**

Chris Brien

**See Also**

`as.alldiffs`, `allDifferences.data.frame`, `print.alldiffs`, `sort.alldiffs`, `facCombine.alldiffs`, `facRecast.alldiffs`, `renewClassify.alldiffs`; `fac.recast` in package **dae**.

**Examples**

```
data("Ladybird.dat")

## Use asreml to get predictions and associated statistics

## Not run:
m1.asr <- asreml(logitP ~ Host*Cadavers*Ladybird,
                random = ~ Run,
                data = Ladybird.dat)
current.asrt <- as.asrtests(m1.asr)
HCL.pred <- asreml::predict.asreml(m1.asr, classify="Host:Cadavers:Ladybird",
                                  sed=TRUE)

HCL.preds <- HCL.pred$pvals
HCL.sed <- HCL.pred$sed
HCL.vcov <- NULL
wald.tab <- current.asrt$wald.tab
den.df <- wald.tab[match("Host:Cadavers:Ladybird", rownames(wald.tab)), "denDF"]

## End(Not run)

## Use lmeTest and emmeans to get predictions and associated statistics
if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(logitP ~ Host*Cadavers*Ladybird + (1|Run),
```

```

                                data=Ladybird.dat)
HCL.emm <- emmeans::emmeans(m1.lmer, specs = ~ Host:Cadavers:Ladybird)
HCL.preds <- summary(HCL.emm)
den.df <- min(HCL.preds$df)
## Modify HCL.preds to be compatible with a predictions.frame
HCL.preds <- as.predictions.frame(HCL.preds, predictions = "emmean",
                                se = "SE", interval.type = "CI",
                                interval.names = c("lower.CL", "upper.CL"))

HCL.vcov <- vcov(HCL.emm)
HCL.sed <- NULL
}

## Use the predictions obtained with either asreml or lmerTest
if (exists("HCL.preds"))
{
  ## Form an all.diffs object
  HCL.diffs <- allDifferences(predictions = HCL.preds,
                            classify = "Host:Cadavers:Ladybird",
                            sed = HCL.sed, vcov = HCL.vcov, tdf = den.df)

  ## Check the class and validity of the alldiffs object
  is.alldiffs(HCL.diffs)
  validAlldiffs(HCL.diffs)

  ## Rename Cadavers
  HCL.diffs <- facRename(HCL.diffs, factor.names = "Cadavers", newnames = "Cadaver.nos")

  ## Check the validity of HCL.diffs
  validAlldiffs(HCL.diffs)
}

```

---

findLSDminerrors.alldiffs

*Find LSD values that minimize the number of errors in pairwise comparisons of predictions.*

---

## Description

Given an `alldiffs.object` with an `sed` component, a search is made of a set of equally spaced values between the minimum and maximum values of the LSDs, calculated from the `sed` component of the `alldiffs.object`, to identify LSD values that minimize the number of errors made in deciding on the significance of pairs of predicted values stored in the `alldiffs.object`. If `LSDtype` is set to `overall`, a search is made over the range of LSD values for all pairwise comparisons for a single LSD value; if `LSDtype` is set to `factor.combinations`, a separate search is made over the LSD values for the set of pairwise comparisons for each `factor.combination` in order to identify a single value for each set. The number of values used in the search is controlled by the argument `nvalues`. For each value in the search, the numbers of false positives and false negatives resulting from employing it as the LSD for each set of pairwise comparisons is calculated. A criterion that combines the false positives and negative is calculated using the `false.pos.wt`, the criterion being the number of false positives times the `false.pos.wt` plus the number of false negatives. The value chosen for the LSD is the smallest value from amongst those with the minimum value of the criterion and the least number of false positives. A secondary search with 10 equally spaced values is made of the interval below the chosen value and the search value immediately below it to

check whether the chosen grid value can be further reduced without changing the value of either its criterion or the number of false positives.

The primary options for changing the numbers of errors associated with the values resulting from the searching is to manipulate the LSDby and/or false.pos.wt arguments.

## Usage

```
## S3 method for class 'alldiffs'
findLSDminerrors(alldiffs.obj,
                  LSDtype = "overall", LSDby = NULL,
                  alpha = 0.05,
                  false.pos.wt = 10, nvalues = 100,
                  retain.zeroLSDs = FALSE,
                  zero.tolerance = .Machine$double.eps ^ 0.5,
                  trace = FALSE, ...)
```

## Arguments

- |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| alldiffs.obj    | An <a href="#">alldiffs.object</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| LSDtype         | A <a href="#">character</a> string that can be overall or factor.combinations. It determines whether the minimum LSD values that are obtained are those for (i) all pairwise comparisons of the predicted values, unless there is only one, possibly repeated, prediction, when a notional LSD is calculated, or (ii) for each combination of the values of the <a href="#">factors</a> and <a href="#">numerics</a> named in LSDby, unless there is only one prediction for a combination, when notional LSDs are calculated. The LSDtype specified here does not have to match that used in the creating the <a href="#">alldiffs.object</a> .                                                                                                                                                                                                   |
| LSDby           | A <a href="#">character</a> (vector) of variables names, being the names of some of the <a href="#">factors</a> or <a href="#">numerics</a> in the classify; for each combination of the values of the <a href="#">factors</a> and <a href="#">numerics</a> , the LSD errors are to be computed when LSDtype is factor.combinations. The LSDby specified here does not have to match that used in the creating the <a href="#">alldiffs.object</a> .                                                                                                                                                                                                                                                                                                                                                                                               |
| alpha           | A <a href="#">numeric</a> specifying the significance level for an LSD to compare a pair of predictions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| false.pos.wt    | A <a href="#">numeric</a> that specifies the weight (e.g. 3 so that a false positive is considered to be equivalent to three false negatives) to apply to the number of false positives in calculating the weighted sums of the numbers of false positives and negatives that is used as the criterion to be minimized in selecting the LSD value that results in the minimum number of errors. If, for a particular LSD value, the number of false positives is $p$ , the number of false negatives is $n$ and $w$ the value of false.pos.wt, then then criterion for that LSD value is $(wp) + n$ .<br>The default of 10 for false.pos.wt greatly favours false negatives; a value with one false positive and no false negative will only be chosen over a value with no false positive when the latter value has more than 10 false negatives. |
| nvalues         | A <a href="#">numeric</a> specifying the number of equally spaced LSD values, between the minimum and maximum LSD values for a set of paired comparisons, to be used in the search for the LSD value with the minimum number of errors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| retain.zeroLSDs | A <a href="#">logical</a> indicating whether to retain or omit LSDs that are zero when calculating the numbers of errors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

zero.tolerance A **numeric** specifying the value such that if an LSD is less than it, the LSD will be considered to be zero.

trace A **logical** indicating whether details of the searching are to be output.

... Provision for passing arguments to functions called internally - not used at present.

### Value

A **data.frame** containing the chosen LSD(s), its(their) numbers of false positives and negatives and the value(s) of the false criterion.

### Author(s)

Chris Brien

### See Also

[asremlPlus-package](#), [exploreLSDs.alldiffs](#) [plotLSDs.data.frame](#), [plotLSDs.alldiffs](#), [plotLSDerrors.alldiffs](#), [plotLSDerrors.data.frame](#), [recalcLSD.alldiffs](#), [redoErrorIntervals.alldiffs](#)

### Examples

```
data(WaterRunoff.dat)

##Use asreml to get predictions and associated statistics

## Not run:
asreml.options(keep.order = TRUE) #required for asreml-R4 only
current.asr <- asreml(fixed = pH ~ Benches + (Sources * (Type + Species)),
                    random = ~ Benches:MainPlots,
                    keep.order=TRUE, data= WaterRunoff.dat)
current.asrt <- as.asrtests(current.asr, NULL, NULL)
TS.diffs <- predictPlus(classify = "Sources:Type",
                      asreml.obj = current.asr,
                      wald.tab = current.asrt$wald.tab,
                      present = c("Sources", "Type", "Species"))

## End(Not run)

## Use lmeTest and emmeans to get predictions and associated statistics

if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(pH ~ Benches + (Sources * (Type + Species)) +
                        (1|Benches:MainPlots),
                        data=na.omit(WaterRunoff.dat))
  TS.emm <- emmeans::emmeans(m1.lmer, specs = ~ Sources:Type)
  TS.preds <- summary(TS.emm)
  den.df <- min(TS.preds$df, na.rm = TRUE)
  ## Modify TS.preds to be compatible with a predictions.frame
  TS.preds <- as.predictions.frame(TS.preds, predictions = "emmean",
                                se = "SE", interval.type = "CI",
                                interval.names = c("lower.CL", "upper.CL"))
}
```

```

## Form an all.diffs object and check its validity
TS.vcov <- vcov(TS.emm)
TS.diffs <- allDifferences(predictions = TS.preds, classify = "Sources:Type",
                           vcov = TS.vcov, tdf = den.df)
validAlldiffs(TS.diffs)
}

## Choose LSD values with the minimum number of error for pairwise comparisons of
## the predictions obtained using asreml or lmerTest
if (exists("TS.diffs"))
{
  ##Pick the LSD values for predictions obtained using asreml or lmerTest
  minLSD <- findLSDminerrors(TS.diffs)
  TS.diffs <- redoErrorIntervals(TS.diffs, LSDtype = "supplied", LSDsupplied = minLSD["LSD"])
  TS.diffs$LSD
  minLSDs <- findLSDminerrors(TS.diffs, LSDtype = "factor.combinations",
                              LSDby = "Sources")
  TS.diffs <- redoErrorIntervals(TS.diffs, LSDtype = "supplied",
                                LSDby = "Sources", LSDsupplied = minLSDs["LSD"])
  TS.diffs$LSD
}

```

---

getASRemlVersionLoaded

*Finds the version of asreml that is loaded and returns the initial characters in version.*

---

## Description

Checks that asreml is loaded and, if it is, returns the first nchar characters of the version that is loaded.

## Usage

```
getASRemlVersionLoaded(nchar = NULL, notloaded.fault = FALSE)
```

## Arguments

nchar                    The number of characters in the asreml version to get.

notloaded.fault        A [logical](#) indicating whether a fault is to occur if asreml is not loaded.

## Value

A character, being the first nchar characters of the version of asreml that is loaded.

## Author(s)

Chris Brien

## See Also

[loadASRemlVersion](#).

**Examples**

```
## Not run:
getASremlVersionLoaded()
## End(Not run)
```

---

|                    |                                                 |
|--------------------|-------------------------------------------------|
| getFormulae.asreml | <i>Gets the formulae from an asreml object.</i> |
|--------------------|-------------------------------------------------|

---

**Description**

Gets the [formulae](#) nominated in the which argument from the call stored in an asreml object.

**Usage**

```
## S3 method for class 'asreml'
getFormulae(asreml.obj, which = c("fixed", "random", "residual"),
            expanded = FALSE, envir = parent.frame(), ...)
```

**Arguments**

|            |                                                                                                                                                                                                                                                                                                                   |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| asreml.obj | An object resulting from the fitting of a model using asreml.                                                                                                                                                                                                                                                     |
| which      | A character listing the <a href="#">formula(e)</a> to be extracted from the call stored in asreml.obj. it should be some combination of fixed, random, residual, sparse and all. If all is included then all <a href="#">formula(e)</a> will be returned, those not having been specified in the call being NULL. |
| expanded   | A logical indicating whether terms are to be expanded to the sum of a set of individual terms.                                                                                                                                                                                                                    |
| envir      | The environment in which the <a href="#">formula(e)</a> are to be evaluated. May also be NULL, a list, a data.frame, a pairlist or an integer as specified to sys.call.                                                                                                                                           |
| ...        | Arguments passed on to update.formula and ultimately to terms.formula.                                                                                                                                                                                                                                            |

**Value**

A list containing a component with each of the extracted [formula\(e\)](#), the name of a component being the [formula](#) that it contains.

**Author(s)**

Chris Brien

**See Also**

[printFormulae.asreml](#)

## Examples

```
## Not run:
data(Wheat.dat)
current.asr <- asreml(yield ~ Rep + WithinColPairs + Variety,
                     random = ~ Row + Column + units,
                     residual = ~ ar1(Row):ar1(Column),
                     data=Wheat.dat)
getFormulae(current.asr)

## End(Not run)
```

---

getTestEntry.asrtests *Gets the entry for a test recorded in the test.summary data.frame of an asrtests.object*

---

## Description

Matches the label in the term column of the test.summary data.frame in the supplied [asrtests.object](#) and extracts the line for it. It only matches the last occurrence of label.

## Usage

```
## S3 method for class 'asrtests'
getTestEntry(asrtests.obj, label, error.absent = TRUE, ...)
```

## Arguments

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| asrtests.obj | An <a href="#">asrtests.object</a> containing the components (i) asreml.obj, (ii) wald.tab, and (iii) test.summary.                                                                                                                                                                                                                                                                                                                       |
| label        | A character specifying the label of the test for which the entry is required. If <a href="#">testtranfix.asrtests</a> was used for the test of interest, then the label will be the value of the term argument supplied to <a href="#">testtranfix.asrtests</a> . For <a href="#">changeModelOnIC.asrtests</a> , the label will be the value of the label argument. Other arguments will be relevant for other test and change functions. |
| error.absent | A logical indicating whether the absence of the supplied label is to result in an error. If set to FALSE, NULL is returned.                                                                                                                                                                                                                                                                                                               |
| ...          | provision for passing arguments to functions called internally - not used at present.                                                                                                                                                                                                                                                                                                                                                     |

## Value

A one-line data.frame containing the entry or, error.absent is NULL, NULL.

## Author(s)

Chris Brien

## See Also

[getTestPvalue.asrtests](#), [as.asrtests](#),  
[testtranfix.asrtests](#), [testswapan.asrtests](#), [testresidual.asrtests](#),  
[changeModelOnIC.asrtests](#), [changeTerms.asrtests](#), [chooseModel.asrtests](#)



## Examples

```
## Not run:
data(Wheat.dat)
current.asr <- asreml(yield ~ Rep + WithinColPairs + Variety,
                     random = ~ Row + Column + units,
                     residual = ~ ar1(Row):ar1(Column),
                     data=Wheat.dat)
current.asrt <- as.srtests(current.asr, NULL, NULL)
current.asrt <- rmboundary(current.asrt)
# Test nugget term
current.asrt <- testranfix(current.asrt, "units", positive=TRUE)
getTestEntry(current.asrt, label = "units")

## End(Not run)
```

---

```
getTestPvalue.asrtests
```

*Gets the p-value for a test recorded in the test.summary data.frame of an [asrtests.object](#)*

---

## Description

Matches the label in the term column of the test.summary data.frame in the supplied [asrtests.object](#) and extracts its p-value. It only matches the last occurrence of label.

## Usage

```
## S3 method for class 'asrtests'
getTestPvalue(asrtests.obj, label, ...)
```

## Arguments

|              |                                                                                                                                                                                                                                                                                                                              |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| asrtests.obj | An <a href="#">asrtests.object</a> containing the components (i) asreml.obj, (ii) wald.tab, and (iii) test.summary.                                                                                                                                                                                                          |
| label        | A character specifying the label of the test for which the p-value is required. If <a href="#">testranfix.asrtests</a> was used for the test of interest, then the label will be the value of the term argument supplied to <a href="#">testranfix.asrtests</a> . Other arguments will be relevant for other test functions. |
| ...          | provision for passing arguments to functions called internally - not used at present.                                                                                                                                                                                                                                        |

## Value

An numeric containing the p-value. It can be NA, for example when a p-value could not be calculated.

## Author(s)

Chris Brien

**See Also**

[getTestEntry.asrtests](#), [as.asrtests](#),  
[testtranfix.asrtests](#), [testswapran.asrtests](#), [testresidual.asrtests](#),  
[changeTerms.asrtests](#), [chooseModel.asrtests](#)

**Examples**

```
## Not run:
data(Wheat.dat)
current.asr <- asreml(yield ~ Rep + WithinColPairs + Variety,
  random = ~ Row + Column + units,
  residual = ~ ar1(Row):ar1(Column),
  data=Wheat.dat)
current.asrt <- as.asrtests(current.asr, NULL, NULL)
current.asrt <- rmboundary(current.asrt)
# Test nugget term
current.asrt <- testtranfix(current.asrt, "units", positive=TRUE)
getTestPvalue(current.asrt, label = "units")

## End(Not run)
```

---

infoCriteria

---

*Computes AIC and BIC for models.*


---

**Description**

Computes Akaike and Bayesian (Schwarz) Information Criteria for models. Either the Restricted Maximum likelihood (REML) or the full likelihood (full) can be used. The full likelihood, evaluated using REML estimates is used when it is desired to compare models that differ in their fixed models.

**Usage**

```
## S3 method for class 'asreml'
infoCriteria(object, DF = NULL,
  bound.exclusions = c("F","B","S","C"),
  ICl likelihood = "REML", fixedDF = NULL, varDF = NULL, ...)
## S3 method for class 'list'
infoCriteria(object, bound.exclusions = c("F","B","S","C"),
  ICl likelihood = "REML", fixedDF = NULL, varDF = NULL, ...)
```

**Arguments**

|        |                                                                                                                                                                                                                                                                       |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object | An asreml object resulting from the fitting of a model using REML or a list of asreml objects. If the components of the list are named, then those names will be used as the rownames for the returned data frame.                                                    |
| DF     | A numeric giving the number of estimated variance parameters. If NULL then this is determined from the information in object. This argument has been replaced by varDF, but is retained for compatibility with legacy code. It is not available with the list method. |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>bound.exclusions</code> | A character specifying the bound (constraint) codes that will result in a variance parameter being excluded from the count of estimated variance parameters. If set to <code>NULL</code> then none will be excluded.                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>IClikelihood</code>     | A character specifying whether Restricted Maximum Likelihood (REML) or the full likelihood, evaluated using REML estimates, ( <code>full</code> ) are to be used in calculating the information criteria for family set to <code>asr_gaussian</code> . For family set to <code>asr_binomial</code> or <code>asr_poisson</code> and with dispersion set to 1, the deviance is extracted from object and used to calculate the AIC and BIC (as suggested by Damian Collins); the setting of <code>IClikelihood</code> is ignored and the log-likelihood set to <code>NA</code> . The information criteria are not valid for other settings of family and dispersion. |
| <code>fixedDF</code>          | A numeric giving the number of estimated fixed parameters. If <code>NULL</code> then this is determined from the information in object. For object a list only a single value that is used for all components of the list has been implemented.                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>varDF</code>            | A numeric giving the number of estimated variance parameters. If <code>NULL</code> then this is determined from the information in object. It replaces the <code>DF</code> argument. For object a list only a single value that is used for all components of the list has been implemented.                                                                                                                                                                                                                                                                                                                                                                       |
| <code>...</code>              | Provision for passing arguments to functions called internally - not used at present.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

## Details

The variance degrees of freedom (`varDF`) are the number of number of variance parameters that have been estimated, excluding those whose estimates have a code for bound specified in `bound.exclusions`. If `varDF` is not `NULL`, the supplied value is used. Otherwise `varDF` is determined from the information in object, i.e. if object is an `asreml` object then from it, or if object is a list then from each `asreml` object in the list. Similarly, the fixed degrees of freedom (`fixedDF`) are the number of number of fixed parameters that have been estimated, any coefficients that have the value `NA` being excluded. If `fixedDF` is not `NULL`, the supplied value is used. Otherwise `fixedDF` is determined from the information in object.

If ASReml-R version 4 is being used then the codes specified in `bound.exclusions` are not restricted to a subset of the default codes, but a warning is issued if a code other than these is specified. For ASReml-R version 3, only a subset of the default codes are allowed: F (Fixed), B (Boundary), C (Constrained) and S (Singular).

The calculation of the information criteria is an adaptation of the code supplied in File S1 of Verbyla (2019). The log-likelihood is calculated as  $\text{loglik} = \log(\text{REML}) - \log(|\mathbf{C}|)/2$ , where  $\mathbf{C}$  is the inverse coefficient matrix; the term involving  $\mathbf{C}$  is omitted for REML. The AIC is calculated as  $-2 * \text{loglik} + 2 * (\text{varDF} + \text{fixedDF})$  and the BIC as  $-2 * \text{loglik} + (\text{fixedDF} + \text{varDF}) * \log(n - r + \text{fixedDF})$ , where  $n$  is the number of observations and  $r$  is the rank of the fixed effects design matrix. For REML, `fixedDF` = 0.

## Value

A `data.frame` containing the numbers of estimated fixed (`fixedDF`) and variance (`varDF`) parameters, the number of bound parameters (`NBound`), AIC, BIC and the value of the log-likelihood (`loglik`). All elements of the `data.frame` will be set to `NA` for the invalid combinations of family and dispersion as noted in the `IClikelihood` argument. If object is a list and its components are named, then those names will be used to set the rownames of the `data.frame`.

**Author(s)**

Chris Brien

**References**

Verbyla, A. P. (2019). A note on model selection using information criteria for general linear models estimated using REML. *Australian & New Zealand Journal of Statistics*, **61**, 39–50. doi:10.1111/anzs.12254.

**See Also**

[REMLRT.asreml](#), [changeTerms.asrtests](#), [changeModelOnIC.asrtests](#)

**Examples**

```
## Not run:
data(Wheat.dat)
## Fit several models to the wheat data and calculate their ICs
# Fit initial model
m.max <- asreml(yield ~ Rep + WithinColPairs + Variety,
               random = ~ Row + Column + units,
               residual = ~ ar1(Row):ar1(Column),
               data=Wheat.dat)
infoCriteria(m.max.asr, IClikelihood = "full")

#Drop term for within Column pairs
m1 <- asreml(yield ~ Rep + Variety,
            random = ~ Row + Column + units,
            residual = ~ ar1(Row):ar1(Column),
            data=Wheat.dat)

#Drop nugget term
m2 <- asreml(yield ~ Rep + WithinColPairs + Variety,
            random = ~ Row + Column,
            residual = ~ ar1(Row):ar1(Column),
            data=Wheat.dat)

#Drop Row autocorrelation
m3 <- asreml(yield ~ Rep + WithinColPairs + Variety,
            random = ~ Row + Column + units,
            residual = ~ Row:ar1(Column),
            data=Wheat.dat)

#Drop Col autocorrelation
m4 <- asreml(yield ~ Rep + WithinColPairs + Variety,
            random = ~ Row + Column + units,
            residual = ~ ar1(Row):Column,
            data=Wheat.dat)

mods.asr <- list(m.max, m1, m2, m3, m4)
infoCriteria(mods.asr, IClikelihood = "full")

## End(Not run)
```

---

|             |                                                     |
|-------------|-----------------------------------------------------|
| is.alldiffs | <i>Tests whether an object is of class alldiffs</i> |
|-------------|-----------------------------------------------------|

---

**Description**

A single-line function that tests whether an object is of class alldiffs.

**Usage**

```
is.alldiffs(object)
```

**Arguments**

object            An object to be tested.

**Value**

A logical.

**Author(s)**

Chris Brien

**See Also**

[asremlPlus-package](#), [alldiffs.object](#), [is.alldiffs](#), [as.alldiffs](#)

**Examples**

```
data(Oats.dat)

## Use lmerTest and emmeans to get predictions and associated statistics
if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(Yield ~ Nitrogen*Variety + (1|Blocks/Wplots),
                           data=Oats.dat)
  Var.emm <- emmeans::emmeans(m1.lmer, specs = ~ Nitrogen:Variety)
  Var.preds <- summary(Var.emm)
  den.df <- min(Var.preds$df)
  ## Modify Var.preds to be compatible with a predictions.frame
  Var.preds <- as.predictions.frame(Var.preds, predictions = "emmean",
                                   se = "SE", interval.type = "CI",
                                   interval.names = c("lower.CL", "upper.CL"))

  Var.vcov <- vcov(Var.emm)
  Var.sed <- NULL

  ## Form an all.diffs object
  Var.diffs <- as.alldiffs(predictions = Var.preds, classify = "Nitrogen:Variety",
                           sed = Var.sed, vcov = Var.vcov, tdf = den.df)

  ## check the class of Var.diffs
  is.alldiffs(Var.diffs)
}
```

---

`is.asrtests`*Tests whether an object is of class asrtests*

---

**Description**

A single-line function that tests whether an object is of class asrtests.

**Usage**

```
is.asrtests(object)
```

**Arguments**

`object`            An object to be tested.

**Value**

A logical.

**Author(s)**

Chris Brien

**See Also**

[asremlPlus-package](#), [asrtests.object](#), [is.asrtests](#), [as.asrtests](#)

**Examples**

```
## Not run:
library(dae)
library(asreml)
library(asremlPlus)
## use ?Wheat.dat for data set details
data(Wheat.dat)

# Fit initial model
current.asr <- asreml(yield ~ Rep + WithinColPairs + Variety,
                     random = ~ Row + Column + units,
                     residual = ~ ar1(Row):ar1(Column),
                     data=Wheat.dat)

# Load current fit into an asrtests object
current.asrt <- as.asrtests(current.asr, NULL, NULL)

# check the class of current.asrt
is.asrtests(current.asrt)

## End(Not run)
```

---

is.predictions.frame    *Tests whether an object is of class predictions.frame*

---

## Description

A single-line function that tests whether an object is of class predictions.frame.

## Usage

```
is.predictions.frame(object)
```

## Arguments

object                      An object to be tested.

## Value

A logical.

## Author(s)

Chris Brien

## See Also

[asremlPlus-package](#), [predictions.frame](#), [validPredictionsFrame](#), [as.predictions.frame](#)

## Examples

```
data(Oats.dat)

## Use asreml to get predictions and associated statistics

## Not run:
m1.asr <- asreml(Yield ~ Nitrogen*Variety,
                 random=~Blocks/Wplots,
                 data=Oats.dat)
current.asrt <- as.asrtests(m1.asr)
Var.pred <- asreml::predict.asreml(m1.asr, classify="Nitrogen:Variety",
                                   sed=TRUE)
if (getASReMLVersionLoaded(nchar = 1) == "3")
  Var.pred <- Var.pred$predictions
Var.preds <- as.predictions.frame(Var.pred$pvals, se = "std.error",
                                  est.status = "status")

## End(Not run)

## Use lmerTest and emmeans to get predictions and associated statistics
if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(Yield ~ Nitrogen*Variety + (1|Blocks/Wplots),
                           data=Oats.dat)
  Var.emm <- emmeans::emmeans(m1.lmer, specs = ~ Nitrogen:Variety)
```

```

Var.preds <- summary(Var.emm)
Var.preds <- as.predictions.frame(Var.preds, predictions = "emmean",
                                se = "SE", interval.type = "CI",
                                interval.names = c("lower.CL", "upper.CL"))
}

if (exists("Var.preds"))
{
  ## Check the class and validity of the alldiffs object
  is.predictions.frame(Var.preds)
}

```

---

isCompoundSymmetric.matrix

*Tests whether an object of class matrix is compound symmetric*


---

## Description

Tests whether an object of class `matrix` is compound symmetric by checking whether all diagonal elements are equal and all off-diagonal elements are equal.

## Usage

```

## S3 method for class 'matrix'
isCompoundSymmetric(object, tol = 100 * .Machine$double.eps, ...)

```

## Arguments

|                     |                                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------|
| <code>object</code> | An object to be tested.                                                                             |
| <code>tol</code>    | a <code>numeric</code> scalar > 0 specifying that values smaller than it are considered to be zero. |
| <code>...</code>    | Provision for passing arguments to functions called internally - not used at present.               |

## Value

A logical.

## Author(s)

Chris Brien

## See Also

`isSymmetric`



## Examples

```
data(Oats.dat)

## Not run:
## Use asreml to get the variance matrix of a set of predictions

m1.asr <- asreml(Yield ~ Nitrogen*Variety,
                random=~Blocks/Wplots,
                data=Oats.dat)
current.asrt <- as.asrtests(m1.asr)
Var.pred <- asreml::predict.asreml(m1.asr, classify="Nitrogen:Variety",
                                   vcov = TRUE)
                                   est.status = "status")
isCompoundSymmetrix(Var.pred$vcov)

## End(Not run)

## Use lmerTest and emmeans to get the variance matrix of a set of predictions
if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(Yield ~ Nitrogen*Variety + (1|Blocks/Wplots),
                           data=Oats.dat)
  Var.emm <- emmeans::emmeans(m1.lmer, specs = ~ Nitrogen:Variety)
  Var.vcov <- vcov(Var.emm)
}

if (exists("Var.vcov"))
{
  ## Check the variance matrix of the predictions is compound symmetric
  isCompoundSymmetric(Var.vcov)
}
```

---

|                  |                                                                                                                 |
|------------------|-----------------------------------------------------------------------------------------------------------------|
| iterate.asrtests | <i>Subject the fitted asreml.obj stored in an asrtests.object to further iterations of the fitting process.</i> |
|------------------|-----------------------------------------------------------------------------------------------------------------|

---

## Description

In an effort to improve convergence, subject the fitted `asreml.obj` stored in an [asrtests.object](#) to further iterations of the fitting process; the model specification is not changed. While no change is made to the `test.summary`, the `wald.tab` is updated.

## Usage

```
## S3 method for class 'asrtests'
iterate(asrtests.obj, denDF="numeric", trace = FALSE, ...)
```

## Arguments

`asrtests.obj` an [asrtests.object](#) containing the components (i) `asreml.obj`, (ii) `wald.tab`, and (iii) `test.summary`.

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| denDF | Specifies the method to use in computing approximate denominator degrees of freedom when <code>wald.asreml</code> is called. Can be <code>none</code> to suppress the computations, <code>numeric</code> for numerical methods, <code>algebraic</code> for algebraic methods or <code>default</code> , the default, to automatically choose numeric or algebraic computations depending on problem size. The denominator degrees of freedom are calculated according to Kenward and Roger (1997) for fixed terms in the dense part of the model. |
| trace | If <code>TRUE</code> then partial iteration details are displayed when <code>ASReml-R</code> functions are invoked; if <code>FALSE</code> then no output is displayed.                                                                                                                                                                                                                                                                                                                                                                           |
| ...   | further arguments passed to <code>update.asreml</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

**Value**

An object of S3-class `asrtests`.

**Author(s)**

Chris Brien

**References**

Kenward, M. G., & Roger, J. H. (1997). Small sample inference for fixed effects from restricted maximum likelihood. *Biometrics*, **53**, 983-997.

**See Also**

[as.asrtests](#), [asrtests.object](#), [newfit.asreml](#)

**Examples**

```
## Not run:
current.asrt <- iterate(current.asrt)

## End(Not run)
```

---

|              |                                                                                |
|--------------|--------------------------------------------------------------------------------|
| Ladybird.dat | <i>Data for an experiment to investigate whether ladybirds transfer aphids</i> |
|--------------|--------------------------------------------------------------------------------|

---

**Description**

Welham et al. (2015, Example 8.2) describe a three-factor factorial experiment to investigate whether ladybirds transfer fungus to live aphids on plants. The three factors are Host plant (beans, trefoil), infected Cadavers (5, 10, 20), and Ladybird (-, +). A generalized randomized complete-block design is used to assign the three factors to 2 Runs, each of which involves 36 containers with a plant and live aphids. The response to be analyzed is the logit of the proportion of live aphids that were infected.

The columns in the data frame are: ID, Run, Plant, Host, Ladybird, Cadavers, Live, Infected, logitP, Prop. The column ID numbers the observations. Live, Infected, logitP, Prop are response variables.

**Usage**

```
data(Ladybird.dat)
```

**Format**

A data.frame containing 72 observations of 10 variables.

**Author(s)**

Chris Brien

**Source**

Welham, S. J., Gezan, S. A., Clark, S. J., & Mead, A. (2015). *Statistical Methods in Biology: Design and Analysis of Experiments and Regression*. Boca Raton: Chapman and Hall/CRC..

---

linTransform.alldiffs *Calculates a linear transformation of the predictions stored in an [alldiffs.object](#).*

---

**Description**

Effects the linear transformation of the predictions in the supplied [alldiffs.object](#), the transformation being specified by a [matrix](#) or a [formula](#). The values of the transformed values are stored in an [alldiffs.object](#). A [matrix](#) might be a contrast [matrix](#) or a [matrix](#) of weights for the levels of a [factor](#) used to obtain the weighted average over the levels of that [factor](#). A [formula](#) gives rise to a projection [matrix](#) that linearly transforms the predictions so that they conform to the model specified by the [formula](#), this model being a submodel of that inherent in the [classify](#).

If pairwise = TRUE, all pairwise differences between the linear transforms of the predictions, their standard errors, p-values and LSD statistics are computed as using [allDifferences.data.frame](#). This adds them to the [alldiffs.object](#) as additional list components named differences, sed, p.differences and LSD.

If a transformation has been applied (any one of transform.power is not one, scale is not one and offset is nonzero), the backtransforms of the transformed values and of the lower and upper limits of their error.intervals are added to a data.frame that is consistent with a [predictions.frame](#). If transform.power is other than one, the standard.error column of the data.frame is set to NA. This data.frame is added to the [alldiffs.object](#) as a list component called backtransforms.

The printing of the components produced is controlled by the tables argument. The order of plotting the levels of one of the factors indexing the predictions can be modified and is achieved using [sort.alldiffs](#).

**Usage**

```
## S3 method for class 'alldiffs'
linTransform(alldiffs.obj, classify = NULL, term = NULL,
             linear.transformation = NULL, EGLS.linTransform = TRUE,
             Vmatrix = FALSE, error.intervals = "Confidence",
             avsed.tolerance = 0.25, accuracy.threshold = NA,
             LSDtype = "overall", LSDsupplied = NULL,
             LSDby = NULL, LSDstatistic = "mean",
```

```

LSDaccuracy = "maxAbsDeviation",
zero.tolerance = .Machine$double.eps ^ 0.5,
response = NULL, response.title = NULL,
x.num = NULL, x.fac = NULL,
tables = "all", level.length = NA,
pairwise = TRUE, alpha = 0.05,
inestimable.rm = TRUE, ...)

```

## Arguments

- |                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| alldiffs.obj          | An <code>alldiffs.object</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| classify              | A <code>character</code> string giving the variables that define the margins of the multiway table corresponding to the predictions in <code>alldiffs.obj</code> . Multiway tables are specified by forming an interaction type term from the classifying variables, that is, separating the variable names with the <code>:</code> operator.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| term                  | A <code>character</code> string giving the variables that define the term that was fitted using <code>asreml</code> and that corresponds to <code>classify</code> . It only needs to be specified when it is different to <code>classify</code> ; it is stored as an attribute of the <code>alldiffs.object</code> . It is likely to be needed when the fitted model includes terms that involve both a <code>numeric</code> covariate and a <code>factor</code> that parallel each other; the <code>classify</code> would include the covariate and the term would include the factor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| linear.transformation | <p>A <code>formula</code> or a <code>matrix</code>. If a <code>formula</code> is given then it is taken to be a submodel of a model term corresponding to the <code>classify</code>. The projection matrix that transforms the predictions so that they conform to the submodel is obtained; the submodel does not have to involve variables in the <code>classify</code>, but the variables must be columns in the predictions component of <code>alldiffs.obj</code> and the space for the submodel must be a subspace of the space for the term specified by the <code>classify</code>. For example, for <code>classify</code> set to "A:B", the submodel <code>~ A + B</code> will result in the predictions for the combinations of A and B being made additive for the <code>factors</code> A and B. The submodel space corresponding to <code>A + B</code> is a subspace of the space A:B. In this case both the submodel and the <code>classify</code> involve only the factors A and B. To fit an intercept-only submodel, specify <code>linear.transformation</code> to be the formula <code>~1</code>.</p> <p>If a <code>matrix</code> is provided then it will be used to apply the linear transformation to the predictions. The number of rows in the <code>matrix</code> should equal the number of linear combinations of the predictions desired and the number of columns should equal the number of predictions.</p> <p>In either case, as well as the values of the linear combinations, their standard errors, pairwise differences and associated statistics are returned.</p> |
| EGLS.linTransform     | A <code>logical</code> indicating whether or not the <code>linear.transformation</code> of the predictions stored in an <code>alldiffs.object</code> by fitting a submodel supplied in a <code>formula</code> is to take into account the variance of the predictions using a Estimated Generalized Least Squares (EGLS) approach. This is likely to be appropriate when the variance matrix of the predictions is not compound symmetric i.e. when not all the variances are equal or not all the covariances are equal. If the variance matrix is compound symmetric, then the setting of <code>EGLS.linTransform</code> will not affect the transformed predictions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Vmatrix               | A <code>logical</code> indicating whether the variance matrix of the predictions will be stored as a component of the <code>alldiffs.object</code> that is returned. If <code>linear.transformation</code> is set, it will be stored irrespective of the value of <code>Vmatrix</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

`error.intervals`

A [character](#) string indicating the type of error interval, if any, to calculate in order to indicate uncertainty in the results. Possible values are "none", "StandardError", "Confidence" and "halfLeastSignificant". The default is for confidence limits to be used. The "halfLeastSignificant" option results in half the Least Significant Difference (LSD) being added and subtracted to the predictions, the LSD being calculated using the square root of the mean of the variances of all or a subset of pairwise differences between the predictions. If the LSD is zero, as can happen when predictions are constrained to be equal, then the limits of the error intervals are set to NA. If LSDtype is set to overall, the avsed.tolerance is not NA and the range of the SEDs divided by the average of the SEDs exceeds avsed.tolerance then the error.intervals calculations and the plotting will revert to confidence intervals.

`avsed.tolerance`

A [numeric](#) giving the value of the SED range, the range of the SEDs divided by the square root of the mean of the variances of all or a subset of the pairwise differences, that is considered reasonable in calculating error.intervals. To have it ignored, set it to NA. It should be a value between 0 and 1. The following rules apply:

1. If avsed.tolerance is NA then mean LSDs of the type specified by LSDtype are calculated and used in error.intervals and plots.
2. Irrespective of the setting of LSDtype, if avsed.tolerance is not exceeded then the mean LSDs are used in error.intervals and plots.
3. If LSDtype is set to overall, avsed.tolerance is not NA, and avsed.tolerance is exceeded then error.intervals and plotting revert to confidence intervals.
4. If LSDtype is set to factor.combinations and avsed.tolerance is not exceeded for any factor combination then the half LSDs are used in error.intervals and plots; otherwise, error.intervals and plotting revert to confidence intervals.
5. If LSDtype is set to per.prediction and avsed.tolerance is not exceeded for any prediction then the half LSDs are used in error.intervals and plots; otherwise, error.intervals and plotting revert to confidence intervals.

`accuracy.threshold`

A [numeric](#) specifying the value of the LSD accuracy measure, which measure is specified by LSDaccuracy, as a threshold value in determining whether the halfLeastSignificant error.interval for a predicted value is a reasonable approximation; this will be the case if the LSDs across all pairwise comparisons for which the interval's LSD was computed, as specified by LSDtype and LSDby, are similar enough to the interval's LSD, as measured by LSDaccuracy. If it is NA, it will be ignored. If it is not NA, a column of [logicals](#) named LSDwarning will be added to the predictions component of the [alldiffs.object](#). The value of LSDwarning for a predicted.value will be TRUE if the value of the LSDaccuracy measure computed from the LSDs for differences between this predicted.value and the other predicted.values as compared to its assignedLSD exceeds the value of accuracy.threshold. Otherwise, the value of LSDwarning for a predicted.value will be FALSE.

`LSDtype`

A [character](#) string that can be overall, factor.combinations, per.prediction or supplied. It determines whether the values stored in a row of a [LSD.frame](#) are the values calculated (i) overall from the LSD values for all pairwise com-

parisons, unless there is only one, possibly repeated, prediction, when a notional LSD is calculated, (ii) the values calculated from the pairwise LSDs for the levels of each factor.combination, unless there is only one prediction for a level of the factor.combination, when a notional LSD is calculated, (iii) per.prediction, being based, for each prediction, on all pairwise differences involving that prediction, or (iv) as supplied values of the LSD, specified with the LSDsupplied argument; these supplied values are to be placed in the assignedLSD column of the `LSD.frame` stored in an `alldiffs.object` so that they can be used in LSD calculations.

See `LSD.frame` for further information on the values in a row of this data.frame and how they are calculated.

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LSDsupplied    | A <code>data.frame</code> or a named <code>numeric</code> containing a set of LSD values that correspond to the observed combinations of the values of the LSDby variables in the <code>predictions.frame</code> or a single LSD value that is an overall LSD. If a <code>data.frame</code> , it may have (i) a column for the LSDby variable and a column of LSD values or (ii) a single column of LSD values with rownames being the combinations of the observed values of the LSDby variables. Any name can be used for the column of LSD values; assignedLSD is sensible, but not obligatory. Otherwise, a <code>numeric</code> containing the LSD values, each of which is named for the observed combination of the values of the LSDby variables to which it corresponds. (Applying the function <code>dae::fac.combine</code> to the predictions component is one way of forming the required combinations for the (row) names.) The values supplied will be incorporated into assignedLSD column of the <code>LSD.frame</code> stored as the LSD component of the <code>alldiffs.object</code> . |
| LSDby          | A <code>character</code> (vector) of variables names, being the names of the <code>factors</code> or <code>numerics</code> in the <code>classify</code> ; for each combination of their levels and values, there will be or is a row in the <code>LSD.frame</code> stored in the LSD component of the <code>alldiffs.object</code> when LSDtype is <code>factor.combinatons</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| LSDstatistic   | A <code>character</code> nominating one or more of minimum, q10, q25, mean, median, q75, q90 or maximum as the value(s) to be stored in the assignedLSD column in an <code>LSD.frame</code> ; the values in the assignedLSD column are used in computing halfLeastSignificant error.intervals. Here q10, q25, q75 and q90 indicate the sample quantiles corresponding to probabilities of 0.1, 0.25, 0.75 and 0.9 for the group of LSDs from which a single LSD value is calculated. The function <code>quantile</code> is used to obtain them. The mean LSD is calculated as the square root of the mean of the squares of the LSDs for the group. The median is calculated using the <code>median</code> function. Multiple values are only produced for LSDtype set to <code>factor.combination</code> , in which case LSDby must not be NULL and the number of values must equal the number of observed combinations of the values of the variables specified by LSDby. If LSDstatistic is NULL, it is reset to mean.                                                                                  |
| LSDaccuracy    | A <code>character</code> nominating one of maxAbsDeviation, maxDeviation, q90Deviation or RootMeanSqDeviation as the statistic to be calculated as a measure of the accuracy of assignedLSD. The option q90Deviation produces the sample quantile corresponding to a probability of 0.90. The deviations are the differences between the LSDs used in calculating the LSD statistics and each assigned LSD and the accuracy is expressed as a proportion of the assigned LSD value. The calculated values are stored in the column named accuracyLSD in an <code>LSD.frame</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| zero.tolerance | A <code>numeric</code> specifying the value such that if a predicted.value, its variance-covariance, or an LSD is less than it, the LSD will be considered to be zero.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| response       | A character specifying the response variable for the predictions. It is stored as an attribute to the <code>alldiffs.object</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| response.title | A character specifying the title for the response variable for the predictions. It is stored as an attribute to the <code>alldiffs.object</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| x.num          | A <code>character</code> string giving the name of the numeric covariate that (i) is potentially included in terms in the fitted model and (ii) is the x-axis variable for plots. Its values will not be converted to a <code>factor</code> .                                                                                                                                                                                                                                                                                                                                                 |
| x.fac          | A <code>character</code> string giving the name of the factor that (i) corresponds to <code>x.num</code> and (ii) is potentially included in terms in the fitted model. It should have the same number of levels as the number of unique values in <code>x.num</code> . The levels of <code>x.fac</code> must be in the order in which they are to be plotted - if they are dates, then they should be in the form <code>yyyymmdd</code> , which can be achieved using <code>as.Date</code> . However, the levels can be non-numeric in nature, provided that <code>x.num</code> is also set. |
| tables         | A <code>character</code> vector containing a combination of <code>none</code> , <code>predictions</code> , <code>vcov</code> , <code>backtransforms</code> , <code>differences</code> , <code>p.differences</code> , <code>sed</code> , <code>LSD</code> and <code>all</code> . These nominate which components of the <code>alldiffs.object</code> to print.                                                                                                                                                                                                                                 |
| level.length   | The maximum number of characters from the levels of factors to use in the row and column labels of the tables of pairwise differences and their p-values and standard errors.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| pairwise       | A <code>logical</code> indicating whether all pairwise differences of the predictions and their standard errors and p-values are to be computed and stored. If <code>tables</code> is equal to <code>"differences"</code> or <code>"all"</code> or <code>error.intervals</code> is equal to <code>"halfLeastSignificant"</code> , they will be stored irrespective of the value of <code>pairwise</code> .                                                                                                                                                                                    |
| alpha          | A <code>numeric</code> giving the significance level for LSDs or one minus the confidence level for confidence intervals. It is stored as an attribute to the <code>alldiffs.object</code> .                                                                                                                                                                                                                                                                                                                                                                                                  |
| inestimable.rm | A <code>logical</code> indicating whether rows for predictions that are not estimable are to be removed from the components of the <code>alldiffs.object</code> .                                                                                                                                                                                                                                                                                                                                                                                                                             |
| ...            | further arguments passed to <code>redoErrorIntervals.alldiffs</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## Details

For a matrix  $\mathbf{L}$ , vector of predictions  $\mathbf{p}$  and variance matrix of the predictions  $\mathbf{V}_p$ , the linear transformed predictions are given by  $\mathbf{Lp}$  with variance matrix  $\mathbf{LV}_p\mathbf{L}^T$ . The last matrix is used to compute the variance of pairwise differences between the transformed values.

If `linear.transformation` is a `matrix`,  $\mathbf{M}$  say, then the linear-transformation `matrix`,  $\mathbf{L}$ , is just the supplied `matrix`  $\mathbf{M}$ .

If `linear.transformation` is a `formula` and `EGLS.linTransform` is `TRUE`, then a matrix  $\mathbf{M}$  is obtained that is the design matrix for all of the terms in the `formula`. Using  $\mathbf{M}$ , the linear-transformation `matrix`,  $\mathbf{L}$ , is formed as  $\mathbf{M}(\mathbf{M}^T\hat{\mathbf{V}}-\mathbf{M})-(\mathbf{M}^T\hat{\mathbf{V}}-)$ .

On the other hand, for `linear.transformation` a `formula` and `EGLS.linTransform` set to `FALSE`,  $\mathbf{L}$  is formed as the sum of the orthogonal projection matrices obtained using `pstructure.formula` from the package `dae`; `grandMean` is set to `TRUE` and `orthogonalize` to `"eigenmethods"`.

## Value

A `alldiffs.object` with the linear transformation of the predictions and their standard errors and all pairwise differences between the linear transforms of their predictions, their standard errors and p-values and LSD statistics.





```

SS.emm <- emmeans::emmeans(m1.lmer, specs = ~ Sources:Species)
SS.preds <- summary(SS.emm)
den.df <- min(SS.preds$df, na.rm = TRUE)
## Modify SS.preds to be compatible with a predictions.frame
SS.preds <- as.predictions.frame(SS.preds, predictions = "emmean",
                                se = "SE", interval.type = "CI",
                                interval.names = c("lower.CL", "upper.CL"))

## Form an all.diffs object and check its validity
SS.vcov <- vcov(SS.emm)
SS.diffs <- allDifferences(predictions = SS.preds, classify = "Sources:Species",
                          vcov = SS.vcov, tdf = den.df)

validAlldiffs(SS.diffs)

#Get additive predictions
diffs.sub <- linTransform(SS.diffs, classify = "Sources:Species",
                          linear.transformation = ~ Sources + Species,
                          Vmatrix = TRUE, tables = "none")
}

##Calculate contrasts from prediction obtained using asreml or lmerTest
if (exists("diffs.sub"))
{
  #Contrast matrix for differences between each species and non-planted for the last source
  L <- cbind(matrix(rep(0,7*32), nrow = 7, ncol = 32),
             diag(1, nrow = 7),
             matrix(rep(-1, 7), ncol = 1))
  rownames(L) <- as.character(diffs.sub$predictions$Species[33:39])
  diffs.L <- linTransform(diffs.sub,
                          classify = "Sources:Species",
                          linear.transformation = L,
                          tables = "predictions")
}

```

---

|                   |                                                             |
|-------------------|-------------------------------------------------------------|
| loadASRemlVersion | <i>Ensures that a specific version of asreml is loaded.</i> |
|-------------------|-------------------------------------------------------------|

---

## Description

Loads the specified version of asreml, provided that it is not already loaded. If the version of asreml is not the required version, then the loaded version is unloaded first.

## Usage

```
loadASRemlVersion(version = 4, ...)
```

## Arguments

|         |                                                                                                                                                                                                                                                                                                              |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| version | The version that is to be loaded, the version consisting of just the initial characters that are significant in the version that should be loaded. For example, the default value of 4 implies that any version that begins with "4" is acceptable. It is used to check that the required version is loaded. |
| ...     | Other library/require arguments that are needed to load the specified version of asreml.                                                                                                                                                                                                                     |

**Value**

A character, being all characters in the version of asreml that is loaded on exit from the function.

**Author(s)**

Chris Brien

**See Also**

[getASRemlVersionLoaded](#).

**Examples**

```
## Not run:
loadASRemlVersion(3, lib.loc = "D:\Analyses\R asreml3")
## End(Not run)
```

---

|           |                                    |
|-----------|------------------------------------|
| LSD.frame | <i>Description of an LSD frame</i> |
|-----------|------------------------------------|

---

**Description**

A data.frame that stores Least Significant differences (LSDs) for predictions for a fitted model.

**Value**

A data.frame that can be a component of an [alldiffs.object](#) and that contains LSD values and statistics to be used in determining the significance of the pairwise differences. In particular, they are used in calculating halfLeastSignificant limits to be included in a predictions.frame.

Exactly what an LSD.frame contains is determined by the following arguments to functions that return an [alldiffs.object](#): LSDtype, LSDby, LSDstatistic, LSDaccuracy and LSDsupplied. The rownames of the LSD.frame indicate, for each of its rows, for what group of predictions the entries in the row were calculated, this being controlled by the LSDtype and LSDby arguments. The values for all of the LSD arguments are stored as attributes to the [alldiffs.object](#) and the predictions and, if present backtransforms, components of the [alldiffs.object](#).

An LSD.frame always has the eight columns c, minimumLSD, meanLSD, maximumLSD, assignedLSD, accuracyLSD, falsePos and falseNeg.

1. c: This gives the number of pairwise comparison of predictions for the combinations of the factor levels given by the row name. If the row name is overall then it is for all predictions.
2. minimumLSD, meanLSD, maximumLSD: These are computed for either overall, factor.combinations, per.prediction or supplied LSD values, as specified by the LSDtype argument. The meanLSD is calculated using the square root of the mean of the variances of set of pairwise differences appropriate to the specific LSDtype argument.

For overall, the mean, minimum and maximum of the LSDs for all pairwise comparisons are computed.

If factor.combinations was specified for LSDtype when the LSDs were being calculated, then the LSD.frame contains a row for each combination of the values of the [factors](#) and [numerics](#) specified by LSDby. The values in a row are calculated from the LSD values for the pairwise differences for each combination of the [factor](#)s and [numeric](#)s values, unless there

is only one prediction for a combination, when notional LSDs are calculated that are based on the standard error of the prediction multiplied by the square root of two. Notional LSDs indicate the LSD that would apply to the difference between the prediction and, if it existed, another prediction with the same standard error.

For `per.prediction`, the minimum, mean and maximum LSD, based, for each prediction, on the LSD values for all pairwise differences involving that prediction are computed.

For `supplied`, the `LSD.frame` is set up based on the setting of `LSDby`: a single row with name `overall` if `LSDby` is `NULL` or, if `LSDby` is a vector of `factor` and `numeric` names, rows for each observed combinations of the values of the named `factors` and `numerics`. The `LSDsupplied` argument is used to provide the values to be stored in the column `assignedLSD`.

3. `assignedLSD`: The `assignedLSD` column contains the values that are assigned for use in calculating `halfLeastSignificant.error.intervals`. Its contents are determined by `LSDstatistic` and `LSDsupplied` arguments. The `LSDsupplied` argument allows the direct specification of values to be placed in the `assignedLSD` column of the `LSD.frame`. The default is to use the values in the `meanLSD` column.
4. `LSDaccuracy`: The `LSDaccuracy` gives an indication of the proportion that the correct LSD for a single predicted value might deviate from its `assignedLSD` value. The contents of the `accuracyLSD` column is controlled by the `LSDaccuracy` argument.
5. `falsePos` and `falseNeg`: These columns contain the number of false positives and negatives if the `assignedLSD` value(s) is(are) used to determine the significance of the pairwise predictions differences. Each LSD value in the `assignedLSD` column is used to determine the significance of pairwise differences that involve predictions for the combination of values given by the row name for the LSD value.

See [recalcLSD.alldiffs](#) for more information.

### Author(s)

Chris Brien

### See Also

[recalcLSD.alldiffs](#), [redoErrorIntervals.alldiffs](#), [predictPresent.asrem1](#), [predictPlus.asrem1](#)

### Examples

```
data(Oats.dat)

## Use asrem1 to get predictions and associated statistics

## Not run:
m1.asr <- asrem1(Yield ~ Nitrogen*Variety,
  random=~Blocks/Wplots,
  data=Oats.dat)
current.asrt <- as.asrtests(m1.asr)
Var.diffs <- predictPlus(m1.asr, classify="Nitrogen:Variety",
  wald.tab = current.asrt$wald.tab,
  tables = "none")

## End(Not run)

## Use lmerTest and emmeans to get predictions and associated statistics
if (requireNamespace("lmerTest", quietly = TRUE) &
```

```

    requireNamespace("emmeans", quietly = TRUE))
  {
    m1.lmer <- lmerTest::lmer(Yield ~ Nitrogen*Variety + (1|Blocks/Wplots),
                             data=Oats.dat)

    #Get predictions
    Var.emm <- emmeans::emmeans(m1.lmer, specs = ~ Nitrogen:Variety)
    Var.preds <- summary(Var.emm)
    ## Modify Var.preds to be compatible with a predictions.frame
    Var.preds <- as.predictions.frame(Var.preds, predictions = "emmean",
                                     se = "SE", interval.type = "CI",
                                     interval.names = c("lower.CL", "upper.CL"))

    Var.vcov <- vcov(Var.emm)
    Var.sed <- NULL

    #Set up an alldiffs object, which includes overall LSDs
    Var.diffs <- allDifferences(predictions = Var.preds, classify = "Variety:Nitrogen",
                               sed = Var.sed, vcov = Var.vcov, tdf = 45)
  }

  if (exists("Var.diffs"))
  {
    ## Use recalLSD to get LSDs for within Variety differences
    Var.LSD.diffs <- recalLSD(Var.diffs,
                             LSDtype = "factor.combinations", LSDby = "Variety")
    print(Var.LSD.diffs$LSD)
  }
}

```

---

```
makeTPPSplineMats.data.frame
```

*Make the spline basis matrices and data needed to fit Tensor Product P-splines.*

---

## Description

Prepares the fixed and random P-spline basis matrices, and associated information, that are needed for fitting of Tensor Product P-splines (TPPS) as described by Rodriguez-Alvarez et al. (2018). This function is called internally by [addSpatialModelOnIC.asrtests](#), [addSpatialModelOnIC.asrtests](#) and [chooseSpatialModelOnIC.asrtests](#) when fitting TPPS models for local spatial variation. There are two methods available, controlled by `asreml.option` for creating and storing the basis functions. This function is most likely to be called directly when `mbf` has been used in creating an `asreml` object and it is desired to use the object in a session subsequent to the session in which the object was created.

## Usage

```
## S3 method for class 'data.frame'
makeTPPSplineMats(data, sections = NULL,
                  row.covar, col.covar,
                  nsegs = NULL, nestorder = c(1,1),
                  degree = c(3,3), difforder = c(2,2),
                  rotateX = FALSE, theta = c(0,0),
                  asreml.option = "grp", mbf.env = sys.frame(),
                  ...)
```

**Arguments**

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data          | An <code>data.frame</code> that holds the spline bases for a section. It is indexed by columns named <code>col</code> and <code>row</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| sections      | A single character string that species the name of the column in the <code>data.frame</code> that contains the <code>factor</code> that identifies different sections of the data to which separate spatial models are to be fitted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| row.covar     | A single <code>character</code> string nominating a <code>numeric</code> column in the <code>data.frame</code> that contains the values of a covariate indexing the rows of the grid.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| col.covar     | A single <code>character</code> string nominating a <code>numeric</code> column in the <code>data.frame</code> that contains the values of a covariate indexing the columns of the grid.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| nsegs         | A pair of <code>numeric</code> values giving the number of segments into which the column and row ranges are to be split, respectively, for fitting a P-spline model (TPPS) (each value specifies the number of internal knots + 1). If not specified, then (number of unique values - 1) is used in each dimension; for a grid layout with equal spacing, this gives a knot at each data value. If sections is not NULL and the grid differs between the sections, then nsegs will differ between the sections.                                                                                                                                                                                                                                                                                                                                                                                               |
| nestorder     | A <code>numeric</code> of length 2. The order of nesting for column and row dimensions, respectively, in fitting a P-spline model (TPPS). A value of 1 specifies no nesting, a value of 2 generates a spline with half the number of segments in that dimension, etc. The number of segments in each direction must be a multiple of the order of nesting.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| degree        | A <code>numeric</code> of length 2. The degree of polynomial spline to be used for column and row dimensions respectively, in fitting a P-spline (TPPS).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| difforder     | A <code>numeric</code> of length 2. The order of differencing for column and row dimensions, respectively, in fitting a P-spline (TPPS).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| rotateX       | A <code>logical</code> indicating whether to rotate the eigenvectors of the penalty matrix, as described by Piepho, Boer and Williams (2022), when fitting a P-spline (TPPS). Setting rotateX to TRUE results in a search for an optimized rotation under a model that omits the random spline interaction terms. If ngridangles is set to NULL, the optimal rotation is found using an optimizer ( <code>nloptr::bobyqa</code> ). Otherwise, the optimal rotation is found by exploring the fit over a two-dimensional grid of rotation angle pairs. The optimization seeks to optimize the criterion nominated in <code>which.rotacriterion</code> . Rotation of the eigenvectors is only relevant for difforder values greater than 1 and has only been implemented for difforder equal to 2.                                                                                                               |
| theta         | A <code>numeric</code> of length 2. The angle (in degrees) to be used in rotating the eigenvectors of the penalty matrix of a P-spline (TPPS).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| asreml.option | A single character string specifying whether the <code>grp</code> or <code>mbf</code> methods are to be used to supply externally formed covariate matrices to <code>asreml</code> when fitting a P-spline (TPPS). Compared to the <code>mbf</code> method, the <code>grp</code> method is somewhat faster, but creates large <code>asrtests.objects</code> for which the time it takes to save them can exceed any gains in execution speed. The <code>grp</code> method adds columns to the <code>data.frame</code> containing the data. On the other hand, the <code>mbf</code> method adds only the fixed covariates to data and stores the random covariates in the environment of the internal function that calls the spline-fitting function; there are three smaller <code>data.frames</code> for each section that are not stored in the <code>asreml.object</code> resulting from the fitted model. |

|                      |                                                                                                                                                                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>mbf.env</code> | A <b>environment</b> specifying the environment to which the <b>data.frames</b> containing the spline bases are to be assigned. If <code>mbf.env</code> is <code>NULL</code> , the <code>data.frames</code> will not be assigned. |
| <code>...</code>     | Further arguments passed to <code>tpsmmb</code> from package <code>TPSbits</code> .                                                                                                                                               |

## Details

The objects are formed using the function `tpsmmb` from the R package `TPSbits` authored by Sue Welham (2022). This function has been extended to allow for sections (see below) and to allow rotation of the penalty matrix for the linear component of the interaction terms in TPPCS models (for more information about rotation see Piepho, Boer and Williams, 2022).

Each combination of a `row.covar` and a `col.covar` does not have to specify a single observation; for example, to fit a local spatial variation model to the main units of a split-unit design, each combination would correspond to a main unit and all subunits of the main unit would have the same combination.

The data for experiment can be divided sections and the spline bases and associated data will be produced for each section. If there is more than one sections, then a **list** is returned that has a component for each section. The component for a section contains:

## Value

A **list** of length equal to the number of sections is produced. Each of these components is a **list** with 8 or 9 components. The component named `data.plus`, being the input **data.frame** to which has been added the columns required to fit the TPPS model (the **data.frame** stored in the `data` component holds only the covariates from data).

List of length 8 or 9 (according to the `asreml.option`).

1. `data` = the input data frame augmented with structures required to fit tensor product splines in `asreml-R`. This data frame can be used to fit the TPS model.

Added columns:

- `TP.col`, `TP.row` = column and row coordinates
- `TP.CxR` = combined index for use with smooth x smooth term
- `TP.C.n` for  $n=1:\text{diff.c}$  =  $X$  parts of column spline for use in random model (where `diff.c` is the order of column differencing)
- `TP.R.n` for  $n=1:\text{diff.r}$  =  $X$  parts of row spline for use in random model (where `diff.r` is the order of row differencing)
- `TP.CR.n` for  $n=1:(\text{diff.c}*\text{diff.r})$  = interaction between the two  $X$  parts for use in fixed model. The first variate is a constant term which should be omitted from the model when the constant (1) is present. If all elements are included in the model then the constant term should be omitted, eg.  $y \sim -1 + \text{TP.CR.1} + \text{TP.CR.2} + \text{TP.CR.3} + \text{TP.CR.4} + \text{other terms} \dots$
- when `asreml="grp"` or `"sepgrp"`, the spline basis functions are also added into the data frame. Column numbers for each term are given in the `grp` list structure.

2. `mbflist` = list that can be used in call to `asreml` (so long as `Z` matrix data frames extracted with right names, eg `BcZ<stub>.df`)
3. `BcZ.df` = `mbf` data frame mapping onto smooth part of column spline, last column (labelled `TP.col`) gives column index
4. `BrZ.df` = `mbf` data frame mapping onto smooth part of row spline, last column (labelled `TP.row`) gives row index

5. BcrZ.df = mbf data frame mapping onto smooth x smooth term, last column (labelled TP.CxR) maps onto col x row combined index
6. dim = list structure, holding dimension values relating to the model:
  - (a) "diff.c" = order of differencing used in column dimension
  - (b) "nbc" = number of random basis functions in column dimension
  - (c) "nbcn" = number of nested random basis functions in column dimension used in smooth x smooth term
  - (d) "diff.r" = order of differencing used in row dimension
  - (e) "nbr" = number of random basis functions in row dimension
  - (f) "nbrn" = number of nested random basis functions in row dimension used in smooth x smooth term
7. trace = list of trace values for ZGZ' for the random TPSpline terms, where Z is the design matrix and G is the known diagonal variance matrix derived from eigenvalues. This can be used to rescale the spline design matrix (or equivalently variance components).
8. grp = list structure, only added for setting asreml="grp". For asreml="grp", provides column indexes for each of the 5 random components of the 2D splines in data.plus. Dimensions of the components can be derived from the values in the dim item.
9. data.plus = the input `data.frame` to which has been added the columns required to fit tensor product splines in asreml-R. This `data.frame` can be used to fit the TPS model. For multiple sections, this `data.frame` will occur in the component for each section. If asreml.option is set to mbf, then this component will have the attribute mbf.env that specifies the environment to which the `data.frames` containing the spline bases have been assigned.

### Author(s)

Chris Brien

### References

- Piepho, H-P, Boer, M. P. & Williams, E. R. (2022) Two-dimensional P-spline smoothing for spatial analysis of plant breeding trials. *Biometrical Journal*, **64**, 835-857.)
- Rodriguez-Alvarez, M. X., Boer, M. P., van Eeuwijk, F. A., & Eilers, P. H. C. (2018). Correcting for spatial heterogeneity in plant breeding experiments with P-splines. *Spatial Statistics*, **23**, 52-71.
- Welham, S. J. (2022) TPSbits: *Creates Structures to Enable Fitting and Examination of 2D Tensor-Product Splines using ASReml-R*. Version 1.0.0.

### See Also

`addSpatialModel.asrtests`, `addSpatialModelOnIC.asrtests`,  
`chooseSpatialModelOnIC.asrtests`, `tpsmmb` from TPSbits

### Examples

```
## Not run:

data(Wheat.dat)

#Add row and column covariates
Wheat.dat <- within(Wheat.dat,
  {
    cColumn <- dae::as.numfac(Column)
```

```

        cColumn <- cColumn - mean(unique(cColumn))
        cRow <- dae::as.numfac(Row)
        cRow <- cRow - mean(unique(cRow))
    })

#Set up the matrices
tps.XZmat <- makeTPPSplineMats(wheat.dat,
                               row.covar = "cRow", col.covar = "cColumn")

## End(Not run)

```

---

|               |                                                                                                                                        |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------|
| newfit.asreml | <i>Refits an asreml model with changed arguments by extracting, modifying and evaluating its call - an alternate to update.asreml.</i> |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------|

---

## Description

Extracts the call from the `asreml.obj` and evaluates that call, replacing supplied `asreml` arguments with changed values. If `update` is `TRUE` and `set.terms` is `NULL`, the call is evaluated using the variance parameter estimates from the call stored in `asreml.obj`; if some variance terms in the newly fitted model are singular (S) or bound (B), a refit of the model will be tried in which the variance parameter estimates in `asreml.obj` are not used and will become the fitted model if its boundary terms are a subset of those in the fitted model stored in `asreml.obj`. If `update` is `FALSE` or `set.terms` is not `NULL`, the previous values of the variance parameters are not used as initial values for fitting the new model; `G.param` and `R.param` are set to `NULL` or to values as specified for `set.terms`. The `...` argument can be used to pass `G.param` and/or `R.param`, provided `update` is `FALSE` and `set.terms` is `NULL`.

Irrespective of whether `set.terms` is `NULL`, variance parameter names, bounds and initial values stored in the `setvparameters` `data.frame` located in the call component of the `asreml.obj` are added to any `set.terms` supplied in the current call. except for those in `setvparameters` that are specified in the `set.terms` argument. In the process of fitting the model, the `setvparameters` `data.frame` stored in the supplied `asreml.obj` is updated to include the prior and current `set.terms`. Then, just before exiting `newfit.asreml`, a search for changes in the bounds of the stored terms is made. If any have changed, an attempt is made to force the values back to their values on entry.

## Usage

```

## S3 method for class 'asreml'
newfit(asreml.obj, fixed., random., sparse.,
       residual., rcov., update = TRUE, trace = FALSE,
       allow.unconverged = TRUE, allow.fixedcorrelation = TRUE,
       keep.order = TRUE,
       set.terms = NULL, ignore.suffices = TRUE,
       bounds = "P", initial.values = NA, ...)

```

## Arguments

|                         |                                                                                                                                                                  |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>asreml.obj</code> | A valid <code>asreml</code> object with a component named <code>call</code> (from a previous call to either <code>asreml</code> or <code>update.asreml</code> ). |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|



|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>fixed.</code>                 | A character or formula specifying changes to the fixed formula. This is a two-sided formula where "." is substituted for existing components in the fixed component of <code>asreml.obj\$call</code> . If changes are specified, the fixed terms will be re-ordered so that single-variable terms come first, followed by two-variable terms and so on.                                                                                                                                                                                                                                                                                                                                                             |
| <code>random.</code>                | A character or formula specifying changes to the random formula. This is a one-sided formula where "." is substituted for existing components in the random component of <code>asreml.obj\$call</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>sparse.</code>                | A character or formula specifying changes to the sparse formula. This is a one-sided formula where "." is substituted for existing components in the sparse component of <code>asreml.obj\$call</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>residual.</code>              | A character or formula specifying changes to the error formula, used when version 4 or later of ASReml-R is loaded. This is a one-sided formula where "." is substituted for existing components in the residual component of <code>asreml.obj\$call</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>rcov.</code>                  | A character or formula specifying changes to the error formula, used when version 3 of ASReml-R is loaded. This is a one-sided formula where "." is substituted for existing components in the residual component of <code>asreml.obj\$call</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <code>update</code>                 | A logical indicating whether to use the variance parameter estimates in fitting a new model using <code>newfit.asreml</code> . If <code>update</code> is TRUE and <code>set.terms</code> is NULL, the call stored in the <code>asreml.obj</code> is evaluated using the variance parameter estimates stored in <code>R.param</code> and <code>G.param</code> , as well as the changes specified in the arguments to <code>newfit.asreml</code> . If FALSE or <code>set.terms</code> is not NULL, then the use of previous values of the variance parameters as initial values is not enforced; <code>G.param</code> and <code>R.param</code> are set to NULL or to values as specified for <code>set.terms</code> . |
| <code>trace</code>                  | A <a href="#">logical</a> that control output from ASReml-R. If TRUE then partial iteration details are displayed when ASReml-R functions are invoked; if FALSE then no output is displayed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>allow.unconverged</code>      | A logical indicating whether to accept a new model even when it does not converge. If FALSE and the fit does not converge, the supplied <code>asreml.obj</code> is returned.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>allow.fixedcorrelation</code> | A logical indicating whether to accept a new model even when it contains correlations in the model whose values have been designated as fixed, bound or singular. If FALSE and the new model contains correlations whose values have not been able to be estimated, the supplied <code>asreml.obj</code> is returned. The fit in the supplied the <code>asreml.obj</code> will also be tested and a warning issued if both fixed correlations are found in it and <code>allow.fixedcorrelation</code> is FALSE.                                                                                                                                                                                                     |
| <code>keep.order</code>             | A logical value indicating whether the terms should keep their positions. If FALSE the terms are reordered so that main effects come first, followed by the interactions, all second-order, all third-order and so on. Effects of a given order are kept in the order specified.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>set.terms</code>              | A character vector specifying the terms that are to have bounds and/or initial values set prior to fitting. The names must match those in the <code>vparameters</code> component of the new <code>asreml.obj</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>ignore.suffices</code>        | A logical vector specifying whether the suffices of the <code>asreml</code> -assigned names of the variance terms (i.e. the information to the right of an "!", other than "R!") is to be ignored in matching elements of terms. If TRUE for an element of terms, the suffices are stripped from the <code>asreml</code> -assigned names. If FALSE                                                                                                                                                                                                                                                                                                                                                                  |

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | for an element of <code>terms</code> , the element must exactly match an <code>asreml</code> -assigned name for a variance term. This vector must be of length one or the same length as <code>terms</code> . If it is of length one then the same action is applied to the <code>asreml</code> -assigned suffices for all the terms in <code>terms</code> .                                                    |
| <code>bounds</code>         | A <code>character</code> vector specifying the bounds to be applied to the terms specified in <code>set.terms</code> . This vector must be of length one or the same length as <code>set.terms</code> . If it is of length one then the same constraint is applied to all the terms in <code>set.terms</code> . If any of the bounds are equal to <code>NA</code> then they are left unchanged for those terms. |
| <code>initial.values</code> | A character vector specifying the initial values for the terms specified in <code>terms</code> . This vector must be of length one or the same length as <code>terms</code> . If it is of length one then the same initial value is applied to all the terms in <code>terms</code> . If any of the <code>initial.values</code> are equal to <code>NA</code> then they are left unchanged for those terms.       |
| <code>...</code>            | additional arguments to the call, or arguments with changed values.                                                                                                                                                                                                                                                                                                                                             |

**Value**

An `asreml` object.

**Author(s)**

Chris Brien

**References**

Butler, D. G., Cullis, B. R., Gilmour, A. R., Gogel, B. J. and Thompson, R. (2023). *ASReml-R Reference Manual Version 4.2*. VSN International Ltd, <https://asreml.kb.vsnr.co.uk/>.

**See Also**

`convAsremlobj.asreml`, `update.asreml`, `setvarianceterms.call`

**Examples**

```
## Not run:
m2.asreml <- newfit(m1.asreml, random. = "~ . - Blocks:Plots", maxiter=75)

## End(Not run)
```

---

|            |                                                                                |
|------------|--------------------------------------------------------------------------------|
| num.recode | <i>Recodes the unique values of a vector using the values in a new vector.</i> |
|------------|--------------------------------------------------------------------------------|

---

**Description**

Recodes the unique values of a variate using the value in position `i` of the `new.values` vector to replace the `i`th sorted unique values of `x`. The new levels do not have to be unique.

**Usage**

```
num.recode(x, new.values)
```

**Arguments**

`x`                      The vector to be recoded.  
`new.values`            A vector of length `unique(x)` containing values to use in the recoding.

**Value**

A vector.

**Author(s)**

Chris Brien

**See Also**

`dae::fac.recast`.

**Examples**

```
## set up a factor with labels
x <- rep(c(-42, -14, 14, 42), 4)

## recode x
b <- num.recode(x, c(0, 28, 56, 84))
```

---

Oats.dat

*Data for an experiment to investigate nitrogen response of 3 oats varieties*

---

**Description**

Yates (1937) describes a split-plot experiment that investigates the effects of three varieties of oats and four levels of Nitrogen fertilizer. The varieties are assigned to the main plots using a randomized complete block design with 6 blocks and the nitrogen levels are randomly assigned to the subplots in each main plot.

The columns in the data frame are: Blocks, Wplots, Subplots, Variety, Nitrogen, xNitrogen, Yield. The column xNitrogen is a numeric version of the factor Nitrogen. The response variable is Yield.

**Usage**

```
data(Oats.dat)
```

**Format**

A data.frame containing 72 observations of 7 variables.

**Author(s)**

Chris Brien

**Source**

Yates, F. (1937). The Design and Analysis of Factorial Experiments. *Imperial Bureau of Soil Science, Technical Communication*, **35**, 1-95.

---

pairediffsTransform.alldiffs

*Calculates the differences between nominated pairs of predictions stored in an `alldiffs.object`.*

---

## Description

Predictions of differences and their error intervals are formed for two levels of a factor, the `pairs.factor`. For each pair of a level of the `pairs.factor` in `numerator.levels` with a level in `denominator.levels`, an `alldiffs.object` is formed that contains the differences between predictions with this pair of levels for all of the combinations of the levels of the other factors in the `classify` of the `alldiffs.object`. These prediction differences are obtained using `linTransform` by forming a suitable contrast matrix to specify the `linear.transformation`. This function has the advantage that the factors indexing the differences are included in the components of the `alldiffs.objects`.

If `pairwise = TRUE`, all pairwise differences between the linear transforms of the predictions, their standard errors, p-values and LSD statistics are computed as using `allDifferences.data.frame`. This adds them to the `alldiffs.object` as additional list components named `differences`, `sed`, `p.differences` and `LSD`.

The printing of the components produced is controlled by the `tables` argument. The order of plotting the levels of one of the factors indexing the predictions can be modified and is achieved using `sort.alldiffs`.

## Usage

```
## S3 method for class 'alldiffs'
pairediffsTransform(alldiffs.obj, pairs.factor,
                    first.levels, second.levels, sep.levs.combine = ",",
                    Vmatrix = FALSE, error.intervals = "Confidence",
                    avsed.tolerance = 0.25, accuracy.threshold = NA,
                    LSDtype = "overall", LSDsupplied = NULL, LSDby = NULL,
                    LSDstatistic = "mean", LSDaccuracy = "maxAbsDeviation",
                    response = NULL, response.title = NULL, tables = "all",
                    pairwise = TRUE, alpha = 0.05, ...)
```

## Arguments

|                               |                                                                                                                                                                                                                                                                               |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>alldiffs.obj</code>     | An <code>alldiffs.object</code> .                                                                                                                                                                                                                                             |
| <code>pairs.factor</code>     | A <code>character</code> string giving the name of the factor for whose levels the differences are to be calculated.                                                                                                                                                          |
| <code>first.levels</code>     | A <code>character</code> string containing the levels of the <code>pairs.factor</code> whose predictions are those subtracted from.                                                                                                                                           |
| <code>second.levels</code>    | A <code>character</code> string containing the levels of the <code>pairs.factor</code> whose predictions are those that are subtracted.                                                                                                                                       |
| <code>sep.levs.combine</code> | A <code>character</code> string specifying the separator to use internally when combining the levels of the <code>factor(s)</code> in the <code>classify</code> of the <code>alldiffs.obj</code> . It should not occur in the levels of the <code>classify factor(s)</code> . |
| <code>Vmatrix</code>          | A <code>logical</code> indicating whether the variance matrix of the predictions will be stored as a component of the <code>alldiffs.object</code> that is returned.                                                                                                          |

`error.intervals`

A [character](#) string indicating the type of error interval, if any, to calculate in order to indicate uncertainty in the results. Possible values are "none", "StandardError", "Confidence" and "halfLeastSignificant". The default is for confidence limits to be used. The "halfLeastSignificant" option results in half the Least Significant Difference (LSD) being added and subtracted to the predictions, the LSD being calculated using the square root of the mean of the variances of all or a subset of pairwise differences between the predictions. If the LSD is zero, as can happen when predictions are constrained to be equal, then the limits of the error intervals are set to NA. If LSDtype is set to overall, the avsed.tolerance is not NA and the range of the SEDs divided by the average of the SEDs exceeds avsed.tolerance then the error.intervals calculations and the plotting will revert to confidence intervals.

`avsed.tolerance`

A [numeric](#) giving the value of the SED range, the range of the SEDs divided by the square root of the mean of the variances of all or a subset of the pairwise differences, that is considered reasonable in calculating error.intervals. To have it ignored, set it to NA. It should be a value between 0 and 1. The following rules apply:

1. If avsed.tolerance is NA then mean LSDs of the type specified by LSDtype are calculated and used in error.intervals and plots.
2. Irrespective of the setting of LSDtype, if avsed.tolerance is not exceeded then the mean LSDs are used in error.intervals and plots.
3. If LSDtype is set to overall, avsed.tolerance is not NA, and avsed.tolerance is exceeded then error.intervals and plotting revert to confidence intervals.
4. If LSDtype is set to factor.combinations and avsed.tolerance is not exceeded for any factor combination then the half LSDs are used in error.intervals and plots; otherwise, error.intervals and plotting revert to confidence intervals.
5. If LSDtype is set to per.prediction and avsed.tolerance is not exceeded for any prediction then the half LSDs are used in error.intervals and plots; otherwise, error.intervals and plotting revert to confidence intervals.

`accuracy.threshold`

A [numeric](#) specifying the value of the LSD accuracy measure, which measure is specified by LSDaccuracy, as a threshold value in determining whether the halfLeastSignificant error.interval for a predicted value is a reasonable approximation; this will be the case if the LSDs across all pairwise comparisons for which the interval's LSD was computed, as specified by LSDtype and LSDby, are similar enough to the interval's LSD, as measured by LSDaccuracy. If it is NA, it will be ignored. If it is not NA, a column of [logicals](#) named LSDwarning will be added to the predictions component of the [alldiffs.object](#). The value of LSDwarning for a predicted.value will be TRUE if the value of the LSDaccuracy measure computed from the LSDs for differences between this predicted.value and the other predicted.values as compared to its assignedLSD exceeds the value of accuracy.threshold. Otherwise, the value of LSDwarning for a predicted.value will be FALSE.

`LSDtype`

A [character](#) string that can be overall, factor.combinations, per.prediction or supplied. It determines whether the values stored in a row of a [LSD.frame](#) are the values calculated (i) overall from the LSD values for all pairwise com-

parisons, unless there is only one, possibly repeated, prediction, when a notional LSD is calculated, (ii) the values calculated from the pairwise LSDs for the levels of each factor.combination, unless there is only one prediction for a level of the factor.combination, when a notional LSD is calculated, (iii) per.prediction, being based, for each prediction, on all pairwise differences involving that prediction, or (iv) as supplied values of the LSD, specified with the LSDsupplied argument; these supplied values are to be placed in the assignedLSD column of the `LSD.frame` stored in an `alldiffs.object` so that they can be used in LSD calculations.

See `LSD.frame` for further information on the values in a row of this data.frame and how they are calculated.

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LSDsupplied  | A <code>data.frame</code> or a named <code>numeric</code> containing a set of LSD values that correspond to the observed combinations of the values of the LSDby variables in the <code>predictions.frame</code> or a single LSD value that is an overall LSD. If a <code>data.frame</code> , it may have (i) a column for the LSDby variable and a column of LSD values or (ii) a single column of LSD values with rownames being the combinations of the observed values of the LSDby variables. Any name can be used for the column of LSD values; assignedLSD is sensible, but not obligatory. Otherwise, a <code>numeric</code> containing the LSD values, each of which is named for the observed combination of the values of the LSDby variables to which it corresponds. (Applying the function <code>dae::fac.combine</code> to the predictions component is one way of forming the required combinations for the (row) names.) The values supplied will be incorporated into assignedLSD column of the <code>LSD.frame</code> stored as the LSD component of the <code>alldiffs.object</code> . |
| LSDby        | A <code>character</code> (vector) of variables names, being the names of the <code>factors</code> or <code>numerics</code> in the <code>classify</code> ; for each combination of their levels and values, there will be or is a row in the <code>LSD.frame</code> stored in the LSD component of the <code>alldiffs.object</code> when LSDtype is <code>factor.combinatons</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| LSDstatistic | A <code>character</code> nominating one or more of minimum, q10, q25, mean, median, q75, q90 or maximum as the value(s) to be stored in the assignedLSD column in an <code>LSD.frame</code> ; the values in the assignedLSD column are used in computing halfLeastSignificant error.intervals. Here q10, q25, q75 and q90 indicate the sample quantiles corresponding to probabilities of 0.1, 0.25, 0.75 and 0.9 for the group of LSDs from which a single LSD value is calculated. The function <code>quantile</code> is used to obtain them. The mean LSD is calculated as the square root of the mean of the squares of the LSDs for the group. The median is calculated using the <code>median</code> function. Multiple values are only produced for LSDtype set to <code>factor.combination</code> , in which case LSDby must not be <code>NULL</code> and the number of values must equal the number of observed combinations of the values of the variables specified by LSDby. If LSDstatistic is <code>NULL</code> , it is reset to mean.                                                       |
| LSDaccuracy  | A <code>character</code> nominating one of <code>maxAbsDeviation</code> , <code>maxDeviation</code> , <code>q90Deviation</code> or <code>RootMeanSqDeviation</code> as the statistic to be calculated as a measure of the accuracy of assignedLSD. The option <code>q90Deviation</code> produces the sample quantile corresponding to a probability of 0.90. The deviations are the differences between the LSDs used in calculating the LSD statistics and each assigned LSD and the accuracy is expressed as a proportion of the assigned LSD value. The calculated values are stored in the column named <code>accuracyLSD</code> in an <code>LSD.frame</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| response     | A character specifying the response variable for the predictions. It is stored as an attribute to the <code>alldiffs.object</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|                |                                                                                                                                                                                                                                                                                                                            |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| response.title | A character specifying the title for the response variable for the predictions. It is stored as an attribute to the <code>alldiffs.object</code> .                                                                                                                                                                         |
| tables         | A <code>character</code> vector containing a combination of none, predictions, vcov, backtransforms, differences, p.differences, sed, LSD and all. These nominate which components of the <code>alldiffs.object</code> to print.                                                                                           |
| pairwise       | A <code>logical</code> indicating whether all pairwise differences of the predictions and their standard errors and p-values are to be computed and stored. If tables is equal to "differences" or "all" or error.intervals is equal to "halfLeastSignificant", they will be stored irrespective of the value of pairwise. |
| alpha          | A <code>numeric</code> giving the significance level for LSDs or one minus the confidence level for confidence intervals. It is stored as an attribute to the <code>alldiffs.object</code> .                                                                                                                               |
| ...            | further arguments passed to <code>linTransform.alldiffs</code> .                                                                                                                                                                                                                                                           |

## Value

A list of `alldiffs.objects` with a component for each combination of a `first.levels` with a `second.levels`. The name of a component will be a level from `first.levels` combined with a level from `second.levels`, separated by a comma. If the predictions in the supplied `alldiffs.object` are based on a response that was transformed, each `alldiffs.object` in the list will include a `backtransforms` component that contains a column labelled `backtransformed.predictions`, along with the backtransforms of the nominated `error.intervals`. The predictions and backtransforms components in an `alldiffs.object` will be indexed by the variables in the `classify` of `alldiffs.obj`, except that the `pairs.factor` is omitted. If the transformation was the logarithmic transformation, these `backtransformed.predictions` are predicted ratios of the untransformed response.

If `sortFactor` attribute is set and is not the `ratio.factor`, the predictions and, if present, their backtransforms will be sorted using the `sortOrder` attribute of the `alldiffs.object`, and both `sortFactor` and `sortOrder` will be set as attributes to the object.

## Author(s)

Chris Brien

## See Also

`linTransform`, `ratioTransform`, `predictPlus.asreml`, `as.alldiffs`, `print.alldiffs`, `sort.alldiffs`, `subset.alldiffs`, `allDifferences.data.frame`, `redoErrorIntervals.alldiffs`, `recalcLSD.alldiffs`, `pickLSDstatistics.alldiffs`, `predictPresent.asreml`, `plotPredictions.data.frame`, `as.Date`, `predict.asreml`

## Examples

```
#### Form the differences for log(RGR) for Salinity
load(system.file("extdata", "testDiffs.rda", package = "asremlPlus", mustWork = TRUE))
#### For the ratios for Cl per WU Temperature - use backtransforms of log-predictions
Preds.ratio.ClUp <- pairediffsTransform(diffs.ClUp,
                                         pairs.factor = "Temperature",
                                         first.levels = "Hot",
                                         second.levels = "Cool",
                                         error.intervals = "halfLeast",
                                         tables = "backtransforms") #Backtransforms are ratios

#### Form the differences for Nitrogen compared to no Nitrogen
```

```

data("Oats.dat")
## Not run:
m1.asr <- asreml(Yield ~ Nitrogen*Variety,
                random=~Blocks/Wplots,
                data=Oats.dat)
current.asrt <- as.asrtests(m1.asr)
wald.tab <- current.asrt$wald.tab
Var.diffs <- predictPlus(m1.asr, classify="Nitrogen:Variety", pairwise = TRUE,
                        Vmatrix = TRUE, error.intervals = "halfLeast",
                        LSDtype = "factor", LSDby = "Variety",
                        wald.tab = wald.tab)

## End(Not run)

## Use lme4 and emmeans to get predictions and associated statistics
if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(Yield ~ Nitrogen*Variety + (1|Blocks/Wplots),
                          data=Oats.dat)

  ## Set up a wald.tab
  int <- as.data.frame(rbind(rep(NA,4)))
  rownames(int) <- "(Intercept)"
  wald.tab <- anova(m1.lmer, ddf = "Kenward", type = 1)[,3:6]
  names(wald.tab) <- names(int) <- c("Df", "denDF", "F.inc", "Pr")
  wald.tab <- rbind(int, wald.tab)
  #Get predictions
  Var.emm <- emmeans::emmeans(m1.lmer, specs = ~ Nitrogen:Variety)
  Var.preds <- summary(Var.emm)
  ## Modify Var.preds to be compatible with a predictions.frame
  Var.preds <- as.predictions.frame(Var.preds, predictions = "emmmean",
                                   se = "SE", interval.type = "CI",
                                   interval.names = c("lower.CL", "upper.CL"))

  Var.vcov <- vcov(Var.emm)
  Var.sed <- NULL
  den.df <- wald.tab[match("Variety", rownames(wald.tab)), "denDF"]

  #Create alldiffs object
  Var.diffs <- as.alldiffs(predictions = Var.preds,
                          sed = Var.sed, vcov = Var.vcov,
                          classify = "Nitrogen:Variety", response = "Yield", tdf = den.df)
}

if (exists("Var.diffs"))
  Preds.diffs.OatsN <- pairdiffsTransform(alldiffs.obj = Var.diffs,
                                         pairs.factor = "Nitrogen",
                                         first.levels = c("0.2", "0.4", "0.6"),
                                         second.levels = "0", error.intervals = "halfLeast",
                                         tables = "none")

```

---

permute.square

*Permutes the rows and columns of a square matrix.*


---

## Description

Permutes the rows and columns of a square matrix.



**Usage**

```
permute.square(x, permutation)
```

**Arguments**

**x**                      A square matrix.

**permutation**        A vector specifying the new order of rows and columns.

**Value**

A square matrix.

**Author(s)**

Chris Brien

**See Also**

[permute.to.zero.lowertri](#)

**Examples**

```
terms.marginality <- matrix(c(1,0,0,0,0, 0,1,0,0,0, 0,1,1,0,0,
                             1,1,1,1,0, 1,1,1,1,1), nrow=5)
permtn <- c(1,3,2,4,5)
terms.marginality <- permute.square(terms.marginality, permtn)
```

---

```
permute.to.zero.lowertri
```

*Permutes a square matrix until all the lower triangular elements are zero.*

---

**Description**

Permutes a square matrix until all the lower triangular elements are zero.

**Usage**

```
permute.to.zero.lowertri(x)
```

**Arguments**

**x**                      A square matrix of order  $n$  with at least  $n*(n-1)/2$  zero elements.

**Value**

A square matrix.

**Author(s)**

Chris Brien

**See Also**

[permute.square](#)

**Examples**

```
terms.marginality <- matrix(c(1,0,0,0,0, 0,1,0,0,0, 0,1,1,0,0,
                             1,1,1,1,0, 1,1,1,1,1), nrow=5)
terms.marginality <- permute.to.zero.lowertri(terms.marginality)
```

---

pickLSDstatistics.alldiffs

*Pick LSDstatistics whose values minimize the number of errors in pair-wise comparisons of predictions.*

---

**Description**

Given an [alldiffs.object](#) with an sed component, [exploreLSDs.alldiffs](#) is used to calculate the LSD values for each set of prediction comparisons specified by LSDtype and LSDby using each of the statistics minimum, q10, q25, mean, median, q75, q90 and maximum. Then the numbers of false positives and false negatives resulting from employing each of the calculated LSDs is obtained. For each set of comparisons, the LSD value(s) with the lowest number of false positives are identified and, from these, the smallest value with the lowest number of false negatives. That is, a conservative approach is taken to picking LSD values by putting the priority on avoiding false positives. Before using the LSDstatistics that this function suggests, the number of false positives and negatives generated by them should be checked. For example, it may be that there are too many false negatives and a better balance between the numbers of false positives and negatives can be identified using [exploreLSDs.alldiffs](#),

**Usage**

```
## S3 method for class 'alldiffs'
pickLSDstatistics(alldiffs.obj,
                  LSDtype = "overall", LSDby = NULL,
                  alpha = 0.05, digits = 3,
                  false.pos.wt = NULL, retain.zeroLSDs = FALSE,
                  zero.tolerance = .Machine$double.eps ^ 0.5,
                  ...)
```

**Arguments**

**alldiffs.obj** An [alldiffs.object](#).

**LSDtype** A [character](#) string that can be overall or factor.combinations. It determines whether the LSD values that are investigated are the overall minimum, quantile10, quantile25, mean, median, quantile75, quantile90, or maximum of (i) all pairwise LSDs, or (ii) the pairwise LSDs for each combination of the values of the [factors](#) and [numerics](#) named in LSDby, unless there is only one prediction for a combination, when notional LSDs are calculated. The LSDtype specified here does not have to match that used in the creating the [alldiffs.object](#).

See [LSD.frame](#) for further information on how the LSD statistics are calculated.

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LSDby           | A <a href="#">character</a> (vector) of variables names, being the names of some of the <a href="#">factors</a> or <a href="#">numerics</a> in the <code>classify</code> ; for each combination of the values of the <a href="#">factors</a> and <a href="#">numerics</a> , the LSD errors are to be computed when <code>LSDtype</code> is <code>factor.combinatons</code> . The <code>LSDby</code> specified here does not have to match that used in the creating the <a href="#">alldiffs.object</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| alpha           | A <a href="#">numeric</a> specifying the significance level for an LSD to compare a pair of predictions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| digits          | A <a href="#">numeric</a> specifying the number of significant digits to retain in rounding the LSDs before determining the distinct rounded LSDs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| false.pos.wt    | <p>When it is not NULL, it should be a <a href="#">numeric</a> that specifies the weight (e.g. 3 so that a false positive is considered to be equivalent to three false negatives) to apply to the number of false positives in calculating the weighted sums of the numbers of false positives and negatives to use in comparing different LSD statistics, one being the weight for the number false negatives. The <code>LSDstatistic</code> that is chosen for making comparisons will be the one that minimizes the weighted sum, has the smallest number of false positives and, amongst these, has the smallest LSD value. If it is NULL, the <code>LSDstatistic</code> that will be chosen is the one that minimizes the number of false negatives from amongst those that minimize the number of false positives.</p> <p>If, amongst the LSD statistics, the least number of false negatives that occurs is <math>m</math>, then for a particular statistic with <math>p</math> the number of false positives, <math>n</math> the number of false negatives and <math>w</math> the value of <code>false.pos.wt</code>, that statistic will be a candidate LSD value if <math>(wp) + n &lt; m</math> and i.e. if <math>p &lt; (m - n)/w</math>.</p> |
| retain.zeroLSDs | A <a href="#">logical</a> indicating whether to retain or omit LSDs that are zero when calculating the summaries of LSDs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| zero.tolerance  | A <a href="#">numeric</a> specifying the value such that if an LSD is less than it, the LSD will be considered to be zero.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| ...             | Provision for passing arguments to functions called internally - not used at present.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## Value

A [character](#) of length one for `LSDby` set to `overall` or of length equal to the number of observed combinations of the values of the [factors](#) and [numerics](#) in `LSDby`. Each element of the returned [character](#) is one of `minimum`, `q10`, `q25`, `mean`, `median`, `q75`, `q90` or `maximum`, reflecting the value(s) of the LSD from amongst those calculated that minimizes the number of false positives; if there is more than one such value, then the element will be correspond to the value of the LSD from amongst those with the minimum number of false positives that minimizes the number of false negatives.

## Author(s)

Chris Brien

## See Also

[asremlPlus-package](#), [exploreLSDs.alldiffs](#) [plotLSDs.data.frame](#), [plotLSDs.alldiffs](#), [plotLSDerrors.alldiffs](#), [plotLSDerrors.data.frame](#), [recalcLSD.alldiffs](#), [redoErrorIntervals.alldiffs](#)

**Examples**

```

data(WaterRunoff.dat)

##Use asreml to get predictions and associated statistics

## Not run:
asreml.options(keep.order = TRUE) #required for asreml-R4 only
current.asr <- asreml(fixed = pH ~ Benches + (Sources * (Type + Species)),
                     random = ~ Benches:MainPlots,
                     keep.order=TRUE, data= WaterRunoff.dat)
current.asrt <- as.asrtests(current.asr, NULL, NULL)
TS.diffs <- predictPlus(classify = "Sources:Type",
                       asreml.obj = current.asr,
                       wald.tab = current.asrt$wald.tab,
                       present = c("Sources", "Type", "Species"))

## End(Not run)

## Use lmeTest and emmeans to get predictions and associated statistics

if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(pH ~ Benches + (Sources * (Type + Species)) +
                          (1|Benches:MainPlots),
                          data=na.omit(WaterRunoff.dat))
  TS.emm <- emmeans::emmeans(m1.lmer, specs = ~ Sources:Type)
  TS.preds <- summary(TS.emm)
  den.df <- min(TS.preds$df, na.rm = TRUE)
  ## Modify TS.preds to be compatible with a predictions.frame
  TS.preds <- as.predictions.frame(TS.preds, predictions = "emmean",
                                  se = "SE", interval.type = "CI",
                                  interval.names = c("lower.CL", "upper.CL"))

  ## Form an all.diffs object and check its validity
  TS.vcov <- vcov(TS.emm)
  TS.diffs <- allDifferences(predictions = TS.preds, classify = "Sources:Type",
                           vcov = TS.vcov, tdf = den.df)
  validAlldiffs(TS.diffs)
}

## Pick LSD statistics for calculating LSD values for pairwise comparisons of
## the predictions obtained using asreml or lmerTest
if (exists("TS.diffs"))
{
  ##Pick the LSD values for predictions obtained using asreml or lmerTest
  LSDstat <- pickLSDstatistics(TS.diffs)
  TS.diffs <- redoErrorIntervals(TS.diffs, LSDstatistic = LSDstat)
  TS.diffs$LSD
  LSDstat <- pickLSDstatistics(TS.diffs, LSDtype = "factor.combinations",
                              LSDby = "Sources")
  TS.diffs <- redoErrorIntervals(TS.diffs, LSDtype = "factor.combinations",
                              LSDby = "Sources", LSDstatistic = LSDstat)
  TS.diffs$LSD
}

```

---

plotLSDerrors.alldiffs

*Plots a map of the errors that occur in using the computed LSD values for pairwise differences between predictions.*

---

## Description

Produces a plot of the errors that occur in using the computed LSD values for pairwise differences predictions by comparing the result obtained from using the LSDs stored in the assignedLSD column of the LSD component of the `alldiffs.object` with those computed from the sed component using the t-value for the df stored in the tdf attribute of the `alldiffs.object`. The sed component is generally a matrix whose rows and columns are labelled by the levels of one or more factors, the set of labels being the same for rows and columns. The sections argument allows multiple plots to be produced, one for each combination of the levels of the factors listed in sections. Otherwise, a single plot is produced for all observed combinations of the levels of the factors in the classify attribute for the `alldiffs.object`. The plots are produced using `plotLSDerrors.data.frame`. The order of plotting the levels of one of the factors indexing the predictions can be modified using `sort.alldiffs`.

## Usage

```
plotLSDerrors(object, ...)
## S3 method for class 'alldiffs'
plotLSDerrors(object, alpha = 0.05, useIntervals = FALSE,
               sections = NULL, gridspacing = 0, factors.per.grid = 0,
               triangles = "both", title = NULL,
               axis.labels = TRUE, axis.text.size = 12,
               sep=",", colours = c("white","blue","red","grey"),
               ggplotFuncs = NULL, printPlot = TRUE,
               sortFactor = NULL, sortParallelToCombo = NULL,
               sortNestingFactor = NULL, sortOrder = NULL,
               decreasing = FALSE, ...)
```

## Arguments

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object       | An <code>alldiffs.object</code> with both LSD and sed components that are not NULL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| alpha        | A <code>numeric</code> giving the significance level for the LSD.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| useIntervals | A <code>logical</code> indicating whether to use the interval limits stored in the predictions component of object, instead of the LSDs stored in the LSD component, for determining whether pairs of predictions are significantly different. It allows a check of how the <code>error.intervals</code> in the predictions component will perform if they are used for all pairwise predictions comparisons, whereas the comparisons to which the LSDs apply may be restricted by the setting of the <code>LSDby</code> attribute of object. There is no restriction on the <code>error.intervals</code> that can be used. However, the limits for them must be in columns in the predictions component of object and their names must end with <code>.limits</code> and begin with <code>lower.</code> and <code>upper.</code> |
| sections     | A character listing the names of the factors that are to be used to break the plot into sections. A separate plot will be produced for each observed combination of the levels of these factors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gridspacing         | A numeric specifying the number(s) of rows and columns that form groups in the grid of differences. An alternative is to specify the <code>factors.per.grid</code> argument to have the grid spacings automatically calculated. Grids are most useful when two or more factors index the rows and columns. If a single, nonzero number, $k$ say, is given then a grid line is placed after every $k$ th row and column. If a vector of values is given then the number of grid lines is the length of the vector and the spacing between each is specified by the elements of the vector.                                                                                                                                                                                                                                                                                                                                                                                                                               |
| factors.per.grid    | A numeric specifying the number of factors to include within each grid of differences. The gridspacing will then be computed based on the numbers of combinations observed within the levels of the remaining factors in a single plot. The gridspacing argument to this function will be ignored if <code>factors.per.grid</code> is greater than zero. Grids are most useful when two or more factors index the rows and columns of each plot.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| triangles           | A character indicating whether the plot should include the lower, upper or both triangle(s).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| title               | A character string giving the main title for the plot and to which is appended the levels combination of the sectioning factors, if any, for each plot.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| axis.labels         | A logical indicating whether a label is to be added to the x- and y-axes. If TRUE, the label is the comma-separated list of factors whose levels combinations are involved in the prediction differences for which the LSD values are calculated.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| axis.text.size      | A numeric giving the size of the labels on the axes of the heatmap.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| sep                 | A character giving the characters separating the levels of different factors in the row and column names of the sed component.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| colours             | A vector of colours to be passed to the ggplot function <code>scale\_colour\_gradientn</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| ggplotFuncs         | A list, each element of which contains the results of evaluating a ggplot2 function. It is created by calling the <code>list</code> function with a ggplot2 function call for each element. It is passed to ggplot via <code>plotLSDerrors.data.frame</code> to be applied in creating the ggplot object.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| printPlot           | A logical indicating whether or not the a plot is to be printed. This would be used when just the returned <code>data.frame</code> is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| sortFactor          | A character containing the name of the factor that indexes the set of predicted values that determines the sorting of the components. If there is only one variable in the <code>classify</code> term then <code>sortFactor</code> can be NULL and the order is defined by the complete set of predicted values. If there is more than one variable in the <code>classify</code> term then <code>sortFactor</code> must be set. In this case the <code>sortFactor</code> is sorted in the same order within each combination of the values of the <code>sortParallelToCombo</code> variables: the <code>classify</code> variables, excluding the <code>sortFactor</code> . There should be only one predicted value for each unique value of <code>sortFactor</code> within each set defined by a combination of the values of the <code>classify</code> variables, excluding the <code>sortFactor</code> factor. The order to use is determined by either <code>sortParallelToCombo</code> or <code>sortOrder</code> . |
| sortParallelToCombo | A list that specifies a combination of the values of the factors and numerics, excluding <code>sortFactor</code> , that are in <code>classify</code> . Each of the components of the supplied list is named for a <code>classify</code> variable and specifies a single value for it. The combination of this set of values will be used to define a subset of the predicted values whose order will define the order of <code>sortFactor</code> . Each of the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

other combinations of the values of the factors and numerics will be sorted in parallel. If sortParallelToCombo is NULL then the first value of each classify variable, except for the sortFactor factor, in the predictions component is used to define sortParallelToCombo. If there is only one variable in the classify then sortParallelToCombo is ignored.

sortNestingFactor

A [character](#) containing the name of the factor that defines groups of the sortFactor within which the predicted values are to be ordered. If there is only one variable in the classify then sortNestingFactor is ignored.

sortOrder

A character vector whose length is the same as the number of levels for sortFactor in the predictions component of the [alldiffs.object](#). It specifies the desired order of the levels in the reordered components of the [alldiffs.object](#). The argument sortParallelToCombo is ignored.

The following creates a sortOrder vector levs for factor f based on the values in x: levs <- levels(f)[order(x)].

decreasing

A logical passed to order that determines whether the order for sorting the [alldiffs.object](#) components is for increasing or decreasing magnitude of the predicted values.

...

Provision for passing arguments to functions called internally - not used at present.

## Value

A [list](#) with components named LSDresults and plots. The LSDresults component contains the data.frame with the columns Rows, Columns, LSDresults, sections1 and sections2. This data.frame is formed using the LSD and sed components of object and is used by [plotLSDerrors.data.frame](#) in producing the plots. The plots component contains a list of ggplot objects, one for each plot produced. Multiple plots are stored in the plots component if the sections argument is set and the plots are named for the levels combinations of the sections.

## Author(s)

Chris Brien

## See Also

[plotLSDerrors.alldiffs](#), [plotLSDerrors.data.frame](#), [plotLSDs.data.frame](#), [exploreLSDs](#), [sort.alldiffs](#), [subset.alldiffs](#), [ggplot](#)

## Examples

```
##Subset WaterRunoff data to reduce time to execute
data(WaterRunoff.dat)
tmp <- subset(WaterRunoff.dat, Date == "05-18" & Benches != "3")

##Use asreml to get predictions and associated statistics

## Not run:
asreml.options(keep.order = TRUE) #required for asreml-R4 only
current.asr <- asreml(fixed = pH ~ Benches + (Sources * (Type + Species)),
                    random = ~ Benches:MainPlots,
                    keep.order=TRUE, data= tmp)
current.asrt <- as.asrtests(current.asr, NULL, NULL)
```

```

TS.diffs <- predictPlus.asreml(classify = "Sources:Type",
                              asreml.obj = current.asr, tables = "none",
                              wald.tab = current.asrt$wald.tab,
                              present = c("Type", "Species", "Sources"))

## End(Not run)

## Use lmeTest and emmeans to get predictions and associated statistics

if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(pH ~ Benches + (Sources * (Type + Species)) +
                           (1|Benches:MainPlots),
                           data=na.omit(WaterRunoff.dat))
  TS.emm <- emmeans::emmeans(m1.lmer, specs = ~ Sources:Type)
  TS.preds <- summary(TS.emm)
  den.df <- min(TS.preds$df, na.rm = TRUE)
  ## Modify TS.preds to be compatible with a predictions.frame
  TS.preds <- as.predictions.frame(TS.preds, predictions = "emmean",
                                   se = "SE", interval.type = "CI",
                                   interval.names = c("lower.CL", "upper.CL"))

  ## Form an all.diffs object and check its validity
  TS.vcov <- vcov(TS.emm)
  TS.diffs <- allDifferences(predictions = TS.preds, classify = "Sources:Type",
                           vcov = TS.vcov, tdf = den.df)
  validAlldiffs(TS.diffs)
}

## Plot LSD values for predictions obtained using asreml or lmerTest
if (exists("TS.diffs"))
{
  plotLSDerrors(TS.diffs, gridspacing = rep(c(3,4), c(4,2)))

  plotLSDerrors(TS.diffs, sections = "Sources", axis.labels = TRUE)
}

```

---

plotLSDerrors.data.frame

*Plots a map of the supplied errors that occur in using the computed LSD values for pairwise differences between predictions.*

---

## Description

Produces a plot of the errors that have been supplied in a data.frame. The data.frame includes two factors whose levels specify, for each LSD result, which combinations of factor levels are being compared. The function `plotLSDerrors.alldiffs` produces such data.frames.

## Usage

```

## S3 method for class 'data.frame'
plotLSDerrors(object, LSDresults = "LSDresults", x, y,

```



```
alpha = 0.05, triangles = "both",
gridspacing = 0, title = NULL,
axis.labels = NULL, axis.text.size = 12,
colours = c("white", "blue", "red", "grey"),
ggplotFuncs = NULL, printPlot = TRUE, ...)
```

### Arguments

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object         | A <a href="#">data.frame</a> containing the three columns specified by LSDresults, x and y.                                                                                                                                                                                                                                                                                                                                                                           |
| LSDresults     | A <a href="#">character</a> giving the name of the column in object that contains the LS-Dresults values to be plotted. The column should be a <a href="#">character</a> or <a href="#">factor</a> with values or levels that are a subset of Ok, FN, FP and na.                                                                                                                                                                                                      |
| x              | A <a href="#">character</a> giving the name of the column in object that contains the factor whose levels index the LSD values that are to be plotted in the same column.                                                                                                                                                                                                                                                                                             |
| y              | A character giving the name of the column in object that contains the labels of the LSD values that are to be plotted as the rows.                                                                                                                                                                                                                                                                                                                                    |
| alpha          | A <a href="#">numeric</a> giving the significance level for the LSD.                                                                                                                                                                                                                                                                                                                                                                                                  |
| triangles      | A character indicating whether the plot should include the lower, upper or both triangle(s). Here it is only used to adjust gridlines for the omission of the diagonal.                                                                                                                                                                                                                                                                                               |
| gridspacing    | A numeric specifying the number(s) of rows and columns that form groups in the grid of differences. This is most useful when two or more factors index the rows and columns. If a single, nonzero number, <i>k</i> say, is given then a grid line is placed after every <i>k</i> th row and column. If a vector of values is given then the number of grid lines is the length of the vector and the spacing between each is specified by the elements of the vector. |
| title          | A character string giving the main title for the plot.                                                                                                                                                                                                                                                                                                                                                                                                                |
| axis.labels    | A character string giving the label to use for both the x- and y-axis.                                                                                                                                                                                                                                                                                                                                                                                                |
| axis.text.size | A <a href="#">numeric</a> giving the size of the labels on the axes of the heatmap.                                                                                                                                                                                                                                                                                                                                                                                   |
| colours        | A vector of colours to be passed to the ggplot function <code>scale\colour\gradientn</code> .                                                                                                                                                                                                                                                                                                                                                                         |
| ggplotFuncs    | A <a href="#">list</a> , each element of which contains the results of evaluating a ggplot2 function. It is created by calling the <a href="#">list</a> function with a ggplot2 function call for each element. These functions are applied in creating the ggplot object.                                                                                                                                                                                            |
| printPlot      | A logical indicating whether or not the a plot is to be printed. This would be used when just the returned ggplot object is required.                                                                                                                                                                                                                                                                                                                                 |
| ...            | Provision for passing arguments to functions called internally - not used at present.                                                                                                                                                                                                                                                                                                                                                                                 |

### Value

An object of class "ggplot", which can be plotted using `print` or otherwise manipulated.

### Author(s)

Chris Brien

### See Also

[plotLSDs.data.frame](#), [plotLSDs.alldiffs](#), [exploreLSDs](#), [ggplot](#)

**Examples**

```

##Subset WaterRunoff data to reduce time to execute
data(WaterRunoff.dat)
tmp <- subset(WaterRunoff.dat, Date == "05-18")

##Use asreml to get predictions and associated statistics

## Not run:
asreml.options(keep.order = TRUE) #required for asreml-R4 only
current.asr <- asreml(fixed = pH ~ Benches + (Sources * (Type + Species)),
                     random = ~ Benches:MainPlots,
                     keep.order=TRUE, data= tmp))
current.asrt <- as.asrtests(current.asr, NULL, NULL)
TS.diffs <- predictPlus.asreml(classify = "Sources:Type",
                              asreml.obj = current.asr, tables = "none",
                              wald.tab = current.asrt$wald.tab,
                              present = c("Type","Species","Sources"))

## End(Not run)

## Use lmeTest and emmeans to get predictions and associated statistics

if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(pH ~ Benches + (Sources * (Type + Species)) +
                          (1|Benches:MainPlots),
                          data=na.omit(WaterRunoff.dat))
  TS.emm <- emmeans::emmeans(m1.lmer, specs = ~ Sources:Type)
  TS.preds <- summary(TS.emm)
  den.df <- min(TS.preds$df, na.rm = TRUE)
  ## Modify TS.preds to be compatible with a predictions.frame
  TS.preds <- as.predictions.frame(TS.preds, predictions = "emmean",
                                  se = "SE", interval.type = "CI",
                                  interval.names = c("lower.CL", "upper.CL"))

  ## Form an all.diffs object and check its validity
  TS.vcov <- vcov(TS.emm)
  TS.diffs <- allDifferences(predictions = TS.preds, classify = "Sources:Type",
                           vcov = TS.vcov, tdf = den.df)

  validAlldiffs(TS.diffs)
}

## Plot LSD values for predictions obtained using asreml or lmerTest
if (exists("TS.diffs"))
{
  LSDresults <- within(reshape2::melt(TS.diffs$p.differences),
                      {
                        Var1 <- factor(Var1, levels=dimnames(TS.diffs$p.differences)[[1]])
                        Var2 <- factor(Var2, levels=levels(Var1))
                      })
  names(LSDresults) <- c("Rows","Columns","LSDresults")
  plotLSDerrors(LSDresults, x = "Rows", y = "Columns", gridspacing = rep(c(3,4), c(4,2)))
}

```

---

|                   |                                                                                              |
|-------------------|----------------------------------------------------------------------------------------------|
| plotLSDs.alldiffs | <i>Plots a heat map of computed LSD values for pairwise differences between predictions.</i> |
|-------------------|----------------------------------------------------------------------------------------------|

---

## Description

Produces a heat-map plot of the computed LSD values for pairwise differences between predictions by multiplying the values stored in the `sed` component of an `alldiffs` object by the `t`-value for the `df` stored in the `tdf` attribute of the object. This component is generally a matrix whose rows and columns are labelled by the levels of one or more factors, the set of labels being the same for rows and columns. The `sections` argument allows multiple plots to be produced, one for each combination of the levels of the factors listed in `sections`. Otherwise, a single plot is produced for all observed combinations of the levels of the factors in the `classify` attribute for the `alldiffs` object. The plots are produced using `plotLSDs.data.frame`. The order of plotting the levels of one of the factors indexing the predictions can be modified using `sort.alldiffs`.

## Usage

```
plotLSDs(object, ...)
## S3 method for class 'alldiffs'
plotLSDs(object, alpha = 0.05,
          sections = NULL, gridspacing = 0, factors.per.grid = 0,
          triangles = "both",
          title = NULL, axis.labels = TRUE, axis.text.size = 12,
          sep=",", colours = RColorBrewer::brewer.pal(3, "Set2"),
          ggplotFuncs = NULL, printPlot = TRUE,
          sortFactor = NULL, sortParallelToCombo = NULL,
          sortNestingFactor = NULL, sortOrder = NULL,
          decreasing = FALSE, ...)
```

## Arguments

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>object</code>           | An <code>alldiffs</code> object with an <code>sed</code> component that is not <code>NULL</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>alpha</code>            | A numeric giving the significance level for the LSD.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>sections</code>         | A character listing the names of the factors that are to be used to break the plot into sections. A separate plot will be produced for each observed combination of the levels of these factors.                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>gridspacing</code>      | A numeric specifying the number(s) of rows and columns that form groups in the grid of differences. An alternative is to specify the <code>factors.per.grid</code> argument to have the grid spacings automatically calculated. Grids are most useful when two or more factors index the rows and columns. If a single, nonzero number, $k$ say, is given then a grid line is placed after every $k$ th row and column. If a vector of values is given then the number of grid lines is the length of the vector and the spacing between each is specified by the elements of the vector. |
| <code>factors.per.grid</code> | A numeric specifying the number of factors to include within each grid of differences. The <code>gridspacing</code> will then be computed based on the numbers of combinations observed within the levels of the remaining factors in a single plot. The <code>gridspacing</code> argument to this function will be ignored if <code>factors.per.grid</code> is greater than zero. Grids are most useful when two or more factors index the rows and columns of each plot.                                                                                                                |

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| triangles           | A character indicating whether the plot should include the lower, upper or both triangle(s).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| title               | A character string giving the main title for the plot and to which is appended the levels combination of the sectioning factors, if any, for each plot.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| axis.labels         | A logical indicating whether a label is to be added to the x- and y-axes. If TRUE, the label is the comma-separated list of factors whose levels combinations are involved in the prediction differences for which the LSD values are calculated.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| axis.text.size      | A <a href="#">numeric</a> giving the size of the labels on the axes of the heatmap.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| sep                 | A character giving the characters separating the levels of different factors in the row and column names of the sed component.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| colours             | A vector of colours to be passed to the ggplot function <code>scale\_colour\_gradientn</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| ggplotFuncs         | A <a href="#">list</a> , each element of which contains the results of evaluating a ggplot2 function. It is created by calling the <a href="#">list</a> function with a ggplot2 function call for each element. It is passed to ggplot via <code>plotLSDs.data.frame</code> to be applied in creating the ggplot object.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| printPlot           | A logical indicating whether or not the a plot is to be printed. This would be used when just the returned <code>data.frame</code> is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| sortFactor          | A <a href="#">character</a> containing the name of the factor that indexes the set of predicted values that determines the sorting of the components. If there is only one variable in the <code>classify</code> term then <code>sortFactor</code> can be NULL and the order is defined by the complete set of predicted values. If there is more than one variable in the <code>classify</code> term then <code>sortFactor</code> must be set. In this case the <code>sortFactor</code> is sorted in the same order within each combination of the values of the <code>sortParallelToCombo</code> variables: the <code>classify</code> variables, excluding the <code>sortFactor</code> . There should be only one predicted value for each unique value of <code>sortFactor</code> within each set defined by a combination of the values of the <code>classify</code> variables, excluding the <code>sortFactor</code> factor. The order to use is determined by either <code>sortParallelToCombo</code> or <code>sortOrder</code> . |
| sortParallelToCombo | A <a href="#">list</a> that specifies a combination of the values of the factors and numerics, excluding <code>sortFactor</code> , that are in <code>classify</code> . Each of the components of the supplied <a href="#">list</a> is named for a <code>classify</code> variable and specifies a single value for it. The combination of this set of values will be used to define a subset of the predicted values whose order will define the order of <code>sortFactor</code> . Each of the other combinations of the values of the factors and numerics will be sorted in parallel. If <code>sortParallelToCombo</code> is NULL then the first value of each <code>classify</code> variable, except for the <code>sortFactor</code> factor, in the predictions component is used to define <code>sortParallelToCombo</code> . If there is only one variable in the <code>classify</code> then <code>sortParallelToCombo</code> is ignored.                                                                                          |
| sortNestingFactor   | A <a href="#">character</a> containing the name of the factor that defines groups of the <code>sortFactor</code> within which the predicted values are to be ordered. If there is only one variable in the <code>classify</code> then <code>sortNestingFactor</code> is ignored.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| sortOrder           | A character vector whose length is the same as the number of levels for <code>sortFactor</code> in the predictions component of the <code>alldiffs.object</code> . It specifies the desired order of the levels in the reordered components of the <code>alldiffs.object</code> . The argument <code>sortParallelToCombo</code> is ignored.<br><br>The following creates a <code>sortOrder</code> vector <code>levs</code> for factor <code>f</code> based on the values in <code>x</code> : <code>levs &lt;- levels(f)[order(x)]</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|            |                                                                                                                                                                                        |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| decreasing | A logical passed to order that determines whether the order for sorting the <code>alldiffs.object</code> components is for increasing or decreasing magnitude of the predicted values. |
| ...        | Provision for passing arguments to functions called internally - not used at present.                                                                                                  |

### Value

A `list` with components named `LSDs` and `plots`. The `LSDs` component contains the `data.frame` with the columns `Rows`, `Columns`, `LSDs`, `sections1` and `sections2`. This `data.frame` is formed using the `sed` component of object and is used by `plotLSDs.data.frame` in producing the plot. The `plots` component contains a list of `ggplot` objects, one for each plot produced. Multiple plots are stored in the `plots` component if the `sections` argument is set and the plots are named for the levels combinations of the sections.

### Author(s)

Chris Brien

### See Also

`plotLSDs.data.frame`, `exploreLSDs`, `sort.alldiffs`, `subset.alldiffs`, `ggplot`

### Examples

```
##Subset WaterRunoff data to reduce time to execute
data(WaterRunoff.dat)
tmp <- subset(WaterRunoff.dat, Date == "05-18" & Benches != "3")

##Use asreml to get predictions and associated statistics

## Not run:
asreml.options(keep.order = TRUE) #required for asreml-R4 only
current.asr <- asreml(fixed = pH ~ Benches + (Sources * (Type + Species)),
  random = ~ Benches:MainPlots,
  keep.order=TRUE, data= tmp)
current.asrt <- as.asrtests(current.asr, NULL, NULL)
TS.diffs <- predictPlus.asreml(classify = "Sources:Type",
  asreml.obj = current.asr, tables = "none",
  wald.tab = current.asrt$wald.tab,
  present = c("Type","Species","Sources"))

## End(Not run)

## Use lmeTest and emmeans to get predictions and associated statistics

if (requireNamespace("lmerTest", quietly = TRUE) &
  requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(pH ~ Benches + (Sources * (Type + Species)) +
    (1|Benches:MainPlots),
    data=na.omit(WaterRunoff.dat))
  TS.emm <- emmeans::emmeans(m1.lmer, specs = ~ Sources:Type)
  TS.preds <- summary(TS.emm)
  den.df <- min(TS.preds$df, na.rm = TRUE)
  ## Modify TS.preds to be compatible with a predictions.frame
```

```

TS.preds <- as.predictions.frame(TS.preds, predictions = "emmean",
                                se = "SE", interval.type = "CI",
                                interval.names = c("lower.CL", "upper.CL"))

## Form an all.diffs object and check its validity
TS.vcov <- vcov(TS.emm)
TS.diffs <- allDifferences(predictions = TS.preds, classify = "Sources:Type",
                           vcov = TS.vcov, tdf = den.df)
validAlldiffs(TS.diffs)
}

## Plot LSD values for predictions obtained using asreml or lmerTest
if (exists("TS.diffs"))
{
  plotLSDs(TS.diffs, gridspacing = rep(c(3,4), c(4,2)))

  plotLSDs(TS.diffs, sections = "Sources", axis.labels = TRUE)
}

```

---

|                     |                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------|
| plotLSDs.data.frame | <i>Plots a heat map of computed LSD-values for pairwise differences between predictions.</i> |
|---------------------|----------------------------------------------------------------------------------------------|

---

## Description

Produces a heat-map plot of the computed LSD values for pairwise differences between predictions that are stored in a data.frame. The data.frame includes two factors whose levels specify, for each LSD value, which combinations of factor levels are being compared.

## Usage

```

## S3 method for class 'data.frame'
plotLSDs(object, LSD = "LSDs", x, y, alpha = 0.05,
          triangles = "both", gridspacing = 0,
          title = NULL, axis.labels = NULL, axis.text.size = 12,
          colours = RColorBrewer::brewer.pal(3, "Set2"),
          ggplotFuncs = NULL, printPlot = TRUE, ...)

```

## Arguments

|           |                                                                                                                                                                         |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object    | A data.frame containing the three columns specified by LSD, x and y.                                                                                                    |
| LSD       | A character giving the name of the column in object that contains the LSD values to be plotted.                                                                         |
| x         | A character giving the name of the column in object that contains the factor whose levels index the LSD values that are to be plotted in the same column.               |
| y         | A character giving the name of the column in object that contains the labels of the LSD values that are to be plotted as the rows.                                      |
| alpha     | A <a href="#">numeric</a> giving the significance level for the LSD.                                                                                                    |
| triangles | A character indicating whether the plot should include the lower, upper or both triangle(s). Here it is only used to adjust gridlines for the omission of the diagonal. |

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gridspacing    | A numeric specifying the number(s) of rows and columns that form groups in the grid of differences. This is most useful when two or more factors index the rows and columns. If a single, nonzero number, $k$ say, is given then a grid line is placed after every $k$ th row and column. If a vector of values is given then the number of grid lines is the length of the vector and the spacing between each is specified by the elements of the vector. |
| title          | A character string giving the main title for the plot.                                                                                                                                                                                                                                                                                                                                                                                                      |
| axis.labels    | A character string giving the label to use for both the x- and y-axis.                                                                                                                                                                                                                                                                                                                                                                                      |
| axis.text.size | A numeric giving the size of the labels on the axes of the heatmap.                                                                                                                                                                                                                                                                                                                                                                                         |
| colours        | A vector of colours to be passed to the ggplot function <code>scale\_colour\_gradientn</code> .                                                                                                                                                                                                                                                                                                                                                             |
| ggplotFuncs    | A list, each element of which contains the results of evaluating a ggplot2 function. It is created by calling the list function with a ggplot2 function call for each element. These functions are applied in creating the ggplot object.                                                                                                                                                                                                                   |
| printPlot      | A logical indicating whether or not the a plot is to be printed. This would be used when just the returned ggplot object is required.                                                                                                                                                                                                                                                                                                                       |
| ...            | Provision for passing arguments to functions called internally - not used at present.                                                                                                                                                                                                                                                                                                                                                                       |

**Value**

An object of class "ggplot", which can be plotted using `print` or otherwise manipulated.

**Author(s)**

Chris Brien

**See Also**

[plotLSDs.alldiffs](#), [plotLSDerrors.alldiffs](#), [plotLSDerrors.data.frame](#), [exploreLSDs](#), [ggplot](#)

**Examples**

```
##Subset WaterRunoff data to reduce time to execute
data(WaterRunoff.dat)
tmp <- subset(WaterRunoff.dat, Date == "05-18")

##Use asreml to get predictions and associated statistics

## Not run:
asreml.options(keep.order = TRUE) #required for asreml-R4 only
current.asr <- asreml(fixed = pH ~ Benches + (Sources * (Type + Species)),
                     random = ~ Benches:MainPlots,
                     keep.order=TRUE, data= tmp))
current.asrt <- as.asrttests(current.asr, NULL, NULL)
TS.diffs <- predictPlus.asreml(classify = "Sources:Type",
                              asreml.obj = current.asr, tables = "none",
                              wald.tab = current.asrt$wald.tab,
                              present = c("Type","Species","Sources"))

## End(Not run)

## Use lmeTest and emmeans to get predictions and associated statistics
```

```

if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(pH ~ Benches + (Sources * (Type + Species)) +
                           (1|Benches:MainPlots),
                           data=na.omit(WaterRunoff.dat))
  TS.emm <- emmeans::emmeans(m1.lmer, specs = ~ Sources:Type)
  TS.preds <- summary(TS.emm)
  den.df <- min(TS.preds$df, na.rm = TRUE)
  ## Modify TS.preds to be compatible with a predictions.frame
  TS.preds <- as.predictions.frame(TS.preds, predictions = "emmean",
                                   se = "SE", interval.type = "CI",
                                   interval.names = c("lower.CL", "upper.CL"))

  ## Form an all.diffs object and check its validity
  TS.vcov <- vcov(TS.emm)
  TS.diffs <- allDifferences(predictions = TS.preds, classify = "Sources:Type",
                            vcov = TS.vcov, tdf = den.df)
  validAlldiffs(TS.diffs)
}

## Plot LSD values for predictions obtained using asreml or lmerTest
if (exists("TS.diffs"))
{
  LSD <- within(reshape2::melt(TS.diffs$p.differences),
               {
                 Var1 <- factor(Var1, levels=dimnames(TS.diffs$p.differences)[[1]])
                 Var2 <- factor(Var2, levels=levels(Var1))
               })
  names(LSD) <- c("Rows", "Columns", "LSDs")
  plotLSDs(LSD, x = "Rows", y = "Columns", gridspacing = rep(c(3,4), c(4,2)))
}

```

---

plotPredictions.data.frame

*Plots the predictions for a term, possibly with error bars.*

---

## Description

This function plots the predictions  $y$  that are based on `classify` and stored in the `data.frame` `data`. The package `ggplot2` is used to produce the plots. Line plots are produced when variables involving `x.num` or `x.fac` are involved in `classify` for the predictions; otherwise, bar charts are produced. Further, for line charts, the argument `panels` determines whether a single plot or multiple plots in a single window are produced; for bar charts, the argument `panels` is ignored.

## Usage

```

## S3 method for class 'data.frame'
plotPredictions(data, classify, y,
                x.num = NULL, x.fac = NULL, nonx.fac.order = NULL,
                colour.scheme = "colour", panels = "multiple",
                graphics.device = NULL,
                error.intervals = "Confidence", interval.annotate = TRUE,

```



```
titles = NULL, y.title = NULL,
filestem = NULL, printPlot = TRUE, ggplotFuncs = NULL, ...)
```

## Arguments

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data           | <p>A <a href="#">predictions.frame</a>, or data.frame, containing the values of the variables to be plotted. Generally, it should contain the variables classifying the predictions and include a column with the name specified in the y argument, usually predicted.value or backtransformed.predictions; each row contains a single predicted value. It should also include columns for the standard.error and est.status. The number of rows should equal the number of unique combinations of the classifying variables. While such a data.frame can be constructed from the beginning, the pvals component of the value produced by predict.asreml is a suitable value to supply for this argument. Note that the names standard.error and est.status have been changed to std.error and status in the pvals component produced by asreml-R4; if the new names are in the data.frame supplied to predictions, they will be returned to the previous names.</p> <p>If error.intervals is not "none", then the predictions component and, if present, the backtransforms component should contain columns for the lower and upper values of the limits for the interval with names that begin with lower and upper, respectively. The second part of the name must be one of Confidence, StandardError or halfLeastSignificant. The last part needs to be consistent between the lower and upper limits.</p> |
| classify       | <p>A character string giving the combinations of the independent variables on which the predictions are based. It is an interaction type term formed from the independent variables, that is, separating the variable names with the : operator. To predict the overall mean, set the classify to "(Intercept)".</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| y              | <p>A character string giving the name of the variable that is to be plotted on the Y axis.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| x.num          | <p>A <a href="#">character</a> string giving the name of the numeric covariate that (i) is potentially included in terms in the fitted model and (ii) is the x-axis variable for plots. Its values will not be converted to a <a href="#">factor</a>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| x.fac          | <p>A character string giving the name of the factor that corresponds to x.num, is potentially included in terms in the fitted model and which corresponds to the x-axis variable. It should have the same number of levels as the number of unique values in x.num. The levels of x.fac must be in the order in which they are to be plotted - if they are dates, then they should be in the form yyyyymmdd, which can be achieved using as.Date. However, the levels can be non-numeric in nature, provided that x.num is also set.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| nonx.fac.order | <p>A character vector giving the order in which factors other than x.fac are to be plotted in faceted plots (i.e. where the number of non x factors is greater than 1). The first factor in the vector will be plotted on the X axis (if there is no x.num or x.fac. Otherwise, the order of plotting the factors is in columns (X facets) and then rows (Y facets). By default the order is in decreasing order for the numbers of levels of the non x factors.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| colour.scheme  | <p>A character string specifying the colour scheme for the plots. The default is "colour" which produces coloured lines and bars, a grey background and white gridlines. A value of "black" results in black lines, grey bars and gridlines and a white background.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>panels</code>            | Possible values are "single" and "multiple". When line plots are to be produced, because variables involving <code>x.num</code> or <code>x.fac</code> are involved in <code>classify</code> for the predictions, <code>panels</code> determines whether or not a single panel or multiple panels in a single window are produced. The <code>panels</code> argument is ignored for bar charts.                                                                    |
| <code>graphics.device</code>   | A character specifying a graphics device for plotting. The default is <code>graphics.device = NULL</code> , which will result in plots being produced on the current graphics device. Setting it to "windows", for example, will result in a windows graphics device being opened.                                                                                                                                                                               |
| <code>error.intervals</code>   | A character string indicating the type of error interval, if any, to plot in order to indicate uncertainty in the results. Possible values are "none", "StandardError", "Confidence" and "halfLeastSignificant". Here, any option other than "none" will result in the interval limits contained in data being plotted.                                                                                                                                          |
| <code>interval.annotate</code> | A logical indicating whether the plot annotation indicating the type of <code>error.interval</code> is to be included in the plot.                                                                                                                                                                                                                                                                                                                               |
| <code>titles</code>            | A list, each component of which is named for a column in the <code>data.frame</code> for the <code>asreml.obj</code> used in making the predictions and contains a character string giving a title to use in output (e.g. tables and graphs). Here they will be used for axis labels for nonresponse variables. For response variable labels see <code>y.title</code> .                                                                                          |
| <code>y.title</code>           | The title to be displayed on the y axis of any plot.                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>filestem</code>          | A character sting giving the beginning of the name of the file in which to save the plot. If <code>filestem = NULL</code> , the plot is not saved. The remainder of the file name will be generated automatically and consists of the following elements separated by full stops: the classify term, Bar or Line and, if <code>error.intervals</code> is not "none", one of SE, CI or LSI. The file will be saved as a 'png' file in the current work directory. |
| <code>printPlot</code>         | A logical indicating whether or not the a plot is to be printed. This would be used when just the returned <code>ggplot</code> object is required.                                                                                                                                                                                                                                                                                                               |
| <code>ggplotFuncs</code>       | A <a href="#">list</a> , each element of which contains the results of evaluating a <code>ggplot2</code> function. It is created by calling the <a href="#">list</a> function with a <code>ggplot2</code> function call for each element. These functions are applied in creating the <code>ggplot</code> object for plotting.                                                                                                                                   |
| <code>...</code>               | further arguments passed to <code>ggplot</code> .                                                                                                                                                                                                                                                                                                                                                                                                                |

**Value**

An object of class `ggplot`, which can be plotted using `print` or otherwise manipulated.

**Author(s)**

Chris Brien

**See Also**

[allDifferences.data.frame](#), [predictPresent.asreml](#), [redoErrorIntervals.alldiffs](#), [recalcLSD.alldiffs](#), `ggplot`, `Devices`

**Examples**

```

## Not run:
data(WaterRunoff.dat)
asreml.options(keep.order = TRUE) #required for asreml-R4 only
current.asr <- asreml(fixed = log.Turbidity ~ Benches + Sources + Type + Species +
                      Sources:Type + Sources:Species +
                      Sources:xDay + Species:xDay + Species:Date,
                      data = WaterRunoff.dat, keep.order = TRUE)
current.asrt <- as.asrttests(current.asr, NULL, NULL)

#### Get the observed combinations of the factors and variables in classify
class.facs <- c("Species","Date","xDay")
levs <- as.data.frame(table(WaterRunoff.dat[class.facs]))
levs <- as.list(levs[levs$Freq != 0, class.facs])
levs$xDay <- as.numfac(levs$xDay)

predictions <- predict(current.asr, classify="Species:Date:xDay",
                        parallel = TRUE, levels = levs,
                        present = c("Type","Species","Sources"))

#### for asreml-R3
predictions <- predictions$predictions$pvals
predictions <- predictions[predictions$est.status == "Estimable",]
#### for asreml-R4
predictions <- predictions$pvals
predictions <- predictions[predictions$status == "Estimable",]
#### end
plotPredictions(classify="Species:Date:xDay", y = "predicted.value",
                 data = predictions,
                 x.num = "xDay", x.fac = "Date",
                 x.title = "Days since first observation",
                 y.title = "Predicted log(Turbidity)",
                 present = c("Type","Species","Sources"),
                 error.intervals = "none",
                 ggplotFuncs = list(ggtitle("Transformed turbidity over time")))

diffs <- predictPlus(classify="Species:Date:xDay",
                      present=c("Type","Species","Sources"),
                      asreml.obj = current.asr, tables = "none",
                      x.num = "xDay", x.fac = "Date",
                      parallel = TRUE, levels = levs,
                      x.plot.values=c(0,28,56,84),
                      wald.tab = current.asrt$wald.tab)
x.title <- "Days since first observation"
names(x.title) <- "xDay"
plotPredictions(classify="Species:Date:xDay", y = "predicted.value",
                 data = diffs$predictions,
                 x.num = "xDay", x.fac = "Date",
                 titles = x.title,
                 y.title = "Predicted log(Turbidity)")

## End(Not run)

## Use lmerTest and emmeans to get predictions and associated statistics
if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{

```

```

data(Ladybird.dat)
m1.lmer <- lmerTest::lmer(logitP ~ Host*Cadavers*Ladybird + (1|Run),
                        data=Ladybird.dat)
HCL.emm <- emmeans::emmeans(m1.lmer, specs = ~ Host:Cadavers:Ladybird)
HCL.preds <- summary(HCL.emm)
den.df <- min(HCL.preds$df)
## Modify HCL.preds to be compatible with a predictions.frame
HCL.preds <- as.predictions.frame(HCL.preds, predictions = "emmean",
                                se = "SE", interval.type = "CI",
                                interval.names = c("lower.CL", "upper.CL"))

## Plot the predictions
plotPredictions(HCL.preds, y = "predicted.value", "Host:Cadavers:Ladybird")
}

```

---

|                      |                                                                                   |
|----------------------|-----------------------------------------------------------------------------------|
| plotPvalues.alldiffs | <i>Plots a heat map of p-values for pairwise differences between predictions.</i> |
|----------------------|-----------------------------------------------------------------------------------|

---

## Description

Produces a heat-map plot of the p-values for pairwise differences between predictions that is stored in the `p.differences` component of an `alldiffs` object. This is generally a matrix whose rows and columns are labelled by the levels of one or more factors, the set of labels being the same for rows and columns. The `sections` argument allows multiple plots to be produced, one for each combination of the levels of the factors listed in `sections`. Otherwise, a single plot is produced for all observed combinations of the levels of the factors in the `classify` for the `alldiffs` object. The plots are produced using `plotPvalues.data.frame`. The order of plotting the levels of one of the factors indexing the predictions can be modified using `sort.alldiffs`.

## Usage

```

plotPvalues(object, ...)
## S3 method for class 'alldiffs'
plotPvalues(object, sections = NULL,
            gridspacing = 0, factors.per.grid = 0,
            show.sig = FALSE, alpha = 0.10,
            sig.size = 3, sig.colour = "black",
            sig.face = "plain", sig.family = "",
            triangles = "both",
            title = NULL, axis.labels = TRUE, axis.text.size = 12,
            sep=",", colours = RColorBrewer::brewer.pal(3, "Set2"),
            ggplotFuncs = NULL, printPlot = TRUE,
            sortFactor = NULL, sortParallelToCombo = NULL,
            sortNestingFactor = NULL, sortOrder = NULL,
            decreasing = FALSE, ...)

```

## Arguments

|                       |                                                                                                                                                                                                  |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>object</code>   | An <code>alldiffs</code> object with a <code>p.differences</code> component that is not <code>NULL</code> .                                                                                      |
| <code>sections</code> | A character listing the names of the factors that are to be used to break the plot into sections. A separate plot will be produced for each observed combination of the levels of these factors. |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gridspacing      | A numeric specifying the number(s) of rows and columns that form groups in the grid of differences. An alternative is to specify the <code>factors.per.grid</code> argument to have the grid spacings automatically calculated. Grids are most useful when two or more factors index the rows and columns. If a single, nonzero number, $k$ say, is given then a grid line is placed after every $k$ th row and column. If a vector of values is given then the number of grid lines is the length of the vector and the spacing between each is specified by the elements of the vector. |
| factors.per.grid | A numeric specifying the number of factors to include within each grid of differences. The gridspacing will then be computed based on the numbers of combinations observed within the levels of the remaining factors in a single plot. The gridspacing argument to this function will be ignored if <code>factors.per.grid</code> is greater than zero. Grids are most useful when two or more factors index the rows and columns of each plot.                                                                                                                                          |
| show.sig         | A logical that specifies whether asterisks indicating the level of significance are to be added to the plot. If they are then '***' indicates that $p \leq 0.001$ , '**' that $0.001 < p \leq 0.01$ , '*' that $0.01 < p \leq 0.05$ '.' that $0.05 < p \leq 0.10$ . The last is only included for $\alpha = 0.10$ .                                                                                                                                                                                                                                                                       |
| alpha            | A numeric giving the significance level for testing pairwise differences; must be 0.05 or 0.10.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| sig.size         | A numeric specifying the size, in pts, of the significance asterisks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| sig.colour       | A character specifying the colour to use for the significance asterisks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| sig.face         | A character specifying the font face for the significance asterisks ("plain", "italic", "bold", "bold.italic").                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| sig.family       | A character specifying the font family for the significance asterisks. The font families that are available depends on the system. For font families other than the basic Postscript fonts, see the <code>extrafont</code> package.                                                                                                                                                                                                                                                                                                                                                       |
| triangles        | A character indicating whether the plot should include the lower, upper or both triangle(s).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| title            | A character string giving the main title for the plot and to which is appended the levels combination of the sectioning factors, if any, for each plot.                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| axis.labels      | A logical indicating whether a label is to be added to the x- and y-axes. If TRUE, the label is the comma-separated list of factors whose levels combinations are involved in the prediction differences for which the p-values are calculated.                                                                                                                                                                                                                                                                                                                                           |
| axis.text.size   | A numeric giving the size of the labels on the axes of the heatmap.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| sep              | A character giving the characters separating the levels of different factors in the row and column names of the p.differences component.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| colours          | A vector of colours to be passed to the ggplot function <code>scale\_colour\_gradientn</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| ggplotFuncs      | A list, each element of which contains the results of evaluating a ggplot2 function. It is created by calling the <code>list</code> function with a ggplot2 function call for each element. It is passed to ggplot via <code>plotPvalues.data.frame</code> to be applied in creating the ggplot object.                                                                                                                                                                                                                                                                                   |
| printPlot        | A logical indicating whether or not the a plot is to be printed. This would be used when just the returned data.frame is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| sortFactor       | A character containing the name of the factor that indexes the set of predicted values that determines the sorting of the components. If there is only one variable in the <code>classify</code> term then <code>sortFactor</code> can be NULL and the order                                                                                                                                                                                                                                                                                                                              |

is defined by the complete set of predicted values. If there is more than one variable in the classify term then sortFactor must be set. In this case the sortFactor is sorted in the same order within each combination of the values of the sortParallelToCombo variables: the classify variables, excluding the sortFactor. There should be only one predicted value for each unique value of sortFactor within each set defined by a combination of the values of the classify variables, excluding the sortFactor factor. The order to use is determined by either sortParallelToCombo or sortOrder.

#### sortParallelToCombo

A [list](#) that specifies a combination of the values of the factors and numerics, excluding sortFactor, that are in classify. Each of the components of the supplied [list](#) is named for a classify variable and specifies a single value for it. The combination of this set of values will be used to define a subset of the predicted values whose order will define the order of sortFactor. Each of the other combinations of the values of the factors and numerics will be sorted in parallel. If sortParallelToCombo is NULL then the first value of each classify variable, except for the sortFactor factor, in the predictions component is used to define sortParallelToCombo. If there is only one variable in the classify then sortParallelToCombo is ignored.

#### sortNestingFactor

A [character](#) containing the name of the factor that defines groups of the sortFactor within which the predicted values are to be ordered. If there is only one variable in the classify then sortNestingFactor is ignored.

#### sortOrder

A character vector whose length is the same as the number of levels for sortFactor in the predictions component of the [alldiffs.object](#). It specifies the desired order of the levels in the reordered components of the [alldiffs.object](#). The argument sortParallelToCombo is ignored.

The following creates a sortOrder vector levs for factor f based on the values in x: levs <- levels(f)[order(x)].

#### decreasing

A logical passed to order that determines whether the order for sorting the [alldiffs.object](#) components is for increasing or decreasing magnitude of the predicted values.

#### ...

Provision for passing arguments to functions called internally - not used at present.

### Value

A [list](#) with components named pvalues and plots. The pvalues component contains the data.frame with the columns Rows, Columns, p, sections1 and sections2. This data.frame is formed using the sed component of object and is used by [plotPvalues.data.frame](#) in producing the plot. The plots component contains a list of ggplot objects, one for each plot produced. Multiple plots are stored in the plots component if the sections argument is set and the plots are named for the levels combinations of the sections.

### Author(s)

Chris Brien

### See Also

[plotPvalues.data.frame](#), [allDifferences.data.frame](#), [sort.alldiffs](#), [subset.alldiffs](#), [ggplot](#)

**Examples**

```

##Subset WaterRunoff data to reduce time to execute
data(WaterRunoff.dat)
tmp <- subset(WaterRunoff.dat, Date == "05-18" & Benches != "3")

##Use asreml to get predictions and associated statistics

## Not run:
asreml.options(keep.order = TRUE) #required for asreml-R4 only
current.asr <- asreml(fixed = pH ~ Benches + (Sources * (Type + Species)),
                     random = ~ Benches:MainPlots,
                     keep.order=TRUE, data= tmp)
current.asrt <- as.asrtests(current.asr, NULL, NULL)
TS.diffs <- predictPlus.asreml(classify = "Sources:Type",
                              asreml.obj = current.asr, tables = "none",
                              wald.tab = current.asrt$wald.tab,
                              present = c("Type","Species","Sources"))

## End(Not run)

## Use lmeTest and emmeans to get predictions and associated statistics

if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(pH ~ Benches + (Sources * (Type + Species)) +
                          (1|Benches:MainPlots),
                          data=na.omit(WaterRunoff.dat))
  TS.emm <- emmeans::emmeans(m1.lmer, specs = ~ Sources:Type)
  TS.preds <- summary(TS.emm)
  den.df <- min(TS.preds$df, na.rm = TRUE)
  ## Modify TS.preds to be compatible with a predictions.frame
  TS.preds <- as.predictions.frame(TS.preds, predictions = "emmean",
                                  se = "SE", interval.type = "CI",
                                  interval.names = c("lower.CL", "upper.CL"))

  ## Form an all.diffs object and check its validity
  TS.vcov <- vcov(TS.emm)
  TS.diffs <- allDifferences(predictions = TS.preds, classify = "Sources:Type",
                           vcov = TS.vcov, tdf = den.df)
  validAlldiffs(TS.diffs)
}

## Plot p-values for predictions obtained using asreml or lmerTest
if (exists("TS.diffs"))
{
  plotPvalues(TS.diffs, gridspacing = rep(c(3,4), c(4,2)), show.sig = TRUE)

  plotPvalues(TS.diffs, sections = "Sources", show.sig = TRUE, axis.labels = TRUE)
}

```

---

plotPvalues.data.frame

*Plots a heat map of p-values for pairwise differences between predictions.*

## Description

Produces a heat-map plot of the p-values for pairwise differences between predictions that is in a data.frame. The data.frame includes two factors whose levels specify, for each p-value, which factor levels are being compared.

## Usage

```
## S3 method for class 'data.frame'
plotPvalues(object, p = "p", x, y,
            gridspacing = 0, show.sig = FALSE, alpha = 0.10,
            sig.size = 3, sig.colour = "black",
            sig.face = "plain", sig.family = "",
            triangles = "both",
            title = NULL, axis.labels = NULL, axis.text.size = 12,
            colours = RColorBrewer::brewer.pal(3, "Set2"),
            ggplotFuncs = NULL, printPlot = TRUE, ...)
```

## Arguments

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object      | A data.frame containing the three columns specified by p, x and y.                                                                                                                                                                                                                                                                                                                                                                                          |
| p           | A character giving the name of the column in object that contains the p-values to be plotted.                                                                                                                                                                                                                                                                                                                                                               |
| x           | A character giving the name of the column in object that contains the factor whose levels index the p-values that are to be plotted in the same column.                                                                                                                                                                                                                                                                                                     |
| y           | A character giving the name of the column in object that contains the labels of the p-values that are to be plotted as the rows.                                                                                                                                                                                                                                                                                                                            |
| gridspacing | A numeric specifying the number(s) of rows and columns that form groups in the grid of differences. This is most useful when two or more factors index the rows and columns. If a single, nonzero number, $k$ say, is given then a grid line is placed after every $k$ th row and column. If a vector of values is given then the number of grid lines is the length of the vector and the spacing between each is specified by the elements of the vector. |
| show.sig    | A logical that specifies whether asterisks indicating the level of significance are to be added to the plot. If they are then '****' indicates that $p \leq 0.001$ , '***' that $0.001 < p \leq 0.01$ , '**' that $0.01 < p \leq 0.05$ , '*' that $0.05 < p \leq 0.10$ . The last is only included for $\alpha = 0.10$ .                                                                                                                                    |
| alpha       | A numeric giving the significance level for testing pairwise differences; must be 0.05 or 0.10.                                                                                                                                                                                                                                                                                                                                                             |
| sig.size    | A numeric specifying the size, in pts, of the significance asterisks.                                                                                                                                                                                                                                                                                                                                                                                       |
| sig.colour  | A character specifying the colour to use for the significance asterisks.                                                                                                                                                                                                                                                                                                                                                                                    |
| sig.face    | A character specifying the font face for the significance asterisks ("plain", "italic", "bold", "bold.italic").                                                                                                                                                                                                                                                                                                                                             |



|                |                                                                                                                                                                                                                                                                            |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| sig.family     | A <a href="#">character</a> specifying the font family for the significance asterisks. The font families that are available depends on the system. For font families other than the basic Postscript fonts, see the <a href="#">extrafont</a> package.                     |
| triangles      | A character indicating whether the plot should include the lower, upper or both triangle(s). Here it is only used to adjust gridlines for the omission of the diagonal.                                                                                                    |
| title          | A character string giving the main title for the plot.                                                                                                                                                                                                                     |
| axis.labels    | A character string giving the label to use for both the x- and y-axis.                                                                                                                                                                                                     |
| axis.text.size | A <a href="#">numeric</a> giving the size of the labels on the axes of the heatmap.                                                                                                                                                                                        |
| colours        | A vector of colours to be passed to the ggplot function <code>scale\_colour\_gradientn</code> .                                                                                                                                                                            |
| ggplotFuncs    | A <a href="#">list</a> , each element of which contains the results of evaluating a ggplot2 function. It is created by calling the <a href="#">list</a> function with a ggplot2 function call for each element. These functions are applied in creating the ggplot object. |
| printPlot      | A logical indicating whether or not the a plot is to be printed. This would be used when just the returned ggplot object is required.                                                                                                                                      |
| ...            | Provision for passing arguments to functions called internally - not used at present.                                                                                                                                                                                      |

**Value**

An object of class "ggplot", which can be plotted using `print` or otherwise manipulated.

**Author(s)**

Chris Brien

**See Also**

[plotPvalues.alldiffs](#), [allDifferences.data.frame](#), [ggplot](#)

**Examples**

```
##Subset WaterRunoff data to reduce time to execute
data(WaterRunoff.dat)
tmp <- subset(WaterRunoff.dat, Date == "05-18")

##Use asreml to get predictions and associated statistics

## Not run:
asreml.options(keep.order = TRUE) #required for asreml-R4 only
current.asr <- asreml(fixed = pH ~ Benches + (Sources * (Type + Species)),
  random = ~ Benches:MainPlots,
  keep.order=TRUE, data= tmp))
current.asrt <- as.asrtests(current.asr, NULL, NULL)
SS.diffs <- predictPlus.asreml(classify = "Sources:Type",
  asreml.obj = current.asr, tables = "none",
  wald.tab = current.asrt$wald.tab,
  present = c("Type","Species","Sources"))

## End(Not run)

## Use lmeTest and emmeans to get predictions and associated statistics
```

```

if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(pH ~ Benches + (Sources * (Type + Species)) +
                           (1|Benches:MainPlots),
                           data=na.omit(WaterRunoff.dat))
  TS.emm <- emmeans::emmeans(m1.lmer, specs = ~ Sources:Type)
  TS.preds <- summary(TS.emm)
  den.df <- min(TS.preds$df, na.rm = TRUE)
  ## Modify TS.preds to be compatible with a predictions.frame
  TS.preds <- as.predictions.frame(TS.preds, predictions = "emmean",
                                   se = "SE", interval.type = "CI",
                                   interval.names = c("lower.CL", "upper.CL"))

  ## Form an all.diffs object and check its validity
  TS.vcov <- vcov(TS.emm)
  TS.diffs <- allDifferences(predictions = TS.preds, classify = "Sources:Type",
                            vcov = TS.vcov, tdf = den.df)

  validAlldiffs(TS.diffs)
}

## Plot p-values for predictions obtained using asreml or lmerTest
if (exists("TS.diffs"))
{
  p <- within(reshape2::melt(TS.diffs$p.differences),
              {
                Var1 <- factor(Var1, levels=dimnames(TS.diffs$p.differences)[[1]])
                Var2 <- factor(Var2, levels=levels(Var1))
              })
  names(p) <- c("Rows", "Columns", "p")
  plotPvalues(p, x = "Rows", y = "Columns",
              gridspacing = rep(c(3,4), c(4,2)), show.sig = TRUE)
}

```

---

plotVariofaces.data.frame

*Plots empirical variogram faces, including envelopes, from supplied residuals as described by Stefanova, Smith & Cullis (2009).*

---

## Description

Produces a plot for each face of an empirical 2D variogram based on supplied residuals from both an observed data set and simulated data sets. Those from simulated data sets are used to produce confidence envelopes. If the data consists of sections, such as separate experiments, the two variogram faces are produced for each section. This function is less efficient in storage terms than [variofaces.asreml](#), because here the residuals from all simulated data sets must be saved, in addition to the values for the variogram faces; in [variofaces.asreml](#), the residuals for each simulated data set are discarded after the variogram has been calculated. On the other hand, the present function is more flexible, because there is no restriction on how the residuals are obtained.

## Usage

```

## S3 method for class 'data.frame'
plotVariofaces(data, residuals, restype="Residuals", ...)

```

**Arguments**

|           |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data      | A data.frame with either 3 or 4 columns. Only if there are 4 columns, the first should be a factor indexing sections for which separate variogram plots are to be produced. In either case, the other 3 columns should be, in order, (i) a factor indexing the x-direction, (ii) a factor indexing the y-direction, and (iii) the residuals for the observed response. |
| residuals | A data.frame, with either 2 or 3 initial columns followed by columns, each of which are the residuals from a simulated data set.                                                                                                                                                                                                                                       |
| restype   | A character describing the type of residuals that have been supplied. It will be used in the plot titles.                                                                                                                                                                                                                                                              |
| ...       | Other arguments that are passed down to the function asreml.variogram.                                                                                                                                                                                                                                                                                                 |

**Details**

For each set of residuals, asreml.variogram is used to obtain the empirical variogram, from which the values for its faces are obtained. Plots are produced for each face and include the observed residuals and the 2.5%, 50% & 97.5% quantiles.

**Value**

A list with the following components:

1. **face1**: a data.frame containing the variogram values on which the plot for the first dimension is based.
2. **face2**: a data.frame containing the variogram values on which the plot for the second dimension is based.

**Author(s)**

Chris Brien

**References**

Stefanova, K. T., Smith, A. B. & Cullis, B. R. (2009) Enhanced diagnostics for the spatial analysis of field trials. *Journal of Agricultural, Biological, and Environmental Statistics*, **14**, 392–410.

**See Also**

[asremlPlus-package](#), [asreml](#), [asreml.variogram](#), [variofaces.asreml](#), [simulate.asreml](#).

**Examples**

```
## Not run:
data(Wheat.dat)
current.asr <- asreml(yield ~ Rep + WithinColPairs + Variety,
                    random = ~ Row + Column + units,
                    residual = ~ ar1(Row):ar1(Column),
                    data=Wheat.dat)
current.asrt <- as.asrtests(current.asr, NULL, NULL)
current.asrt <- rmboundary.asrtests(current.asrt)
# Form variance matrix based on estimated variance parameters
s2 <- current.asr$sigma2
gamma.Row <- current.asr$gammas[1]
```

```

gamma.unit <- current.asr$gammas[2]
rho.r <- current.asr$gammas[4]
rho.c <- current.asr$gammas[5]
row.ar1 <- mat.ar1(order=10, rho=rho.r)
col.ar1 <- mat.ar1(order=15, rho=rho.c)
V <- gamma.Row * fac.sumop(Wheat.dat$Row) +
  gamma.unit * diag(1, nrow=150, ncol=150) +
  mat.dirprod(col.ar1, row.ar1)
V <- s2*V

#Produce variogram faces plot (Stefanaova et al, 2009)
resid <- simulate(current.asr, V=V, which="residuals")
resid$residuals <- cbind(resid$observed[c("Row", "Column")],
  resid$residuals)
plotVariofaces(data=resid$observed[c("Row", "Column", "residuals")],
  residuals=resid$residuals,
  restype="Standardized conditional residuals")

## End(Not run)

```

---

|                |                                                                                                                                                |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| powerTransform | <i>Performs a combination of a linear and a power transformation on a variable. The transformed variable is stored in the data.frame data.</i> |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------|

---

## Description

Perform a combination of a linear and a power transformation on a variable whose name is given as a character string in `var.name`. The transformed variable is stored in the `data.frame data`. The name of the transformed variable is made by prepending to the original `var.name` a combination of (i) `.offset`, if `offset` is nonzero, (ii) `neg.`, if `scale` is -1, or `scaled.`, if `abs(scale)` is other than one, and (iii) either `log.`, `sqrt.`, `recip.` or `power.`, if `power` is other than one. No action is taken if there is no transformation (i.e. `offset = 0`, `scale = 1` and `power = 1`). Also, the `titles` list is extended to include a component with a generated title for the transformed variable with text indicating the transformation prepended to the title for the `var.name` obtained from the `titles` list. For nonzero `offset`, 'Offset ' is prepended, For `scaled` not equal to one, the possible prepends are 'Negative of ' and 'Scaled '. The possible prepended texts for `power` not equal to one are 'Logarithm of', 'Square root of', 'Reciprocal of ' and 'Power nnnn of ', where `nnn` is the power used.

## Usage

```
powerTransform(var.name, power = 1, offset = 0, scale = 1, titles = NULL, data)
```

## Arguments

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>var.name</code> | A character string specifying the name of the variable in the <code>data.frame data</code> that is to be transformed.                                                                                                                                                                                                                                                                                                                         |
| <code>power</code>    | A number specifying the power to be used in the transformation. If equal to 1, the default, no power transformation is applied. Otherwise, the variable is raised to the specified power, after scaling and applying any nonzero <code>offset</code> . If <code>power = 0</code> , the natural logarithm is used to transform the response; however, if the smallest value to be log-transformed is less than 1e-04, an error is generated. A |

|        |                                                                                                                                                                                                                                                                   |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | log-transformation in this situation may be possible if a nonzero offset and/or a scale not equal to one is used.                                                                                                                                                 |
| offset | A number to be added to each value of the variable, after any scaling and before applying any power transformation.                                                                                                                                               |
| scale  | A number to multiply each value of the variable, before adding any offset and applying any power transformation.                                                                                                                                                  |
| titles | A character vector, each element of which is named for a variable in data and is a character string giving a title to use in output (e.g. tables and graphs) involving the variable. If titles are not supplied, the column name of the variable in data is used. |
| data   | A data.frame containing the variable to be transformed and to which the transformed variable is to be appended.                                                                                                                                                   |

**Value**

A list with a component named `data` that is the data.frame containing the transformed variable, a component named `tvar.name` that is a character string that is the name of the transformed variable in data, and a component named `titles` that extends the list supplied in the `titles` argument to include a generated title for the transformed title, the name of the new component being `tvar.name`.

**Author(s)**

Chris Brien

**See Also**

[angular](#), [angular.mod](#).

**Examples**

```
## set up a factor with labels
x.dat <- data.frame(y = c(14, 42, 120, 150))

## transform y to logarithms
trans <- powerTransform("y", power = 0, titles=list(y = "Length (cm)"), data = x.dat)
x.dat <- trans$data
tvar.name <- trans$tvar.name

## transform y to logarithms after multiplying by -1 and adding 1.
z.dat <- data.frame( y = c(-5.25, -4.29, -1.22, 0.05))
trans <- powerTransform("y", power = 0, scale = -1, offset = 1 ,
                        titles=list(y = "Potential"), data = z.dat)
z.dat <- trans$data
tvar.name <- trans$tvar.name
```

---

|                   |                                            |
|-------------------|--------------------------------------------|
| predictions.frame | <i>Description of a predictions object</i> |
|-------------------|--------------------------------------------|

---

### Description

A data.frame of S3-class predictions.frame that stores the predictions for a fitted model.

[as.predictions.frame](#) is function that converts a [data.frame](#) to an object of this class.

[is.predictions.frame](#) is the membership function for this class; it tests that an object has class predictions.frame.

[validPredictionsFrame](#) can be used to test the validity of a predictions.frame.

### Value

A data.frame that begins with the variables classifying the predictions, in the same order as in the classify, followed by a column of predictions that is named either predicted.value or backtransformed.predictions; it also contains columns named standard.error and est.status. The number of rows should equal the number of unique combinations of the classifying variables. While such a data.frame can be constructed from the beginning, the pvals component of the value produced by predict.asreml is a suitable value to supply for this argument. Note that the names standard.error and est.status have been changed to std.error and status in the pvals component produced by asreml-R4; if the new names are in the data.frame supplied to predictions, they will be returned to the previous names.

The data.frame may also include columns for the lower and upper values of error intervals, either standard error, confidence or half-LSD intervals. The names of these columns will consist of three parts separated by full stops: 1) the first part will be lower or upper; 2) the second part will be one of Confidence, StandardError or halfLeastSignificant; 3) the third component will be limits.

IF accuracy.threshold is set to a numeric value at the time the prediction.frame is formed, it will also include a column logical values named LSDwarning.

When halfLeastSignificant limits have been included in a predictions.frame, its attributes will include those that are not NULL of LSDtype, LSDby, LSDstatistic, LSDaccuracy and LSDvalues. LSDvalues are the LSD values used to calculate the halfLeastSignificant error.intervals and are an expanded version of the values stored in the assignedLSD column of the [LSD.frame](#).

See [predictPlus.asreml](#) for more information.

### Author(s)

Chris Brien

### See Also

[predictPlus.asreml](#), [is.predictions.frame](#), [as.predictions.frame](#), [validPredictionsFrame](#)

### Examples

```
data(Oats.dat)

## Use asreml to get predictions and associated statistics
```

```

## Not run:
m1.asr <- asreml(Yield ~ Nitrogen*Variety,
               random=~Blocks/Wplots,
               data=Oats.dat)
current.asrt <- as.asrtests(m1.asr)
Var.pred <- asreml::predict.asreml(m1.asr, classify="Nitrogen:Variety",
                                sed=TRUE)
if (getASRemlVersionLoaded(nchar = 1) == "3")
  Var.pred <- Var.pred$predictions
Var.preds <- as.predictions.frame(Var.pred$pvals, se = "std.error",
                                est.status = "status")

## End(Not run)

## Use lmerTest and emmeans to get predictions and associated statistics
if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(Yield ~ Nitrogen*Variety + (1|Blocks/Wplots),
                          data=Oats.dat)
  Var.emm <- emmeans::emmeans(m1.lmer, specs = ~ Nitrogen:Variety)
  Var.preds <- summary(Var.emm)
  Var.preds <- as.predictions.frame(Var.preds, predictions = "emmean",
                                  se = "SE", interval.type = "CI",
                                  interval.names = c("lower.CL", "upper.CL"))
}

if (exists("Var.preds"))
{
  ## Check the class and validity of the alldiffs object
  is.predictions.frame(Var.preds)
  validPredictionsFrame(Var.preds)
}

```

---

|                    |                                                                                                                                                                                                                                                                   |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| predictPlus.asreml | <i>Forms the predictions for a term, their pairwise differences and associated statistics. A factor having parallel values may occur in the model and a linear transformation of the predictions can be specified. It results in an object of class alldiffs.</i> |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

---

## Description

This function forms the predictions for term using `classify` and the supplied `asreml` object and stores them in an `alldiffs.object`. If `x.num` is supplied, the predictions will be obtained for the values supplied in `x.pred.values` and, if supplied, `x.plot.values` will replace them in the `alldiffs.object` that is returned. If `x.fac`, but not `x.num`, is specified, predictions will involve it and, if supplied, `x.plot.values` will replace the levels of `x.fac` in the `alldiffs.object` that is returned. In order to get the correct predictions you may need to supply additional arguments to `predict.asreml` through ... e.g. `present`, `parallel`, `levels`. Any aliased predictions will be removed, as will any standard error of pairwise differences involving them.

Also calculated are the approximate degrees of freedom of the standard errors of the predictions. If the denominator degrees of freedom for term are available in `wald.tab`, they are used. Otherwise the residual degrees of freedom or the maximum of the denominator degrees in `wald.tab`, excluding

the Intercept, are used. Which is used depends on the setting of `dDF.fault`. These degrees of freedom are used for the t-distribution on which p-values and confidence intervals are based. It is stored as an attribute to the `alldiffs.object`. The degrees of freedom are also used in calculating the minimum, mean and maximum LSD for comparing pairs of predictions, which are also stored in the `alldiffs.object`.

If `pairwise = TRUE`, all pairwise differences between the predictions, their standard errors, p-values and LSD statistics are computed using `allDifferences.data.frame`. This adds them to the `alldiffs.object` as additional list components named `differences`, `sed`, `p.differences` and `LSD`.

If a linear transformation of the predictions is specified then the values of this linear transformation are returned, instead of the original predictions, along with their standard errors and the pairwise differences and associated statistics.

If a transformation has been applied in the analysis (any one of `transform.power` is not one, `scale` is not one and `offset` is nonzero), the backtransforms of the transformed values and their lower and upper error intervals are added to a `data.frame` that is consistent with the predictions `data.frame`. If `transform.power` is other than one, the `standard.error` column of the `data.frame` is set to NA. This `data.frame` is added to the `alldiffs.object` as a list component called `backtransforms`.

The printing of the components produced is controlled by the `tables` argument. The order of plotting the levels of one of the factors indexing the predictions can be modified and is achieved using `sort.alldiffs`.

## Usage

```
## S3 method for class 'asreml'
predictPlus(asreml.obj, classify, term = NULL,
            inestimable.rm = TRUE,
            linear.transformation = NULL, EGLS.linTransform = TRUE,
            error.intervals = "Confidence", alpha = 0.05,
            wald.tab = NULL, dDF.fault = "residual", dDF.values = NULL,
            pairwise = TRUE, Vmatrix = FALSE,
            avsed.tolerance = 0.25, accuracy.threshold = NA,
            LSDtype = "overall", LSDsupplied = NULL, LSDby = NULL,
            LSDstatistic = "mean", LSDaccuracy = "maxAbsDeviation",
            x.num = NULL, x.fac = NULL,
            x.pred.values = NULL, x.plot.values = NULL,
            titles = NULL, tables = "all", level.length = NA,
            transform.power = 1, offset = 0, scale = 1,
            transform.function = "identity",
            sortFactor = NULL, sortParallelToCombo = NULL,
            sortNestingFactor = NULL, sortOrder = NULL,
            decreasing = FALSE, trace = FALSE, ...)
```

## Arguments

|                         |                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>asreml.obj</code> | <code>asreml</code> object for a fitted model.                                                                                                                                                                                                                                                                                                                                               |
| <code>classify</code>   | A <a href="#">character</a> string giving the variables that define the margins of the multiway table to be predicted. Multiway tables are specified by forming an interaction type term from the classifying variables, that is, separating the variable names with the <code>:</code> operator. To predict the overall mean, set the <code>classify</code> to <code>"(Intercept)"</code> . |



- term** A **character** string giving the variables that define the term that was fitted using `asreml` and that corresponds to `classify`. It only needs to be specified when it is different to `classify`; it is stored as an attribute of the `alldiffs.object`. It is likely to be needed when the fitted model includes terms that involve both a **numeric** covariate and a **factor** that parallel each other; the `classify` would include the covariate and the term would include the factor.
- inestimable.rm** A logical indicating whether rows for predictions that are not estimable are to be removed from the components of the `alldiffs.object`.
- linear.transformation** A **formula** or a **matrix**. If a **formula** is given then it is taken to be a submodel of a model term corresponding to the `classify`. The projection matrix that transforms the predictions so that they conform to the submodel is obtained; the submodel does not have to involve variables in the `classify`, but the variables must be columns in the predictions component of `alldiffs.obj` and the space for the submodel must be a subspace of the space for the term specified by the `classify`. For example, for `classify` set to "A:B", the submodel `~ A + B` will result in the predictions for the combinations of A and B being made additive for the **factors** A and B. The submodel space corresponding to `A + B` is a subspace of the space `A:B`. In this case both the submodel and the `classify` involve only the factors A and B. To fit an intercept-only submodel, specify `linear.transformation` to be the formula `~1`.
- If a **matrix** is provided then it will be used to apply the linear transformation to the predictions. It might be a contrast **matrix** or a **matrix** of weights for a factor used to obtain the weighted average over that factor. The number of rows in the **matrix** should equal the number of linear combinations of the predictions desired and the number of columns should equal the number of predictions.
- In either case, as well as the values of the linear combinations, their standard errors, pairwise differences and associated statistics are returned.
- EGLS.linTransform** A **logical** indicating whether or not the `linear.transformation` of the predictions stored in an `alldiffs.object` by fitting a submodel supplied in a **formula** is to take into account the variance of the predictions using a Estimated Generalized Least Squares (EGLS) approach. This is likely to be appropriate when the variance matrix of the predictions is not compound symmetric i.e. when not all the variances are equal or not all the covariances are equal. If the variance matrix is compound symmetric, then the setting of `EGLS.linTransform` will not affect the transformed predictions.
- error.intervals** A **character** string indicating the type of error interval, if any, to calculate in order to indicate uncertainty in the results. Possible values are "none", "StandardError", "Confidence" and "halfLeastSignificant". The default is for confidence limits to be used. The "halfLeastSignificant" option results in half the Least Significant Difference (LSD) being added and subtracted to the predictions, the LSD being calculated using the square root of the mean of the variances of all or a subset of pairwise differences between the predictions. If the LSD is zero, as can happen when predictions are constrained to be equal, then the limits of the error intervals are set to NA. If `LSDtype` is set to `overall`, the `avsed.tolerance` is not NA and the range of the SEDs divided by the average of the SEDs exceeds `avsed.tolerance` then the `error.intervals` calculations and the plotting will revert to confidence intervals.

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| alpha           | A <b>numeric</b> giving the significance level for LSDs or one minus the confidence level for confidence intervals. It is stored as an attribute to the <code>alldiffs.object</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| wald.tab        | A <b>data.frame</b> containing the pseudo-anova table for the fixed terms produced by a call to <code>wald.asreml</code> . The main use of it here is in determining the degrees of freedom for calculating confidence or half-LSD error.intervals and p-values, the latter to be stored in the <code>p.differences</code> component of the <code>alldiffs.object</code> that is created.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| dDF.fault       | A <b>character</b> specifying the method to use to obtain substitute denominator degrees of freedom. when the numeric or algebraic methods produce faulty values, viz. NA, Inf or less than 0.01. Consistent with when no <code>denDF</code> are available, the default is "residual" and so the residual degrees of freedom from <code>asreml.obj\$nedf</code> are used. If <code>dDF.fault = "none"</code> , no substitute denominator degrees of freedom are employed; if <code>dDF.fault = "residual"</code> ; if <code>dDF.fault = "maximum"</code> , the maximum of those <code>denDF</code> that are available, excluding that for the Intercept, is used; if all <code>denDF</code> are faulty, <code>asreml.obj\$nedf</code> is used. If <code>dDF.fault = "supplied"</code> , a vector of values for the denominator degrees of freedom is to be supplied in <code>dDF.values</code> . Any other setting is ignored and a warning message produced. Generally, substituting these degrees of freedom is anticonservative in that it is likely that the degrees of freedom used will be too large.                                                                                                                                                                                                                                                                                                                      |
| dDF.values      | A <b>vector</b> of values to be used when <code>dDF.fault = "supplied"</code> . Its values will be used when <code>denDF</code> in a test for a fixed effect is NA. This vector must be the same length as the number of fixed terms, including (Intercept) whose value could be NA.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| pairwise        | A <b>logical</b> indicating whether all pairwise differences of the predictions and their standard errors and p-values are to be computed and stored. If tables is equal to "differences" or "all" or error.intervals is equal to "halfLeastSignificant", they will be stored irrespective of the value of pairwise.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Vmatrix         | A <b>logical</b> indicating whether the variance matrix of the predictions will be stored as a component of the <code>alldiffs.object</code> that is returned. If <code>linear.transformation</code> is set, it will be stored irrespective of the value of Vmatrix.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| avsed.tolerance | <p>A <b>numeric</b> giving the value of the SED range, the range of the SEDs divided by the square root of the mean of the variances of all or a subset of the pairwise differences, that is considered reasonable in calculating error.intervals. To have it ignored, set it to NA. It should be a value between 0 and 1. The following rules apply:</p> <ol style="list-style-type: none"> <li>1. If <code>avsed.tolerance</code> is NA then mean LSDs of the type specified by <code>LSDtype</code> are calculated and used in error.intervals and plots.</li> <li>2. Irrespective of the setting of <code>LSDtype</code>, if <code>avsed.tolerance</code> is not exceeded then the mean LSDs are used in error.intervals and plots.</li> <li>3. If <code>LSDtype</code> is set to overall, <code>avsed.tolerance</code> is not NA, and <code>avsed.tolerance</code> is exceeded then error.intervals and plotting revert to confidence intervals.</li> <li>4. If <code>LSDtype</code> is set to factor.combinations and <code>avsed.tolerance</code> is not exceeded for any factor combination then the half LSDs are used in error.intervals and plots; otherwise, error.intervals and plotting revert to confidence intervals.</li> <li>5. If <code>LSDtype</code> is set to per.prediction and <code>avsed.tolerance</code> is not exceeded for any prediction then the half LSDs are used in error.intervals</li> </ol> |

and plots; otherwise, `error.intervals` and plotting revert to confidence intervals.

`accuracy.threshold`

A **numeric** specifying the value of the LSD accuracy measure, which measure is specified by `LSDaccuracy`, as a threshold value in determining whether the halfLeastSignificant `error.interval` for a predicted value is a reasonable approximation; this will be the case if the LSDs across all pairwise comparisons for which the interval's LSD was computed, as specified by `LSDtype` and `LSDby`, are similar enough to the interval's LSD, as measured by `LSDaccuracy`. If it is NA, it will be ignored. If it is not NA, a column of **logicals** named `LSDwarning` will be added to the `predictions` component of the `alldiffs.object`. The value of `LSDwarning` for a `predicted.value` will be TRUE if the value of the `LSDaccuracy` measure computed from the LSDs for differences between this `predicted.value` and the other `predicted.values` as compared to its `assignedLSD` exceeds the value of `accuracy.threshold`. Otherwise, the value of `LSDwarning` for a `predicted.value` will be FALSE.

`LSDtype`

A **character** string that can be overall, `factor.combinations`, `per.prediction` or supplied. It determines whether the values stored in a row of a `LSD.frame` are the values calculated (i) overall from the LSD values for all pairwise comparisons, unless there is only one, possibly repeated, prediction, when a notional LSD is calculated, (ii) the values calculated from the pairwise LSDs for the levels of each `factor.combination`, unless there is only one prediction for a level of the `factor.combination`, when a notional LSD is calculated, (iii) `per.prediction`, being based, for each prediction, on all pairwise differences involving that prediction, or (iv) as supplied values of the LSD, specified with the `LSDsupplied` argument; these supplied values are to be placed in the `assignedLSD` column of the `LSD.frame` stored in an `alldiffs.object` so that they can be used in LSD calculations.

See `LSD.frame` for further information on the values in a row of this `data.frame` and how they are calculated.

`LSDsupplied`

A `data.frame` or a named **numeric** containing a set of LSD values that correspond to the observed combinations of the values of the `LSDby` variables in the `predictions.frame` or a single LSD value that is an overall LSD. If a `data.frame`, it may have (i) a column for the `LSDby` variable and a column of LSD values or (ii) a single column of LSD values with rownames being the combinations of the observed values of the `LSDby` variables. Any name can be used for the column of LSD values; `assignedLSD` is sensible, but not obligatory. Otherwise, a **numeric** containing the LSD values, each of which is named for the observed combination of the values of the `LSDby` variables to which it corresponds. (Applying the function `dae::fac.combine` to the `predictions` component is one way of forming the required combinations for the (row) names.) The values supplied will be incorporated into `assignedLSD` column of the `LSD.frame` stored as the LSD component of the `alldiffs.object`.

`LSDby`

A **character** (vector) of variables names, being the names of the **factors** or **numerics** in the `classify`; for each combination of their levels and values, there will be or is a row in the `LSD.frame` stored in the LSD component of the `alldiffs.object` when `LSDtype` is `factor.combinations`.

`LSDstatistic`

A **character** nominating one or more of minimum, q10, q25, mean, median, q75, q90 or maximum as the value(s) to be stored in the `assignedLSD` column in an `LSD.frame`; the values in the `assignedLSD` column are used in computing halfLeastSignificant `error.intervals`. Here q10, q25, q75 and q90 indicate the sample quantiles corresponding to probabilities of 0.1, 0.25, 0.75 and

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | 0.9 for the group of LSDs from which a single LSD value is calculated. The function <code>quantile</code> is used to obtain them. The mean LSD is calculated as the square root of the mean of the squares of the LSDs for the group. The median is calculated using the <code>median</code> function. Multiple values are only produced for <code>LSDtype</code> set to <code>factor.combination</code> , in which case <code>LSDby</code> must not be <code>NULL</code> and the number of values must equal the number of observed combinations of the values of the variables specified by <code>LSDby</code> . If <code>LSDstatistic</code> is <code>NULL</code> , it is reset to mean. |
| <code>LSDaccuracy</code>     | A <code>character</code> nominating one of <code>maxAbsDeviation</code> , <code>maxDeviation</code> , <code>q90Deviation</code> or <code>RootMeanSqDeviation</code> as the statistic to be calculated as a measure of the accuracy of assignedLSD. The option <code>q90Deviation</code> produces the sample quantile corresponding to a probability of 0.90. The deviations are the differences between the LSDs used in calculating the LSD statistics and each assigned LSD and the accuracy is expressed as a proportion of the assigned LSD value. The calculated values are stored in the column named <code>accuracyLSD</code> in an <code>LSD.frame</code> .                         |
| <code>titles</code>          | A <code>list</code> , each component of which is named for a column in the <code>data.frame</code> for <code>asreml.obj</code> and contains a character string giving a title to use in output (e.g. tables and graphs). Here they will be used for table headings.                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <code>tables</code>          | A <code>character</code> vector containing a combination of <code>none</code> , <code>predictions</code> , <code>vcov</code> , <code>backtransforms</code> , <code>differences</code> , <code>p.differences</code> , <code>sed</code> , <code>LSD</code> and <code>all</code> . These nominate which components of the <code>alldiffs.object</code> to print.                                                                                                                                                                                                                                                                                                                               |
| <code>x.num</code>           | A <code>character</code> string giving the name of the numeric covariate that (i) is potentially included in terms in the fitted model and (ii) is the x-axis variable for plots. Its values will not be converted to a <code>factor</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>x.fac</code>           | A <code>character</code> string giving the name of the factor that (i) corresponds to <code>x.num</code> and (ii) is potentially included in terms in the fitted model. It should have the same number of levels as the number of unique values in <code>x.num</code> . The levels of <code>x.fac</code> must be in the order in which they are to be plotted - if they are dates, then they should be in the form <code>yyyymmdd</code> , which can be achieved using <code>as.Date</code> . However, the levels can be non-numeric in nature, provided that <code>x.num</code> is also set.                                                                                               |
| <code>x.pred.values</code>   | The values of <code>x.num</code> for which predicted values are required. If <code>levels</code> is set for passing to <code>predict.asreml</code> , <code>x.pred.values</code> is ignored. Note that while <code>levels</code> is an alternative to <code>x.pred.values</code> , <code>x.pred.values</code> allows more general setting of the levels to be predicted.                                                                                                                                                                                                                                                                                                                     |
| <code>x.plot.values</code>   | The actual values to be plotted on the x axis. They are needed when values different to those in <code>x.num</code> are to be plotted or <code>x.fac</code> is to be plotted because there is no <code>x.num</code> term corresponding to the same term with <code>x.fac</code> .                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>level.length</code>    | The maximum number of characters from the levels of factors to use in the row and column labels of the tables of pairwise differences and their p-values and standard errors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>transform.power</code> | A <code>numeric</code> specifying the power of a transformation, if one has been applied to the response variable. Unless it is equal to 1, the default, back-transforms of the predictions will be obtained and presented in tables or graphs as appropriate. The back-transformation raises the predictions to the power equal to the reciprocal of <code>transform.power</code> , unless it equals 0 in which case the exponential of the predictions is taken. Confidence limits and Least Significant limits will be similarly backtransformed. Approximate standard errors of the backtransformed predictions will also be calculated.                                                |

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| offset              | A <b>numeric</b> that has been added to each value of the response after any scaling and before applying any power transformation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| scale               | A <b>numeric</b> by which each value of the response has been multiplied before adding any offset and applying any power transformation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| transform.function  | A <b>character</b> giving the name of a function that specifies the scale on which the predicted values are defined. This may be the result of a transformation of the data using the function or the use of the function as a link function in the fitting of a generalized linear (mixed) model (GL(M)M). The possible transform.functions are identity, log, inverse, sqrt, logit, probit, and cloglog. The predicted.values and error.intervals, if not StandardError intervals, will be back-transformed using the inverse function of the transform.function. The standard.error column will be set to NA, unless (i) asreml returns columns named transformed.value and approx.se, as well as those called predicted.values and standard.error (such as when a GLM is fitted) and (ii) the values in transformed.value are equal to those obtained by backtransforming the predicted.values using the inverse function of the transform.function. Then, the approx.se values will be saved in the standard.error column of the backtransforms component of the returned alldiffs.obj. Also, the transformed.value and approx.se columns are removed from both the predictions and backtransforms components of the alldiffs.obj. Note that the values that end up in the standard errors column are approximate for the backtransformed values and are not used in calculating error.intervals. |
| sortFactor          | A <b>character</b> containing the name of the factor that indexes the set of predicted values that determines the sorting of the components. If there is only one variable in the classify term then sortFactor can be NULL and the order is defined by the complete set of predicted values. If there is more than one variable in the classify term then sortFactor must be set. In this case the sortFactor is sorted in the same order within each combination of the values of the sortParallelToCombo variables: the classify variables, excluding the sortFactor. There should be only one predicted value for each unique value of sortFactor within each set defined by a combination of the values of the classify variables, excluding the sortFactor factor. The order to use is determined by either sortParallelToCombo or sortOrder.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| sortParallelToCombo | A <b>list</b> that specifies a combination of the values of the factors and numerics, excluding sortFactor, that are in classify. Each of the components of the supplied list is named for a classify variable and specifies a single value for it. The combination of this set of values will be used to define a subset of the predicted values whose order will define the order of sortFactor. Each of the other combinations of the values of the factors and numerics will be sorted in parallel. If sortParallelToCombo is NULL then the first value of each classify variable, except for the sortFactor factor, in the predictions component is used to define sortParallelToCombo. If there is only one variable in the classify then sortParallelToCombo is ignored.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| sortNestingFactor   | A <b>character</b> containing the name of the factor that defines groups of the sortFactor within which the predicted values are to be ordered. If there is only one variable in the classify then sortNestingFactor is ignored.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| sortOrder           | A character vector whose length is the same as the number of levels for sortFactor in the predictions component of the alldiffs.object. It speci-                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|            |                                                                                                                                                                                                                                                                                                                                                                     |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | <p>fies the desired order of the levels in the reordered components of the <code>alldiffs.object</code>. The argument <code>sortParallelToCombo</code> is ignored.</p> <p>The following creates a <code>sortOrder</code> vector <code>levs</code> for factor <code>f</code> based on the values in <code>x</code>: <code>levs &lt;- levels(f)[order(x)]</code>.</p> |
| decreasing | A logical passed to <code>order</code> that determines whether the order for sorting the components of the <code>alldiffs.object</code> is for increasing or decreasing magnitude of the predicted values.                                                                                                                                                          |
| trace      | A <code>logical</code> that control output from ASReml-R. If TRUE then partial iteration details are displayed when ASReml-R functions are invoked; if FALSE then no output is displayed.                                                                                                                                                                           |
| ...        | further arguments passed to <code>predict.asreml</code> .                                                                                                                                                                                                                                                                                                           |

### Value

For `linear.transformations` set to NULL, an S3-class `alldiffs.object` with predictions and their standard errors and, depending on the settings of the arguments, all pairwise differences between predictions, their standard errors and p-values and LSD statistics. Also, unless the `sortFactor` or `sortOrder` arguments are invoked, the rows of predictions component are ordered so that they are in standard order for the variables in the `classify`. That is, the values of the last variable change with every row, those of the second-last variable only change after all the values of the last variable have been traversed; in general, the values of a variable are the same for all the combinations of the values to the variables to its right in the `classify`. In addition, if necessary, the order of the columns of the variables in the predictions component are changed to match their order in the `classify`.

If `transform.power` or `scale` is not one or `offset` is not zero, it will contain a `data.frame` with the backtransformed linear transformation of the predictions. The backtransformation will, after backtransforming for any power transformation, subtract the `offset` and then divide by the `scale`.

If `error.intervals` is not "none", then the predictions component and, if present, the backtransforms component will contain columns for the lower and upper values of the limits for the interval.

The name of the response, the `response.title`, the term, the `classify`, `tdf`, `sortFactor` and the `sortOrder` will be set as attributes to the object. Also, if `error.intervals` is "halfLeastSignificant", then those of `LSDtype`, `LSDby` and `LSDstatistic` that are not NULL will be added as attributes of the object and of the predictions frame; additionally, `LSDvalues` will be added as attribute of the predictions frame, `LSDvalues` being the LSD values used in calculating the `error.intervals`. Note that the `classify` in an `alldiffs.object` is based on the variables indexing the predictions, which may differ from the `classify` used to obtain the original predictions (for example, when the `alldiffs.objects` stores a linear transformation of predictions).

For `linear.transformations` set to other than NULL, an `alldiffs.object` with the `linear.transformation` applied to the predictions and their standard errors and, depending on the settings of the arguments, all pairwise differences between the linearly transformed predictions, their standard errors and p-values and LSD statistics. (See also `linTransform.alldiffs`.)

### Author(s)

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### See Also

`alldiffs.object`, `as.alldiffs`, `print.alldiffs`, `linTransform.alldiffs`, `sort.alldiffs`, `subset.alldiffs`, `allDifferences.data.frame`, `redoErrorIntervals.alldiffs`, `recalcLSD.alldiffs`, `exploreLSDs.alldiffs`, `pickLSDstatistics.alldiffs`, `predictPresent.asreml`, `plotPredictions.data.frame`, `as.Date`, `predict.asreml`



## Examples

```
## Not run:
data(WaterRunoff.dat)
asreml.options(keep.order = TRUE) #required for asreml-R4 only
current.asr <- asreml(fixed = pH ~ Benches + (Sources * (Type + Species)),
                     random = ~ Benches:MainPlots,
                     keep.order=TRUE, data= WaterRunoff.dat)
current.asrt <- as.asrtests(current.asr, NULL, NULL)
diffs <- predictPlus(classify = "Sources:Type",
                     asreml.obj = current.asr,
                     wald.tab = current.asrt$wald.tab,
                     present = c("Sources", "Type", "Species"))

## End(Not run)
```

---

`predictPresent.asreml` *Forms the predictions for each of one or more terms and presents them in tables and/or graphs.*

---

## Description

This function forms the predictions for each term in terms using a supplied `asreml` object and `predictPlus.asreml`. Tables are produced using `predictPlus.asreml`, in conjunction with `allDifferences.data.frame`, with the argument `tables` specifying which tables are printed. The argument `plots`, along with `transform.power`, controls which plots are produced. The plots are produced using `plotPredictions.data.frame`, with line plots produced when variables involving `x.num` or `x.fac` are involved in `classify` for the predictions and bar charts otherwise. In order to get the correct predictions you may need to supply additional arguments to `predict.asreml` through ... e.g. `present`, `parallel`, `levels`.

The order of plotting the levels of one of the factors indexing the predictions can be modified and is achieved using `sort.alldiffs`.

## Usage

```
## S3 method for class 'asreml'
predictPresent(asreml.obj, terms, inestimable.rm = TRUE,
               linear.transformation = NULL, EGLS.linTransform,
               error.intervals = "Confidence", alpha = 0.05,
               wald.tab = NULL, dDF.fault = "residual", dDF.values = NULL,
               pairwise = TRUE, Vmatrix = FALSE,
               avsed.tolerance = 0.25, accuracy.threshold = NA,
               LSDtype = "overall", LSDsupplied = NULL, LSDby = NULL,
               LSDstatistic = "mean", LSDaccuracy = "maxAbsDeviation",
               x.num = NULL, x.fac = NULL, nonx.fac.order = NULL,
               x.pred.values = NULL, x.plot.values = NULL,
               plots = "predictions", panels = "multiple",
               graphics.device = NULL, interval.annotate = TRUE,
               titles = NULL, colour.scheme = "colour", save.plots = FALSE,
               transform.power = 1, offset = 0, scale = 1,
               transform.function = "identity",
               tables = "all", level.length = NA,
```

```
sortFactor = NULL, sortParallelToCombo = NULL,
sortNestingFactor = NULL, sortOrder = NULL,
decreasing = FALSE,
trace = FALSE, ggplotFuncs = NULL, ...)
```

## Arguments

- `asreml.obj`      `asreml` object for a fitted model.
- `terms`            A character vector giving the terms for which predictions are required.
- `inestimable.rm`   A logical indicating whether rows for predictions that are not estimable are to be removed from the components of the `alldiffs.object`.
- `linear.transformation`
- A `formula` or a `matrix`. If a `formula` is given then it is taken to be a submodel of a model term corresponding to the `classify`. The projection matrix that transforms the predictions so that they conform to the submodel is obtained; the submodel does not have to involve variables in the `classify`, but the variables must be columns in the predictions component of `alldiffs.obj` and the space for the submodel must be a subspace of the space for the term specified by the `classify`. For example, for `classify` set to "A:B", the submodel  $\sim A + B$  will result in the predictions for the combinations of A and B being made additive for the `factors` A and B. The submodel space corresponding to  $A + B$  is a subspace of the space A:B. In this case both the submodel and the `classify` involve only the factors A and B. To fit an intercept-only submodel, specify `linear.transformation` to be the formula  $\sim 1$ .
- If a `matrix` is provided then it will be used to apply the linear transformation to the predictions. It might be a contrast `matrix` or a `matrix` of weights for a factor used to obtain the weighted average over that factor. The number of rows in the `matrix` should equal the number of linear combinations of the predictions desired and the number of columns should equal the number of predictions.
- In either case, as well as the values of the linear combinations, their standard errors, pairwise differences and associated statistics are returned in the `alldiffs.object`.
- `EGLS.linTransform`
- A `logical` indicating whether or not the `linear.transformation` of the predictions stored in an `alldiffs.object` by fitting a submodel supplied in a `formula` is to take into account the variance of the predictions using a Estimated Generalized Least Squares (EGLS) approach. This is likely to be appropriate when the variance matrix of the predictions is not compound symmetric i.e. when not all the variances are equal or not all the covariances are equal. If the variance matrix is compound symmetric, then the setting of `EGLS.linTransform` will not affect the transformed predictions.
- `error.intervals`
- A character string indicating the type of error interval, if any, to calculate in order to indicate uncertainty in the results. Possible values are "none", "StandardError", "Confidence" and "halfLeastSignificant". The default is for confidence limits to be used. The "halfLeastSignificant" option results in half the Least Significant Difference (LSD) being added and subtracted to the predictions, the LSD being calculated using the square root of the mean of the variances of all or a subset of pairwise differences between the predictions. If the LSD is zero, as can happen when predictions are constrained to be equal, then the limits of the error intervals are set to NA. If `LSDtype` is set to



|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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|                              | overall, the <code>avsed.tolerance</code> is not NA and the range of the SEDs divided by the average of the SEDs exceeds <code>avsed.tolerance</code> then the <code>error.intervals</code> calculations and the plotting will revert to confidence intervals.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>alpha</code>           | A <a href="#">numeric</a> giving the significance level for LSDs or one minus the confidence level for confidence intervals. It is stored as an attribute to the <code>alldiffs.object</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>wald.tab</code>        | A <a href="#">data.frame</a> containing the pseudo-anova table for the fixed terms produced by a call to <code>wald.asreml</code> . The main use of it here is in determining the degrees of freedom for calculating confidence or half-LSD <code>error.intervals</code> and <code>p-values</code> , the latter to be stored in the <code>p.differences</code> component of the <code>alldiffs.object</code> that is created.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>dDF.fault</code>       | A <a href="#">character</a> specifying the method to use to obtain substitute denominator degrees of freedom. when the numeric or algebraic methods produce faulty values, viz. NA, Inf or less than 0.01. Consistent with when no <code>denDF</code> are available, the default is "residual" and so the residual degrees of freedom from <code>asreml.obj\$nedf</code> are used. If <code>dDF.fault = "none"</code> , no substitute denominator degrees of freedom are employed; if <code>dDF.fault = "residual"</code> ; if <code>dDF.fault = "maximum"</code> , the maximum of those <code>denDF</code> that are available, excluding that for the Intercept, is used; if all <code>denDF</code> are faulty, <code>asreml.obj\$nedf</code> is used. If <code>dDF.fault = "supplied"</code> , a vector of values for the denominator degrees of freedom is to be supplied in <code>dDF.values</code> . Any other setting is ignored and a warning message produced. Generally, substituting these degrees of freedom is anticonservative in that it is likely that the degrees of freedom used will be too large.                                                                                                                                                                                               |
| <code>dDF.values</code>      | A vector of values to be used when <code>dDF.fault = "supplied"</code> . Its values will be used when <code>denDF</code> in a test for a fixed effect is NA. This vector must be the same length as the number of fixed terms, including (Intercept) whose value could be NA.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>pairwise</code>        | A logical indicating whether all pairwise differences of the predictions and their standard errors and <code>p-values</code> are to be computed and stored. If <code>tables</code> is equal to "differences" or "all" or <code>error.intervals</code> is equal to "halfLeastSignificant", they will be stored irrespective of the value of <code>pairwise</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>Vmatrix</code>         | A <a href="#">logical</a> indicating whether the variance matrix of the predictions will be stored as a component of the <code>alldiffs.object</code> that is returned. If <code>linear.transformation</code> is set, it will be stored irrespective of the value of <code>Vmatrix</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>avsed.tolerance</code> | <p>A <a href="#">numeric</a> giving the value of the SED range, the range of the SEDs divided by the square root of the mean of the variances of all or a subset of the pairwise differences, that is considered reasonable in calculating <code>error.intervals</code>. It should be a value between 0 and 1. The following rules apply:</p> <ol style="list-style-type: none"> <li>1. If <code>avsed.tolerance</code> is NA then mean LSDs of the type specified by <code>LSDtype</code> are calculated and used in <code>error.intervals</code> and plots.</li> <li>2. Irrespective of the setting of <code>LSDtype</code>, if <code>avsed.tolerance</code> is not exceeded then the mean LSDs are used in <code>error.intervals</code> and plots.</li> <li>3. If <code>LSDtype</code> is set to overall, <code>avsed.tolerance</code> is not NA, and <code>avsed.tolerance</code> is exceeded then <code>error.intervals</code> and plotting revert to confidence intervals.</li> <li>4. If <code>LSDtype</code> is set to <code>factor.combinations</code> and <code>avsed.tolerance</code> is not exceeded for any factor combination then the half LSDs are used in <code>error.intervals</code> and plots; otherwise, <code>error.intervals</code> and plotting revert to confidence intervals.</li> </ol> |

5. If `LSDtype` is set to `per.prediction` and `avsed.tolerance` is not exceeded for any prediction then the half LSDs are used in `error.intervals` and plots; otherwise, `error.intervals` and plotting revert to confidence intervals.

#### `accuracy.threshold`

A **numeric** specifying the value of the LSD accuracy measure, which measure is specified by `LSDaccuracy`, as a threshold value in determining whether the halfLeastSignificant `error.interval` for a predicted value is a reasonable approximation; this will be the case if the LSDs across all pairwise comparisons for which the interval's LSD was computed, as specified by `LSDtype` and `LSDby`, are similar enough to the interval's LSD, as measured by `LSDaccuracy`. If it is NA, it will be ignored. If it is not NA, a column of **logicals** named `LSDwarning` will be added to the predictions component of the **alldiffs.object**. The value of `LSDwarning` for a predicted.value will be TRUE if the value of the `LSDaccuracy` measure computed from the LSDs for differences between this predicted.value and the other predicted.values as compared to its assignedLSD exceeds the value of `accuracy.threshold`. Otherwise, the value of `LSDwarning` for a predicted.value will be FALSE.

#### `LSDtype`

A **character** string that can be overall, factor.combinations, per.prediction or supplied. It determines whether the values stored in a row of a **LSD.frame** are the values calculated (i) overall from the LSD values for all pairwise comparisons, unless there is only one, possibly repeated, prediction, when a notional LSD is calculated, (ii) the values calculated from the pairwise LSDs for the levels of each factor.combination, unless there is only one prediction for a level of the factor.combination, when a notional LSD is calculated, (iii) per.prediction, being based, for each prediction, on all pairwise differences involving that prediction, or (iv) as supplied values of the LSD, specified with the `LSDsupplied` argument; these supplied values are to be placed in the assignedLSD column of the **LSD.frame** stored in an **alldiffs.object** so that they can be used in LSD calculations.

See **LSD.frame** for further information on the values in a row of this data.frame and how they are calculated.

#### `LSDsupplied`

A **data.frame** or a named **numeric** containing a set of LSD values that correspond to the observed combinations of the values of the `LSDby` variables in the **predictions.frame** or a single LSD value that is an overall LSD. If a **data.frame**, it may have (i) a column for the `LSDby` variable and a column of LSD values or (ii) a single column of LSD values with rownames being the combinations of the observed values of the `LSDby` variables. Any name can be used for the column of LSD values; assignedLSD is sensible, but not obligatory. Otherwise, a **numeric** containing the LSD values, each of which is named for the observed combination of the values of the `LSDby` variables to which it corresponds. (Applying the function `dae::fac.combine` to the predictions component is one way of forming the required combinations for the (row) names.) The values supplied will be incorporated into assignedLSD column of the **LSD.frame** stored as the LSD component of the **alldiffs.object**.

#### `LSDby`

A **character** (vector) of variables names, being the names of the **factors** or **numerics** in the classify; for each combination of their levels and values, there will be or is a row in the **LSD.frame** stored in the LSD component of the **alldiffs.object** when `LSDtype` is factor.combinatons.

#### `LSDstatistic`

A **character** nominating one or more of minimum, q10, q25, mean, median, q75, q90 or maximum as the value(s) to be stored in the assignedLSD column

in an `LSD.frame`; the values in the assignedLSD column are used in computing halfLeastSignificant error.intervals. Here q10, q25, q75 and q90 indicate the sample quantiles corresponding to probabilities of 0.1, 0.25, 0.75 and 0.9 for the group of LSDs from which a single LSD value is calculated. The function `quantile` is used to obtain them. The mean LSD is calculated as the square root of the mean of the squares of the LSDs for the group. The median is calculated using the `median` function. Multiple values are only produced for LSDtype set to `factor.combination`, in which case LSDby must not be NULL and the number of values must equal the number of observed combinations of the values of the variables specified by LSDby. If LSDstatistic is NULL, it is reset to mean.

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LSDaccuracy    | A <code>character</code> nominating one of <code>maxAbsDeviation</code> , <code>maxDeviation</code> , <code>q90Deviation</code> or <code>RootMeanSqDeviation</code> as the statistic to be calculated as a measure of the accuracy of assignedLSD. The option <code>q90Deviation</code> produces the sample quantile corresponding to a probability of 0.90. The deviations are the differences between the LSDs used in calculating the LSD statistics and each assigned LSD and the accuracy is expressed as a proportion of the assigned LSD value. The calculated values are stored in the column named <code>accuracyLSD</code> in an <code>LSD.frame</code> . |
| x.num          | A <code>character</code> string giving the name of the numeric covariate that (i) is potentially included in terms in the fitted model and (ii) is the x-axis variable for plots. Its values will not be converted to a <code>factor</code> .                                                                                                                                                                                                                                                                                                                                                                                                                       |
| x.fac          | A <code>character</code> string giving the name of the factor that (i) corresponds to x.num and (ii) is potentially included in terms in the fitted model. It should have the same number of levels as the number of unique values in x.num. The levels of x.fac must be in the order in which they are to be plotted - if they are dates, then they should be in the form <code>yyyymmdd</code> , which can be achieved using <code>as.Date</code> . However, the levels can be non-numeric in nature, provided that x.num is also set.                                                                                                                            |
| nonx.fac.order | A character vector giving the order in which factors other than x.fac are to be plotted in plots with multiple panels (i.e. where the number of non-x factors is greater than 1). The first factor in the vector will be plotted on the X axis (if there is no x.num or x.fac. Otherwise, the order of plotting the factors is in columns (X facets) and then rows (Y facets). By default the order is in decreasing order for the numbers of levels of the non x factors.                                                                                                                                                                                          |
| x.pred.values  | The values of x.num for which predicted values are required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| x.plot.values  | The actual values to be plotted on the x axis or in the labels of tables. They are needed when values different to those in x.num are to be plotted or x.fac is to be plotted because there is no x.num term corresponding to the same term with x.fac.                                                                                                                                                                                                                                                                                                                                                                                                             |
| plots          | Possible values are "none", "predictions", "backtransforms" and "both". Plots are not produced if the value is "none". If data are not transformed for analysis ( <code>transform.power = 1</code> ), a plot of the predictions is produced provided plots is not "none". If the data are transformed, the value of plots determines what is produced.                                                                                                                                                                                                                                                                                                              |
| panels         | Possible values are "single" and "multiple". When line plots are to be produced, because variables involving x.num or x.fac are involved in classify for the predictions, panels determines whether or not a single panel or multiple panels in a single window are produced. The panels argument is ignored for bar charts.                                                                                                                                                                                                                                                                                                                                        |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| graphics.device    | A character specifying a graphics device for plotting. The default is <code>graphics.device = NULL</code> , which will result in plots being produced on the current graphics device. Setting it to "windows", for example, will result in a windows graphics device being opened.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| interval.annotate  | A logical indicating whether the plot annotation indicating the type of error . interval is to be included in the plot.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| titles             | A list, each component of which is named for a column in the <code>data.frame</code> for <code>asreml.obj</code> and contains a character string giving a title to use in output (e.g. tables and graphs). Here they will be used for axis labels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| colour.scheme      | A character string specifying the colour scheme for the plots. The default is "colour" which produces coloured lines and bars, a grey background and white gridlines. A value of "black" results in black lines, grey bars and gridlines and a white background.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| save.plots         | A logical that determines whether any plots will be saved. If they are to be saved, a file name will be generated that consists of the following elements separated by full stops: the response variable name with .back if backtransformed values are being plotted, the classify term, Bar or Line and, if error . intervals is not "none", one of SE, CI or LSI. The file will be saved as a 'png' file in the current work directory.                                                                                                                                                                                                                                                                                                                                 |
| transform.power    | A <a href="#">numeric</a> specifying the power of a transformation, if one has been applied to the response variable. Unless it is equal to 1, the default, back-transforms of the predictions will be obtained and stored in the backtransforms component of the <a href="#">alldiffs.object</a> . The plots and tables arguments control the plotting and output of the predictions and backtransforms. The back-transformation raises the predictions to the power equal to the reciprocal of transform.power, unless it equals 0 in which case the exponential of the predictions is taken. Confidence limits and Least Significant limits will be similarly backtransformed. Approximate standard errors of the backtransformed predictions will also be calculated. |
| offset             | A number that has been added to each value of the response after any scaling and before applying any power transformation. Unless it is equal to 0, the default, back-transforms of the predictions will be obtained and stored in the backtransforms component of the <a href="#">alldiffs.object</a> . The plots and tables arguments control the plotting and output of the predictions and backtransforms. The backtransformation will, after backtransforming for any power transformation, subtract the offset.                                                                                                                                                                                                                                                     |
| scale              | A number by which each value of the response has been multiply before adding any offset and applying any power transformation. Unless it is equal to 1, the default, back-transforms of the predictions will be obtained and stored in the backtransforms component of the <a href="#">alldiffs.object</a> . The plots and tables arguments control the plotting and output of the predictions and backtransforms. The backtransformation will, after backtransforming for any power transformation and then subtracting the offset, divide by the scale.                                                                                                                                                                                                                 |
| transform.function | A <a href="#">character</a> giving the name of a function that specifies the scale on which the predicted values are defined. This may be the result of a transformation of the data using the function or the use of the function as a link function in the fitting of a generalized linear (mixed) model (GL(M)M). The possible                                                                                                                                                                                                                                                                                                                                                                                                                                         |

transform.functions are identity, log, inverse, sqrt, logit, probit, and cloglog. The predicted.values and error.intervals, if not StandardError intervals, will be back-transformed using the inverse function of the transform.function. The standard.error column will be set to NA, unless (i) asreml returns columns named transformed.value and approx.se, as well as those called predicted.values and standard.error (such as when a GLM is fitted) and (ii) the values in transformed.value are equal to those obtained by backtransforming the predicted.values using the inverse function of the transform.function. Then, the approx.se values will be saved in the standard.error column of the backtransforms component of the returned alldiffs.obj. Also, the transformed.value and approx.se columns are removed from both the predictions and backtransforms components of the alldiffs.obj. Note that the values that end up in the standard errors column are approximate for the backtransformed values and are not used in calculating error.intervals.

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tables              | A character vector containing a combination of predictions, vcov, backtransforms, differences, p.differences, sed, LSD and all. These nominate which components of the <a href="#">alldiffs.object</a> to print.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| level.length        | The maximum number of characters from the levels of factors to use in the row and column labels of the tables produced by <a href="#">allDifferences.data.frame</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| sortFactor          | A <a href="#">character</a> containing the name of the factor that indexes the set of predicted values that determines the sorting of the components. If there is only one variable in the classify term then sortFactor can be NULL and the order is defined by the complete set of predicted values. If there is more than one variable in the classify term then sortFactor must be set. In this case the sortFactor is sorted in the same order within each combination of the values of the sortParallelToCombo variables: the classify variables, excluding the sortFactor. There should be only one predicted value for each unique value of sortFactor within each set defined by a combination of the values of the classify variables, excluding the sortFactor factor. The order to use is determined by either sortParallelToCombo or sortOrder. |
| sortParallelToCombo | A <a href="#">list</a> that specifies a combination of the values of the factors and numerics, excluding sortFactor, that are in classify. Each of the components of the supplied <a href="#">list</a> is named for a classify variable and specifies a single value for it. The combination of this set of values will be used to define a subset of the predicted values whose order will define the order of sortFactor. Each of the other combinations of the values of the factors and numerics will be sorted in parallel. If sortParallelToCombo is NULL then the first value of each classify variable, except for the sortFactor factor, in the predictions component is used to define sortParallelToCombo. If there is only one variable in the classify then sortParallelToCombo is ignored.                                                     |
| sortNestingFactor   | A <a href="#">character</a> containing the name of the factor that defines groups of the sortFactor within which the predicted values are to be ordered. If there is only one variable in the classify then sortNestingFactor is ignored.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| sortOrder           | A character vector whose length is the same as the number of levels for sortFactor in the predictions component of the <a href="#">alldiffs.object</a> . It specifies the desired order of the levels in the reordered components of the <a href="#">alldiffs.object</a> . The argument sortParallelToCombo is ignored.<br><br>The following creates a sortOrder vector levs for factor f based on the values in x: <code>levs &lt;- levels(f)[order(x)]</code> .                                                                                                                                                                                                                                                                                                                                                                                            |



```

error.intervals = "StandardError",
titles = titles,
transform.power = 0,
present = c("Type", "Species", "Sources"),
tables = "none",
level.length = 6)

## End(Not run)

```

---

|                |                                                                                  |
|----------------|----------------------------------------------------------------------------------|
| print.alldiffs | <i>Prints the values in an <a href="#">alldiffs.object</a> in a nice format.</i> |
|----------------|----------------------------------------------------------------------------------|

---

## Description

Prints the predictions and standard errors from a fitted model, including the attributes of the [predictions.frame](#). Also prints out all pairwise differences between the predictions to 2 significant figures, along with their p-values and standard errors to 4 decimal places. If LSDs are requested the mean, minimum and maximum LSDs will be printed.

## Usage

```

## S3 method for class 'alldiffs'
print(x, which = "all", colourise = FALSE, ...)

```

## Arguments

|           |                                                                                                                                                                                                                                                                                                  |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| x         | An <a href="#">alldiffs.object</a> .                                                                                                                                                                                                                                                             |
| which     | A character vector containing a combination of predictions, vcov, backtransforms, differences, p.differences, sed, LSD and all. These nominate which components of the <a href="#">alldiffs.object</a> to print.                                                                                 |
| colourise | A <a href="#">logical</a> which, if TRUE, results in the header text produced by <code>predict.asreml</code> being displayed in a different colour, if supported by the output terminal device. It overrides the TRUE setting of the colourise argument of <code>asreml::asreml.options</code> . |
| ...       | further arguments passed to <code>print.predictions.frame</code> .                                                                                                                                                                                                                               |

## Value

No value is returned, but the components of x are printed.

## Author(s)

Chris Brien

## See Also

[print.predictions.frame](#), [as.alldiffs](#), [allDifferences.data.frame](#)

## Examples

```

## Not run:
print.alldiffs(diffs, which = "predictions")

## End(Not run)

```



---

|                |                                                                |
|----------------|----------------------------------------------------------------|
| print.asrtests | <i>Prints the values in an <a href="#">asrtests.object</a></i> |
|----------------|----------------------------------------------------------------|

---

## Description

Prints a summary of the asreml object, the pseudoanova and the test.summary data. frame that are stored in the [asrtests.object](#).

## Usage

```
## S3 method for class 'asrtests'
print(x, which = "key", colourise = FALSE, ...)
```

## Arguments

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| x         | An <a href="#">asrtests.object</a> .                                                                                                                                                                                                                                                                                                                                                                                                             |
| which     | Which elements of the <a href="#">asrtests.object</a> to print. Possible values are some combination of asremlsummary, vparametersummary, pseudoanova, wald.tab, testsummary and key or all. The option wald.tab is a synonym for pseudoanova. The options key and all are mutually exclusive; key includes vparametersummary, but not the rest of asremlsummary, while all includes the full asremlsummary that includes the vparametersummary. |
| colourise | A <a href="#">logical</a> which, if TRUE, results in the header text produced by wald.asreml being displayed in a different colour, if supported by the output terminal device. It overrides the TRUE setting of the colourise argument of asreml::asreml.options.                                                                                                                                                                               |
| ...       | further arguments passed to print and print.wald.tab.                                                                                                                                                                                                                                                                                                                                                                                            |

## Value

No value is returned, but the elements of the list in x are printed.

## Author(s)

Chris Brien

## See Also

[print.wald.tab](#), [as.asrtests](#), [asremlPlus-package](#)

## Examples

```
## Not run:
data(Wheat.dat)
current.asr <- asreml(yield ~ Rep + WithinColPairs + Variety,
                    random = ~ Row + Column + units,
                    residual = ~ ar1(Row):ar1(Column),
                    data=Wheat.dat)
current.asrt <- as.asrtests(current.asr, NULL, NULL)
current.asrt <- rmboundary.asrtests(current.asrt)
# Test Row autocorrelation
current.asrt <- testresidual(current.asrt, , "~ Row:ar1(Column)",
                           label="Row autocorrelation", simpler=TRUE)
```



```
print(current.asrt)

## End(Not run)
```

---

|               |                                                                                                                 |
|---------------|-----------------------------------------------------------------------------------------------------------------|
| print.LSDdata | <i>Prints the components of a list containing data on the LSDs for all pairwise differences of predictions.</i> |
|---------------|-----------------------------------------------------------------------------------------------------------------|

---

## Description

Prints the components of an LSDdata [list](#) created by [exploreLSDs](#), that contains data on the LSDs for all pairwise differences of predictions stored in an [alldiffs.object](#).

## Usage

```
## S3 method for class 'LSDdata'
print(x, which.print = c("statistics", "false.pos", "false.neg"), ...)
```

## Arguments

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| x           | An object that, ideally, is of class LSDdata.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| which.print | Which components of the LSDdata <a href="#">list</a> to print. Possible values are any combination of frequencies, distinct.vals, statistics, accuracy, false.pos, false.neg, per.pred.accuracy, LSD, summary and all, except that summary and all cannot occur together. For a description of the components, see <a href="#">alldiffs.object</a> . The default is to print statistics, false.pos, false.neg. The option summary results in the printing of distinct.vals, statistics, false.pos, false.neg. |
| ...         | further arguments passed to print.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## Value

No value is returned, but components of x are printed as specified in which.print.

## Author(s)

Chris Brien

## See Also

[exploreLSDs.alldiffs](#), [alldiffs.object](#)

## Examples

```
## Not run:
data(WaterRunoff.dat)
asreml.options(keep.order = TRUE) #required for asreml-R4 only
current.asr <- asreml(fixed = pH ~ Benches + (Sources * (Type + Species)),
                     random = ~ Benches:MainPlots,
                     keep.order=TRUE, data= WaterRunoff.dat)
current.asrt <- as.asrtests(current.asr, NULL, NULL)
diffs <- predictPlus(classify = "Sources:Type",
                     asreml.obj = current.asr,
```

```

        wald.tab = current.asrt$wald.tab,
        present = c("Sources", "Type", "Species"))
LSDdata <- exploreLSDs(diffs, LSDtype = "factor.combinations", LSDby = "Sources")
print(LSDdata)

## End(Not run)

```

---

```
print.predictions.frame
```

*Prints the values in a `predictions.frame`, with or without title and heading.*

---

## Description

Prints the predictions from a fitted model, along with their standard errors and, if present, their error intervals, with or without title and headings.

## Usage

```

## S3 method for class 'predictions.frame'
print(x, title = NULL,
      which.predictions = c("title", "heading", "table"),
      colourise = FALSE, ...)

```

## Arguments

|                                |                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>x</code>                 | An object that, ideally, is of class <code>predictions.frame</code> .                                                                                                                                                                                                                                                                                              |
| <code>title</code>             | A <code>character</code> giving a title to be printed out before the heading and table for the <code>predictions.frame</code> ,                                                                                                                                                                                                                                    |
| <code>which.predictions</code> | what Which elements of the <code>predictions.frame</code> to print. Possible values are some combination of title, heading, table and all. The heading is an attribute of <code>x</code> .                                                                                                                                                                         |
| <code>colourise</code>         | A <code>logical</code> which, if TRUE, results in the header text produced by <code>predict.asreml</code> being displayed in a different colour, if supported by the output terminal device. It overrides the TRUE setting of the <code>colourise</code> argument of <code>asreml::asreml.options</code> , but is only operational when the table is also printed. |
| <code>...</code>               | further arguments passed to <code>print.predictions.frame</code> .                                                                                                                                                                                                                                                                                                 |

## Value

No value is returned, but the components of `x` are printed.

## Author(s)

Chris Brien

## See Also

`print.alldiffs`, `as.alldiffs`, `allDifferences.data.frame`

**Examples**

```
## Not run:
print.predictions.frame(diffs$predictions, which = "all")

## End(Not run)
```

---

```
print.test.summary
```

*Prints a data.frame containing a test.summary.*

---

**Description**

Prints a test.summary (also a choose.summary) with or without a title and with p-values limited to 4-digits.

**Usage**

```
## S3 method for class 'test.summary'
print(x, which.print = c("title", "table"),
      omit.columns = NULL, response = NULL, ...)
```

**Arguments**

|              |                                                                                                                                                                                  |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| x            | A object that, ideally, is of class test.summary.                                                                                                                                |
| which.print  | A character specifying the aspects of the test.summary to print. Possible values are some combination of title, table and all.                                                   |
| omit.columns | A character specifying the columns of the test.summary table to be omitted from the print. If NULL, none are omitted.                                                            |
| response     | A character specifying the name of the response that the test.summary table is based on and is to be printed in the heading for the table. If NULL, no response name is printed. |
| ...          | further arguments passed to print, but is only operational when the table is also printed.                                                                                       |

**Value**

No value is returned, but x is printed, possibly with a title.

**Author(s)**

Chris Brien

**See Also**

[print.wald.tab](#), [print.asrtests](#), [as.asrtests](#), [asremlPlus-package](#)

**Examples**

```
## Not run:
data(Wheat.dat)
current.asr <- asreml(yield ~ Rep + WithinColPairs + Variety,
                     random = ~ Row + Column + units,
                     residual = ~ ar1(Row):ar1(Column),
                     data=Wheat.dat)
current.asrt <- as.asrtests(current.asr, NULL, NULL)
current.asrt <- rmboundary.asrtests(current.asrt)
# Test Row autocorrelation
current.asrt <- testresidual(current.asrt, "~ Row:ar1(Column)",
                             label="Row autocorrelation", simpler=TRUE)
print(current.asrt$test.summary)

## End(Not run)
```

---

|                |                                                                    |
|----------------|--------------------------------------------------------------------|
| print.wald.tab | <i>Prints a data.frame containing a Wald or pseudoanova table.</i> |
|----------------|--------------------------------------------------------------------|

---

**Description**

Prints a wald.tab with or without title and/or heading. The printing of the p-values is limited to 4 digits.

**Usage**

```
## S3 method for class 'wald.tab'
print(x, which.wald = c("title", "heading", "table"),
      colourise = FALSE, ...)
```

**Arguments**

|            |                                                                                                                                                                                                                                                                    |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| x          | An object that, ideally, is of class wald.tab.                                                                                                                                                                                                                     |
| which.wald | Which elements of the wald.tab to print. Possible values are some combination of title, heading, table and all. The heading is an attribute of x.                                                                                                                  |
| colourise  | A <a href="#">logical</a> which, if TRUE, results in the header text produced by wald.asreml being displayed in a different colour, if supported by the output terminal device. It overrides the TRUE setting of the colourise argument of asreml::asreml.options. |
| ...        | further arguments passed to print and print.wald.tab, but is only operational when the table is also printed.                                                                                                                                                      |

**Value**

No value is returned, but x is printed as specified in which.wald.

**Author(s)**

Chris Brien

**See Also**

[print.test.summary](#), [print.asrtests](#), [as.asrtests](#), [asremlPlus-package](#)

## Examples

```
## Not run:
data(Wheat.dat)
current.asr <- asreml(yield ~ Rep + WithinColPairs + Variety,
                     random = ~ Row + Column + units,
                     residual = ~ ar1(Row):ar1(Column),
                     data=Wheat.dat)
current.asrt <- as.asrtests(current.asr, NULL, NULL)
current.asrt <- rmboundary.asrtests(current.asrt)
# Test Row autocorrelation
current.asrt <- testresidual(current.asrt, "~ Row:ar1(Column)",
                             label="Row autocorrelation", simpler=TRUE)
print(current.asrt$wald.tab)

## End(Not run)
```

---

```
printFormulae.asreml  Prints the formulae from an asreml object.
```

---

## Description

Prints the [formulae](#) nominated in the which argument from the call stored in an asreml object. The output is broken into lines with print width equal to the option width.

## Usage

```
## S3 method for class 'asreml'
printFormulae(asreml.obj, which = c("fixed", "random", "residual"),
              expanded = FALSE, envir = parent.frame(), ...)
```

## Arguments

|            |                                                                                                                                                                                                                                                          |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| asreml.obj | An asreml object resulting from the fitting of a model using REML.                                                                                                                                                                                       |
| which      | A character listing the <a href="#">formula</a> (e) to be printed from the call stored in asreml.obj. It should be some combination of fixed, random, residual, sparse and all. If all is included then all <a href="#">formula</a> (e) will be printed. |
| expanded   | A logical indicating whether terms are to be expanded to the sum of a set of individual terms.                                                                                                                                                           |
| envir      | The environment in which the <a href="#">formula</a> (e) are to be evaluated. May also be NULL, a list, a data.frame, a pairlist or an integer as specified to sys.call.                                                                                 |
| ...        | Arguments passed on to getFormulae.asreml and ultimately to update.formula and terms.formula.                                                                                                                                                            |

## Value

Invisibly returns a character, each element of which contains one of the extracted [formulae](#).

## Author(s)

Chris Brien

**See Also**

[printFormulae.asreml](#)

**Examples**

```
## Not run:
data(Wheat.dat)
current.asr <- asreml(yield ~ Rep + WithinColPairs + Variety,
  random = ~ Row + Column + units,
  residual = ~ ar1(Row):ar1(Column),
  data=Wheat.dat)
printFormulae(current.asr)

## End(Not run)
```

---

R2adj.asreml

*Calculates the adjusted coefficient of determination for a specified combination of fixed and random terms.*

---

**Description**

Calculates the adjusted coefficient of determination (R<sup>2</sup>) that measures the contributions to the total variance exhibited by the observations of a specified combination of fixed and random terms in a fitted linear mixed model.

Note that the adjusted R<sup>2</sup> can be negative, which indicates that the contribution of the terms to the total variance is very small relative to the sum of the degrees of freedom of the terms.

Piepho's (2023) method for GLMMs has not been implemented. This function is not available for ASReml-R version 3.

**Usage**

```
## S3 method for class 'asreml'
R2adj(asreml.obj,
  include.which.fixed = ~ ., orthogonalize = "hybrid",
  include.which.random = NULL,
  bound.exclusions = c("F", "B", "S", "C"), ...)
```

**Arguments**

**asreml.obj**      An asreml object returned from a call to asreml.

**include.which.fixed**

A [formula](#) specifying the fixed terms whose joint contribution to the total variance is to be measured. If it is NULL, no fixed term is to be included in the terms whose joint contribution is to be assessed. The [formula](#) `~ .` indicates that the joint contribution of all fixed terms are to be measured. Otherwise, the joint contribution of the set of terms specified by the [formula](#) will be assessed. The [formula](#) can include a `"."`, which means all fixed terms currently fitted, and is most likely followed by a `"-"` with a bracketed set of terms to be removed that can be specified using [formula](#) operators. The names of the resulting terms must be the same as those in either the terms attribute of the fixed component

of the coefficient component of the supplied `asreml.obj`, or the Wald table produced by `wald.asreml`.

Note that the contribution of a subset of the fixed terms is only unique if the effects for the fixed terms are orthogonal; if the effects are not orthogonal then the contributions will depend on the order of the terms in the formula. Also, determining the joint contribution of a subset of the fixed terms in the model may be computationally demanding because the projection matrices have to be formed for all fixed terms and these projections matrices have to be orthogonalized. A heavy computational burden is most likely when the effects for the fixed terms are not orthogonal, for example, when numeric covariates are included amongst the terms.

- `orthogonalize` A [character](#) vector indicating the method for orthogonalizing a projector to those for terms that occurred previously in the [formula](#) for `include.which.fixed`. Orthogonalizing the projectors of fixed terms is not performed for the default setting of `.~`. When required, two options are available for orthogonalizing: `hybrid` and `eigenmethods`. The `hybrid` option is the most general and uses the relationships between the projection operators for the terms in the formula to decide which projectors to subtract and which to orthogonalize using `eigenmethods`. The `eigenmethods` option recursively orthogonalizes the projectors using an eigenanalysis of each projector with previously orthogonalized projectors. See the documentation for `porthogonalize.list` from the R package `dae` for more information.
- `include.which.random` A [formula](#) specifying the random terms whose joint contribution to the total variance is to be measured. If it is `NULL`, no random term is to be included in the terms whose joint contribution is to be assessed. The formula `~ .` indicates that the joint contribution of all random terms is to be measured. Otherwise, the joint contribution of the set of terms specified by the [formula](#) will be assessed. The [formula](#) can include a `"."`, which means all random terms currently fitted, and is most likely followed by a `"-"` with a bracketed set of terms to be removed that can be specified using [formula](#) operators. The resulting terms must be one of those occurring in either the `vparameters` component of the supplied `asreml.obj`, or in the terms attribute of the random component of the coefficient component of the supplied `asreml.obj`.
- `bound.exclusions` A character specifying one or more bound codes that will result in a variance parameter in the random model being excluded from contributing to the variance. If set to `NULL` then none will be excluded.
- `...` Provision for passing arguments to functions called internally - not used at present.

## Details

The method used to compute the adjusted R2 under a linear mixed model (LMM) is that described by Piepho (2023). Here, the method has been extended to allow computation of the adjusted R2 for a subset of the fixed terms. A set of orthogonalized projectors for all of the fixed terms in the model (a set of  $\mathbf{Q}_i$ s) is obtained and the combined contribution of the fixed terms nominated in `include.which.fixed` is obtained by computing the average semisquared bias, ASSB, for the nominated fixed terms as:

$$\sum_i \{ (\mathbf{Q}_i \mathbf{X} \boldsymbol{\beta})^T \mathbf{Q}_i \mathbf{X} \boldsymbol{\beta} + \text{trace}(\mathbf{X}^T \mathbf{Q}_i \mathbf{X} \text{var}(\boldsymbol{\beta})) \} / (n - 1)$$

Of the two methods, eigenmethods is least likely to fail, but it does not establish the marginality between the terms. It is often needed when there is nonorthogonality between terms, such as when there are several linear covariates. It can also be more efficient in these circumstances.

The process can be computationally expensive, particularly for a large data set (500 or more observations) and/or when many terms are to be orthogonalized, particularly if they are not orthogonal.

If the error "Matrix is not idempotent" should occur then, especially if there are many terms, one might try using `set.daeTolerance` from the `dae` package to reduce the tolerance used in determining if values are either the same or are zero; it may be necessary to lower the tolerance to as low as 0.001. Also, setting `orthogonalize` to `eigenmethods` is worth a try.

In doing the computations, no changes are made to the fitted model, nor is the `formula` stored in `asreml.obj` referred to. Instead, the names of the terms referred to are those stored in the coefficients component of the `asreml.obj`. Use `attr(asreml.obj$coefficients$fixed, which = "terms")` to access the attribute for fixed terms; substitute `random` for fixed to see the names of the random terms. For fixed terms, the term names are the same as those in the Wald table produced by `wald.asreml`, and, for random terms, the same as those in the `vparameters` component of the `asreml.obj`. Two `asreml formula` functions whose terms can differ from their formulation in a model `formula` are `at` and `str`.)

The function `estimateV.asreml` is used to calculate the variance matrices required in calculating the adjusted R2.

## Value

A numeric that is the adjusted R2, expressed as a percentage. It has attributes `include.which.fixed`, `include.which.random` and `missing.termmatrix` (use `attr(x, which = "name")` to access the attribute name). The `missing.termmatrix` attribute will be `NULL`, unless the design matrix could not be obtained for one or more model terms. If it is not `NULL`, it will be a list of terms whose design matrices could not be produced and so are not included in the variance matrix estimate. An NA will be returned for the adjusted R2 if `missing.termmatrix` is not `NULL` or a generalized inverse could not be computed for the variance matrix estimate.

## Author(s)

Chris Brien

## References

Piepho, H.-P. (2023). An adjusted coefficient of determination (R2) for generalized linear mixed models in one go. *Biometrical Journal*, **65**(7), 2200290. doi:10.1002/bimj.202200290.

## See Also

`asreml`, `estimateV.asreml`.

## Examples

```
## Not run:
data(Oats.dat)

current.asr <- asreml(Yield ~ Nitrogen*Variety,
                     random=~Blocks/Wplots,
                     data=Oats.dat)
R2.adj.fix <- R2adj.asreml(current.asr)
R2.adj.ran <- R2adj.asreml(current.asr,
```



```

                                include.which.fixed = NULL, include.which.random = ~ .)
R2.adj.tot <- R2adj.asreml(current.asr, include.which.random = ~ .)
R2.adj.tot <- R2adj.asreml(current.asr, include.which.random = ~ Blocks)
R2.adj.add <- R2adj.asreml(current.asr, include.which.fixed = ~ Nitrogen + Variety)
R2.adj.int <- R2adj.asreml(current.asr,
                            include.which.fixed = ~ . - (Nitrogen + Variety))
R2.adj.int <- R2adj.asreml(current.asr, include.which.fixed = ~ Nitrogen:Variety)

## End(Not run)

```

---

ratioTransform.alldiffs

*Calculates the ratios of nominated pairs of predictions stored in an [alldiffs.object](#).*

---

## Description

Ratio predictions and error intervals are formed for two levels of a factor, the `ratio.factor`. For each pair of a level of the `ratio.factor` in `numerator.levels` with a level in `denominator.levels`, the ratio predictions are formed from all combinations of the other factors as the ratio of the two predictions for each combination, along with confidence intervals for the ratio predictions computed using the Fieller (1954) method.

The printing of the components produced is controlled by the `tables` argument.

## Usage

```

## S3 method for class 'alldiffs'
ratioTransform(alldiffs.obj, ratio.factor,
               numerator.levels, denominator.levels,
               sep.levs.combine = ", ",
               method = "Fieller", alpha = 0.05,
               response = NULL, response.title = NULL,
               tables = "predictions", ...)

```

## Arguments

|                                 |                                                                                                                                                                                                                                                                                      |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>alldiffs.obj</code>       | An <a href="#">alldiffs.object</a> .                                                                                                                                                                                                                                                 |
| <code>ratio.factor</code>       | A <a href="#">character</a> string giving the name of the factor for whose levels the ratios are to be calculated.                                                                                                                                                                   |
| <code>numerator.levels</code>   | A <a href="#">character</a> string containing the levels of <code>ratio.factor</code> to be used as numerators of the ratio.                                                                                                                                                         |
| <code>denominator.levels</code> | A <a href="#">character</a> string containing the levels of <code>ratio.factor</code> to be used as denominators of the ratio.                                                                                                                                                       |
| <code>sep.levs.combine</code>   | A <a href="#">character</a> string specifying the separator to use internally when combining the levels of the <a href="#">factor</a> (s) in the <code>classify</code> of the <code>alldiffs.obj</code> . It should not occur in the levels of the <code>classify factor</code> (s). |

|                |                                                                                                                                                                                                                                                                                              |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| method         | A <a href="#">character</a> string specifying the method to use in calculating the ratios and their error.intervals. At present only Fieller is available. For the Fieller method, ratios of predictions are formed and confidence intervals formed for them using Fieller's (1954) theorem. |
| alpha          | A <a href="#">numeric</a> giving the significance level for LSDs or one minus the confidence level for confidence intervals.                                                                                                                                                                 |
| response       | A character specifying the response variable for the predictions. It is stored as an attribute to the <a href="#">alldiffs.object</a> .                                                                                                                                                      |
| response.title | A character specifying the title for the response variable for the predictions. It is stored as an attribute to the <a href="#">alldiffs.object</a> .                                                                                                                                        |
| tables         | A <a href="#">character</a> vector containing either none or predictions                                                                                                                                                                                                                     |
| ...            | further arguments passed to <a href="#">linTransform.alldiffs</a> .                                                                                                                                                                                                                          |

### Value

A list of [predictions.frames](#), each containing the ratio predictions and their confidence limits for a combination of the numerator.levels with the denominator.levels. It will also contain the values of the variables in the classify of [alldiffs.obj](#) that index the ratio predictions, except that the ratio.factor is omitted.

If sortFactor attribute of the [alldiffs.object](#) is set and is not the ratio.factor, the predictions and their backtransforms will be sorted using the sortOrder attribute of the [alldiffs.object](#).

### Author(s)

Chris Brien

### References

Fieller, E. C. (1954). Some Problems in Interval Estimation. *Journal of the Royal Statistical Society.Series B (Methodological)*, **16**, 175-185.

### See Also

[pairdiffsTransform](#), [linTransform](#), [predictPlus.asreml](#), [as.alldiffs](#),  
[print.alldiffs](#), [sort.alldiffs](#), [subset.alldiffs](#), [allDifferences.data.frame](#),  
[redoErrorIntervals.alldiffs](#), [recalcLSD.alldiffs](#), [predictPresent.asreml](#),  
[plotPredictions.data.frame](#),  
[as.Date](#), [predict.asreml](#)

### Examples

```
#### Form the ratios and Fieller CIs for RGR Salinity
load(system.file("extdata", "testDiffs.rda", package = "asremlPlus", mustWork = TRUE))
Preds.ratio.RGR <- ratioTransform(diffs.RGR,
                                ratio.factor = "Salinity",
                                numerator.levels = "Salt",
                                denominator.levels = "Control")

#### Form the ratios and Fieller CIs for Nitrogen compared to no Nitrogen
data("Oats.dat")
## Not run:
m1.asr <- asreml(Yield ~ Nitrogen*Variety,
                 random=~Blocks/Wplots,
```

```

        data=Oats.dat)
current.asrt <- as.asrtests(m1.asr)
wald.tab <- current.asrt$wald.tab
Var.diffs <- predictPlus(m1.asr, classify="Nitrogen:Variety", pairwise = TRUE,
                        Vmatrix = TRUE, error.intervals = "halfLeast",
                        LSDtype = "factor", LSDby = "Variety",
                        wald.tab = wald.tab)

## End(Not run)

## Use lme4 and emmeans to get predictions and associated statistics
if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(Yield ~ Nitrogen*Variety + (1|Blocks/Wplots),
                          data=Oats.dat)

  ## Set up a wald.tab
  int <- as.data.frame(rbind(rep(NA,4)))
  rownames(int) <- "(Intercept)"
  wald.tab <- anova(m1.lmer, ddf = "Kenward", type = 1)[,3:6]
  names(wald.tab) <- names(int) <- c("Df", "denDF", "F.inc", "Pr")
  wald.tab <- rbind(int, wald.tab)
  #Get predictions
  Var.emm <- emmeans::emmeans(m1.lmer, specs = ~ Nitrogen:Variety)
  Var.preds <- summary(Var.emm)
  ## Modify Var.preds to be compatible with a predictions.frame
  Var.preds <- as.predictions.frame(Var.preds, predictions = "emmean",
                                   se = "SE", interval.type = "CI",
                                   interval.names = c("lower.CL", "upper.CL"))

  Var.vcov <- vcov(Var.emm)
  Var.sed <- NULL
  den.df <- wald.tab[match("Variety", rownames(wald.tab)), "denDF"]

  #Create alldiffs object
  Var.diffs <- as.alldiffs(predictions = Var.preds,
                          sed = Var.sed, vcov = Var.vcov,
                          classify = "Nitrogen:Variety", response = "Yield", tdf = den.df)
}

if (exists("Var.diffs"))
  Preds.ratio.OatsN <- ratioTransform(alldiffs.obj = Var.diffs,
                                     ratio.factor = "Nitrogen",
                                     numerator.levels = c("0.2", "0.4", "0.6"),
                                     denominator.levels = "0.2")

```

---

|                    |                                                                                                                |
|--------------------|----------------------------------------------------------------------------------------------------------------|
| recalcLSD.alldiffs | Adds or recalculates the <a href="#">LSD.frame</a> that is a component of an <a href="#">alldiffs.object</a> . |
|--------------------|----------------------------------------------------------------------------------------------------------------|

---

## Description

Given an [alldiffs.object](#), adds or recalculate its [LSD.frame](#). **N.B. No changes are made to the [error.intervals](#) — use [redoErrorIntervals.alldiffs](#) to modify both the [error.intervals](#) and the [LSD.frame](#).**

## Usage

```
## S3 method for class 'alldiffs'
recalcLSD(alldiffs.obj, LSDtype = "overall", LSDsupplied = NULL,
          LSDby = NULL, LSDstatistic = "mean", LSDaccuracy = "maxAbsDeviation",
          alpha = 0.05, ...)
```

## Arguments

- |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| alldiffs.obj | An <a href="#">alldiffs.object</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| LSDtype      | <p>A <a href="#">character</a> string that can be overall, factor.combinations, per.prediction or supplied. It determines whether the values stored in a row of a <a href="#">LSD.frame</a> are the values calculated (i) overall from the LSD values for all pairwise comparisons, unless there is only one, possibly repeated, prediction, when a notional LSD is calculated, (ii) the values calculated from the pairwise LSDs for the levels of each factor.combination, unless there is only one prediction for a level of the factor.combination, when a notional LSD is calculated, (iii) per.prediction, being based, for each prediction, on all pairwise differences involving that prediction, or (iv) as supplied values of the LSD, specified with the LSDsupplied argument; these supplied values are to be placed in the assignedLSD column of the <a href="#">LSD.frame</a> stored in an <a href="#">alldiffs.object</a> so that they can be used in LSD calculations.</p> <p>See <a href="#">LSD.frame</a> for further information on the values in a row of this data.frame and how they are calculated.</p>       |
| LSDsupplied  | <p>A <a href="#">data.frame</a> or a named <a href="#">numeric</a> containing a set of LSD values that correspond to the observed combinations of the values of the LSDby variables in the <a href="#">predictions.frame</a> or a single LSD value that is an overall LSD. If a <a href="#">data.frame</a>, it may have (i) a column for the LSDby variable and a column of LSD values or (ii) a single column of LSD values with rownames being the combinations of the observed values of the LSDby variables. Any name can be used for the column of LSD values; assignedLSD is sensible, but not obligatory. Otherwise, a <a href="#">numeric</a> containing the LSD values, each of which is named for the observed combination of the values of the LSDby variables to which it corresponds. (Applying the function <code>dae::fac.combine</code> to the predictions component is one way of forming the required combinations for the (row) names.) The values supplied will be incorporated into assignedLSD column of the <a href="#">LSD.frame</a> stored as the LSD component of the <a href="#">alldiffs.object</a>.</p> |
| LSDby        | <p>A <a href="#">character</a> (vector) of variables names, being the names of the <a href="#">factors</a> or <a href="#">numerics</a> in the <code>classify</code>; for each combination of their levels and values, there will be or is a row in the <a href="#">LSD.frame</a> stored in the LSD component of the <a href="#">alldiffs.object</a> when LSDtype is factor.combinatons.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| LSDstatistic | <p>A <a href="#">character</a> nominating one or more of minimum, q10, q25, mean, median, q75, q90 or maximum as the value(s) to be stored in the assignedLSD column in an <a href="#">LSD.frame</a>; the values in the assignedLSD column are used in computing halfLeastSignificant error.intervals. Here q10, q25, q75 and q90 indicate the sample quantiles corresponding to probabilities of 0.1, 0.25, 0.75 and 0.9 for the group of LSDs from which a single LSD value is calculated. The function <a href="#">quantile</a> is used to obtain them. The mean LSD is calculated as the square root of the mean of the squares of the LSDs for the group. The median is calculated using the <a href="#">median</a> function. Multiple values are only produced for LSDtype set to factor.combination, in which case LSDby must not be NULL and the number of values must equal the number of observed combinations of</p>                                                                                                                                                                                                      |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | the values of the variables specified by LSDby. If LSDstatistic is NULL, it is reset to mean.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| LSDaccuracy | A <a href="#">character</a> nominating one of maxAbsDeviation, maxDeviation, q90Deviation or RootMeanSqDeviation as the statistic to be calculated as a measure of the accuracy of assignedLSD. The option q90Deviation produces the sample quantile corresponding to a probability of 0.90. The deviations are the differences between the LSDs used in calculating the LSD statistics and each assigned LSD and the accuracy is expressed as a proportion of the assigned LSD value. The calculated values are stored in the column named accuracyLSD in an <a href="#">LSD.frame</a> . |
| alpha       | The significance level for an LSD to compare a pair of predictions. It is stored as an attribute to the <a href="#">alldiffs.object</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| ...         | further arguments passed to <a href="#">allDifferences.data.frame</a> ; attributes transform.power, offset and scale cannot be passed.                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

### Value

An [alldiffs.object](#) with components predictions, vcov, differences, p.differences sed, LSD and, if present in alldiffs.obj, backtransforms.

### Author(s)

Chris Brien

### See Also

[asremlPlus-package](#), [as.alldiffs](#), [sort.alldiffs](#), [subset.alldiffs](#), [print.alldiffs](#), [renewClassify.alldiffs](#), [exploreLSDs.alldiffs](#), [pickLSDstatistics.alldiffs](#), [redoErrorIntervals.alldiffs](#), [plotPredictions.data.frame](#), [predictPlus.asreml](#), [predictPresent.asreml](#)

### Examples

```
data(WaterRunoff.dat)

##Use asreml to get predictions and associated statistics

## Not run:
asreml.options(keep.order = TRUE) #required for asreml-R4 only
current.asr <- asreml(fixed = pH ~ Benches + (Sources * (Type + Species)),
                     random = ~ Benches:MainPlots,
                     keep.order=TRUE, data= WaterRunoff.dat)
current.asrt <- as.asrttests(current.asr, NULL, NULL)
TS.diffs <- predictPlus(classify = "Sources:Type",
                       asreml.obj = current.asr,
                       wald.tab = current.asrt$wald.tab,
                       present = c("Sources", "Type", "Species"))

## End(Not run)

## Use lmeTest and emmeans to get predictions and associated statistics

if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
```

```

m1.lmer <- lmerTest::lmer(pH ~ Benches + (Sources * (Type + Species)) +
                        (1|Benches:MainPlots),
                        data=na.omit(WaterRunoff.dat))
TS.emm <- emmeans::emmeans(m1.lmer, specs = ~ Sources:Species)
TS.preds <- summary(TS.emm)
den.df <- min(TS.preds$df, na.rm = TRUE)
## Modify TS.preds to be compatible with a predictions.frame
TS.preds <- as.predictions.frame(TS.preds, predictions = "emmean",
                                se = "SE", interval.type = "CI",
                                interval.names = c("lower.CL", "upper.CL"))

## Form an all.diffs object and check its validity
TS.vcov <- vcov(TS.emm)
TS.diffs <- allDifferences(predictions = TS.preds, classify = "Sources:Species",
                          vcov = TS.vcov, tdf = den.df)
validAlldiffs(TS.diffs)
}

## Plot p-values for predictions obtained using asreml or lmerTest
if (exists("TS.diffs"))
{
  ##Recalculate the LSD values for predictions obtained using asreml or lmerTest
  TS.diffs <- recalcLSD.alldiffs(TS.diffs, LSDtype = "factor.combinations",
                                LSDby = "Sources")
}

```

---

```
recalcWaldTab.asrtests
```

*Recalculates the denDF, F.inc and P values for a table of Wald test statistics obtained using wald.asreml*

---

## Description

If some or all denDF are not available, either because they are NA or because F.inc values were not calculated, this function allows the user to specify how approximate denDF values are to be obtained. This is done through the dDF.fault and dDF.values arguments. Note that if denDF values are available in the Wald table then only those that are NA will be replaced. The P values are recalculated using F.con, if present in the wald.tab, otherwise F.inc is used. It is noted that, as of asreml version 4, wald.asreml has a kenadj argument.

## Usage

```

## S3 method for class 'asrtests'
recalcWaldTab(asrtests.obj, recalc.wald = FALSE,
              denDF="numeric", dDF.fault = "none",
              dDF.values = NULL, trace = FALSE, ...)

```

## Arguments

|              |                                                                                                                                                                |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| asrtests.obj | an <a href="#">asrtests.object</a> containing the components (i) asreml.obj, (ii) wald.tab, and (iii) test.summary.                                            |
| recalc.wald  | A logical indicating whether to call wald.asreml to recalculate the pseudoanova table for the model fit stored in the asreml object contained in asrtests.obj. |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| denDF      | Specifies the method to use in computing approximate denominator degrees of freedom when <code>wald.asreml</code> is called. Can be none to suppress the computations, numeric for numerical methods, algebraic for algebraic methods or default, the default, to automatically choose numeric or algebraic computations depending on problem size. The denominator degrees of freedom are calculated according to Kenward and Roger (1997) for fixed terms in the dense part of the model.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| dDF.fault  | A <a href="#">character</a> specifying the method to use to obtain substitute denominator degrees of freedom. when the numeric or algebraic methods produce faulty values, viz. NA, Inf or less than 0.01. Consistent with when no denDF are available, the default is "residual" and so the residual degrees of freedom from <code>asreml.obj\$nedf</code> are used. If <code>dDF.fault = "none"</code> , no substitute denominator degrees of freedom are employed; if <code>dDF.fault = "residual"</code> ; if <code>dDF.fault = "maximum"</code> , the maximum of those denDF that are available, excluding that for the Intercept, is used; if all denDF are faulty, <code>asreml.obj\$nedf</code> is used. If <code>dDF.fault = "supplied"</code> , a vector of values for the denominator degrees of freedom is to be supplied in <code>dDF.values</code> . Any other setting is ignored and a warning message produced. Generally, substituting these degrees of freedom is anticonservative in that it is likely that the degrees of freedom used will be too large. |
| dDF.values | A vector of values to be used when <code>dDF.fault = "supplied"</code> . Its values will be used when denDF in a test for a fixed effect is NA. This vector must be the same length as the number of fixed terms, including (Intercept) whose value could be NA.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| trace      | If TRUE then partial iteration details are displayed when ASReml-R functions are invoked; if FALSE then no output is displayed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| ...        | further arguments passed to <code>asreml</code> and to <code>wald.asreml</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

**Value**

A `wald.tab`: a 4- or 6-column data.frame containing a pseudo-anova table for the fixed terms produced by `wald.asreml`.

**Author(s)**

Chris Brien

**References**

Kenward, M. G., & Roger, J. H. (1997). Small sample inference for fixed effects from restricted maximum likelihood. *Biometrics*, **53**, 983-997.

**See Also**

[as.asrtests](#), [testtranfix.asrtests](#)

**Examples**

```
## Not run:
wald.tab <- recalcWaldTab(current.asrt,
  dDF.fault = "supplied",
  dDF.values = c(NA,rep(c(330,346), c(4,3))))
```

```
## End(Not run)
```

---

```
redoErrorIntervals.alldiffs
```

*Adds or replaces the error intervals stored in a prediction component of an `alldiffs.object`.*

---

## Description

Given an `alldiffs.object`, adds or replaces `error.intervals` for its prediction component. If the `backtransforms` component is present, the `transform.power`, `offset` and `scale` will be retrieved from the `backtransforms` attributes, ignoring the values for the function's arguments, and the backtransformed `error.intervals` will also be calculated.

## Usage

```
## S3 method for class 'alldiffs'
redoErrorIntervals(alldiffs.obj, error.intervals = "Confidence",
  alpha = 0.05,
  avsed.tolerance = 0.25, accuracy.threshold = NA,
  LSDtype = NULL, LSDsupplied = NULL,
  LSDby = NULL, LSDstatistic = "mean",
  LSDaccuracy = "maxAbsDeviation",
  retain.zeroLSDs = FALSE,
  zero.tolerance = .Machine$double.eps ^ 0.5, ...)
```

## Arguments

`alldiffs.obj` An `alldiffs.object`.  
`error.intervals`

A `character` string indicating the type of error interval, if any, to calculate in order to indicate uncertainty in the predicted values. Possible values are "none", "StandardError", "Confidence" and "halfLeastSignificant". The default is for confidence limits to be used. The "halfLeastSignificant" option results in half the Least Significant Difference (LSD) being added and subtracted to the predictions, the LSD being calculated using the square root of the mean of the variances of all or a subset of pairwise differences between the predictions. If the LSD is zero, as can happen when predictions are constrained to be equal, then the limits of the error intervals are set to NA. If `LSDtype` is set to overall, the `avsed.tolerance` is not NA and the range of the SEDs divided by the average of the SEDs exceeds `avsed.tolerance` then the `error.intervals` calculations and the plotting will revert to confidence intervals.

`alpha` A `numeric` giving the significance level for LSDs or one minus the confidence level for confidence intervals. It is stored as an attribute to the `alldiffs.object`.

`avsed.tolerance`

A `numeric` giving the value of the SED range, the range of the SEDs divided by the square root of the mean of the variances of all or a subset of the pairwise differences, that is considered reasonable in calculating `error.intervals`. To have it ignored, set it to NA. It should be a value between 0 and 1. The following rules apply:



1. If `avsed.tolerance` is NA then mean LSDs of the type specified by `LSDtype` are calculated and used in `error.intervals` and plots.
2. Irrespective of the setting of `LSDtype`, if `avsed.tolerance` is not exceeded then the mean LSDs are used in `error.intervals` and plots.
3. If `LSDtype` is set to `overall`, `avsed.tolerance` is not NA, and `avsed.tolerance` is exceeded then `error.intervals` and plotting revert to confidence intervals.
4. If `LSDtype` is set to `factor.combinations` and `avsed.tolerance` is not exceeded for any factor combination then the half LSDs are used in `error.intervals` and plots; otherwise, `error.intervals` and plotting revert to confidence intervals.
5. If `LSDtype` is set to `per.prediction` and `avsed.tolerance` is not exceeded for any prediction then the half LSDs are used in `error.intervals` and plots; otherwise, `error.intervals` and plotting revert to confidence intervals.

#### `accuracy.threshold`

A **numeric** specifying the value of the LSD accuracy measure, which measure is specified by `LSDaccuracy`, as a threshold value in determining whether the halfLeastSignificant `error.interval` for a predicted value is a reasonable approximation; this will be the case if the LSDs across all pairwise comparisons for which the interval's LSD was computed, as specified by `LSDtype` and `LSDby`, are similar enough to the interval's LSD, as measured by `LSDaccuracy`. If it is NA, it will be ignored. If it is not NA, a column of **logicals** named `LSDwarning` will be added to the `predictions` component of the `alldiffs.object`. The value of `LSDwarning` for a `predicted.value` will be TRUE if the value of the `LSDaccuracy` measure computed from the LSDs for differences between this `predicted.value` and the other `predicted.values` as compared to its `assignedLSD` exceeds the value of `accuracy.threshold`. Otherwise, the value of `LSDwarning` for a `predicted.value` will be FALSE.

#### `LSDtype`

A **character** string that can be `overall`, `factor.combinations`, `per.prediction` or supplied. It determines whether the values stored in a row of a **LSD.frame** are the values calculated (i) `overall` from the LSD values for all pairwise comparisons, unless there is only one, possibly repeated, prediction, when a notional LSD is calculated, (ii) the values calculated from the pairwise LSDs for the levels of each `factor.combination`, unless there is only one prediction for a level of the `factor.combination`, when a notional LSD is calculated, (iii) `per.prediction`, being based, for each prediction, on all pairwise differences involving that prediction, or (iv) as supplied values of the LSD, specified with the `LSDsupplied` argument; these supplied values are to be placed in the `assignedLSD` column of the **LSD.frame** stored in an `alldiffs.object` so that they can be used in LSD calculations.

If `LSDtype` is NULL (the default), the `LSDtype` attribute of the `alldiffs.obj` will be used; it is also NULL, then the `LSDtype` will be set to `overall`.

See **LSD.frame** for further information on the values in a row of this `data.frame` and how they are calculated.

#### `LSDsupplied`

A **data.frame** or a named **numeric** containing a set of LSD values that correspond to the observed combinations of the values of the `LSDby` variables in the `predictions.frame` or a single LSD value that is an overall LSD. If a **data.frame**, it may have (i) a column for the `LSDby` variable and a column of LSD values or (ii) a single column of LSD values with rownames being the combinations of the observed values of the `LSDby` variables. Any name can be used

for the column of LSD values; assignedLSD is sensible, but not obligatory. Otherwise, a `numeric` containing the LSD values, each of which is named for the observed combination of the values of the LSDby variables to which it corresponds. (Applying the function `dae::fac.combine` to the predictions component is one way of forming the required combinations for the (row) names.) The values supplied will be incorporated into assignedLSD column of the `LSD.frame` stored as the LSD component of the `alldiffs.object`.

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LSDby           | A <code>character</code> (vector) of variables names, being the names of the <code>factors</code> or <code>numerics</code> in the <code>classify</code> ; for each combination of their levels and values, there will be or is a row in the <code>LSD.frame</code> stored in the LSD component of the <code>alldiffs.object</code> when LSDtype is <code>factor.combinations</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| LSDstatistic    | A <code>character</code> nominating one or more of minimum, q10, q25, mean, median, q75, q90 or maximum as the value(s) to be stored in the assignedLSD column in an <code>LSD.frame</code> ; the values in the assignedLSD column are used in computing halfLeastSignificant error.intervals. Here q10, q25, q75 and q90 indicate the sample quantiles corresponding to probabilities of 0.1, 0.25, 0.75 and 0.9 for the group of LSDs from which a single LSD value is calculated. The function <code>quantile</code> is used to obtain them. The mean LSD is calculated as the square root of the mean of the squares of the LSDs for the group. The median is calculated using the <code>median</code> function. Multiple values are only produced for LSDtype set to <code>factor.combination</code> , in which case LSDby must not be <code>NULL</code> and the number of values must equal the number of observed combinations of the values of the variables specified by LSDby. If LSDstatistic is <code>NULL</code> , it is reset to mean. |
| LSDaccuracy     | A <code>character</code> nominating one of <code>maxAbsDeviation</code> , <code>maxDeviation</code> , <code>q90Deviation</code> or <code>RootMeanSqDeviation</code> as the statistic to be calculated as a measure of the accuracy of assignedLSD. The option <code>q90Deviation</code> produces the sample quantile corresponding to a probability of 0.90. The deviations are the differences between the LSDs used in calculating the LSD statistics and each assigned LSD and the accuracy is expressed as a proportion of the assigned LSD value. The calculated values are stored in the column named <code>accuracyLSD</code> in an <code>LSD.frame</code> .                                                                                                                                                                                                                                                                                                                                                                                  |
| retain.zeroLSDs | A <code>logical</code> indicating whether to retain or omit LSDs that are zero when calculating the summaries of LSDs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| zero.tolerance  | A <code>numeric</code> specifying the value such that if an LSD is less than it, the LSD will be considered to be zero.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| ...             | further arguments passed to <code>recalcLSD.alldiffs</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## Value

An `alldiffs.object` with components `predictions`, `vcov`, `differences`, `p.differences` `sed`, `LSD` and, if present in `alldiffs.obj`, `backtransforms`.

If `error.intervals` is not "none", then the `predictions` component and, if present, the `backtransforms` component will contain columns for the lower and upper values of the limits for the interval. The names of these columns will consist of three parts separated by full stops: 1) the first part will be lower or upper; 2) the second part will be one of `Confidence`, `StandardError` or `halfLeastSignificant`; 3) the third component will be `limits`.

The name of the response, the term, the `classify` and `tdf`, as well as the degrees of freedom of the standard error, will be set as attributes to the object. Also, if `error.intervals` is "halfLeastSignificant", then those of `LSDtype`, `LSDby` and `LSDstatistic` that are not `NULL`

will be added as attributes of the object and of the predictions frame; additionally, LSDvalues will be added as attribute of the predictions frame, LSDvalues being the LSD values used in calculating the error.intervals.

### Author(s)

Chris Brien

### See Also

[recalcLSD.alldiffs](#), [exploreLSDs.alldiffs](#), [pickLSDstatistics.alldiffs](#),  
[predictPresent.asreml](#), [plotPredictions.data.frame](#), [allDifferences.data.frame](#),  
[as.alldiffs](#), [print.alldiffs](#), [sort.alldiffs](#), [subset.alldiffs](#),  
[as.Date](#), [predict.asreml](#)

### Examples

```
data(WaterRunoff.dat)

##Use asreml to get predictions and associated statistics

## Not run:
asreml.options(keep.order = TRUE) #required for asreml-R4 only
current.asr <- asreml(fixed = pH ~ Benches + (Sources * (Type + Species)),
                     random = ~ Benches:MainPlots,
                     keep.order=TRUE, data= WaterRunoff.dat)
current.asrt <- as.asrttests(current.asr, NULL, NULL)
TS.diffs <- predictPlus(classify = "Sources:Type",
                       asreml.obj = current.asr,
                       wald.tab = current.asrt$wald.tab,
                       present = c("Sources", "Type", "Species"))

## End(Not run)

## Use lmeTest and emmeans to get predictions and associated statistics

if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(pH ~ Benches + (Sources * (Type + Species)) +
                          (1|Benches:MainPlots),
                          data=na.omit(WaterRunoff.dat))
  TS.emm <- emmeans::emmeans(m1.lmer, specs = ~ Sources:Species)
  TS.preds <- summary(TS.emm)
  den.df <- min(TS.preds$df, na.rm = TRUE)
  ## Modify TS.preds to be compatible with a predictions.frame
  TS.preds <- as.predictions.frame(TS.preds, predictions = "emmean",
                                  se = "SE", interval.type = "CI",
                                  interval.names = c("lower.CL", "upper.CL"))

  ## Form an all.diffs object and check its validity
  TS.vcov <- vcov(TS.emm)
  TS.diffs <- allDifferences(predictions = TS.preds,
                           classify = "Sources:Species",
                           vcov = TS.vcov, tdf = den.df)

  validAlldiffs(TS.diffs)
```

```

}

## Plot p-values for predictions obtained using asreml or lmerTest
if (exists("TS.diffs"))
{
  ##Recalculate the LSD values for predictions obtained using asreml or lmerTest
  TS.diffs <- redoErrorIntervals.alldiffs(TS.diffs,
                                         error.intervals = "halfLeastSignificant")
}

```

REMLRT.asreml

*Performs a REML ratio test to compare two models.*

## Description

Extracts the REML log likelihood and the number of variance parameters from two `asreml` objects. It assumes that the first `asreml` object corresponds to the null hypothesis and the second `asreml` object to the alternative hypothesis for the test being conducted. That is, the second `asreml` object is the result of fitting a model that is a reduced version of the model for the first object. In the case where the reduced model is obtained by setting positively-constrained variance parameters in the full model to zero, the `positive.zero` argument should be set to `TRUE` so that the p-value is computed using a mixture of chi-square distributions as described in Self and Liang (1987).

The function checks that the models do not differ in either their fixed or sparse models.

## Usage

```

## S3 method for class 'asreml'
REMLRT(h0.asreml.obj, h1.asreml.obj,
       positive.zero = FALSE, bound.test.parameters = "none",
       DF = NULL, bound.exclusions = c("F", "B", "S", "C"), ...)

```

## Arguments

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>h0.asreml.obj</code>         | <code>asreml</code> object containing the fit under the model for the null hypothesis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>h1.asreml.obj</code>         | <code>asreml</code> object containing the fit under the model for the alternative hypothesis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>positive.zero</code>         | Indicates whether the hypothesized values for the variance components being tested are on the boundary of the parameter space. For example, this is true for positively-constrained variance components that, under the reduced model, are zero. This argument does not need to be set if <code>bound.test.parameters</code> is set.                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>bound.test.parameters</code> | Indicates whether for the variance components being tested, at least some of the hypothesized values are on the boundary of the parameter space. The possibilities are "none", "onlybound" and "one-and-one". The default is "none", although if it is set to "none" and <code>positive.zero</code> is <code>TRUE</code> then <code>bound.test.parameters</code> is taken to be "onlybound". When <code>bound.test.parameters</code> is set to "one-and-one", it signifies that there are two parameters being tested, one of which is bound and the other is not. For example, the latter is true for testing a covariance and a positively-constrained variance component that, under the reduced model, are zero. |

|                               |                                                                                                                                                                                                                                                                                                         |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DF                            | A numeric giving the difference between the two models in the number of variance parameters whose estimates are not of the type specified in <code>bound.exclusions</code> . If NULL then this is determined from the information in <code>full.asreml.obj</code> and <code>reduced.asreml.obj</code> . |
| <code>bound.exclusions</code> | A character specifying one or more bound (constraint) codes that will result in a variance parameter being excluded from the count of estimated variance parameters. If set to NULL then none will be excluded.                                                                                         |
| ...                           | Provision for passing arguments to functions called internally - not used at present.                                                                                                                                                                                                                   |

### Value

A data.frame containing the log of the likelihood ratio, its degrees of freedom, its p-value and the number of bound parameters in each of the two models being compared.

### Note

If DF is not NULL, the supplied value is used. Otherwise DF is determined from the information in `h1.asreml.obj` and `h0.asreml.obj`. In this case, the degrees of freedom for the test are computed as the difference between the two models in the number of variance parameters whose estimates do not have a code for bound specified in `bound.exclusions`.

If ASReml-R version 4 is being used then the codes specified in `bound.exclusions` are not restricted to a subset of the default codes, but a warning is issued if a code other than these is specified. For ASReml-R version 3, only a subset of the default codes are allowed: F (Fixed), B (Boundary), C (Constrained) and S (Singular).

The test statistic is calculated as  $2(\log(REML)_1 - \log(REML)_0)$ .

This procedure is only appropriate when the null hypothesis is that (i) all parameters are on the boundary of the parameter space (ii) all parameters are in the interior of the parameter space, or (iii) there are two parameters, one of which is on the boundary and the other is not. Other cases have been discussed by Self and Liang (1987), but are not implemented here.

### Author(s)

Chris Brien

### References

Self, S.G., and Liang, K-Y. (1987) Asymptotic Properties of Maximum Likelihood Estimators and Likelihood Ratio Tests Under Nonstandard Conditions. *Journal of the American Statistical Association*, **82**, 605-10.

### See Also

[infoCriteria.asreml](#), [testranfix.asrtests](#)

### Examples

```
## Not run:
  REMLRT(ICV.max, ICV.red, bound.test.parameters = "onlybound")

## End(Not run)
```

---

renewClassify.alldiffs

*Renews the components in an [alldiffs.object](#) according to a new classify.*


---

## Description

The classify is an attribute of an [alldiffs.object](#) and determines the order within the components of an unsorted [alldiffs.object](#). This function resets the classify attribute and re-orders the components of [alldiffs.object](#) to be in standard order for the variables in a newclassify, using [allDifferences.data.frame](#). The newclassify may be just a re-ordering of the variable names in the previous classify, or be based on a new set of variable names. The latter is particularly useful when [linTransform.alldiffs](#) has been used with a [matrix](#) and it is desired to replace the resulting Combination classify with a newclassify comprised of a more meaningful set of variables; first replace Combination in the predictions component with the new set of variables and then call renewClassify.

## Usage

```
## S3 method for class 'alldiffs'
renewClassify(alldiffs.obj, newclassify,
              sortFactor = NULL, sortParallelToCombo = NULL,
              sortNestingFactor = NULL, sortOrder = NULL, decreasing = FALSE, ...)
```

## Arguments

- |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| alldiffs.obj        | An <a href="#">alldiffs.object</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| newclassify         | A <a href="#">character</a> string giving the variables that define the margins of the multiway table that was predicted, but ordered so that the predictions are in the desired order when they are arranged in standard order for the newclassify. Multiway tables are specified by forming an interaction type term from the classifying variables, that is, separating the variable names with the : operator. The number of combined values of the set of variable name(s) must equal the number of rows in the predictions component.                                                                                                                                                                                                                                                                                                                  |
| sortFactor          | A <a href="#">character</a> containing the name of the factor that indexes the set of predicted values that determines the sorting of the components. If there is only one variable in the classify term then sortFactor can be NULL and the order is defined by the complete set of predicted values. If there is more than one variable in the classify term then sortFactor must be set. In this case the sortFactor is sorted in the same order within each combination of the values of the sortParallelToCombo variables: the classify variables, excluding the sortFactor. There should be only one predicted value for each unique value of sortFactor within each set defined by a combination of the values of the classify variables, excluding the sortFactor factor. The order to use is determined by either sortParallelToCombo or sortOrder. |
| sortParallelToCombo | A <a href="#">list</a> that specifies a combination of the values of the factors and numerics, excluding sortFactor, that are in classify. Each of the components of the supplied <a href="#">list</a> is named for a classify variable and specifies a single value for it. The combination of this set of values will be used to define a subset of the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

predicted values whose order will define the order of sortFactor. Each of the other combinations of the values of the factors and numerics will be sorted in parallel. If sortParallelToCombo is NULL then the first value of each classify variable, except for the sortFactor factor, in the predictions component is used to define sortParallelToCombo. If there is only one variable in the classify then sortParallelToCombo is ignored.

sortNestingFactor

A **character** containing the name of the factor that defines groups of the sortFactor within which the predicted values are to be ordered. If there is only one variable in the classify then sortNestingFactor is ignored.

sortOrder

A **character** vector whose length is the same as the number of levels for sortFactor in the predictions component of the **alldiffs.object**. It specifies the desired order of the levels in the reordered components of the **alldiffs.object**. The argument sortParallelToCombo is ignored.

The following creates a sortOrder vector levs for factor f based on the values in x: levs <- levels(f)[order(x)].

decreasing

A **logical** passed to order that determines whether the order is for increasing or decreasing magnitude of the predicted values.

...

further arguments passed to **allDifferences.data.frame**; attributes tranform.power, offset and scale cannot be passed.

## Details

First, the components of the **alldiffs.object** is arranged in standard order for the newclassify. Then predictions are reordered according to the settings of sortFactor, sortParallelToCombo, sortOrder and decreasing (see **sort.alldiffs** for details).

## Value

The **alldiffs.object** supplied with the following components, if present, sorted: predictions, vcov, backtransforms, differences, p.differences and sed. Also, the sortFactor and sortOrder attributes are set.

## Author(s)

Chris Brien

## See Also

**as.alldiffs**, **allDifferences.data.frame**, **print.alldiffs**, **sort.alldiffs**, **redoErrorIntervals.alldiffs**, **recalcLSD.alldiffs**, **predictPlus.asreml**, **predictPresent.asreml**

## Examples

```
data(WaterRunoff.dat)

##Use asreml to get predictions and associated statistics

## Not run:
#Analyse pH
m1.asr <- asreml(fixed = pH ~ Benches + (Sources * (Type + Species)),
                 random = ~ Benches:MainPlots,
```

```

        keep.order=TRUE, data= WaterRunoff.dat)
current.asrt <- as.asrtests(m1.asr, NULL, NULL)
current.asrt <- as.asrtests(m1.asr)
current.asrt <- rmboundary(current.asrt)
m1.asr <- current.asrt$asreml.obj

#Get predictions and associated statistics
TS.diffs <- predictPlus.asreml(classify = "Sources:Type",
                              asreml.obj = m1.asr, tables = "none",
                              wald.tab = current.asrt$wald.tab,
                              present = c("Type", "Species", "Sources"))

## End(Not run)

## Use lmeTest and emmeans to get predictions and associated statistics

if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  #Analyse pH
  m1.lmer <- lmerTest::lmer(pH ~ Benches + (Sources * (Type + Species)) +
                          (1|Benches:MainPlots),
                          data=na.omit(WaterRunoff.dat))
  TS.emm <- emmeans::emmeans(m1.lmer, specs = ~ Sources:Type)
  TS.preds <- summary(TS.emm)
  den.df <- min(TS.preds$df, na.rm = TRUE)
  ## Modify TS.preds to be compatible with a predictions.frame
  TS.preds <- as.predictions.frame(TS.preds, predictions = "emmean",
                                  se = "SE", interval.type = "CI",
                                  interval.names = c("lower.CL", "upper.CL"))

  ## Form an all.diffs object and check its validity
  TS.vcov <- vcov(TS.emm)
  TS.diffs <- allDifferences(predictions = TS.preds,
                           classify = "Sources:Type",
                           vcov = TS.vcov, tdf = den.df)

  validAlldiffs(TS.diffs)
}

#Re-order predictions from asreml or lmerTest so all Sources for the same Type are together
#for each combination of A and B
if (exists("TS.diffs"))
{
  TS.diffs.reord <- renewClassify(TS.diffs, newclassify = "Type:Sources")
  validAlldiffs(TS.diffs.reord)
}

```

---

reparamSigDevn.asrtests

*Reparameterizes each random (deviations) term involving devn.fac to a fixed term and ensures that the same term, with trend.num replacing devn.fac, is included if any other term with trend.num is included in terms.*

---



## Description

This function reparameterizes each random (deviations) term involving devn.fac to a fixed term and ensures that the same term with trend.num replacing devn.fac is included if any other term with trend.num is included in terms. It also ensures that any term with spl{trend.num} replacing devn.fac in a term being reparameterized is removed from the model.

## Usage

```
## S3 method for class 'asrtests'
reparamSigDevn(asrtests.obj, terms = NULL,
               trend.num = NULL, devn.fac = NULL,
               allow.unconverged = TRUE, allow.fixedcorrelation = TRUE,
               checkboundaryonly = FALSE,
               denDF = "numeric", IClikelihood = "none",
               trace = FALSE, update = TRUE,
               set.terms = NULL, ignore.suffices = TRUE,
               bounds = "P", initial.values = NA,...)
```

## Arguments

- |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| asrtests.obj           | an <a href="#">asrtests.object</a> containing the components (i) asreml.obj, (ii) wald.tab, and (iii) test.summary.                                                                                                                                                                                                                                                                                                                                                                                  |
| terms                  | A character string vector giving the random terms that are to be reparameterized.                                                                                                                                                                                                                                                                                                                                                                                                                    |
| trend.num              | A character string giving the name of the numeric covariate that corresponds to devn.fac and is potentially included in terms in the fitted model.                                                                                                                                                                                                                                                                                                                                                   |
| devn.fac               | A character string giving the name of the factor that corresponds to trend.num and is included in terms in the fitted model. The name must match those in the vparameters component of the asreml.obj component in the asrtests.obj.                                                                                                                                                                                                                                                                 |
| allow.unconverged      | A logical indicating whether to accept a new model even when it does not converge. Initially all changes are made with allow.unconverged set to TRUE. If allow.unconverged has been set to FALSE in the call and the final fit does not converge, an attempt is made to achieve convergence by removing any boundary terms. If this is unsuccessful, the supplied asrtests.obj is returned.                                                                                                          |
| allow.fixedcorrelation | A logical indicating whether to accept a new model even when it contains correlations in the model whose values have been designated as fixed, bound or singular. If FALSE and the new model contains correlations whose values have not been able to be estimated, the supplied asrtests.obj is returned. The fit in the asreml.obj component of the supplied asrtests.obj will also be tested and a warning issued if both fixed correlations are found in it and allow.fixedcorrelation is FALSE. |
| checkboundaryonly      | If TRUE then boundary and singular terms are not removed by <a href="#">rmboundary.asrtests</a> ; a warning is issued instead.                                                                                                                                                                                                                                                                                                                                                                       |
| denDF                  | Specifies the method to use in computing approximate denominator degrees of freedom when wald.asreml is called. Can be none to suppress the computations, numeric for numerical methods, algebraic for algebraic methods or default, the default, to automatically choose numeric or algebraic computations depending on problem size. The denominator degrees of freedom are                                                                                                                        |

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | calculated according to Kenward and Roger (1997) for fixed terms in the dense part of the model.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| IClikelihood    | A character that controls both the occurrence and the type of likelihood for information criterion in the test.summary of the new <code>asrtests.object</code> . If none, none are included. Otherwise, if REML, then the AIC and BIC based on the Restricted Maximum Likelihood are included; if full, then the AIC and BIC based on the full likelihood, evaluated using REML estimates, are included. (See also <code>infoCriteria.asreml</code> .)                                                                                                                                                                                                                |
| trace           | If TRUE then partial iteration details are displayed when ASReML-R functions are invoked; if FALSE then no output is displayed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| update          | If TRUE, and <code>set.terms</code> is NULL, then <code>newfit.asreml</code> is called to fit the model to be tested, using the values of the variance parameters stored in the <code>asreml.object</code> , that is stored in <code>asrtests.obj</code> , as starting values. If FALSE or <code>set.terms</code> is not NULL, then <code>newfit.asreml</code> will not use the stored variance parameter values as starting values when fitting the new model, the only modifications being (i) the models are updated and (ii) those specified via . . . .                                                                                                          |
| set.terms       | A character vector specifying the terms that are to have bounds and/or initial values set prior to fitting.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| ignore.suffices | A logical vector specifying whether the suffices of the <code>asreml</code> -assigned names of the variance terms (i.e. the information to the right of an "!", other than "R!") is to be ignored in matching elements of terms. If TRUE for an element of terms, the suffices are stripped from the <code>asreml</code> -assigned names. If FALSE for an element of terms, the element must exactly match an <code>asreml</code> -assigned name for a variance term. This vector must be of length one or the same length as terms. If it is of length one then the same action is applied to the <code>asreml</code> -assigned suffices for all the terms in terms. |
| bounds          | A <code>character</code> vector specifying the bounds to be applied to the terms specified in <code>set.terms</code> . This vector must be of length one or the same length as <code>set.terms</code> . If it is of length one then the same constraint is applied to all the terms in <code>set.terms</code> . If any of the bounds are equal to NA then they are left unchanged for those terms.                                                                                                                                                                                                                                                                    |
| initial.values  | A character vector specifying the initial values for the terms specified in terms. This vector must be of length one or the same length as terms. If it is of length one then the same initial value is applied to all the terms in terms. If any of the initial.values are equal to NA then they are left unchanged for those terms.                                                                                                                                                                                                                                                                                                                                 |
| ...             | further arguments passed to <code>asreml</code> via <code>changeTerms.asrtests</code> and <code>as.asrtests</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

**Value**

An `asrtests.object` containing the components (i) `asreml.obj`, (ii) `wald.tab`, and (iii) `test.summary`.

**Author(s)**

Chris Brien

**References**

Kenward, M. G., & Roger, J. H. (1997). Small sample inference for fixed effects from restricted maximum likelihood. *Biometrics*, **53**, 983-997.

**See Also**

[as.asrtests](#), [changeTerms.asrtests](#), [testranfix.asrtests](#), [testresidual.asrtests](#),  
[newfit.asreml](#), [chooseModel.asrtests](#)

**Examples**

```
## Not run:
data(WaterRunoff.dat)
asreml.options(keep.order = TRUE) #required for asreml-R4 only
current.asr <- asreml(fixed = log.Turbidity ~ Benches + Sources + Type + Species +
                     Sources:Type + Sources:Species + Sources:Species:xDay +
                     Sources:Species:Date,
                     data = WaterRunoff.dat, keep.order = TRUE)
current.asrt <- as.asrtests(current.asr, NULL, NULL)

#Examine terms that describe just the interactions of Date and the treatment factors
terms.treat <- c("Sources", "Type", "Species", "Sources:Type", "Sources:Species")
date.terms <- sapply(terms.treat,
                     FUN=function(term){paste("Date:", term, sep="")},
                     simplify=TRUE)
date.terms <- c("Date", date.terms)
date.terms <- unname(date.terms)
treat.marginality <- matrix(c(1,0,0,0,0,0, 1,1,0,0,0,0, 1,0,1,0,0,0,
                             1,0,1,1,0,0, 1,1,1,0,1,0, 1,1,1,1,1,1), nrow=6)
rownames(treat.marginality) <- date.terms
colnames(treat.marginality) <- date.terms
choose <- chooseModel(current.asrt, treat.marginality, denDF="algebraic")
current.asrt <- choose$asrtests.obj
current.asr <- current.asrt$asreml.obj
sig.date.terms <- choose$sig.terms

#Remove all Date terms left in the fixed model
terms <- "(Date/(Sources * (Type + Species)))"
current.asrt <- changeTerms(current.asrt, dropFixed = terms)
#if there are significant date terms, reparameterize to xDays + spl(xDays) + Date
if (length(sig.date.terms) != 0)
{ #add lin + spl + devn for each to fixed and random models
  trend.date.terms <- sapply(sig.date.terms,
                             FUN=function(term){sub("Date", "xDay", term)},
                             simplify=TRUE)
  trend.date.terms <- paste(trend.date.terms, collapse=" + ")
  current.asrt <- changeTerms(current.asrt, addFixed=trend.date.terms)
  trend.date.terms <- sapply(sig.date.terms,
                             FUN=function(term){sub("Date", "spl(xDay)", term)},
                             simplify=TRUE)
  trend.date.terms <- c(trend.date.terms, sig.date.terms)
  trend.date.terms <- paste(trend.date.terms, collapse=" + ")
  current.asrt <- changeTerms(current.asrt, addRandom = trend.date.terms)
  current.asrt <- rmboundary(current.asrt)
}

#Now test terms for sig date terms
spl.terms <- sapply(terms.treat,
                    FUN=function(term){paste("spl(xDay):", term, sep="")},
                    simplify=TRUE)
spl.terms <- c("spl(xDay)", spl.terms)
```

```

lin.terms <- sapply(terms.treat,
  FUN=function(term){paste(term,":xDay",sep=""")},
  simplify=TRUE)
lin.terms <- c("xDay",lin.terms)
systematic.terms <- c(terms.treat, lin.terms, spl.terms, date.terms)
systematic.terms <- unname(systematic.terms)
treat.marginality <- matrix(c(1,0,0,0,0,0, 1,1,0,0,0,0, 1,0,1,0,0,0,
  1,0,1,1,0,0, 1,1,1,1,1,0, 1,1,1,1,1,1), nrow=6)
systematic.marginality <- kronecker(matrix(c(1,0,0,0, 1,1,0,0,
  1,1,1,0, 1,1,1,1), nrow=4),
  treat.marginality)
systematic.marginality <- systematic.marginality[-1, -1]
rownames(systematic.marginality) <- systematic.terms
colnames(systematic.marginality) <- systematic.terms
choose <- chooseModel(current.asrt, systematic.marginality,
  denDF="algebraic", pos=TRUE)
current.asrt <- choose$asrtests.obj

#Check if any deviations are significant and, for those that are, go back to
#fixed dates
current.asrt <- reparamSigDevn(current.asrt, choose$sig.terms,
  trend.num = "xDay", devn.fac = "Date",
  denDF = "algebraic")

## End(Not run)

```

---

|                     |                                                                                                                                                                  |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rmboundary.asrtests | <i>Removes any boundary or singular variance components from the fit stored in asrtests.obj and records their removal in an <a href="#">asrtests.object</a>.</i> |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|

---

## Description

Any terms specified in the random model that are estimated on the boundary or are singular and can be removed are removed from the fit stored in the `asreml` object stored in the `asrtests.object`. Terms that specify multiple parameters in the random model cannot be removed (e.g. terms specified using the `at` function with more than one level of the factor) and terms in residual model are not removed. Terms that can be removed are selected for removal in the following order based on whether they involve: (i) a dev function, (ii) only factors, (iii) an `spl` function, (iv) a `pol` function and (v) a `lin` function or a variable that is an integer or a numeric. It should be noted that this order of removal presumes that random deviation terms are specified via the dev function rather than via a random factor. Once the earliest of the above classes with a boundary term is identified, a term within this class is selected for removal. For all classes, except for factor-only terms, the smallest term with the largest number of variables/factors is removed. Amongst factor-only terms, the smallest term with the smallest number of variables/factors is removed. After each variance component is removed, a row for it is added to the `test.summary.data.frame` and the model re-fitted. If there are further boundary or singular terms, one is removed using the above strategy. This process continues until there are no further boundary or singular variance components that are removable. Other types of boundary or singular terms, which cannot be removed, are reported in warning messages.

**Usage**

```
## S3 method for class 'asrtests'
rmboundary(asrtests.obj, checkboundaryonly = FALSE,
           ICLikelihood = "none", trace = FALSE, update = TRUE,
           set.terms = NULL, ignore.suffices = TRUE,
           bounds = "P", initial.values = NA, ...)
```

**Arguments**

- `asrtests.obj` an [asrtests.object](#) containing the components (i) `asreml.obj`, (ii) `wald.tab`, and (iii) `test.summary`.
- `checkboundaryonly` If TRUE then boundary and singular terms are not removed by [rmboundary.asrtests](#); a warning is issued instead.
- `ICLikelihood` A character that controls both the occurrence and the type of likelihood for information criterion in the `test.summary` of the new [asrtests.object](#). If none, none are included. Otherwise, if REML, then the AIC and BIC based on the Restricted Maximum Likelihood are included; if full, then the AIC and BIC based on the full likelihood, evaluated using REML estimates, are included. (See also [infoCriteria.asreml](#).)
- `trace` If TRUE then partial iteration details are displayed when ASReml-R functions are invoked; if FALSE then no output is displayed.
- `update` If TRUE, and `set.terms` is NULL, then [newfit.asreml](#) is called to fit the model with any boundary terms removed and using the values of the variance parameters stored in the `asreml.object`, that is stored in `asrtests.obj`, as starting values. If FALSE or `set.terms` is not NULL, then [newfit.asreml](#) will not use the stored variance parameter values as starting values when fitting the model without the boundary terms.
- `set.terms` A character vector specifying the terms that are to have bounds and/or initial values set prior to fitting. The names must match those in the `vparameters` component of the `asreml.obj` component in the [asrtests.object](#).
- `ignore.suffices` A logical vector specifying whether the suffices of the `asreml`-assigned names of the variance terms (i.e. the information to the right of an "!", other than "R!") is to be ignored in matching elements of terms. If TRUE for an element of terms, the suffices are stripped from the `asreml`-assigned names. If FALSE for an element of terms, the element must exactly match an `asreml`-assigned name for a variance term. This vector must be of length one or the same length as terms. If it is of length one then the same action is applied to the `asreml`-assigned suffices for all the terms in terms.
- `bounds` A [character](#) vector specifying the bounds to be applied to the terms specified in `set.terms`. This vector must be of length one or the same length as `set.terms`. If it is of length one then the same constraint is applied to all the terms in `set.terms`. If any of the bounds are equal to NA then they are left unchanged for those terms.
- `initial.values` A character vector specifying the initial values for the terms specified in terms. This vector must be of length one or the same length as terms. If it is of length one then the same initial value is applied to all the terms in terms. If any of the `initial.values` are equal to NA then they are left unchanged for those terms.
- `...` Further arguments, including `asreml` arguments, passed to [newfit.asreml](#).

**Value**

An `asrtests.object` containing the components (i) `asreml.obj`, (ii) `wald.tab`, and (iii) `test.summary`.

**Author(s)**

Chris Brien

**See Also**

`as.asrtests`, `changeTerms.asrtests`, `testranfix.asrtests`, `testresidual.asrtests`,  
`newfit.asreml`, `reparamSigDevn.asrtests`, `chooseModel.asrtests`

**Examples**

```
## Not run:
current.asrt <- rmboundary(current.asrt)

## End(Not run)
```

---

`setvarianceterms.call` *allows the setting of bounds and initial values for terms in the random and residual arguments of an asreml call, with the resulting call being evaluated.*

---

**Description**

Takes an unevaluated call and evaluates the call after setting the bounds and initial values for the terms specified in `terms`. The elements of `terms` are matched with those generated by `asreml` and used, for example, in the `varcomp` component of a `summary.asreml` object. These names generally include descriptive suffices. To match an element of `terms` that includes such a suffix, set `ignore.suffices` to `FALSE` so that a literal match between the element and the assigned names is sought.

Note that the `terms`, `bounds` and `initial.values` are stored on entry in a `data.frame`, named `setvparameters`, as a component in `call` that is itself a component of the `asreml.obj` that is returned. The `data.frame` `setvparameters` contains all of the values of `terms`, `ignore.suffices`, `bounds` and `initial.values` that have been set in this and previous calls to `setvarianceterms.call` and other model modification and selection functions in `asremlPlus`, for example `changeModelOnIC.asrtests`, `testranfix.asrtests` and `changeTerms.asrtests`. It is used in subsequent calls to model modification and selection functions to ensure that the bounds and initial values that have been set are retained in new model fits.

**Usage**

```
## S3 method for class 'call'
setvarianceterms(call, terms, ignore.suffices = TRUE,
                  bounds = "P", initial.values = NA, ...)
```

**Arguments**

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| call            | an unevaluated call to asreml. One way to create such a call is to use the call function with its name argument set to "asreml". Another is to obtain it from the call component of an asreml object (e.g. call <- asreml.obj\$call).                                                                                                                                                                                                                                                                                                                                                                         |
| terms           | A character vector specifying the terms that are to have bounds and/or initial values specified. The names must match those in the vparameters component of the asreml.obj component in the <a href="#">asrtests.object</a> .                                                                                                                                                                                                                                                                                                                                                                                 |
| ignore.suffices | A logical vector specifying whether the suffices of the asreml-assigned names of the variance terms (i.e. the information to the right of an "!", other than "R!") is to be ignored in matching elements of terms. If TRUE for an element of terms, the suffices are stripped from the asreml-assigned names. If FALSE for an element of terms, the element must exactly match an asreml-assigned name for a variance term. This vector must be of length one or the same length as terms. If it is of length one then the same action is applied to the asreml-assigned suffices for all the terms in terms. |
| bounds          | A <a href="#">character</a> vector specifying the bounds to be applied to the terms specified in terms. This vector must be of length one or the same length as terms. If it is of length one then the same constraint is applied to all the terms in terms. If any of the bounds are equal to NA then they are left unchanged for those terms. The codes used by ASReml are: <ul style="list-style-type: none"> <li>• B - fixed at a boundary;</li> <li>• F - fixed by the user;</li> <li>• P - positive definite;</li> <li>• C - Constrained by user;</li> <li>• U - unbounded.</li> </ul>                  |
| initial.values  | A character vector specifying the initial values for the terms specified in terms. This vector must be of length one or the same length as terms. If it is of length one then the same initial value is applied to all the terms in terms. If any of the initial.values are equal to NA then they are left unchanged for those terms.                                                                                                                                                                                                                                                                         |
| ...             | additional arguments to be added to the asreml call, or arguments in the asreml call with changed values.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

**Value**

An asreml object, with the updated setvparameters [data.frame](#), stored in the call component of the asreml object.

**Author(s)**

Chris Brien

**References**

Butler, D. G., Cullis, B. R., Gilmour, A. R., Gogel, B. J. and Thompson, R. (2023). *ASReml-R Reference Manual Version 4.2*. VSN International Ltd, <https://asreml.kb.vsnl.co.uk/>.

**See Also**

[newfit.asreml](#), [update.asreml](#)

## Examples

```
## Not run:
m1.call <- call("asreml",
  fixed = Height ~ (Block + Irrig)*csDay.num,
  random= ~ spl(csDay.num)/(Irrig+Block)
  + dev(csDay.num)
  + str(~Block:Plot/csDay.num, ~us(2):id(20))
  + Block:Plot:spl(csDay.num),
  data=quote(dat)) ##use quote to stop evaluation of dat here
terms <- c("Block:Plot+Block:Plot:csDay.num!us(2).2:1", "R!variance")
m1.asreml <- setvarianceterms(m1.call, terms, bounds=c("U","P"),
  initial=c(NA,3), ignore.suffices=c(FALSE,TRUE))

summary(m1.asreml)

## End(Not run)
```

---

|                 |                                                                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------|
| simulate.asreml | <i>Produce sets of simulated data from a multivariate normal distribution and save quantities related to the simulated data</i> |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------|

---

## Description

Produce in parallel sets of simulated data corresponding to an asreml model, along with its fitted values and residuals. A variance matrix V, corresponding to the random and residual models must be supplied. What to save is specified by the which argument.

## Usage

```
## S3 method for class 'asreml'
simulate(object, nsim=100, seed = NULL, means=NULL, V, tolerance = 1E-10,
  update = TRUE, trace = FALSE, which="data", units = "ignore",
  ncores = 2, ...)
```

## Arguments

|           |                                                                                                                                                                                                                               |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object    | An asreml object from a call to asreml in which the data argument has been set.                                                                                                                                               |
| means     | The vector of means to be used in generating simulated data sets. If it is NULL, the fitted values based on object are used. It must be the same length as the response variable for object.                                  |
| V         | The fitted variance matrix, i.e. having the pattern and values that conform to the model fit stored in the supplied object.                                                                                                   |
| nsim      | The number of data sets to be simulated.                                                                                                                                                                                      |
| seed      | A single value, interpreted as an integer, that specifies the starting value of the random number generator. The "L'Ecuyer-CMRG" random generator is used and nextRNGStream is used to seed each core from the original seed. |
| tolerance | The value such that eigenvalues less than it are considered to be zero.                                                                                                                                                       |



|        |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| update | If TRUE then the arguments R.param and G.param are set to those in the asreml object supplied in object so that the values from the original model are used as starting values. If FALSE then asreml calls are evaluated, the only changes from the previous call being that (i) the model is fitted to simulated data and (ii) modifications specified via ... are made, except that changes cannot be made to any of the models. |
| trace  | If TRUE then partial iteration details are displayed when ASReml-R functions are invoked; if FALSE then no output is displayed.                                                                                                                                                                                                                                                                                                    |
| which  | The quantities from the simulated data set to be stored. Any combination of "response", "residuals" and "fitted", or "all". If residuals and/or fitted is specified, those for the analysis stored in object will be added to the data.frame nominated in the data argument of object and the modified data.frame added as a component named data in the list that is the value returned by the function.                          |
| units  | A character indicating whether the BLUPs for units are added to the residuals when this reserved factor is included in the random model. Possible values are addtoresiduals and ignore.                                                                                                                                                                                                                                            |
| ncores | A numeric specifying the number of cores to use in doing the simulations. In choosing a value for ncores, it is necessary to take into account other processes that are using parallel processing at the same time.                                                                                                                                                                                                                |
| ...    | Other arguments that are passed down to the function asreml. Changes to the models are not allowed. Other changes are dangerous and generally should be avoided.                                                                                                                                                                                                                                                                   |

## Details

Generate nsim sets of data and analyse them using asreml using the model in object, performing the generation and analysis of several sets in parallel. Note, if the analysis for a data set does not converge in maxiter iterations, it is discarded and a replacement data set generated. The value of maxiter can be specified in the call to simulate.asreml. The fitted values and residuals are extracted as required. If aom = TRUE when the simulated data are analysed, standardised conditional residuals are stored. If which includes residuals or fitted, the specified quantities for the observed data are added to the data.frame on which the fit in object is based.

## Value

A list with the following components whose presence depends on the setting of which:

1. **observed:** present if which includes residuals or fitted, in which case it will be the data.frame on which the fit in object is based, with residuals and/or fitted.
2. **data:** present if which includes data, a data.frame containing the simulated data sets.
3. **fitted:** present if which includes fitted, a data.frame containing the fitted values from the analyses of the simulated data sets.
4. **residuals:** present if which includes residuals, a data.frame containing the residuals from the analyses of the simulated data sets.

## Author(s)

Chris Brien

**See Also**

asreml, newfit.asreml, variofaces.asreml, plotVariofaces.data.frame, set.seed.

**Examples**

```
## Not run:
data(Wheat.dat)
current.asr <- asreml(yield ~ Rep + WithinColPairs + Variety,
                      random = ~ Row + Column + units,
                      residual = ~ ar1(Row):ar1(Column),
                      data=Wheat.dat)
current.asrt <- as.asrtests(current.asr, NULL, NULL)
current.asrt <- rmboundary.asrtests(current.asrt)
# Form variance matrix based on estimated variance parameters
s2 <- current.asr$sigma2
gamma.Row <- current.asr$gammas[1]
gamma.unit <- current.asr$gammas[2]
rho.r <- current.asr$gammas[4]
rho.c <- current.asr$gammas[5]
row.ar1 <- mat.ar1(order=10, rho=rho.r)
col.ar1 <- mat.ar1(order=15, rho=rho.c)
V <- gamma.Row * fac.sumop(Wheat.dat$Row) +
    gamma.unit * diag(1, nrow=150, ncol=150) +
    mat.dirprod(col.ar1, row.ar1)
V <- s2*V

#Produce residuals from 100 simulated data sets
resid <- simulate(current.asr, V=V, which="residuals", ncores = parallel::detectCores())

## End(Not run)
```

---

sort.alldiffs

*Sorts the components in an [alldiffs.object](#) according to the predicted values associated with a factor.*

---

**Description**

Sorts the rows of the components in an [alldiffs.object](#) (see [as.alldiffs](#)) that are data.frames and the rows and columns of those that are matrices according to the predicted values in the predictions component. These predicted values are generally obtained using `predict.asreml` by specifying a `classify` term comprised of one or more variables. Generally, the values associated with one variable are sorted in parallel within each combination of values of the other variables. When there is more than one variable in the `classify` term, the sorting is controlled using one or more of `sortFactor`, `sortParallelToCombo` and `sortOrder`. If there is only one variable in the `classify` then all components are sorted according to the order of the complete set of predictions.

Note that [renewClassify.alldiffs](#) is called after sorting to ensure that the order of the rows and columns of the components is in standard order for the new variable order.

**Usage**

```
## S3 method for class 'alldiffs'
sort(x, decreasing = FALSE, classify = NULL, sortFactor = NULL,
     sortParallelToCombo = NULL, sortNestingFactor = NULL,
     sortOrder = NULL, ...)
```

**Arguments**

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| x                   | An <code>alldiffs.object</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| decreasing          | A <code>logical</code> passed to <code>order</code> that determines whether the order is for increasing or decreasing magnitude of the predicted values.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| classify            | A <code>character</code> string giving the variables that define the margins of the multiway table that was predicted. Multiway tables are specified by forming an interaction type term from the classifying variables, that is, separating the variable names with the <code>:</code> operator. If <code>NULL</code> , it will be obtained from the <code>classify</code> attribute of the <code>as.alldiffs</code> object supplied through <code>x</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| sortFactor          | A <code>character</code> containing the name of the factor that indexes the set of predicted values that determines the sorting of the components. If there is only one variable in the <code>classify</code> term then <code>sortFactor</code> can be <code>NULL</code> and the order is defined by the complete set of predicted values. If there is more than one variable in the <code>classify</code> term then <code>sortFactor</code> must be set. In this case the <code>sortFactor</code> is sorted in the same order within each combination of the values of the <code>sortParallelToCombo</code> variables: the <code>classify</code> variables, excluding the <code>sortFactor</code> . There should be only one predicted value for each unique value of <code>sortFactor</code> within each set defined by a combination of the values of the <code>classify</code> variables, excluding the <code>sortFactor</code> factor. The order to use is determined by either <code>sortParallelToCombo</code> or <code>sortOrder</code> . |
| sortParallelToCombo | A <code>list</code> that specifies a combination of the values of the factors and numerics, excluding <code>sortFactor</code> , that are in <code>classify</code> . Each of the components of the supplied <code>list</code> is named for a <code>classify</code> variable and specifies a single value for it. The combination of this set of values will be used to define a subset of the predicted values whose order will define the order of <code>sortFactor</code> . Each of the other combinations of the values of the factors and numerics will be sorted in parallel. If <code>sortParallelToCombo</code> is <code>NULL</code> then the first value of each <code>classify</code> variable, except for the <code>sortFactor</code> factor, in the predictions component is used to define <code>sortParallelToCombo</code> . If there is only one variable in the <code>classify</code> then <code>sortParallelToCombo</code> is ignored.                                                                                             |
| sortNestingFactor   | A <code>character</code> containing the name of the factor that defines groups of the <code>sortFactor</code> within which the predicted values are to be ordered. If there is only one variable in the <code>classify</code> then <code>sortNestingFactor</code> is ignored.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| sortOrder           | A <code>character</code> vector whose length is the same as the number of levels for <code>sortFactor</code> in the predictions component of the <code>alldiffs.object</code> . It specifies the desired order of the levels in the reordered components of the <code>alldiffs.object</code> . The argument <code>sortParallelToCombo</code> is ignored.<br>The following creates a <code>sortOrder</code> vector <code>levs</code> for factor <code>f</code> based on the values in <code>x</code> : <code>levs &lt;- levels(f)[order(x)]</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| ...                 | further arguments passed to or from other methods. Not used at present.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

**Details**

The basic technique is to change the order of the levels of the `sortFactor` within the predictions and, if present, backtransforms components so that they are ordered for a subset of predicted values, one for each levels of the `sortFactor`. When the `classify` term consists of more than one variable then a subset of one combination of the values of variables other than the `sortFactor`, the `sortParallelToCombo` combination, must be chosen for determining the order of the `sortFactor`

levels. Then the sorting of the rows (and columns) will be in parallel within each combination of the values of sortParallelToCombo variables: the classify term, excluding the sortFactor.

### Value

The `alldiffs.object` supplied with the following components, if present, sorted: predictions, vcov, backtransforms, differences, p.differences and sed. Also, the sortFactor and sortOrder attributes are set.

### Author(s)

Chris Brien

### See Also

`as.alldiffs`, `allDifferences.data.frame`, `print.alldiffs`,  
`sort.predictions.frame`, `renewClassify.alldiffs`, `redoErrorIntervals.alldiffs`,  
`recalcLSD.alldiffs`, `predictPlus.asreml`, `predictPresent.asreml`

### Examples

```
##Halve WaterRunoff data to reduce time to execute
data(WaterRunoff.dat)
tmp <- subset(WaterRunoff.dat, Date == "05-18")

##Use asreml to get predictions and associated statistics

## Not run:
#Analyse pH
m1.asr <- asreml(fixed = pH ~ Benches + (Sources * (Type + Species)),
                 random = ~ Benches:MainPlots,
                 keep.order=TRUE, data= tmp)
current.asrt <- as.asrtests(m1.asr, NULL, NULL)
current.asrt <- as.asrtests(m1.asr)
current.asrt <- rmboundary(current.asrt)
m1.asr <- current.asrt$asreml.obj

#Get predictions and associated statistics
TS.diffs <- predictPlus.asreml(classify = "Sources:Type",
                              asreml.obj = m1.asr, tables = "none",
                              wald.tab = current.asrt$wald.tab,
                              present = c("Type", "Species", "Sources"))

#Use sort.alldiffs and save order for use with other response variables
TS.diffs.sort <- sort(TS.diffs, sortFactor = "Sources",
                     sortParallelToCombo = list(Type = "Control"))
sort.order <- attr(TS.diffs.sort, which = "sortOrder")

#Analyse Turbidity
m2.asr <- asreml(fixed = Turbidity ~ Benches + (Sources * (Type + Species)),
                 random = ~ Benches:MainPlots,
                 keep.order=TRUE, data= tmp)
current.asrt <- as.asrtests(m2.asr)
#Use pH sort.order to sort Turbidity alldiffs object
diffs2.sort <- predictPlus(m2.asr, classify = "Sources:Type",
                          pairwise = FALSE, error.intervals = "Stand",
```

```

        tables = "none", present = c("Type", "Species", "Sources"),
        sortFactor = "Sources",
        sortOrder = sort.order)

## End(Not run)

## Use lmeTest and emmeans to get predictions and associated statistics

if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  #Analyse pH
  m1.lmer <- lmerTest::lmer(pH ~ Benches + (Sources * (Type + Species)) +
                           (1|Benches:MainPlots),
                           data=na.omit(tmp))
  TS.emm <- emmeans::emmeans(m1.lmer, specs = ~ Sources:Type)
  TS.preds <- summary(TS.emm)
  den.df <- min(TS.preds$df, na.rm = TRUE)
  ## Modify TS.preds to be compatible with a predictions.frame
  TS.preds <- as.predictions.frame(TS.preds, predictions = "emmean",
                                   se = "SE", interval.type = "CI",
                                   interval.names = c("lower.CL", "upper.CL"))

  ## Form an all.diffs object and check its validity
  TS.vcov <- vcov(TS.emm)
  TS.diffs <- allDifferences(predictions = TS.preds,
                           classify = "Sources:Type",
                           vcov = TS.vcov, tdf = den.df)

  validAlldiffs(TS.diffs)

  #Use sort.alldiffs and save order for use with other response variables
  TS.diffs.sort <- sort(TS.diffs, sortFactor = "Sources",
                       sortParallelToCombo = list(Type = "Control"))
  sort.order <- attr(TS.diffs.sort, which = "sortOrder")

  #Analyse Turbidity
  m2.lmer <- lmerTest::lmer(Turbidity ~ Benches + (Sources * (Type + Species)) +
                           (1|Benches:MainPlots),
                           data=na.omit(tmp))
  TS.emm <- emmeans::emmeans(m2.lmer, specs = ~ Sources:Type)
  TS.preds <- summary(TS.emm)
  den.df <- min(TS.preds$df, na.rm = TRUE)
  ## Modify TS.preds to be compatible with a predictions.frame
  TS.preds <- as.predictions.frame(TS.preds, predictions = "emmean",
                                   se = "SE", interval.type = "CI",
                                   interval.names = c("lower.CL", "upper.CL"))

  ## Form an all.diffs object, sorting it using the pH sort.order and check its validity
  TS.vcov <- vcov(TS.emm)
  TS.diffs2.sort <- allDifferences(predictions = TS.preds,
                                   classify = "Sources:Type",
                                   vcov = TS.vcov, tdf = den.df,
                                   sortFactor = "Sources",
                                   sortOrder = sort.order)

  validAlldiffs(TS.diffs2.sort)
}

```

---

```
sort.predictions.frame
```

*Sorts a `predictions.frame` according to the predicted values associated with a factor.*

---

## Description

Sorts the rows of a `predictions.frame` according to the predicted values in the `predictions.frame`. These predicted values are generally obtained using `predict.asreml` by specifying a `classify` term comprised of one or more variables. Generally, the values associated with one variable are sorted in parallel within each combination of values of the other variables. When there is more than one variable in the `classify` term, the sorting is controlled using one or more of `sortFactor`, `sortParallelToCombo` and `sortOrder`. If there is only one variable in the `classify` then the `predictions.frame` is sorted according to the order of the complete set of predictions.

## Usage

```
## S3 method for class 'predictions.frame'
sort(x, decreasing = FALSE, classify, sortFactor = NULL,
     sortParallelToCombo = NULL, sortNestingFactor = NULL,
     sortOrder = NULL, ...)
```

## Arguments

- |                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>x</code>                   | A <code>predictions.frame</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <code>decreasing</code>          | A <code>logical</code> passed to <code>order</code> that determines whether the order is for increasing or decreasing magnitude of the predicted values.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>classify</code>            | A <code>character</code> string giving the variables that define the margins of the multiway table that was predicted. Multiway tables are specified by forming an interaction type term from the classifying variables, that is, separating the variable names with the <code>:</code> operator.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>sortFactor</code>          | A <code>character</code> containing the name of the factor that indexes the set of predicted values that determines the sorting of the components. If there is only one variable in the <code>classify</code> term then <code>sortFactor</code> can be <code>NULL</code> and the order is defined by the complete set of predicted values. If there is more than one variable in the <code>classify</code> term then <code>sortFactor</code> must be set. In this case the <code>sortFactor</code> is sorted in the same order within each combination of the values of the <code>sortParallelToCombo</code> variables: the <code>classify</code> variables, excluding the <code>sortFactor</code> . There should be only one predicted value for each unique value of <code>sortFactor</code> within each set defined by a combination of the values of the <code>classify</code> variables, excluding the <code>sortFactor</code> factor. The order to use is determined by either <code>sortParallelToCombo</code> or <code>sortOrder</code> . |
| <code>sortParallelToCombo</code> | A <code>list</code> that specifies a combination of the values of the factors and numerics, excluding <code>sortFactor</code> , that are in <code>classify</code> . Each of the components of the supplied <code>list</code> is named for a <code>classify</code> variable and specifies a single value for it. The combination of this set of values will be used to define a subset of the predicted values whose order will define the order of <code>sortFactor</code> . Each of the other combinations of the values of the factors and numerics will be sorted in parallel. If <code>sortParallelToCombo</code> is <code>NULL</code> then the first value of each <code>classify</code>                                                                                                                                                                                                                                                                                                                                                     |

variable, except for the sortFactor factor, in the predictions component is used to define sortParallelToCombo. If there is only one variable in the classify then sortParallelToCombo is ignored.

sortNestingFactor

A **character** containing the name of the factor that defines groups of the sortFactor within which the predicted values are to be ordered. If there is only one variable in the classify then sortNestingFactor is ignored.

sortOrder

A **character** vector whose length is the same as the number of levels for sortFactor in the predictions.frame. It specifies the desired order of the levels in the reordered the **predictions.frame**. The argument sortParallelToCombo is ignored.

The following creates a sortOrder vector levs for factor f based on the values in x: levs <- levels(f)[order(x)].

...

further arguments passed to or from other methods. Not used at present.

## Details

The basic technique is to change the order of the levels of the sortFactor within the predictions.frame so that they are ordered for a subset of predicted values, one for each levels of the sortFactor. When the classify term consists of more than one variable then a subset of one combination of the values of variables other than the sortFactor, the sortParallelToCombo combination, must be chosen for determining the order of the sortFactor levels. Then the sorting of the rows (and columns) will be in parallel within each combination of the values of sortParallelToCombo variables: the classify term, excluding the sortFactor.

## Value

The sorted **predictions.frame**. Also, the sortFactor and sortOrder attributes are set.

## Author(s)

Chris Brien

## See Also

**as.predictions.frame**, **print.predictions.frame**, **sort.alldiffs**, **predictPlus.asreml**, **predictPresent.asreml**

## Examples

```
##Halve WaterRunoff data to reduce time to execute
data(WaterRunoff.dat)
tmp <- subset(WaterRunoff.dat, Date == "05-18")

##Use asreml to get predictions and associated statistics

## Not run:
#Analyse pH
m1.asr <- asreml(fixed = pH ~ Benches + (Sources * (Type + Species)),
  random = ~ Benches:MainPlots,
  keep.order=TRUE, data= tmp)
current.asrt <- as.asrtests(m1.asr, NULL, NULL)
current.asrt <- as.asrtests(m1.asr)
current.asrt <- rmboundary(current.asrt)
```

```

m1.asr <- current.asrt$asreml.obj

#Get predictions and associated statistics
TS.diffs <- predictPlus.asreml(classify = "Sources:Type",
                              asreml.obj = m1.asr, tables = "none",
                              wald.tab = current.asrt$wald.tab,
                              present = c("Type", "Species", "Sources"))

#Use sort.predictions.frame and save order for use with other response variables
TS.preds <- TS.diffs$predictions
TS.preds.sort <- sort(TS.preds, sortFactor = "Sources",
                     sortParallelToCombo = list(Type = "Control"))
sort.order <- attr(TS.preds.sort, which = "sortOrder")

#Analyse Turbidity
m2.asr <- asreml(fixed = Turbidity ~ Benches + (Sources * (Type + Species)),
                random = ~ Benches:MainPlots,
                keep.order=TRUE, data= tmp)
current.asrt <- as.asrtests(m2.asr)
#Use pH sort.order to sort Turbidity alldiffs object
TS.diffs2 <- predictPlus(m2.asr, classify = "Sources:Type",
                        pairwise = FALSE, error.intervals = "Stand",
                        tables = "none", present = c("Type", "Species", "Sources"))
TS.preds2 <- TS.diffs2$predictions
TS.preds2.sort <- sort(TS.preds, sortFactor = "Sources", sortOrder = sort.order)

## End(Not run)

## Use lmeTest and emmeans to get predictions and associated statistics

if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  #Analyse pH
  m1.lmer <- lmerTest::lmer(pH ~ Benches + (Sources * (Type + Species)) +
                          (1|Benches:MainPlots),
                          data=na.omit(tmp))
  TS.emm <- emmeans::emmeans(m1.lmer, specs = ~ Sources:Type)
  TS.preds <- summary(TS.emm)
  den.df <- min(TS.preds$df, na.rm = TRUE)
  ## Modify TS.preds to be compatible with a predictions.frame
  TS.preds <- as.predictions.frame(TS.preds, predictions = "emmean",
                                  se = "SE", interval.type = "CI",
                                  interval.names = c("lower.CL", "upper.CL"))

  #Use sort.predictions.frame and save order for use with other response variables
  TS.preds.sort <- sort(TS.preds, classify = "Sources:Type", sortFactor = "Sources",
                      sortParallelToCombo = list(Type = "Control"))
  sort.order <- attr(TS.preds.sort, which = "sortOrder")

  #Analyse Turbidity
  m2.lmer <- lmerTest::lmer(Turbidity ~ Benches + (Sources * (Type + Species)) +
                          (1|Benches:MainPlots),
                          data=na.omit(tmp))
  TS.emm <- emmeans::emmeans(m2.lmer, specs = ~ Sources:Type)
  TS.preds <- summary(TS.emm)
  den.df <- min(TS.preds$df, na.rm = TRUE)

```



```
## Modify TS.preds to be compatible with a predictions.frame
TS.preds <- as.predictions.frame(TS.preds, predictions = "emmean",
                                se = "SE", interval.type = "CI",
                                interval.names = c("lower.CL", "upper.CL"))
}
```

---

|                 |                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------|
| subset.alldiffs | <i>Subsets the components in an <code>alldiffs.object</code> according to the supplied condition.</i> |
|-----------------|-------------------------------------------------------------------------------------------------------|

---

## Description

Subsets each of the components of an `alldiffs.object`. The subset is determined by applying the condition to the prediction component to determine which of its rows are to be included in the subset. Then, if present, this subset is applied to the rows of backtransforms and to the rows and columns of differences, `p.differences` and `sed` components. In addition, if `sed` is present, `recalcLSD.alldiffs` is called to recalculate the values in the `LSD.frame` stored in the `LSD` component, with any arguments supplied via the `...` argument passed to it.

The `select` argument of `subset` is not implemented, but can be achieved for variables in the `classify` using the `rmClassifyVars` argument.

## Usage

```
## S3 method for class 'alldiffs'
subset(x, subset = rep(TRUE, nrow(x$predictions)),
       rmClassifyVars = NULL, ...)
```

## Arguments

|                             |                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>x</code>              | An <code>alldiffs.object</code> .                                                                                                                                                                                                                                                                                                                                                   |
| <code>subset</code>         | A <code>logical</code> that determines rows of the predictions component of <code>x</code> to be included in the subset. By default all rows are included.                                                                                                                                                                                                                          |
| <code>rmClassifyVars</code> | A <code>character</code> that contains the names of the variables in the <code>classify</code> attribute of <code>x</code> that are to be removed from the predictions data.frame and the names of the dimensions of the other components of <code>x</code> . In doing this, the combinations of the remaining <code>classify</code> variables must uniquely index the predictions. |
| <code>...</code>            | further arguments passed to <code>recalcLSD.alldiffs</code> .                                                                                                                                                                                                                                                                                                                       |

## Value

An `alldiffs.object` with the following components of the supplied `alldiffs.object` subsetting, if present in the original object: `predictions`, `vcov`, `backtransforms`, `differences`, `p.differences` and `sed`. In addition, if `sed` is present, the `LSD.frame` in the `LSD` component will be recalculated.

## Author(s)

Chris Brien

**See Also**

[as.alldiffs](#), [allDifferences.data.frame](#), [print.alldiffs](#), [sort.alldiffs](#),  
[redoErrorIntervals.alldiffs](#), [recalcLSD.alldiffs](#),  
[predictPlus.asreml](#), [predictPresent.asreml](#)

**Examples**

```
data(WaterRunoff.dat)

##Use asreml to get predictions and associated statistics

## Not run:
asreml.options(keep.order = TRUE) #required for asreml-R4 only
current.asr <- asreml(fixed = pH ~ Benches + (Sources * (Type + Species)),
                     random = ~ Benches:MainPlots,
                     keep.order=TRUE, data= WaterRunoff.dat)
current.asrt <- as.asrtests(current.asr, NULL, NULL)
TS.diffs <- predictPlus.asreml(classify = "Sources:Type",
                              asreml.obj = current.asr, tables = "none",
                              wald.tab = current.asrt$wald.tab,
                              present = c("Type", "Species", "Sources"))

## End(Not run)

## Use lmeTest and emmeans to get predictions and associated statistics

if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(pH ~ Benches + (Sources * (Type + Species)) +
                          (1|Benches:MainPlots),
                          data=na.omit(WaterRunoff.dat))
  TS.emm <- emmeans::emmeans(m1.lmer, specs = ~ Sources:Type)
  TS.preds <- summary(TS.emm)
  den.df <- min(TS.preds$df, na.rm = TRUE)
  ## Modify TS.preds to be compatible with a predictions.frame
  TS.preds <- as.predictions.frame(TS.preds, predictions = "emmean",
                                  se = "SE", interval.type = "CI",
                                  interval.names = c("lower.CL", "upper.CL"))

  ## Form an all.diffs object and check its validity
  TS.vcov <- vcov(TS.emm)
  TS.diffs <- allDifferences(predictions = TS.preds, classify = "Sources:Type",
                           vcov = TS.vcov, tdf = den.df)

  validAlldiffs(TS.diffs)
}

## Plot p-values for predictions obtained using asreml or lmerTest
if (exists("TS.diffs"))
{
  ##Use subset.alldiffs to select a subset of the alldiffs object
  TS.diffs.subs <- subset(TS.diffs,
                         subset = grepl("R", Sources, fixed = TRUE) &
                         Type %in% c("Control", "Medicinal"))
}
```

---

|             |                                                                                                              |
|-------------|--------------------------------------------------------------------------------------------------------------|
| subset.list | Forms a <a href="#">list</a> that contains a subset of the components of the supplied <a href="#">list</a> . |
|-------------|--------------------------------------------------------------------------------------------------------------|

---

## Description

Select components of a [list](#) specified by a list of numbers or names, or by a logical indicating for each component of the list whether or not it is to be retained.

## Usage

```
## S3 method for class 'list'
subset(x, select = 1:length(x), ...)
```

## Arguments

|        |                                                                                                                                                                                                                                                                                                  |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| x      | An <a href="#">list</a> object.                                                                                                                                                                                                                                                                  |
| select | A <a href="#">numeric</a> or <a href="#">character</a> that lists or names the components of the <a href="#">list</a> x that are to be retained in the subset. It can also be a <a href="#">logical</a> that is the same length as x and indicates whether or not a component is to be retained. |
| ...    | further arguments passed to or from other methods. Not used at present.                                                                                                                                                                                                                          |

## Value

A [list](#) with just the subset of the components from x. If the components of x are named, then these names are retained in the subset [list](#).

## Author(s)

Chris Brien

## See Also

[subset.alldiffs](#)

## Examples

```
x <- list(1:3, letters[1:3], LETTERS[1:3])
y <- subset.list(x, select = c(1,3))
y <- subset.list(x, select = c(TRUE,FALSE,TRUE))

names(x) <- LETTERS[1:3]
y <- subset.list(x, select = c(1,3))
z <- subset.list(x, select = LETTERS[c(1,3)])
x <- list(1:3, letters[1:3], LETTERS[1:3])
names(x)[c(1,3)] <- LETTERS[c(1,3)]
z <- subset.list(x, select = c(1,2))
v <- subset.list(x)
```

---

|                     |                                                                                                                                           |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| testranfix.asrtests | <i>Tests for a single fixed or random term in model fitted using asreml and records the result in an <a href="#">asrtests.object</a>.</i> |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------|

---

## Description

Tests for a single term, using a REML ratio test (REMLRT) for a random term or based on Wald statistics for a fixed term. The term must be in the fitted model. A random term is removed from the model fit and a REMLRT is performed using [REMLRT.asreml](#). It compares the fit of the model in `asreml.obj` and the newly fitted model without the term. If the newly fitted model is retained, any boundary terms are then removed using [rmboundary.asrtests](#). For a fixed term, the probability of the Wald statistics is extracted from the pseudo-anova table produced by `wald.asreml`. If this is available in the [asrtests.object](#), it is used; otherwise `wald.asreml` is called to add it to the [asrtests.object](#). Whether nonsignificant terms are dropped is controlled by `drop.ran.ns` for random terms and `drop.fix.ns` for fixed terms. A row is added to the `test.summary.data.frame` for the term that is tested.

## Usage

```
## S3 method for class 'asrtests'
testranfix(asrtests.obj, term=NULL, alpha = 0.05,
  allow.unconverged = TRUE, allow.fixedcorrelation = TRUE,
  checkboundaryonly = FALSE,
  drop.ran.ns = TRUE, positive.zero = FALSE,
  bound.test.parameters = "none",
  bound.exclusions = c("F","B","S","C"), REMLDF = NULL,
  drop.fix.ns = FALSE, denDF="numeric", dDF.fault = "none",
  dDF.values = NULL, IClikelihood = "none",
  trace = FALSE, update = TRUE,
  set.terms = NULL, ignore.suffices = TRUE,
  bounds = "P", initial.values = NA, ...)
```

## Arguments

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>asrtests.obj</code>           | An <a href="#">asrtests.object</a> containing the components (i) <code>asreml.obj</code> , (ii) <code>wald.tab</code> , and (iii) <code>test.summary</code> .                                                                                                                                                                                                                                                   |
| <code>term</code>                   | A single model term that is valid in <code>asreml</code> , stored as a character. The names of fixed terms must match those in the <code>wald.tab</code> component of the <code>asrtests.obj</code> , while the names of random terms must match those in the <code>vparameters</code> component of the <code>asreml.obj</code> component in the <code>asrtests.obj</code> .                                    |
| <code>alpha</code>                  | The significance level for the test.                                                                                                                                                                                                                                                                                                                                                                            |
| <code>allow.unconverged</code>      | A logical indicating whether to accept a new model even when it does not converge. If <code>FALSE</code> , it will be checked whether convergence can be achieved with the removal of any boundary random terms; random terms will be retested if terms are removed. Also, if <code>FALSE</code> and the fit of the new model has converged, but that of the old model has not, the new model will be accepted. |
| <code>allow.fixedcorrelation</code> | A logical indicating whether to accept a new model even when it contains correlations in the model whose values have been designated as fixed, bound                                                                                                                                                                                                                                                            |

or singular. If FALSE and the new model contains correlations whose values have not been able to be estimated, the supplied `asrtests.obj` is returned. The fit in the `asreml.obj` component of the supplied `asrtests.obj` will also be tested and a warning issued if both fixed correlations are found in it and `allow.fixedcorrelation` is FALSE.

`checkboundaryonly`

If TRUE then boundary and singular terms are not removed by `rmboundary.asrtests`; a warning is issued instead.

`drop.ran.ns`

A logical indicating whether to drop a random term from the model when it is nonsignificant. Note that multiple terms specified using a single `asreml::at` function will only be dropped as a whole. If the term was specified using an `asreml::at` function with a single level, then it can be removed and either the level itself or its `numeric` position in the levels returned by the `levels` function can be specified in `term`.

`positive.zero`

Indicates whether the hypothesized values for the variance components being tested are on the boundary of the parameter space. For example, this is true for positively-constrained variance components that, under the reduced model, are zero. This argument does not need to be set if `bound.test.parameters` is set.

`bound.test.parameters`

Indicates whether for the variance components being tested, at least some of the hypothesized values are on the boundary of the parameter space. The possibilities are "none", "onlybound" and "one-and-one". The default is "none", although if it is set to "none" and `positive.zero` is TRUE then `bound.test.parameters` is taken to be "onlybound". When `bound.test.parameters` is set to "one-and-one", it signifies that there are two parameters being tested, one of which is bound and the other is not. For example, the latter is true for testing a covariance and a positively-constrained variance component that, under the reduced model, are zero.

`bound.exclusions`

A character specifying one or more bound (constraint) codes that will result in a variance parameter being excluded from the count of estimated variance parameters in using `REMLRT.asreml`. If set to NULL then none will be excluded.

`REMLDF`

A numeric giving the difference in the number of variance parameters whose estimates are not of the type specified in `bound.exclusions` for two models being compared in a REML ratio test using `REMLRT.asreml`. If NULL then this is determined from the information in the `asreml` object for the two models.

`drop.fix.ns`

A logical indicating whether to drop a fixed term from the model when it is nonsignificant. Note that multiple terms specified using a single `asreml::at` function can only be dropped as a whole. If the term was specified using an `asreml::at` function with a single level, then it can be removed and either the level itself or its `numeric` position in the levels returned by the `levels` function can be specified.

`denDF`

Specifies the method to use in computing approximate denominator degrees of freedom when `wald.asreml` is called. Can be none to suppress the computations, numeric for numerical methods, algebraic for algebraic methods or default, the default, to automatically choose numeric or algebraic computations depending on problem size. The denominator degrees of freedom are calculated according to Kenward and Roger (1997) for fixed terms in the dense part of the model.

`dDF.fault`

A character specifying the method to use to obtain substitute denominator degrees of freedom. when the numeric or algebraic methods produce faulty

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | <p>values, viz. NA, Inf or less than 0.01. Consistent with when no denDF are available, the default is "residual" and so the residual degrees of freedom from <code>asreml.obj\$nedf</code> are used. If <code>dDF.fault = "none"</code>, no substitute denominator degrees of freedom are employed; if <code>dDF.fault = "residual"</code>; if <code>dDF.fault = "maximum"</code>, the maximum of those denDF that are available, excluding that for the Intercept, is used; if all denDF are faulty, <code>asreml.obj\$nedf</code> is used. If <code>dDF.fault = "supplied"</code>, a vector of values for the denominator degrees of freedom is to be supplied in <code>dDF.values</code>. Any other setting is ignored and a warning message produced. Generally, substituting these degrees of freedom is anticonservative in that it is likely that the degrees of freedom used will be too large.</p> |
| <code>dDF.values</code>      | <p>A vector of values to be used when <code>dDF.fault = "supplied"</code>. Its values will be used when denDF in a test for a fixed effect is NA. This vector must be the same length as the number of fixed terms, including (Intercept) whose value could be NA.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>IClikelihood</code>    | <p>A character that controls both the occurrence and the type of likelihood for information criterion in the <code>test.summary</code> of the new <a href="#">asrtests.object</a>. If none, none are included. Otherwise, if REML and family is set to <code>asr_guassian</code> (the default), then the AIC and BIC based on the Restricted Maximum Likelihood are included; if full and family is set to <code>asr_guassian</code>, then the AIC and BIC based on the full likelihood, evaluated using REML estimates, are included. If family is <code>asr_binomial</code> or <code>asr_poisson</code>, with dispersion set to 1, the deviance is extracted from object and used to calculate the AIC and BIC. (See also <a href="#">infoCriteria.asreml</a>.)</p>                                                                                                                                        |
| <code>trace</code>           | <p>If TRUE then partial iteration details are displayed when ASReML-R functions are invoked; if FALSE then no output is displayed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>update</code>          | <p>If TRUE, and <code>set.terms</code> is NULL, then <a href="#">newfit.asreml</a> is called to fit the model to be tested, using the values of the variance parameters stored in the <code>asreml.object</code>, that is stored in <code>asrtests.obj</code>, as starting values. If FALSE or <code>set.terms</code> is not NULL, then <a href="#">newfit.asreml</a> will not use the stored variance parameter values as starting values when fitting the new model, the only modifications being (i) those for the supplied terms and (ii) those specified via ...</p>                                                                                                                                                                                                                                                                                                                                    |
| <code>set.terms</code>       | <p>A character vector specifying the terms that are to have bounds and/or initial values set prior to fitting. The names must match those in the <code>vparameters</code> component of the <code>asreml.obj</code> component in the new <a href="#">asrtests.object</a>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>ignore.suffices</code> | <p>A logical vector specifying whether the suffices of the <code>asreml</code>-assigned names of the variance terms (i.e. the information to the right of an "!", other than "R!") is to be ignored in matching elements of terms. If TRUE for an element of terms, the suffices are stripped from the <code>asreml</code>-assigned names. If FALSE for an element of terms, the element must exactly match an <code>asreml</code>-assigned name for a variance term. This vector must be of length one or the same length as terms. If it is of length one then the same action is applied to the <code>asreml</code>-assigned suffices for all the terms in terms.</p>                                                                                                                                                                                                                                     |
| <code>bounds</code>          | <p>A <a href="#">character</a> vector specifying the bounds to be applied to the terms specified in <code>set.terms</code>. This vector must be of length one or the same length as <code>set.terms</code>. If it is of length one then the same constraint is applied to all the terms in <code>set.terms</code>. If any of the bounds are equal to NA then they are left unchanged for those terms.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>initial.values</code>  | <p>A character vector specifying the initial values for the terms specified in terms. This vector must be of length one or the same length as terms. If it is of length</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

one then the same initial value is applied to all the terms in terms. If any of the initial.values are equal to NA then they are left unchanged for those terms.

... Further arguments passed to asreml, wald.asreml and [as.asrtests](#).

## Value

An [asrtests.object](#) containing the components (i) asreml.obj, (ii) wald.tab, and (iii) test.summary. If the term is not in the model, then the supplied asreml object will be returned. Also, reml.test will have the likelihood ratio and the p-value set to NA and the degrees of freedom to zero. Similarly, the row of test.summary for the term will have its name, DF set to NA, p-value set to NA, and action set to Absent.

## Author(s)

Chris Brien

## References

Kenward, M. G., & Roger, J. H. (1997). Small sample inference for fixed effects from restricted maximum likelihood. *Biometrics*, **53**, 983-997.

## See Also

[asremlPlus-package](#), [as.asrtests](#), [chooseModel.asrtests](#), [REMLRT.asreml](#), [rmboundary.asrtests](#), [newfit.asreml](#), [changeModelOnIC.asrtests](#), [changeTerms.asrtests](#), [reparamSigDevn.asrtests](#)

## Examples

```
## Not run:
data(Wheat.dat)
current.asr <- asreml(yield ~ Rep + WithinColPairs + Variety,
                     random = ~ Row + Column + units,
                     residual = ~ ar1(Row):ar1(Column),
                     data=Wheat.dat)
current.asrt <- as.asrtests(current.asr, NULL, NULL)
current.asrt <- rmboundary(current.asrt)
# Test nugget term
current.asrt <- testranfix(current.asrt, "units", positive=TRUE)

## End(Not run)
```

---

`testresidual.asrtests` *Fits a new residual formula, tests whether the change is significant and records the result in an [asrtests.object](#).*

---

## Description

Fits a new residual formula using `asreml-R4` (replaces the `rcov` formula of `asreml-R3`) and tests whether the change is significant. If `simpler = FALSE` the model to be fitted must be more complex than the one whose fit has been stored in `asrtests.obj`. That is, the new model must have more parameters. However, if `simpler = TRUE` the model to be fitted must be simpler than the one whose fit has been stored in `asrtests.obj` in that it must have fewer parameters. Any boundary terms are removed using `rmboundary.asrtests`, which may mean that the models are not nested. The test is a REML likelihood ratio test that is performed using `REMLRT.asreml`, which is only valid if the models are nested. It compares the newly fitted model with the fit of the model in `asrtest.obj`. If the two models have the same number of variance parameters, then no change is made to the residual. A row is added to the `test.summary.data.frame` using the supplied label.

## Usage

```
## S3 method for class 'asrtests'
testresidual(asrtests.obj, terms=NULL, label = "R model",
             simpler = FALSE, alpha = 0.05,
             allow.unconverged = TRUE, allow.fixedcorrelation = TRUE,
             checkboundaryonly = FALSE, positive.zero = FALSE,
             bound.test.parameters = "none",
             bound.exclusions = c("F", "B", "S", "C"), REMLDF = NULL,
             denDF="numeric", IClikelihood = "none",
             update = TRUE, trace = FALSE,
             set.terms = NULL, ignore.suffices = TRUE,
             bounds = "P", initial.values = NA, ...)
```

## Arguments

- |                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>asrtests.obj</code>           | an <a href="#">asrtests.object</a> for a fitted model that is a list containing the componets (i) <code>asreml.obj</code> , (ii) <code>wald.tab</code> (iii) <code>test.summary</code> .                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>terms</code>                  | A model for the residual argument in <code>asreml-R4</code> (the <code>rcov</code> formula in older versions of <code>asreml</code> ), stored as a character. To remove the model, enter <code>"-(.)"</code> .                                                                                                                                                                                                                                                                                                                                                                      |
| <code>label</code>                  | A character string to use as the label in <code>test.summary</code> and which indicates what is being tested.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>simpler</code>                | A logical indicating whether the new model to be fitted is simpler than the already fitted model whose fit is stored in <code>asrtests.obj</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>alpha</code>                  | The significance level for the test.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <code>allow.unconverged</code>      | A logical indicating whether to accept a new model even when it does not converge. If <code>FALSE</code> and the fit of the new model does not converge, the supplied <code>asreml</code> object is returned. Also, if <code>FALSE</code> and the fit of the new model has converged, but that of the old model has not, the new model will be accepted.                                                                                                                                                                                                                            |
| <code>allow.fixedcorrelation</code> | A logical indicating whether to accept a new model even when it contains correlations in the model whose values have been designated as fixed, bound or singular. If <code>FALSE</code> and the new model contains correlations whose values have not been able to be estimated, the supplied <code>asrtests.obj</code> is returned. The fit in the <code>asreml.obj</code> component of the supplied <code>asrtests.obj</code> will also be tested and a warning issued if both fixed correlations are found in it and <code>allow.fixedcorrelation</code> is <code>FALSE</code> . |



|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| checkboundaryonly     | If TRUE then boundary and singular terms are not removed by <code>rmboundary.asrtests</code> ; a warning is issued instead.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| positive.zero         | Indicates whether the hypothesized values for the variance components being tested are on the boundary of the parameter space. For example, this is true for positively-constrained variance components that, under the reduced model, are zero. This argument does not need to be set if <code>bound.test.parameters</code> is set.                                                                                                                                                                                                                                                                                                                                                                    |
| bound.test.parameters | Indicates whether for the variance components being tested, at least some of the hypothesized values are on the boundary of the parameter space. The possibilities are "none", "onlybound" and "one-and-one". The default is "none", although if it is set to "none" and <code>positive.zero</code> is TRUE then <code>bound.test.parameters</code> is taken to be "onlybound". When <code>bound.test.parameters</code> is set to "one-and-one", it signifies that there are two parameters being tested, one of which is bound and the other is not. For example, the latter is true for testing a covariance and a positively-constrained variance component that, under the reduced model, are zero. |
| bound.exclusions      | A <a href="#">character</a> specifying one or more bound (constraint) codes that will result in a variance parameter being excluded from the count of estimated variance parameters in using <code>REMLRT.asreml</code> . If set to NULL then none will be excluded.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| REMLDF                | A numeric giving the difference in the number of variance parameters whose estimates are not of the type specified in <code>bound.exclusions</code> for two models being compared in a REML ratio test using <code>REMLRT.asreml</code> . If NULL then this is determined from the information in the <code>asreml</code> object for the two models.                                                                                                                                                                                                                                                                                                                                                    |
| denDF                 | Specifies the method to use in computing approximate denominator degrees of freedom when <code>wald.asreml</code> is called. Can be none to suppress the computations, numeric for numerical methods, algebraic for algebraic methods or default, the default, to automatically choose numeric or algebraic computations depending on problem size. The denominator degrees of freedom are calculated according to Kenward and Roger (1997) for fixed terms in the dense part of the model.                                                                                                                                                                                                             |
| IClikelihood          | A character that controls both the occurrence and the type of likelihood for information criterion in the <code>test.summary</code> of the new <a href="#">asrtests.object</a> . If none, none are included. Otherwise, if REML, then the AIC and BIC based on the Restricted Maximum Likelihood are included; if full, then the AIC and BIC based on the full likelihood, evaluated using REML estimates, are included. (See also <a href="#">infoCriteria.asreml</a> .)                                                                                                                                                                                                                               |
| update                | If TRUE, and <code>set.terms</code> is NULL, then <code>newfit.asreml</code> is called to fit the model to be tested, using the values of the variance parameters stored in the <code>asreml.object</code> , that is stored in <code>asrtests.obj</code> , as starting values. If FALSE or <code>set.terms</code> is not NULL, then <code>newfit.asreml</code> will not use the stored variance parameter values as starting values when fitting the new model, the only modifications being (i) the residual (rcov) model is that specified in <code>terms</code> (ii) those specified via ...                                                                                                         |
| trace                 | If TRUE then partial iteration details are displayed when ASReML-R functions are invoked; if FALSE then no output is displayed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| set.terms             | A <a href="#">character</a> vector specifying the terms that are to have bounds and/or initial values set prior to fitting. The names must match those in the <code>vparameters</code> component of the <code>asreml.obj</code> component in the new <a href="#">asrtests.object</a> .                                                                                                                                                                                                                                                                                                                                                                                                                  |

`ignore.suffices`

A [logical](#) vector specifying whether the suffices of the `asreml`-assigned names of the variance terms (i.e. the information to the right of an "!", other than "R!") is to be ignored in matching elements of `terms`. If TRUE for an element of `terms`, the suffices are stripped from the `asreml`-assigned names. If FALSE for an element of `terms`, the element must exactly match an `asreml`-assigned name for a variance term. This vector must be of length one or the same length as `terms`. If it is of length one then the same action is applied to the `asreml`-assigned suffices for all the terms in `terms`.

`bounds`

A [character](#) vector specifying the bounds to be applied to the terms specified in `set.terms`. This vector must be of length one or the same length as `set.terms`. If it is of length one then the same constraint is applied to all the terms in `set.terms`. If any of the bounds are equal to NA then they are left unchanged for those terms.

`initial.values`

A character vector specifying the initial values for the terms specified in `terms`. This vector must be of length one or the same length as `terms`. If it is of length one then the same initial value is applied to all the terms in `terms`. If any of the `initial.values` are equal to NA then they are left unchanged for those terms.

...

Further arguments passed to `asreml`, `wald.asreml` and [as.asrtests](#).

## Value

An [asrtests.object](#) containing the components (i) `asreml.obj`, (ii) `wald.tab`, and (iii) `test.summary`. If the term is not in the model, then the supplied `asreml.obj` will be returned. Also, `reml.test` will have the likelihood ratio and the p-value set to NA and the degrees of freedom to zero. Similarly, the row of `test.summary` for the term will have its name, a p-value set to NA, and action set to Absent.

## Author(s)

Chris Brien

## References

Kenward, M. G., & Roger, J. H. (1997). Small sample inference for fixed effects from restricted maximum likelihood. *Biometrics*, **53**, 983-997.

## See Also

[asremlPlus-package](#), [as.asrtests](#), [changeTerms.asrtests](#), [chooseModel.asrtests](#), [REMLRT.asreml](#), [rmboundary.asrtests](#), [newfit.asreml](#), [testswaprان.asrtests](#), [changeModelOnIC.asrtests](#), [changeTerms.asrtests](#), [reparamSigDevn.asrtests](#)

## Examples

```
## Not run:
data(Wheat.dat)
current.asr <- asreml(yield ~ Rep + WithinColPairs + Variety,
                     random = ~ Row + Column + units,
                     residual = ~ ar1(Row):ar1(Column),
                     data=Wheat.dat)
current.asrt <- as.asrtests(current.asr, NULL, NULL)
current.asrt <- rmboundary(current.asrt)
```

```
# Test Row autocorrelation
current.asrt <- testresidual(current.asrt, "~ Row:ar1(Column)",
                           label="Row autocorrelation", simpler=TRUE)
print(current.asrt)

## End(Not run)
```

---

|                      |                                                                                                                                                                                                                               |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| testswapran.asrtests | <i>Tests, using a REMLRT, the significance of the difference between the current random model and one in which oldterms are dropped and newterms are added. The result is recorded in an <a href="#">asrtests.object</a>.</i> |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

---

## Description

Fits a new random model using `asreml` by removing `oldterms` and adding `newterms`. If `simpler = FALSE` the model to be fitted must be more complex than the one whose fit has been stored in `asrtests.obj`. That is, the new model must have more parameters. However, if `simpler = TRUE` the model to be fitted must be simpler than the one whose fit has been stored in `asrtests.obj` in that it must have fewer parameters. The test is a REML ratio test that is performed using [REMLRT.asreml](#), **which is only valid if the models are nested**. It compares the newly fitted model with the fit of the model in `asrtests.obj`. A row is added to the `test.summary` data frame using the supplied label. If the newly fitted model is retained, any boundary terms are then removed using [rmboundary.asrtests](#). If the models are not nested, then using [changeModelOnIC.asrtests](#) may be the more appropriate approach for comparing models.

## Usage

```
## S3 method for class 'asrtests'
testswapran(asrtests.obj, oldterms = NULL, newterms = NULL,
            label = "Swap in random model", simpler = FALSE, alpha = 0.05,
            allow.unconverged = TRUE, allow.fixedcorrelation = TRUE,
            checkboundaryonly = FALSE,
            positive.zero = FALSE, bound.test.parameters = "none",
            bound.exclusions = c("F", "B", "S", "C"), REMLDF = NULL,
            denDF="numeric", IClikelihood = "none",
            trace = FALSE, update = TRUE,
            set.terms = NULL, ignore.suffices = TRUE,
            bounds = "P", initial.values = NA, ...)
```

## Arguments

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>asrtests.obj</code> | an <a href="#">asrtests.object</a> for a fitted model that is a list containing the components (i) <code>asreml.obj</code> , (ii) <code>wald.tab</code> (iii) <code>test.summary</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>oldterms</code>     | Terms, stored as a character, that are to be removed from the random model using <code>asreml</code> . The names of the terms must match those in the <code>vparameters</code> component of the <code>asreml.obj</code> component in <code>asrtests.obj</code> . Note that multiple terms specified using a single <code>asreml::at</code> function can only be dropped as a whole. If the term was specified using an <code>asreml::at</code> function with a single level, then it can be removed and either the level itself or its <a href="#">numeric</a> position in the levels returned by the <a href="#">levels</a> function can be specified. |

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| newterms               | Terms, stored as a character, that are to be added to the random model using <code>asreml</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| simpler                | A logical indicating whether the new model to be fitted. after the changes made as a result of swapping <code>oldterms</code> for <code>newterms</code> , is simpler than the already fitted model whose fit is stored in <code>asrtests.obj</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| alpha                  | The significance level for the test.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| allow.unconverged      | A logical indicating whether to accept a new model even when it does not converge. If FALSE and the fit of the new model does not converge, the supplied <code>asrtests.obj</code> is returned. Also, if FALSE and the fit of the new model has converged, but that of the old model has not, the new model will be accepted.                                                                                                                                                                                                                                                                                                                                                                           |
| allow.fixedcorrelation | A logical indicating whether to accept a new model even when it contains correlations in the model whose values have been designated as fixed, bound or singular. If FALSE and the new model contains correlations whose values have not been able to be estimated, the supplied <code>asrtests.obj</code> is returned. The fit in the <code>asreml.obj</code> component of the supplied <code>asrtests.obj</code> will also be tested and a warning issued if both fixed correlations are found in it and <code>allow.fixedcorrelation</code> is FALSE.                                                                                                                                                |
| checkboundaryonly      | If TRUE then boundary and singular terms are not removed by <code>rmboundary.asrtests</code> ; a warning is issued instead.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| label                  | A character string to use as the label in <code>test.summary</code> and which indicates what is being tested.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| positive.zero          | Indicates whether the hypothesized values for the variance components being tested are on the boundary of the parameter space. For example, this is true for positively-constrained variance components that, under the reduced model, are zero. This argument does not need to be set if <code>bound.test.parameters</code> is set.                                                                                                                                                                                                                                                                                                                                                                    |
| bound.test.parameters  | Indicates whether for the variance components being tested, at least some of the hypothesized values are on the boundary of the parameter space. The possibilities are "none", "onlybound" and "one-and-one". The default is "none", although if it is set to "none" and <code>positive.zero</code> is TRUE then <code>bound.test.parameters</code> is taken to be "onlybound". When <code>bound.test.parameters</code> is set to "one-and-one", it signifies that there are two parameters being tested, one of which is bound and the other is not. For example, the latter is true for testing a covariance and a positively-constrained variance component that, under the reduced model, are zero. |
| bound.exclusions       | A character specifying one or more bound (constraint) codes that will result in a variance parameter being excluded from the count of estimated variance parameters in using <code>REMLRT.asreml</code> . If set to NULL then none will be excluded.                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| REMLDF                 | A numeric giving the difference in the number of variance parameters whose estimates are not of the type specified in <code>bound.exclusions</code> for two models being compared in a REML ratio test using <code>REMLRT.asreml</code> . If NULL then this is determined from the information in the <code>asreml</code> object for the two models.                                                                                                                                                                                                                                                                                                                                                    |
| denDF                  | Specifies the method to use in computing approximate denominator degrees of freedom when <code>wald.asreml</code> is called. Can be none to suppress the computations, numeric for numerical methods, algebraic for algebraic methods or                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | default, the default, to automatically choose numeric or algebraic computations depending on problem size. The denominator degrees of freedom are calculated according to Kenward and Roger (1997) for fixed terms in the dense part of the model.                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| IClikelihood    | A character that controls both the occurrence and the type of likelihood for information criterion in the <code>test.summary</code> of the new <code>asrtests.object</code> . If none, none are included. Otherwise, if REML, then the AIC and BIC based on the Restricted Maximum Likelihood are included; if full, then the AIC and BIC based on the full likelihood, evaluated using REML estimates, are included. (See also <code>infoCriteria.asreml</code> .)                                                                                                                                                                                                                               |
| trace           | If TRUE then partial iteration details are displayed when ASReML-R functions are invoked; if FALSE then no output is displayed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| update          | If TRUE, and <code>set.terms</code> is NULL, then <code>newfit.asreml</code> is called to fit the model to be tested, using the values of the variance parameters stored in the <code>asreml.object</code> , that is stored in <code>asrtests.obj</code> , as starting values. If FALSE or <code>set.terms</code> is not NULL, then <code>newfit.asreml</code> will not use the stored variance parameter values as starting values when fitting the new model, the only modifications being (i) for the supplied <code>oldterms</code> and (ii) those specified via . . .                                                                                                                        |
| set.terms       | A character vector specifying the terms that are to have bounds and/or initial values set prior to fitting. The names must match those in the <code>vparameters</code> component of the <code>asreml.obj</code> component in the <code>asrtests.object</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| ignore.suffices | A logical vector specifying whether the suffices of the <code>asreml</code> -assigned names of the variance terms (i.e. the information to the right of an "!", other than "R!") is to be ignored in matching elements of terms. If TRUE for an element of terms, the suffices are stripped from the <code>asreml</code> -assigned names. If FALSE for an element of terms, the element must exactly match an <code>asreml</code> -assigned name for a variance term. This vector must be of length one or the same length as <code>terms</code> . If it is of length one then the same action is applied to the <code>asreml</code> -assigned suffices for all the terms in <code>terms</code> . |
| bounds          | A <code>character</code> vector specifying the bounds to be applied to the terms specified in <code>set.terms</code> . This vector must be of length one or the same length as <code>set.terms</code> . If it is of length one then the same constraint is applied to all the terms in <code>set.terms</code> . If any of the bounds are equal to NA then they are left unchanged for those terms.                                                                                                                                                                                                                                                                                                |
| initial.values  | A character vector specifying the initial values for the terms specified in <code>terms</code> . This vector must be of length one or the same length as <code>terms</code> . If it is of length one then the same initial value is applied to all the terms in <code>terms</code> . If any of the <code>initial.values</code> are equal to NA then they are left unchanged for those terms.                                                                                                                                                                                                                                                                                                      |
| ...             | Further arguments passed to <code>asreml</code> , <code>wald.asreml</code> and <code>as.asrtests</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

### Value

An `asrtests.object` for a fitted model that is a list containing the components (i) `asreml.obj`, (ii) `wald.tab` (iii) `test.summary`. If the term is not in the model, then the supplied `asreml` object will be returned. Also, `reml.test` will have the likelihood ratio and the p-value set to NA and the degrees of freedom to zero. Similarly, the row of `test.summary` for the term will have its name, a p-value set to NA, and action set to Absent.

### Author(s)

Chris Brien

## References

Kenward, M. G., & Roger, J. H. (1997). Small sample inference for fixed effects from restricted maximum likelihood. *Biometrics*, **53**, 983-997.

## See Also

[as.asrtests](#), [chooseModel.asrtests](#), [REMLRT.asreml](#), [rmboundary.asrtests](#),  
[newfit.asreml](#), [testresidual.asrtests](#), [changeModelOnIC.asrtests](#),  
[changeTerms.asrtests](#), [reparamSigDevn.asrtests](#)

## Examples

```
## Not run:
current.asrt <- testswapran(current.asrt, oldterms = "str(~ Cart/xDays, ~us(2):id(184))",
                           newterms = "Cart/xDays", pos = FALSE,
                           label = "Intercept/Slope correlation",
                           simpler = TRUE)

print(current.asrt)

## End(Not run)
```

---

validAlldiffs

*Checks that an object is a valid alldiffs object.*

---

## Description

Checks that an object is an [alldiffs.object](#) of S3-class alldiffs containing the components asreml.obj, wald.tab and test.summary.

## Usage

```
validAlldiffs(object)
```

## Arguments

object            an [alldiffs.object](#).

## Value

TRUE or a character describing why the object is not a valid [alldiffs.object](#).

## Author(s)

Chris Brien

## See Also

[alldiffs.object](#), [is.alldiffs](#), [as.alldiffs](#),  
[validPredictionsFrame](#), [validAsrtests](#)

## Examples

```
data(Oats.dat)

## Use lmerTest and emmeans to get predictions and associated statistics
if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(Yield ~ Nitrogen*Variety + (1|Blocks/Wplots),
                          data=Oats.dat)
  Var.emm <- emmeans::emmeans(m1.lmer, specs = ~ Nitrogen:Variety)
  Var.preds <- summary(Var.emm)
  den.df <- min(Var.preds$df)
  ## Modify Var.preds to be compatible with a predictions.frame
  Var.preds <- as.predictions.frame(Var.preds, predictions = "emmean",
                                   se = "SE", interval.type = "CI",
                                   interval.names = c("lower.CL", "upper.CL"))

  Var.vcov <- vcov(Var.emm)
  Var.sed <- NULL

  ## Form an all.diffs object
  Var.diffs <- as.alldiffs(predictions = Var.preds, classify = "Nitrogen:Variety",
                          sed = Var.sed, vcov = Var.vcov, tdf = den.df)

  ## check the validity of Var.diffs
  validAlldiffs(Var.diffs)
}
```

---

validAsrtests

*Checks that an object is a valid asrtests object.*


---

## Description

Checks that an object is an [asrtests.object](#) of S3-class asrtests containing the components asreml.obj, wald.tab and test.summary.

## Usage

```
validAsrtests(object)
```

## Arguments

object            an [asrtests.object](#).

## Value

TRUE or a character describing why the object is not a valid [asrtests.object](#).

## Author(s)

Chris Brien

**See Also**

[asrtests.object](#), [is.asrtests](#), [as.asrtests](#),  
[validPredictionsFrame](#), [validAlldiffs](#)

**Examples**

```
## Not run:
library(dae)
library(asreml)
library(asremlPlus)
## use ?Wheat.dat for data set details
data(Wheat.dat)

# Fit initial model
current.asr <- asreml(yield ~ Rep + WithinColPairs + Variety,
                     random = ~ Row + Column + units,
                     residual = ~ ar1(Row):ar1(Column),
                     data=Wheat.dat)

# Load current fit into an asrtests object
current.asrt <- as.asrtests(current.asr, NULL, NULL)

# check validity of current.asrt
validAsrtests(current.asrt)

## End(Not run)
```

---

`validPredictionsFrame` *Checks that an object is a valid predictions.frame.*

---

**Description**

Checks that an object is a [predictions.frame](#) of S3-class `data.frame` that contains the columns `predicted.value`, `standard.error` and `est.status`.

**Usage**

```
validPredictionsFrame(object)
```

**Arguments**

`object`                    an [predictions.frame](#).

**Value**

TRUE or a character describing why the object is not a valid [predictions.frame](#).

**Author(s)**

Chris Brien



**See Also**

[predictions.frame](#), [is.predictions.frame](#), [as.predictions.frame](#),  
[validAsrtests](#), [validAlldiffs](#)

**Examples**

```
data(Oats.dat)

## Use asreml to get predictions and associated statistics

## Not run:
m1.asr <- asreml(Yield ~ Nitrogen*Variety,
                 random=~Blocks/Wplots,
                 data=Oats.dat)
current.asrt <- as.asrtests(m1.asr)
Var.pred <- asreml::predict.asreml(m1.asr, classify="Nitrogen:Variety",
                                  sed=TRUE)
if (getASRemlVersionLoaded(nchar = 1) == "3")
  Var.pred <- Var.pred$predictions
Var.preds <- as.predictions.frame(Var.pred$pvals, se = "std.error",
                                 est.status = "status")

## End(Not run)

## Use lmerTest and emmeans to get predictions and associated statistics
if (requireNamespace("lmerTest", quietly = TRUE) &
    requireNamespace("emmeans", quietly = TRUE))
{
  m1.lmer <- lmerTest::lmer(Yield ~ Nitrogen*Variety + (1|Blocks/Wplots),
                           data=Oats.dat)
  Var.emm <- emmeans::emmeans(m1.lmer, specs = ~ Nitrogen:Variety)
  Var.preds <- summary(Var.emm)
  Var.preds <- as.predictions.frame(Var.preds, predictions = "emmean",
                                   se = "SE", interval.type = "CI",
                                   interval.names = c("lower.CL", "upper.CL"))
}

if (exists("Var.preds"))
{
  ## Check the class and validity of the predictions.frame
  is.predictions.frame(Var.preds)
  validPredictionsFrame(Var.preds)
}
```

---

variofaces.asreml

*Plots empirical variogram faces, including envelopes, as described by  
Stefanova, Smith & Cullis (2009).*

---

**Description**

A function that produces a plot for each face of an empirical 2D variogram based on residuals produced after the fitting of a model using the function `asreml`. It also adds envelopes to the plot by simulating data sets in parallel from a multivariate normal distribution with expectation equal

to the fitted values obtained from the fixed and spline terms and variance matrix equal to the fitted variance matrix (Stefanova, Smith & Cullis, 2009). The plot is controlled by the residual model, which must consist of two factors corresponding to the two physical dimensions underlying the data. It can also have a third term involving the `at` or `dsum` function that defines sections of the data, such as experiments in different environments. In this case, the two variogram faces are produced for each section.

## Usage

```
## S3 method for class 'asreml'
variofaces(asreml.obj, means=NULL, V=NULL,
           sections = NULL, row.factor = NULL, col.factor = NULL,
           nsim=100, seed = NULL,
           extra.matrix = NULL, ignore.terms = NULL, fixed.spline.terms = NULL,
           bound.exclusions = c("F","B","S","C"), tolerance=1E-10,
           units = "ignore", update = TRUE, trace = FALSE,
           graphics.device=NULL, ncores = 2, ...)
```

## Arguments

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>asreml.obj</code>   | An <code>asreml</code> object from a call to <code>asreml</code> in which the data argument has been set.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>means</code>        | The vector of means to be used in generating simulated data sets. If it is <code>NULL</code> , the fitted values based on object are used. It must be the same length as the response variable for object.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>V</code>            | The fitted variance matrix, i.e. having the appropriate pattern and values given the model fitted to the observed data and the estimates of the parameters obtained. If <code>V</code> is <code>NULL</code> then <code>estimateV.asreml</code> is called to obtain it from <code>asreml.obj</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>sections</code>     | A single character string that specifies the name of the column in the <code>data.frame</code> that contains the <code>factor</code> that identifies different sections of the data to which separate spatial models have been fitted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <code>row.factor</code>   | A single character string nominating a <code>factor</code> that indexes the rows of a grid that are one dimension of a spatial correlation model. The <code>factor</code> must a column in the <code>data.frame</code> stored in the <code>asreml.obj</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>col.factor</code>   | A single character string nominating a <code>factor</code> that indexes the columns of a grid that are one dimension of a spatial correlation model. The <code>factor</code> must a column in the <code>data.frame</code> stored in the <code>asreml.obj</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>nsim</code>         | The number of data sets to be simulated in obtaining the envelopes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>seed</code>         | A single value, interpreted as an integer, that specifies the starting value of the random number generator. The "L'Ecuyer-CMRG" random generator is used and <code>nextRNGStream</code> is used to seed each core from the original seed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>extra.matrix</code> | A matrix of order equal to the number of observations that is to be added to the variance matrix, the latter based on the information in <code>asreml.obj</code> . It is assumed that the sigma-parameterized values of the variance parameter estimates, such as is given in the <code>varcomp</code> component of <code>summary.asreml</code> , have been used in calculating <code>extra.matrix</code> ; the values in the <code>vparameters</code> component of <code>G.param</code> and <code>R.param</code> may be either gamma- or sigma-parameterized. The argument <code>extra.matrix</code> can be used in conjunction with <code>ignore.terms</code> as a workaround to include components of the variance matrix for variance functions that have not been implemented in <code>estimateV</code> . |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>ignore.terms</code>       | A character giving terms from either the random or residual models that are to be ignored in that their contributions to the variance is not to be included in the estimated matrix. The term names are those given in the <code>vparameters</code> component of the <code>asreml</code> object or the <code>varcomp</code> component produced by <code>summary.asreml</code> , but only up to the first exclamation mark (!). This can be used in conjunction with <code>estimateV.asreml</code> as a workaround to include components of the variance matrix for variance functions that have not been implemented in <code>estimateV</code> .                                                       |
| <code>fixed.spline.terms</code> | A character vector giving one or more spline terms in the random model that are regarded as fixed and so are to be ignored because they are not regarded as contributing to the variance. The term names are those given in the <code>vparameters</code> component of the <code>asreml</code> object or the <code>varcomp</code> component produced by <code>summary.asreml</code> , but only up to the first exclamation mark (!).                                                                                                                                                                                                                                                                    |
| <code>bound.exclusions</code>   | A character specifying one or more bound codes that will result in a variance parameter in the random model being excluded from contributing to the variance. If set to <code>NULL</code> then none will be excluded.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>tolerance</code>          | The value such that eigenvalues less than it are considered to be zero.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <code>units</code>              | A character indicating whether the BLUPs for <code>units</code> are added to the residuals when this reserved factor is included in the random model. Possible values are <code>addtoresiduals</code> and <code>ignore</code> . If standardized conditional residuals are plotted and the BLUPs for <code>units</code> are to be added then it is the standardized BLUPs that are added.                                                                                                                                                                                                                                                                                                               |
| <code>update</code>             | If <code>TRUE</code> , and <code>set.terms</code> is <code>NULL</code> , then <code>newfit.asreml</code> is called to fit the model to be tested, using the values of the variance parameters stored in the <code>asreml.object</code> , that is stored in <code>asrtests.obj</code> , as starting values. If <code>FALSE</code> or <code>set.terms</code> is not <code>NULL</code> , then <code>newfit.asreml</code> will not use the stored variance parameter values as starting values when fitting the new model, the only modifications being (i) the model is fitted to simulated data and (ii) those specified via <code>...</code> , except that changes cannot be made to any of the models. |
| <code>trace</code>              | If <code>TRUE</code> then partial iteration details are displayed when ASReml-R functions are invoked; if <code>FALSE</code> then no output is displayed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>graphics.device</code>    | A character specifying a graphics device for plotting. The default is <code>graphics.device = NULL</code> , which will result in plots being produced on the current graphics device. Setting it to "windows", for example, will result in a windows graphics device being opened.                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>ncores</code>             | A numeric specifying the number of cores to use in doing the simulations. In choosing a value for <code>ncores</code> , it is necessary to take into account other processes that are using parallel processing at the same time.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>...</code>                | Other arguments that are passed down to the function <code>asreml</code> . Changes to the models are not allowed. Other changes are dangerous and generally should be avoided.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

## Details

The residual model is scanned to ensure that it involves only two factors not included in the `at` function, and to see if it has a third factor in an `at` function. If so, the faces of the 2D variogram, each based on one of the two non-`at` factors, are derived from the residuals in the supplied `asreml` object using `asreml.variogram`, this yielding the observed variogram faces. If `aom` was set to

TRUE for the `asreml` object, the standardized conditional residuals are used. Then `nsim` data sets are generated by adding the `fitted.values`, extracted from the `asreml` object, to a vector of values randomly generated from a normal distribution with expectation zero and variance matrix `V`. Each data set is analyzed using the model in `object` and several sets are generated and analyzed in parallel. The variogram values for the faces are obtained using `asreml.variogram` stored. Note, if the analysis for a data set does not converge in `maxiter` iterations, it is discarded and a replacement data set generated. The value of `maxiter` can be specified in the call to `variofaces.asreml`. Plots are produced for each face and include the observed values and the 2.5%, 50% & 97.5% quantiles.

### Value

A list with the following components:

1. **face1**: a `data.frame` containing the variogram values on which the plot for the first dimension is based.
2. **face2**: a `data.frame` containing the variogram values on which the plot for the second dimension is based.

### Author(s)

Chris Brien

### References

Stefanova, K. T., Smith, A. B. & Cullis, B. R. (2009) Enhanced diagnostics for the spatial analysis of field trials. *Journal of Agricultural, Biological, and Environmental Statistics*, **14**, 392–410.

### See Also

[asremlPlus-package](#), [asreml](#), [newfit.asreml](#), [plotVariofaces.data.frame](#), [simulate.asreml](#), [set.seed](#).

### Examples

```
## Not run:
data(Wheat.dat)
current.asr <- asreml(yield ~ Rep + WithinColPairs + Variety,
                     random = ~ Row + Column + units,
                     residual = ~ ar1(Row):ar1(Column),
                     data=Wheat.dat)
current.asrt <- as.asrttests(current.asr, NULL, NULL)
current.asrt <- rmboundary.asrttests(current.asrt)
# Form variance matrix based on estimated variance parameters
s2 <- current.asr$sigma2
gamma.Row <- current.asr$gammas[1]
gamma.unit <- current.asr$gammas[2]
rho.r <- current.asr$gammas[4]
rho.c <- current.asr$gammas[5]
row.ar1 <- mat.ar1(order=10, rho=rho.r)
col.ar1 <- mat.ar1(order=15, rho=rho.c)
V <- gamma.Row * fac.sumop(Wheat.dat$Row) +
    gamma.unit * diag(1, nrow=150, ncol=150) +
    mat.dirprod(col.ar1, row.ar1)
V <- s2*V
```

```
#Produce variogram faces plot (Stefanaova et al, 2009)
variofaces(current.asr, V=V, ncores = parallel::detectCores())

## End(Not run)
```

---

|                 |                                                                                    |
|-----------------|------------------------------------------------------------------------------------|
| WaterRunoff.dat | <i>Data for an experiment to investigate the quality of water runoff over time</i> |
|-----------------|------------------------------------------------------------------------------------|

---

### Description

This data is from an experiment to investigate the quality of water runoff. However, it has been modified to hide the true identity of the Species and Sources. It is used to provide executable examples of the functions listed under **Examples**.

### Usage

```
data(WaterRunoff.dat)
```

### Format

A data.frame containing 440 observations of 13 variables.

### Author(s)

Chris Brien

### Source

Kazemi, F. (pers. comm.)

### See Also

[chooseModel.asrtests](#), [reparamSigDevn.asrtests](#),  
[plotPredictions.data.frame](#), [predictPlus.asreml](#), [predictPresent.asreml](#)

---

|           |                                                                        |
|-----------|------------------------------------------------------------------------|
| Wheat.dat | <i>Data for a 1976 experiment to investigate 25 varieties of wheat</i> |
|-----------|------------------------------------------------------------------------|

---

### Description

The data appears in Gilmour et al. (1995) and is from a field experiment designed to compare the performance of 25 varieties of spring wheat. An analysis of it using asreml is presented by Butler et al. (2023, Section 7.6), although they suggest that it is a barley experiment. It is used in the Wheat vignettes [Enter `vignette(package = "asremlPlus")`] as an executable example of the use of the asremlPlus to analyse a data set.

The experiment was conducted at Slate Hall Farm, UK, in 1976 and was designed as a balanced lattice square with 6 replicates laid out in a  $10 \times 15$  rectangular grid. The columns in the data frame are: Rep, Row, Column, WithinColPairs, Variety, yield. The response variable is the grain yield.

**Usage**

```
data(Wheat.dat)
```

**Format**

A data.frame containing 150 observations of 6 variables.

**Author(s)**

Chris Brien

**Source**

Butler, D. G., Cullis, B. R., Gilmour, A. R., Gogel, B. J. and Thompson, R. (2023). *ASReml-R Reference Manual Version 4.2*. VSN International Ltd, <https://asreml.kb.vsnl.co.uk/>.

Gilmour, A. R., et al. (1995) Average Information REML: An efficient algorithm for variance parameter estimation in linear mixed models. *Biometrics*, **51**, 1440-1450.

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